

Process Resin Selection Guide

Innovative Resins and Customized Solutions to Advance the Production of Biologics

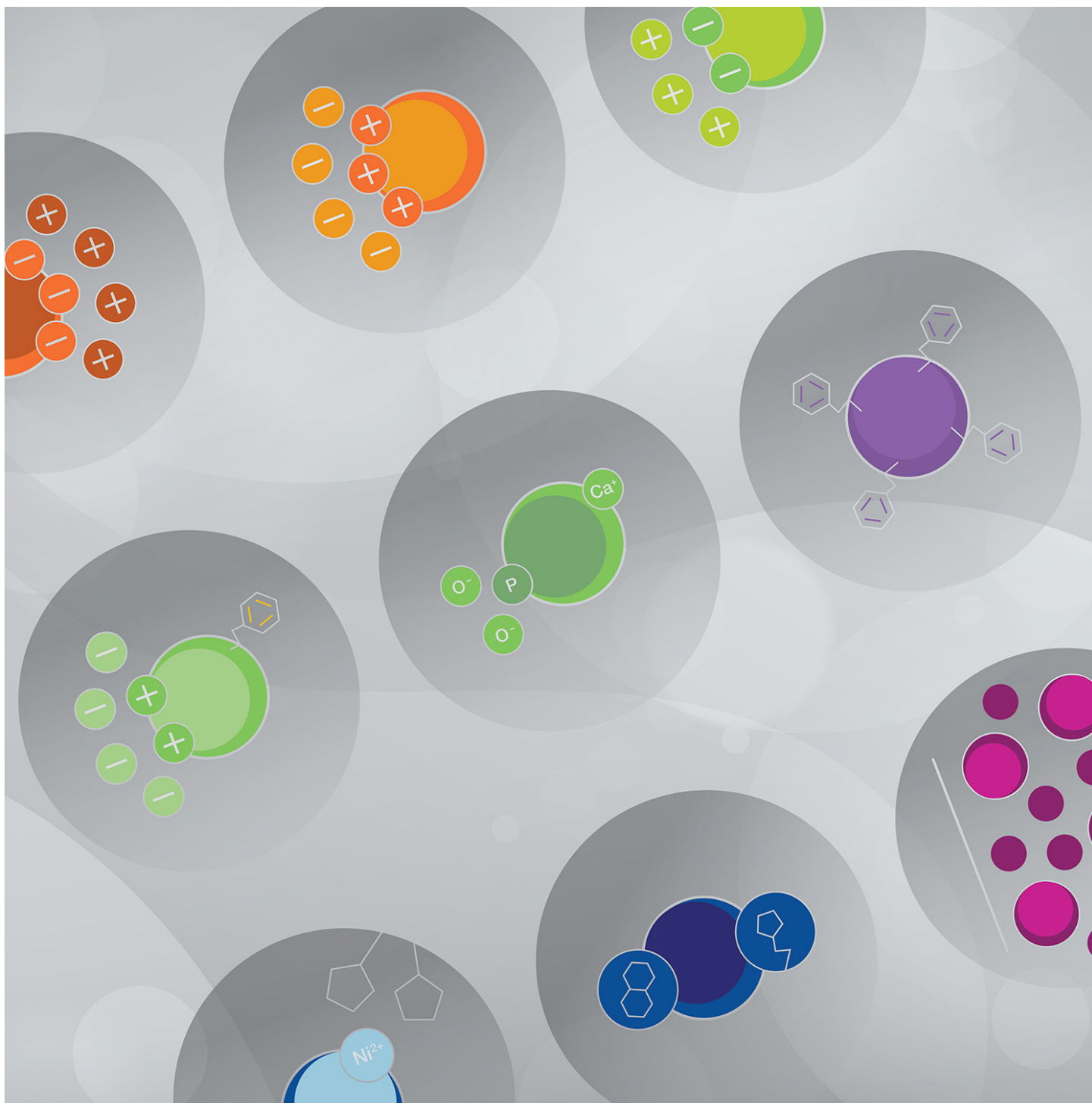


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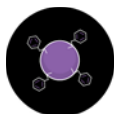
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Resin, Application, and Bead Overview

Type	Resin Name	Applications	Bead Property Highlights	Base Bead Chemistry	Mean Particle Size or Range, μm	Available Formats
ION EXCHANGE	Nuvia Q, page 15	- Capture and polish purification of antibodies, viruses, and proteins - Depletion of endotoxins	Strong AEX, high DBC	Polyacrylamide	85	B, EF, F
	Nuvia HP-Q, page 18	- Capture and polish purification of large biomolecules (viruses, IgM) - Plasma fractionation - High-throughput purification	Strong AEX, large pore size	Polyacrylamide	50	B, EF, F
	UNOsphere Q, page 21	- Capture and polish purification of antibodies, viruses, and proteins - Depletion of endotoxins	Strong AEX	Polyacrylamide	120	B, EF, F
	Macro-Prep High Q and High Q-3HT, page 24	- Capture purification of acidic and neutral proteins - Depletion of endotoxins - Viral clearance	Strong AEX	Polymethacrylate	High Q: 70 High Q-3HT: 75	B, EF
	Macro-Prep DEAE, page 27	- Capture purification of large biomolecules - Plasma purification - Clearance of negatively charged HCP and hcDNA	Weak AEX	Polymethacrylate	65	B, EF
	AG 1, AG MP-1M, AG 2, and AG 4, page 30	Separation of low molecular weight molecules (<3 kD), including organic acids, nucleic acids, and carbohydrates	Strong or weak AEX, available as 2, 4, and 8% crosslinked, and macroporous versions	Multiple	Multiple products from 38–75 to 300–1,180	B, GC
	Bio-Rex 5, page 34	Separate organic acids from neutral sugars	Weak AEX	Polystyrene	45–75, 75–150	B
	Nuvia S, page 36	Alternative to Protein A for capture purification of antibodies, IgMs, viruses, and proteins	Strong CEX, high DBC	Polyacrylamide	85	B, EF, F
	Nuvia HR-S, page 39	- Separation of charge variants - Intermediate and polish purification of proteins and antibodies - Aggregate removal	Strong CEX, high resolution	Polyacrylamide	50	B, EF, F
	UNOsphere S, page 43	- Capture purification of proteins and antibodies - Separation of similarly charged proteins	Strong CEX	Polyacrylamide	80	B, EF, F
	Macro-Prep High S, page 46	- Polish purification of proteins and mAbs - Viral clearance	Strong CEX	Polymethacrylate	60	B, EF
	Macro-Prep CM, page 49	Polish purification of proteins	Weak CEX	Polymethacrylate	60	B
	AG 50W and AG MP-50, page 51	Separation of low molecular weight molecules (<3 kD), including organic acids, nucleic acids, and carbohydrates	Strong CEX, available as 2, 4, 8, 12, and 16% crosslinked, and macroporous versions	Polystyrene	Multiple products from 53–106 to 300–1,180	B, GC
	Bio-Rex 70, page 55	Purification and fractionation of peptides, proteins, antibiotics, and other cationic molecules	Weak CEX	Polystyrene	45–75, 75–150, 150–300, 300–1,180	B
	Chelex 100, page 57	Removal of metal ions from samples and buffers	Strong CEX	Polystyrene	75–150, 150–300, 300–1,180	B
	AG 11 A8, page 59	- Chromatographic desalting and separation of acids from nonionic molecules - Separation of salts from organic materials by absorbing both anions and cations in equivalent amounts while allowing the organic compounds to pass through	Mixed-bed (strong AEX + weak CEX)	Polystyrene	180–425	B
	AG 501-X8, page 61	- Water deionization and softening - Removal of nonionic substances, including urea, acrylamide, formamide, and glyoxal - Removal of radioactive substances from nuclear power plant wastewater and polishing of condensate	Mixed-bed (strong AEX + strong CEX)	Polystyrene	300–1,180	B
	Bio-Rex MSZ 501(D), page 61	- Water deionization and softening - Removal of nonionic substances, including urea, acrylamide, formamide, and glyoxal - Removal of radioactive substances from nuclear power plant wastewater and polishing of condensate	Mixed-bed (strong AEX + strong CEX)	Polystyrene	600 $\pm$ 100 μm (median)	B
	Bio-Rex RG 501-X8, page 64	- Cleanup of power plant deionization systems - Large-scale cleanup of metals from wastewater	Mixed-bed (strong AEX + strong CEX)	Polystyrene	500–1,200	B

B, bulk; EF, EconoFit Prepacked Columns; F, Foresight™ Prepacked Filter Plates and Columns; GC, gravity column.

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Resin, Application, and Bead Overview

Type	Resin Name	Applications	Bead Property Highlights	Base Bead Chemistry	Mean Particle Size or Range, μm	Available Formats
MIXED-MODE / MULTIMODAL	Nuvia cPrime, page 70	Capture and polish purification of proteins, antibodies, and viruses	HI + weak CEX	Polyacrylamide	70	B, EF, F
	Nuvia aPrime 4A, page 74	- Capture and polish purification of proteins, antibodies, and viruses - Provides a balance of high recovery and yield	HI + strong AEX	Polyacrylamide	50	B, EF, F
	CHT Ceramic Hydroxyapatite, page 80	- Intermediate and polish purification of proteins, antibodies, and viruses - Gold standard in aggregate removal - Single-step impurity removal of DNA, endotoxins, and others	Ca^{2+} affinity + CEX	Ceramic hydroxyapatite	40, 80	B, EF, F, FP
	Bio-Gel Hydroxyapatite HT and HTP, page 88	Preparative fractionation of monoclonal antibodies, proteins, enzymes, nucleic acids, and other macromolecules	Ca^{2+} affinity + CEX	Crystalline hydroxyapatite	10–100	B
AFFINITY	Nuvia IMAC, page 92	Ni-charged Nuvia IMAC is used for capture purification of histidine-tagged proteins	Metal affinity	Polyacrylamide	50	B, EF, F
	Affi-Gel Protein A, page 96	Preparation of highly purified immunoglobulins (IgG)	Antibody affinity	Agarose	90–180	B
	Affi-Prep Protein A, page 96	Preparation of highly purified immunoglobulins (IgG) at high linear flow rates; exhibits excellent mechanical strength	Antibody affinity	Polymethacrylate	50	B, EF
	Affi-Gel Blue, page 98	Ideal for albumin removal and enzyme purification	Affinity	Agarose	75–150, 150–300	B, EF
	CM Affi-Gel Blue, page 99	Initial purification of serum proteins to remove albumin and serum proteases	Affinity and CEX	Agarose	75–150	B, EF
	DEAE Affi-Gel Blue, page 100	Single-step for IgG purification from serum	Affinity and AEX	Agarose	150–300	B, EF
	Affi-Gel Boronate, page 101	Ideal for separation of low molecular weight molecules such as nucleotides, nucleosides, catecholamines, and sugars	Affinity	Polyacrylamide gel	45–90	B
	Affi-Gel 10, page 102	Activated affinity media for coupling neutral or basic proteins that contain primary amines	Activated affinity	Agarose	75–300	B
	Affi-Gel 102, page 103	Activated affinity media for immobilizing ligands that contain primary or terminal carboxyl groups	Activated affinity	Agarose	75–300	B
HYDROPHOBIC INTERACTION	Macro-Prep Methyl, page 108	Capture purification of proteins that have strongly hydrophobic regions	Weak HIC	Polymethacrylate	60	B, EF
	Macro-Prep t-Butyl, page 108	Capture purification of proteins that have few or weakly hydrophobic regions	Mild HIC	Polymethacrylate	60	B, EF
	Bio-Beads SM-2, page 111	Adsorption of organics with <2 kD molecular weights	HIC	Polystyrene	300–1,180	B
SIZE EXCLUSION / GEL FILTRATION	Bio-Gel A, page 115	- Ideal for purification of antibodies, aggregates, immune complexes, DNA, chromatin, and extracellular vesicles - Compatible with high-salt buffers without bed volume change - Fractionation range is from 10,000 to 5,000,000 Da	SEC	Agarose	38–75, 75–150, 150–300	B
	Bio-Gel P, page 117	- Efficient, gentle gel filtration of sensitive compounds - Ideal for purification of antibodies and aggregates - Desalting and buffer exchange - Several particle size ranges available with MW exclusion limits ranging from 100 to 100,000 Da	SEC	Polyacrylamide gel	<45, 45–90, 80–140, 90–180	B, EF
	Bio-Beads S-X, page 120	- Separation of lipophilic polymers and other solutes that require organic eluents - Useful for fractionation of low MW organic polymers and other hydrophobic substances such as pesticides, rodenticides, polycyclic aromatic compounds, and unsaturated lipids - Fractionation range is from 400 to 14,000 Da	SEC	Polystyrene	40–80	B

B, bulk; EF, EconoFit Prepacked Columns; F, Foresight Prepacked Filter Plates and Columns; FP, Foresight Pro Prepacked Process-Scale Columns.

List of Abbreviations and Definitions

AAV – adeno-associated virus	HPLC – high performance liquid chromatography
Ab – antibody	HSA – human serum albumin
ADP – adenosine diphosphate	HTP – high-throughput purification
AdV – adenovirus	ICP-MS – inductively coupled plasma mass spectrometry
AEX – anion exchange	ID – inner diameter
AIDS – acquired immunodeficiency syndrome	IDA – iminodiacetic acid
ALA – α -lactalbumin	IEX – ion exchange
AMP – adenosine monophosphate	IgG – immunoglobulin G
ATP – adenosine triphosphate	IgM – immunoglobulin M
AU – absorbance units	IMAC – immobilized metal affinity chromatography
B – bulk	IU – international units
B.A. (also BA) – benzyl alcohol	IVIG – intravenous immunoglobulin
BLA – β -lactoglobulin	JEV – Japanese encephalitis virus
BME – β -mercaptoethanol	LMW – low molecular weight
BSA – bovine serum albumin	LYO – lysozyme
BT – breakthrough	mAb – monoclonal antibody
CCD – central composite design	MALDI-TOF – matrix-assisted laser desorption/ionization–time of flight
CEX – cation exchange	MHV – mouse hepatitis virus
CHO – Chinese hamster ovary	MM – mixed mode (also molecular markers)
CHY – α -chymotrypsinogen	MS – mass spectrometry
CIP – clean-in-place (also cleaning in place)	MW – molecular weight
CM – carboxyl	ND – not determined
CPB – carboxypeptidase B	NTA – nitrilotriacetic acid
CV – column volume	PAH – polycyclic aromatic hydrocarbon
D – diameter	PBS – phosphate buffered saline
DBC – dynamic binding capacity	PCB – polychlorinated biphenyls
DEAE – diethylaminoethyl	PEG – polyethylene glycol
DMSO – dimethyl sulfoxide	PfDyP – <i>Pseudomonas fluorescens</i> –expressed dye-decolorizing peroxidase
DOE – design of experiments	pI – isoelectric point
DTT – dithiothreitol	PV – poliovirus
DyP – dye-decolorizing peroxidase	PVC – polyvinyl chloride
EDTA – ethylenediaminetetraacetic acid	Q – quaternary ammonium
EF – EconoFit Prepacked Columns	QC – quality control
ELISA – enzyme-linked immunosorbent assay	rPA – recombinant Protein A
EP – elution pool	RSDA – raw starch digesting α -amylase
F – Foresight Prepacked Filter Plates and Columns	RT-qPCR – reverse transcription quantitative PCR
FC – fragment crystallizable	S – sulfonfyl
FCV – feline calicivirus	SARS-CoV – severe acute respiratory syndrome coronavirus
FI – fluorescence intensity	SDVB – styrene-divinylbenzene
FP – Foresight Pro Prepacked Process-Scale Columns	SEC – size exclusion chromatography
GC – gravity column	SIP – steam-in-place
GFP – green fluorescent protein	TCEP – tris (2-carboxyethyl) phosphine hydrochloride
GLC-MS – gas–liquid chromatography–mass spectrometry	TCF – tissue culture fluid
HA – hemagglutinin	TCID₅₀ – 50% tissue culture infective dose
hcDNA – host cell DNA	TGY – thyroglobulin
HCP – host cell protein	UV – ultraviolet
HI – hydrophobic interaction	
HIC – hydrophobic interaction chromatography	
hIgG – human immunoglobulin G	

History and Timeline of Bio-Rad Chromatography Resins and Products

Since 1957 Bio-Rad has provided chromatography resins for laboratory- and process-scale biopharmaceutical workflows, including resin screening, preparative, and analytical uses.

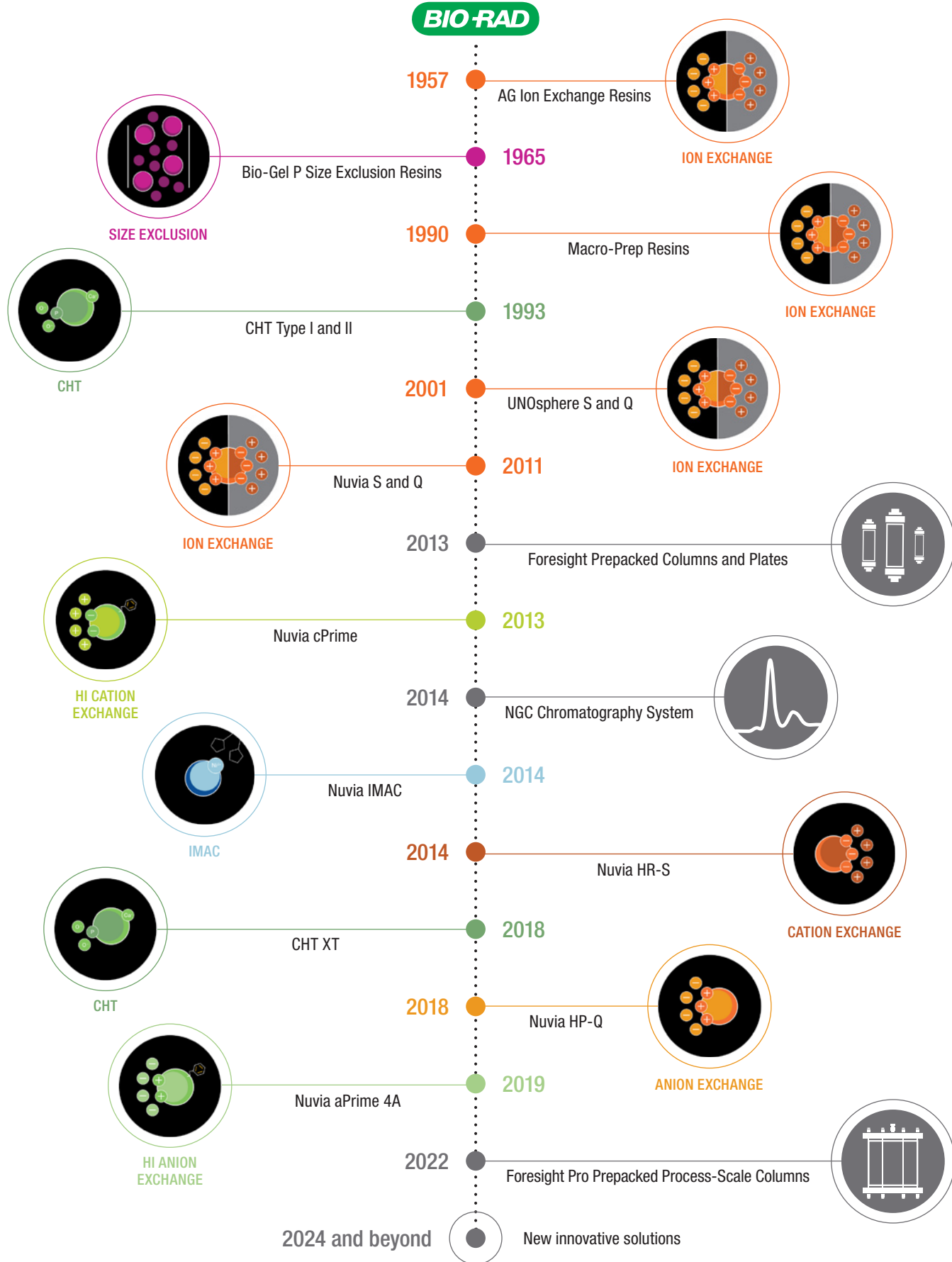
Bio-Rad's initial polymer-based resins included polystyrene ion exchange, size exclusion, and hydrophobic adsorption resins. These are used for the removal of trace impurities from therapeutic proteins, pre-use purification of bioprocessing buffers, and purification of compounds less than 3,000 daltons. During the 1960s and 1970s, Bio-Rad developed polyacrylamide- and agarose-based chromatography resins, which have been used for size exclusion, immunoprecipitation, and affinity chromatography.

The company introduced crystalline hydroxyapatite in 1956 for protein purification and nucleic acid separations. This expertise enabled Bio-Rad to become the exclusive worldwide supplier of CHT Ceramic Hydroxyapatite in the early 1990s. CHT exhibits unique selectivity for a wide variety of biomolecules, which allows it to achieve effective single-step separation of antibodies from antibody aggregates, host cell proteins, DNA, viruses, and endotoxins. CHT has also been demonstrated to be useful in the purification of viruses, plasmid DNA, and various recombinant proteins.

During the 1980s, Bio-Rad introduced methacrylate-based Macro-Prep Ion Exchange, Macro-Prep Hydrophobic Interaction Chromatography (HIC), and affinity resins, which provide high flow rate separations at low and medium pressures. The combination of large pores, good flow properties, and a mild hydrophobic effect makes Macro-Prep Ion Exchange Resins particularly useful for plasma protein fractionation.

Since 2000, to meet the biopharmaceutical industry's need for high binding capacity, fast capture, and high resolution, Bio-Rad has developed UNOsphere and Nuvia Ion Exchange, Affinity, and Mixed-Mode Resins. UNOsphere and Nuvia Resins are available as either high-capacity or high-resolution ion exchange resins. Nuvia cPrime Resin, an HIC-CEX mixed-mode resin, and Nuvia aPrime Resin, an HIC-AEX mixed-mode resin, provide unique selectivities for proteins and can be easily combined with CHT for excellent purification of various types of biomolecules.

BIO-RAD



Chromatography Resins and Media by Technique: **Ion Exchange**



ION EXCHANGE

MIXED-MODE / MULTIMODAL

AFFINITY

HYDROPHOBIC INTERACTION

SIZE EXCLUSION / GEL FILTRATION

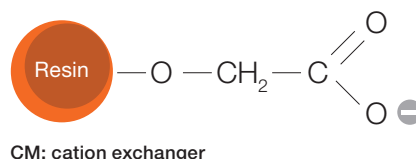
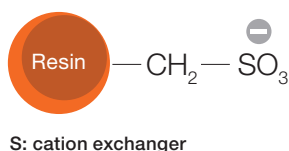
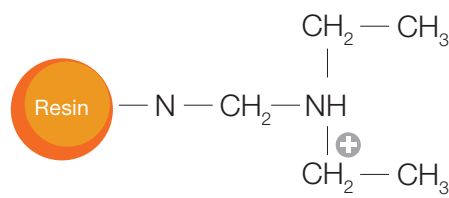
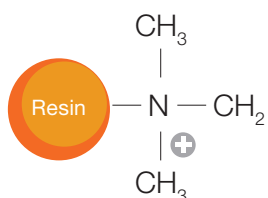
Ion Exchange Chromatography Resins



Ion exchange (IEX) is one of the most frequently used techniques for purification of proteins, peptides, nucleic acids, and other charged biomolecules. It offers high resolution with high loading capacity. It can be used to separate molecular species that have only minor differences in their charge properties, for example two proteins differing by one charged amino acid. These features make IEX well suited for capture and intermediate purification or polishing steps in a purification protocol, and the technique can be used from microscale purification and analysis through to purification of kilograms of product.

IEX resins can be broadly divided into several types, the most common being anion and cation, depending upon the class of charged species they are capable of binding. Anion exchange (AEX) resins contain cationic (positively charged) ligands and hence bind anionic (negatively charged) species; for cation exchange (CEX) resins, the reverse is true. Within each of the anion and cation exchange classes there are primarily two ligand types that are commercially available: for AEX resins, the ligands contain quaternary ammonium (Q) or diethylaminoethyl (DEAE) functionalities; for CEX resins, sulfonyl (S) or carboxyl (CM) groups. Variants of these four basic ligands also exist (for example, phosphate-based CEX resins). Mixed-bed resins are another type of IEX resin and are composed of a combination of anion and cation exchange resins.

Bio-Rad offers the following types of IEX resins — Nuvia, UNOsphere, Macro-Prep, AG, Bio-Rex, and Chelex 100.



NUVIA RESINS

Nuvia beads are produced using controlled polymerization of water-soluble acrylamido and vinylic monomers and exhibit low nonspecific binding due to the hydrophilic nature of the polymers. Nuvia Resins are designed to meet specific purification challenges. For example, Nuvia S provides high binding capacity with salt tolerance and is recommended for nonaffinity capture. Nuvia HR-S provides high-resolution CEX polishing for removal of closely related impurities. Nuvia cPrime provides unique selectivity with both cation and hydrophobic interactions. Nuvia aPrime 4a provides optimal balance of purity and yield, in addition to a unique selectivity, during biomolecule purification. Due to their high binding capacities, these ion exchangers are the resins of choice for therapeutic antibodies and protein purification.

UNOSPHERE RESINS

These resins are unique among most chromatography resins, which are manufactured by first producing a base polymer and then converting it through one or more chemical reactions to a functional ion exchange derivative. In contrast, UNOsphere Resins are produced in a controlled single-step polymerization using matrix, crosslinker, and functional monomers. The use of a single-step manufacturing process increases batch-to-batch consistency. During the formation of spherical beads, quaternary amine or sulfonic acid IEX groups are incorporated directly into the polymer. They are produced from water-soluble acrylamido and vinylic monomers, and therefore are hydrophilic and exhibit low nonspecific adsorption. UNOsphere IEX Resins have excellent pressure/flow properties. These ion exchangers have broad applications and perform well in capture, intermediate, and polishing purification steps.

MACRO-PREP RESINS

These are rigid, macroporous polymethacrylate beads, providing high mechanical strength and low backpressure for bioprocessing applications. The combination of large pores and good flow properties make Macro-Prep useful for a variety of biomolecule purifications. Macro-Prep is produced as a base polymer that is converted to strong and weak IEX derivatives, namely Macro-Prep High Q, Macro-Prep High Q-3HT, Macro-Prep High S, Macro-Prep DEAE, and Macro-Prep CM. The average pore radius ranges from 60 to 90 nm. Macro-Prep ion exchangers have unique performance capabilities for purification of large biomolecules, including plasma proteins and virus-like particles, due to the large pore structure and mild hydrophobic effect of the polymer backbone.

AG RESINS

AG Analytical Grade Resins are a microporous copolymer of styrene and divinylbenzene. They are available as strong and weak cation and anion exchangers, as well as mixed-bed ion exchangers. They are primarily employed for the separation of low molecular weight (<3,000 daltons) molecules such as inorganic ions, and small molecules such as organic acids, nucleic acids, and carbohydrates.

BIO-REX RESINS

Bio-Rex Resins are weak ion exchangers available as Bio-Rex 5 and Bio-Rex 70. Bio-Rex 5 Resins are weak anion exchangers that contain mostly tertiary and some quaternary amines. They are exhaustively sized, purified, and converted to make them suitable for accurate, reproducible analytical techniques. Bio-Rex 5 Resins have an affinity for organic acids that can be used to remove acids from sugars, and they exchange anions such as iodide, bromide, and chloride with weak interactions.

Bio-Rex 70 Resins are weak cation exchangers that contain carboxylic acid exchange groups on a macroreticular acrylic polymer lattice. They are available in both analytical and biotechnology grades. Bio-Rex 70 Resins are used for purification and fractionation of peptides, proteins, antibiotics, and other cationic molecules.

CHELEX 100 RESIN

Chelex 100 Resin is composed of styrene-divinylbenzene copolymers containing iminodiacetate ions. These act as chelating groups in binding polyvalent metal ions with high selectivity and are used to remove metal ions from samples and buffers.

See Tables 1 and 2 for additional details.

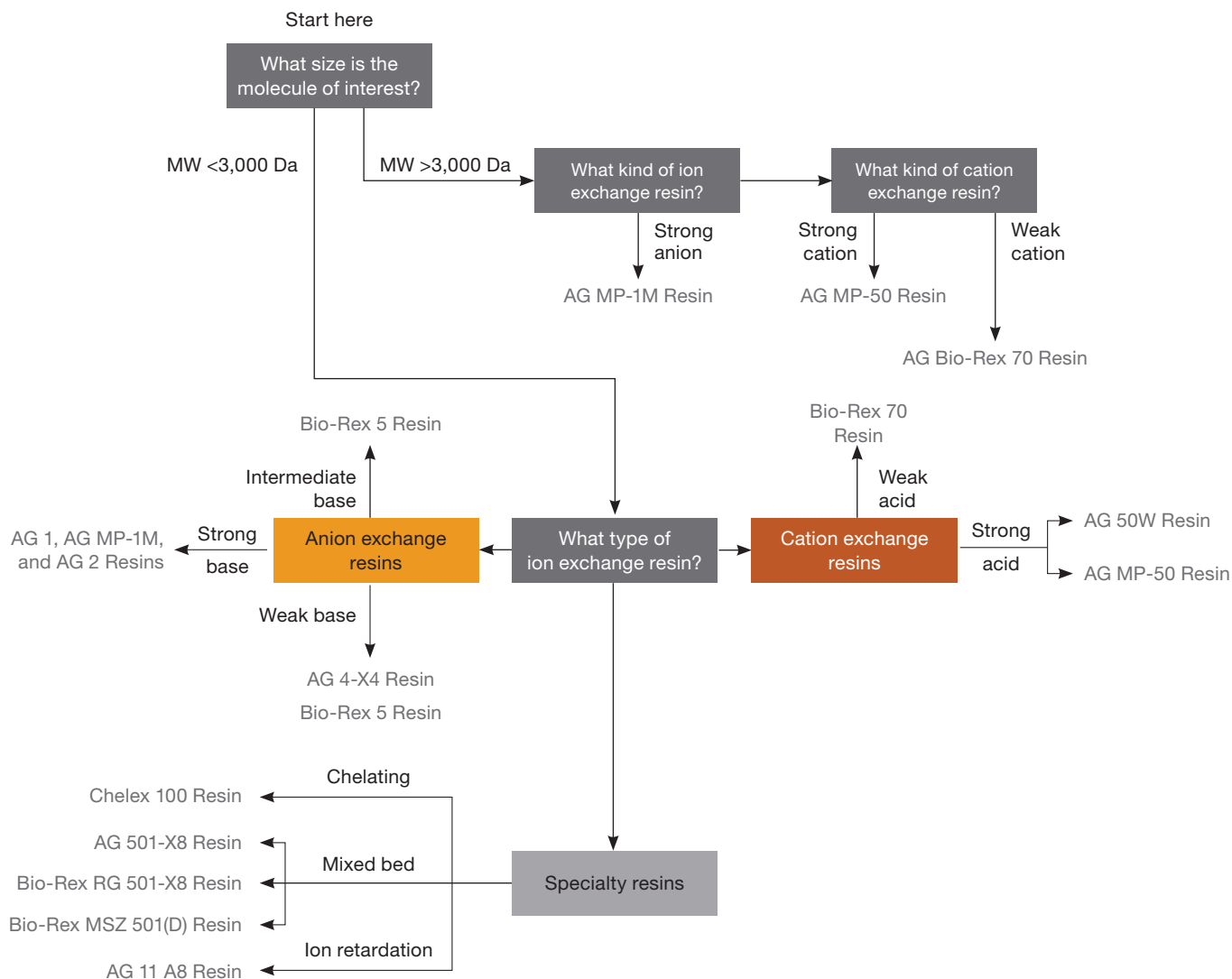
Table 1. Comparison of selected resin types.

	Nuvia Resins	UNOsphere Resins	Macro-Prep Resins
DBC, mg/ml	Mid-High	Low-Mid	Low
Particle size, μm	50–85	80–120	60–70
Constant DBC up to flow rate of, cm/hr	600	Q: 600 S: 300	300
Matrix nature	Hydrophilic	Hydrophilic	Relatively hydrophobic

Table 2. Bio-Rad AEX and CEX offerings.

	Nuvia Resins	UNOsphere Resins	Macro-Prep Resins	AG Analytical Grade Resins
AEX	Nuvia Q	UNOsphere Q	Macro-Prep High Q	AG 1, AG MP-1M, AG 2, and AG 4 AG 501 and Bio-Rex 5
	Nuvia HP-Q		Macro-Prep High Q-3HT	
			Macro-Prep DEAE	
CEX	Nuvia S	UNOsphere S	Macro-Prep High S	AG 50W and AG MP-50 AG 501 and Bio-Rex 70 Chelex 100
	Nuvia HR-S		Macro-Prep CM	

FLOWCHART FOR SELECTING AN AG RESIN



For a comparison of ion exchange resins, see page 176.

Nuvia Q Resin



ANION
EXCHANGE

RESIN FUNCTIONS

- Best-in-class DBC at high flow rates, leading to faster cycle time for downstream processing during capture and polish purifications of antibodies, viruses, and proteins
- High binding performance over a wide operational window (pH and salt) for flexible DOE
- Reduced space requirements and cycle time can significantly improve productivity by contributing to smaller column footprint, decreased capital cost, and less buffer consumption

RESIN CHARACTERISTICS

- Hydrophilic and rigid polyacrylamido-based resins
- Designed with large pore diameter and optimized surface chemistry for binding of large molecules at high flow rates
- Mechanically and chemically stable resins with wide range of operating conditions

SUPPORTING INFORMATION AND APPLICATION EXAMPLES

- Product information sheet, bulletin 6129
- Retrovirus purification, bulletin 7200
- Adenovirus purification, bulletin 6719
- Biosimilar purification, bulletin 7130
- Protein purification, bulletin 6711
- Instruction manual, document 10018215

Bead Properties

Property	Description
Type of ion exchanger	Strong anion
Functional group	$-\text{N}(\text{CH}_3)_3^+$
Base matrix composition	Polyacrylamide
Mean particle size	85 μm
Total ionic capacity	100–170 $\mu\text{eq/ml}$
Dynamic binding capacity*	>170 mg/ml at 300 cm/hr
Recommended linear flow rate	50–600 cm/hr
Compression factor (settled bed volume/packed bed volume)	1.10–1.15
pH stability	Short term: 2–14 Long term: 4–12
Shipping solution	20% ethanol + 1 M NaCl or 2% benzyl alcohol
Regeneration	1–2 M NaCl
Sanitization	0.5–1.0 N NaOH
Storage conditions	20% ethanol or 0.01 N NaOH
Storage temperature	Room temperature
Chemical stability	1.0 N NaOH (20°C), up to 1 week 0.01 N NaOH (20°C), up to 5 years
Shelf life	5 years

* 10% BT capacity determined with 5 mg/ml BSA in 20 mM Tris-HCl, pH 8.5.

High Binding Capacity

With strong mechanical properties and optimized bead size distribution, Nuvia Q consistently delivers high binding capacity at fast linear flow velocities (Figure 1), allowing users the advantage of increased productivity without compromising binding capacity.

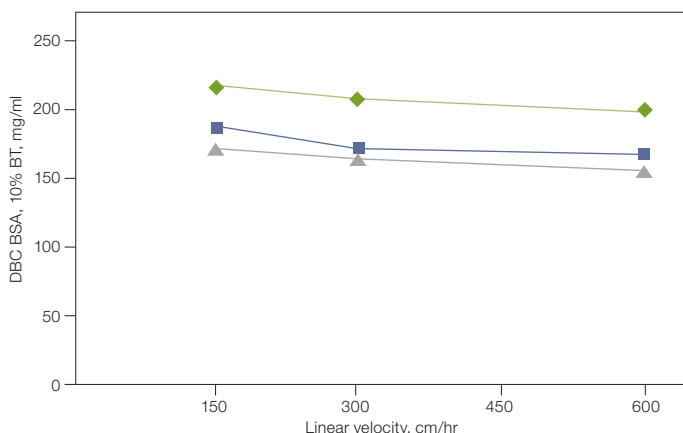


Fig. 1. DBC vs. flow velocity of Nuvia Q Resin. Each 1.1 cm column was packed to a 10.6 cm bed height with Nuvia Q, agarose Q, or polymeric Q resin. BSA (5 mg/ml) in 20 mM Tris-HCl, pH 8.5, was loaded onto each column until 10% BT was observed. Nuvia Q Resin (◆); agarose Q resin (■); polymeric Q resin (▲).

Good Pressure/Flow Performance

Figure 2 shows the effect of flow rate on Nuvia Q Resin backpressure.

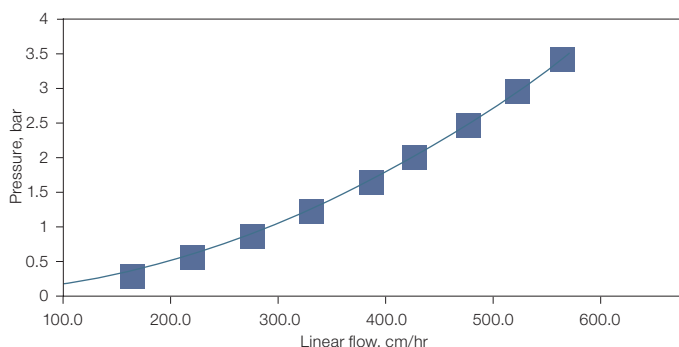


Fig. 2. Nuvia Q Resin pressure/flow performance. A 20 cm diameter x 20 cm bed height column was packed to a compression factor of 1.10.

High Resolution of Biomolecules with Similar pI

Figure 3 shows the purification of α -lactalbumin (ALA) and β -lactoglobulin (BLG), the major proteins of whey, accounting for approximately 75% of the total mass of crude whey. The two major peaks were resolved using a segmented NaCl gradient. SDS-PAGE analysis (Figure 4) of the fractions showed two protein peaks, one containing ALA (Peak 1) and the other containing BLG (Peak 2), with pIs of 4.7–5.1 and 5.2, respectively. Nuvia Q Resin exhibits excellent baseline resolution of these biomolecules with similar pIs. Purities of the two purified proteins were estimated by densitometry to be at least 85%.

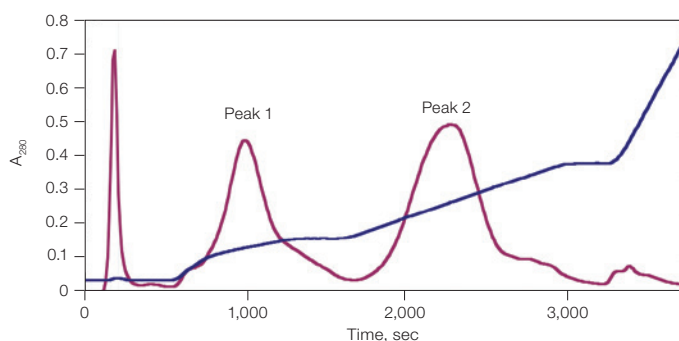


Fig. 3. Purification of whey protein on Nuvia Q Resin. Crude whey feed stock (8 mg) was loaded on a 7 x 27 mm column in 0.02 M sodium phosphate buffer, pH 6.0. The samples were eluted using a segmented gradient of 0–0.1 M NaCl, 0.1–0.3 M NaCl, and 0.3–0.9 M NaCl. Each fraction was 2 ml and analyzed at OD 280. A_{280} (—); conductivity (—).

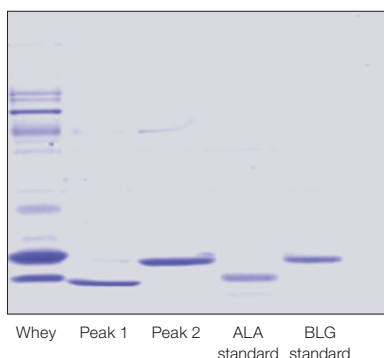


Fig. 4. Analysis of whey purified on Nuvia Q Resin. Fractions collected from the chromatography run shown in Figure 4 were separated by SDS-PAGE.

Comparative Data

Best Capture Recovery of a Retrovirus

Four commercial resins including Nuvia Q were tested for capture and elution of a retrovirus. Quantitation of virus eluted from each medium by RT-qPCR revealed that product recovery was higher with Nuvia Q than with the other resins tested (Table 1).

Table 1. Screening of anion exchange resins.

Anion Exchange Resin	Recovery, %
DEAE Sepharose Fast Flow	34
Q Sepharose Fast Flow	42
Nuvia Q	47
Sartobind Q	20

Purification Workflows

Application	Capture Step Resins	Intermediate Step Resins	Polish Step Resins/Media
Adenovirus	Nuvia cPrime	—	Nuvia Q
mAb	Protein A	Nuvia Q	Nuvia HR-S
Whey protein	One step with Nuvia Q		
Retrovirus	One step with Nuvia Q		

Ordering Information

Catalog #	Description
Bulk Resins	
1560411	Nuvia Q Media, 25 ml
12003708	Nuvia Q Media (in BA), 25 ml
1560413	Nuvia Q Media, 100 ml
12003709	Nuvia Q Media (in BA), 100 ml
156-0415	Nuvia Q Media, 500 ml
12003710	Nuvia Q Media (in BA), 500 ml
12018183	Nuvia Q Media, 5 L
156-0417	Nuvia Q Media, 10 L
12003711	Nuvia Q Media (in BA), 10 L
BA, 2% benzyl alcohol.	

Prepacked Screening Tools

12009311	EconoFit Anionic IEX Kit, includes 1 ml each of Macro-Prep High Q, Macro-Prep High Q-3HT, UNOsphere Q, Nuvia Q, Macro-Prep DEAE, and Nuvia HP-Q Resins
12009290	EconoFit Nuvia Q Column, 1 x 1 ml, 7 x 25 mm
732-4703	Foresight Nuvia Q Plates, 2 x 96-well, 20 µl
732-4804	Foresight Nuvia Q RoboColumn Unit, 200 µl
732-4805	Foresight Nuvia Q RoboColumn Unit, 600 µl
732-4721	Foresight Nuvia Q Column, 1 x 1 ml
732-4741	Foresight Nuvia Q Column, 1 x 5 ml

Case Studies

- Biosimilar purification, page 124
- Adenovirus purification, page 151
- Protein purification, page 173

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Nuvia HP-Q Resin



ANION
EXCHANGE

RESIN FUNCTIONS

- High dynamic binding capacity for purification of large biomolecules, such as plasma proteins, IgMs, and viruses, ensures improved productivity
- Fast mass transfer kinetics helps in high-throughput purification and ensures time efficiency
- The long resin lifetime without loss of DBC ensures repeated use, thereby improving process economics

RESIN CHARACTERISTICS

- Hydrophilic and rigid polyacrylamido-based resins
- Designed with large pore diameter and optimized surface chemistry for binding of large molecules at high flow rates
- Mechanically and chemically stable resins with wide range of operating conditions

SUPPORTING INFORMATION AND APPLICATION EXAMPLES

- Product information sheet, bulletin 7078
- Large biomolecule purification troubleshooting flier, bulletin 7098
- IgM purification, bulletin 7127
- DNA oligonucleotides purification, bulletin 3290
- Virus purification strategy flier, bulletin 7375
- Virus purification troubleshooting flier, bulletin 3415
- Instruction manual, document 10000097131

Bead Properties

Property	Description
Type of ion exchanger	Strong anion
Functional group	$-\text{N}^+(\text{CH}_3)_3$
Base matrix composition	Polyacrylamide
Mean particle size	50 μm
Total ionic capacity	48–88 $\mu\text{eq/ml}$
Dynamic binding capacity*	>50 mg/ml at 100 cm/hr
Recommended linear flow rate	50–300 cm/hr
Compression factor (settled bed volume/packed bed volume)	~1.2
pH stability	2–10
Shipping solution	20% ethanol or 2% benzyl alcohol
Regeneration	1–2 M NaCl
Sanitization	0.5 N NaOH
Storage conditions	20% ethanol
Storage temperature	Room temperature
Chemical stability	1 N NaOH (22°C), up to 1 week 0.01 N NaOH (22°C), up to 5 weeks
Shelf life	5 years

* 10% breakthrough capacity determined with 1.1 mg/ml thyroglobulin in 20 mM Tris HCl, pH 8.0.

High Binding Capacity

Nuvia HP-Q Resin's optimized particle size facilitates high DBC of large biomolecules at high flow rates (Figure 1).

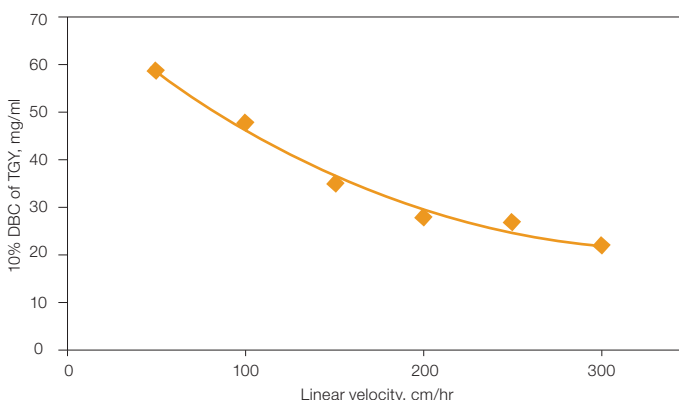


Fig. 1. Dynamic binding capacity (DBC) vs. flow velocity of Nuvia HP-Q. The resin was packed into a 1 ml Bio-Scale Mini Column (0.56 x 4 cm). Thyroglobulin (TYG) solution (1.1 mg/ml) in 20 mM Tris Cl, pH 8.0, was loaded onto the column until 10% breakthrough was observed.

Good Pressure/Flow Performance

Nuvia HP-Q facilitates high flow rates without being limited by column pressure. This leads to an increase in productivity during protein purification. The column pressure remains below 1.5 bar at a linear velocity of 350 cm/hr (Figure 2).

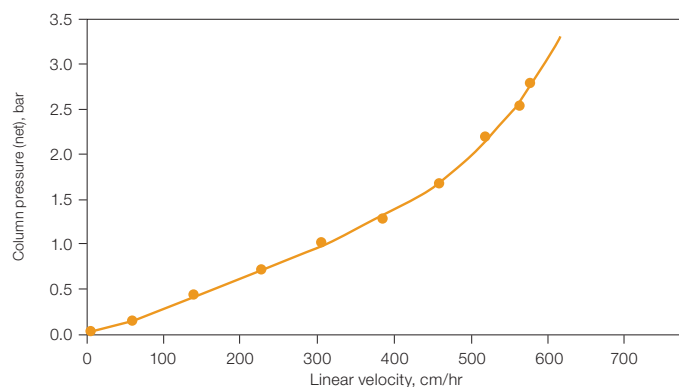


Fig. 2. Pressure/flow performance of Nuvia HP-Q Resin. Nuvia HP-Q slurry prepared in 1x PBS, pH 7.5, was packed into a 20 x 20 cm column by axial compression with a compression factor of 1.2.

Robust Performance and Recovery

Nuvia HP-Q Resin is produced by a validated manufacturing process that ensures batch-to-batch reproducibility. The chemical stability of Nuvia HP-Q allows the resin to perform consistently with minimal changes to DBC or recovery even after prolonged exposure to NaOH (Figure 3).

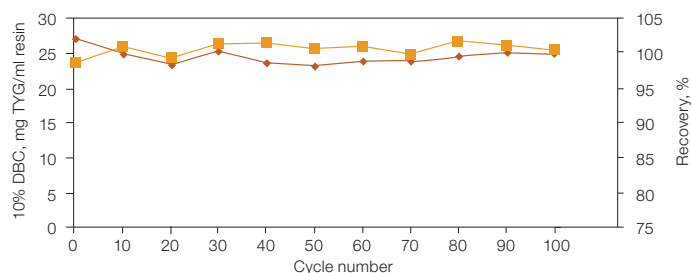


Fig. 3. Stability, reusability, and recovery with Nuvia HP-Q Resin. Thyroglobulin (TYG) solution (1.1 mg/ml) in equilibration buffer (20 mM Tris Cl, pH 8.0) was loaded onto a 1 ml Bio-Scale Mini Column (0.56 x 4 cm) packed with Nuvia HP-Q to a compression factor of 1.2. The column was operated at 300 cm/hr. The protein was eluted in 5 CV of elution buffer (20 mM Tris Cl, 1 M NaCl, pH 8.0) at 300 cm/hr. Cleaning in place (CIP) was performed on the column with 3 CV of 0.5 N NaOH at 300 cm/hr followed by a 40 min hold. The 10% DBC at linear velocity of 300 cm/hr was determined after every 10 cycles. DBC (◆); recovery (■).

Comparative Data

Best-in-Class Binding Capacity

Parallel runs of thyroglobulin on Nuvia HP-Q and other commercially available resins demonstrate that Nuvia HP-Q overcomes the productivity compromise that results from low DBCs at process-scale flow rates (Figure 4).

IgM purification from plasma fractions showed higher DBC at faster flow rates with Nuvia HP-Q relative to other resins from different vendors (Table 1).

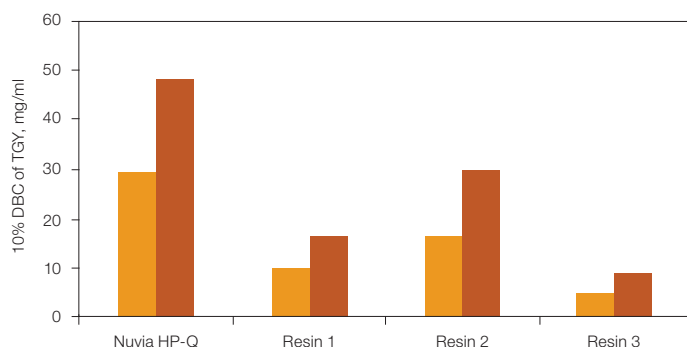


Fig. 4. Dynamic binding capacity (DBC) vs. residence time of Nuvia HP-Q. Comparison of DBCs between Nuvia HP-Q and three other commercially available resins at 1.2 and 2.4 min residence times. The resins were packed into 1 ml columns. Thyroglobulin (TGY) solution (1.1 mg/ml) in 20 mM Tris Cl, pH 8.0, was loaded onto the columns until 10% breakthrough was observed. 1.2 min residence (■); 2.4 min residence (■).

Table 1. Superior DBC of Nuvia HP-Q at a high flow rate relative to other commercially available resins.

Resin	Matrix Material	Mean Particle Size, μm	Pressure, bar	Recommended Flow Rate, cm/hr^*	DBC (IgM)
Nuvia HP-Q	UNOsphere epoxide	50	<3	300	+++**
Resin 1	Dextran beads	50	<3	30	+++**
Resin 2	Agarose	75	<3	300	+++
Resin 3	PS-DVB	50	<3	300	++
Resin 4	PMMA	50	<3	300	++

* Recommended flow rate for industrial scale column ($D > 30 \text{ cm}$).

** DBC data at 10% breakthrough.

*** Data obtained from vendor presentation.

Purification Workflows

Application	Capture Step Resins	Intermediate Step Resins	Polish Step Resins/Media
IgM	Nuvia HP-Q	—	CHT Ceramic Hydroxapatite Type II, $40 \mu\text{m}$
Influenza A virus subtype H1N1	One step with Nuvia HP-Q		

Ordering Information

Catalog # Description

Bulk Resins

12006693	Nuvia HP-Q Media, 25 ml
12007022	Nuvia HP-Q Media (in BA), 25 ml
12006691	Nuvia HP-Q Media, 100 ml
12007018	Nuvia HP-Q Media (in BA), 100 ml
12006660	Nuvia HP-Q Media, 500 ml
12007019	Nuvia HP-Q Media (in BA), 500 ml
12006659	Nuvia HP-Q Media, 5 L
12007033	Nuvia HP-Q Media (in BA), 5 L
12007023	Nuvia HP-Q Media, 10 L
12006994	Nuvia HP-Q Media (in BA), 10 L

BA, 2% benzyl alcohol.

Prepacked Screening Tools

12009314	EconoFit Large Biomolecule Kit, includes 1 ml each of Nuvia Q, Nuvia HP-Q, Nuvia cPrime, Nuvia aPrime 4A, CHT II, $40 \mu\text{m}$, and CHT XT Media
12009311	EconoFit Anionic IEX Kit, includes 1 ml each of Macro-Prep High Q, Macro-Prep High Q-3HT, UNOsphere Q, Nuvia Q, Macro-Prep DEAE, and Nuvia HP-Q Resins
12009282	EconoFit Nuvia HP-Q Column, $1 \times 1 \text{ ml}$, $7 \times 25 \text{ mm}$
12006908	Foresight Nuvia HP-Q Plates, $2 \times 96\text{-well}$, $20 \mu\text{l}$
12007013	Foresight Nuvia HP-Q RoboColumn Unit, $200 \mu\text{l}$
12007014	Foresight Nuvia HP-Q RoboColumn Unit, $600 \mu\text{l}$
12007020	Foresight Nuvia HP-Q Column, 1 ml
12007021	Foresight Nuvia HP-Q Column, 5 ml

Case Studies

- IgM purification, page 128

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UNOsphere Q Resin



ANION
EXCHANGE

RESIN FUNCTIONS

- Ultra-high binding capacity for high productivity and process economics
- Fast capture of impurities like DNA in flow-through mode purification
- Provides optimal balance between resolution and binding capability

RESIN CHARACTERISTICS

- Hydrophilic and rigid polyacrylamide-based resin
- Large pore diameter, large surface area, and optimized functionalities

SUPPORTING INFORMATION AND APPLICATION EXAMPLES

- UNOsphere Q Support Technical Data, bulletin 2729
- Effective Cleaning and Sanitizing of Anion Exchange Chromatography Resins poster, bulletin 5543
- UNOsphere Chromatography Resins Publications List, bulletin 6867
- Instruction manual, document 4110109

Bead Properties

Property	Description
Type of ion exchanger	Strong anion
Functional group	$-N^+(CH_3)_3$
Base matrix composition	Polyacrylamide
Mean particle size	120 μm
Total ionic capacity	120 $\mu eq/ml$
Dynamic binding capacity*	≥ 180 mg/ml at 150 cm/hr ≥ 125 mg/ml at 600 cm/hr
Recommended linear flow rate	50–1,200 cm/hr
Compression factor (settled bed volume/packed bed volume)	1.15–1.20
pH stability	1–14
Shipping solution	20% ethanol + 0.1 M NaCl or 2% benzyl alcohol
Regeneration	1–2 M NaCl
Sanitization	0.5–1.0 M NaOH
Storage conditions	20% ethanol or 0.1 M NaOH
Storage temperature	Room temperature
Chemical stability	1.0 M NaOH (20°C), up to 12 weeks 0.1 M NaOH (20°C), up to 5 years 1.0 M HCl (20°C), up to 200 hr
Shelf life	5 years

* 10% BT capacity determined with 2.0 mg/ml BSA in 1.1 x 10 cm column.

High Binding Capacity

The mean particle size of UNOsphere Q Resin is 120 μm , which generates a low backpressure (Figure 1). Increasing flow rates have minimal impact on binding capacity due to the excellent mass transfer properties of UNOsphere Q.

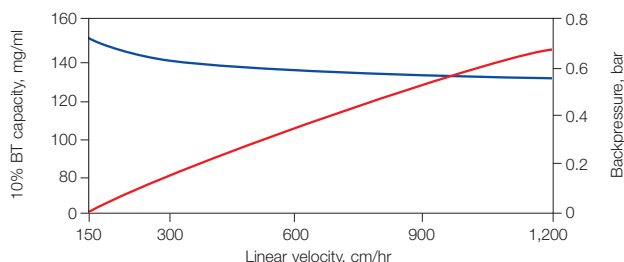


Fig. 1. Binding and backpressure properties of UNOsphere Q Resin.

A 5 mg/ml sample of BSA in 10 mM Tris, pH 8.5, was loaded onto a 1.1 x 20 cm column. Backpressure (—); 10% BT capacity (—).

Good Pressure/Flow Performance

The high productivity exhibited by UNOsphere Q Resin compared to other commercially available resins is due in part to its open architecture and low backpressure at high flow rates (Figure 2).

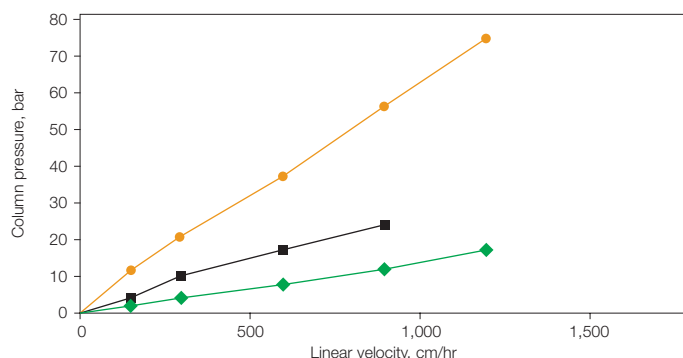


Fig. 2. Pressure/flow comparison for AEX resins. Conditions were as in Table 1. The Q Sepharose Fast Flow was run above the manufacturer's recommended 600 cm/hr. UNOsphere Q (—◆—); Q Sepharose Fast Flow (—■—); Fractogel EMD TMAE (M) (—●—).

High Productivity: Comparative Data

An inverse relationship exists between the DBC and productivity, and the latter continues to increase even at 1,200 cm/hr (Figure 3). Higher productivity for UNOsphere Q Resin is shown compared to other process resins at 1 bar constant pressure (Figure 4 and Table 1).

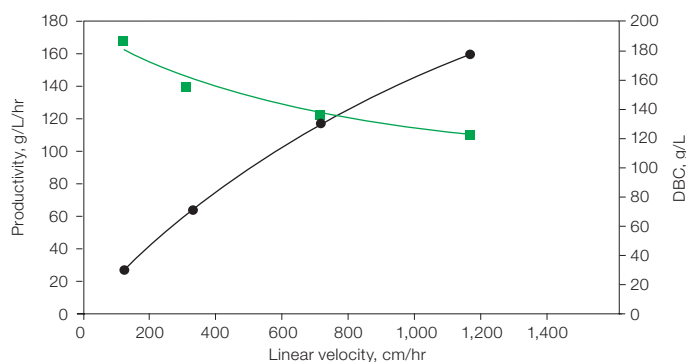


Fig. 3. Protein binding capacity (—) and productivity (—) of UNOsphere Q Resin. 1.1 x 20 cm column; conditions were as in Table 1.

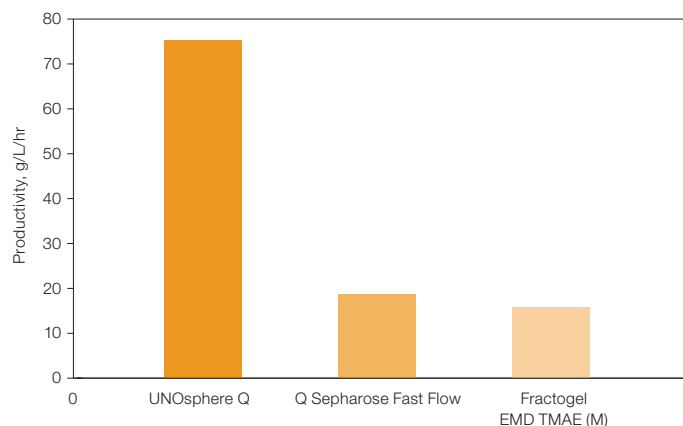


Fig. 4. UNOsphere Q productivity comparison. Conditions were as in Table 1.

Table 1. Comparison of properties of UNOsphere Q, Q Sepharose, and Fractogel TMAE Resins.

Support	Linear Velocity, cm/hr	Recovery, %	BSA Binding Capacity, g/L	Process Time, hr	Productivity, g/L/hr
UNOsphere Q	615	100.0	120.0	1.58	75.0
Q Sepharose Fast Flow	300	99.0	23.0	1.19	19.0
Fractogel EMD TMAE (M)	105	99.0	82.0	5.04	16.0

Properties were evaluated on a 1.1 x 20 cm (20 ml) column equilibrated with 10 mM Tris buffer, pH 8.5 (buffer A). BSA (5.0 mg/ml in buffer A) was loaded until 10% BT occurred. Elution was performed with buffer A containing 0.5 M NaCl.

High concentrations of salt or chaotropic agents are often required for the chromatography of inclusion body proteins and column regeneration, among other things. Table 2 shows that addition of these chemicals does not increase column pressures at high flow rates.

Table 2. Column pressure in various test solvents over a range of linear velocities.*

Solvent	Pressure (psi) at Given Linear Velocity (cm/hr)				
	150	300	600	900	1,200
20 mM Tris, pH 8.5	2	2	4	8	9
1 M NaCl	2	3	6	9	11
1 N NaOH	3	3	6	9	10
6 M guanidine HCl	7	9	15	22	ND**
8 M urea	6	14	28	43	52

* Resin was suspended in 1 M NaCl and packed into a 1.1 x 20 cm column at 1,200 cm/hr. The column was equilibrated with 20 mM Tris buffer, pH 8.5, then with the test solvent, and run at the velocities indicated. Between tests the column was re-equilibrated with Tris buffer.

** ND, not determined.

Purification Workflows

Application	Capture Step Resins	Intermediate Step Resins	Polish Step Resins/Media
mAb	Nuvia S	UNOsphere Q	CHT Ceramic Hydroxapatite

Ordering Information

Catalog # Description

Bulk Resins

1560101	UNOsphere Q Support, 25 ml
1560103	UNOsphere Q Support, 100 ml
156-0105	UNOsphere Q Support, 500 ml
12003698	UNOsphere Q Support (in BA), 500 ml
156-0107	UNOsphere Q Support, 10 L
12003699	UNOsphere Q Support (in BA), 10 L

BA, 2% benzyl alcohol.

Prepacked Screening Tools

12009311	EconoFit Anionic IEX Kit, includes 1 ml each of Macro-Prep High Q, Macro-Prep High Q-3HT, UNOsphere Q, Nuvia Q, Macro-Prep DEAE, and Nuvia HP-Q Resins
12009307	EconoFit UNOsphere Q Column, 1 x 1 ml, 7 x 25 mm
12009302	EconoFit UNOsphere Q Column, 1 x 5 ml, 16 x 25 mm
12009301	EconoFit UNOsphere Q Columns, 5 x 1 ml, 7 x 25 mm
12009303	EconoFit UNOsphere Q Columns, 5 x 5 ml, 16 x 25 mm
732-4714	Foresight UNOsphere Q Plates, 2 x 96-well, 20 µl
732-4819	Foresight UNOsphere Q RoboColumn Unit, 200 µl
732-4820	Foresight UNOsphere Q RoboColumn Unit, 600 µl
732-4732	Foresight UNOsphere Q Column, 1 ml
732-4752	Foresight UNOsphere Q Column, 5 ml

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Macro-Prep High Q and High Q-3HT Resins



ANION
EXCHANGE

RESIN FUNCTIONS

- Strong anion exchangers with unique selectivity for rapid purification of large acidic and neutral proteins and peptides
- Mechanically and chemically stable resins specifically designed to achieve excellent resolution at very high flow rates

RESIN CHARACTERISTICS

- Resins consist of rigid polymethacrylate beads
- Resins exhibit minimal shrinkage and swelling in response to variations in pH or ionic strength of buffers, enabling their effective use in low- and medium-pressure chromatography

SUPPORTING INFORMATION AND APPLICATION EXAMPLES

- Product information sheet, bulletin 5644
- Virus purification, bulletin 1985
- Negatively charged impurities removal, bulletin 2204
- Instruction manual, document LIT271

Bead Properties

Property	Macro-Prep High Q Resin	Macro-Prep High Q-3HT Resin
Type of ion exchanger	Strong anion	Strong anion
Functional group	$-N^+(CH_3)_3$	$-N^+(CH_3)_3$
Base matrix composition	Polymethacrylate	Polymethacrylate
Mean particle size*	70 μm	75 μm
Total ionic capacity	325–475 $\mu eq/ml$	400–520 $\mu eq/ml$
Dynamic binding capacity	≥ 37 mg BSA/ml	≥ 37 mg BSA/ml
Recommended linear flow rate	50–300 cm/hr	50–300 cm/hr
pH stability**	1–10	1–10
Shipping solution	20% ethanol or 2% benzyl alcohol	20% ethanol or 2% benzyl alcohol
Regeneration	1–2 M NaCl	1–2 M NaCl
Sanitization	Acetic acid or phosphoric acid solution, pH ≥ 1.5	Acetic acid or phosphoric acid solution, pH ≥ 1.5
Storage conditions	20% ethanol, 1 M acetic acid, or 50 mM phosphoric acid	20% ethanol, 1 M acetic acid, or 50 mM phosphoric acid
Storage temperature	Room temperature	Room temperature
Chemical stability	1% SDS, 8 M guanidine HCl, 100% ethanol	1% SDS, 8 M guanidine HCl, 100% ethanol
Shelf life	5 years	5 years

* Mean particle size after derivatization. Underivatized bead size is 50 μm .

** The use of basic reagents >pH 10 should be evaluated for each application.

Binding Capacity

The dynamic binding capacity of Macro-Prep High Q Resin is unaffected by the linear flow rate (Figure 1).

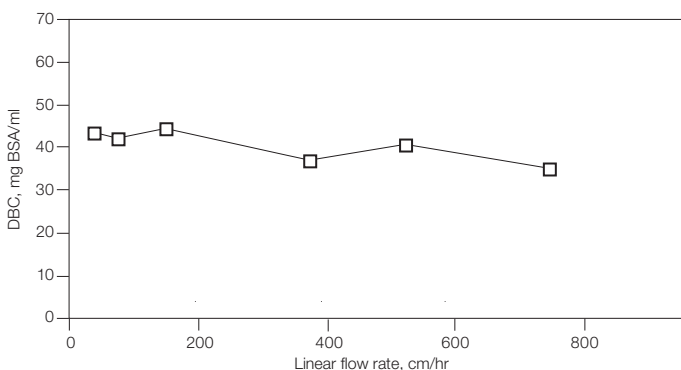


Fig. 1. DBC as a function of linear flow rate. A 100 ml sample (3 mg/ml) of BSA was run on a 1 x 10 cm (8.0 ml) column at the indicated flow rates. The buffer was 50 mM Tris-HCl, pH 8.3. Elution was with 1.0 M NaCl.

Good Pressure/Flow Performance

Figure 2 shows the linear relationship between flow rate and backpressure on the Macro-Prep Resin.

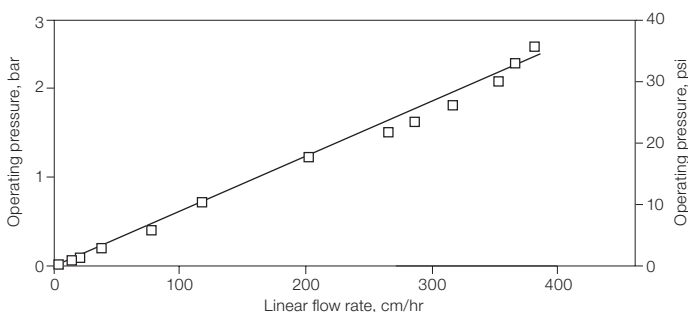


Fig. 2. Operating pressure of 50 μm Macro-Prep Resin in a 14 x 17 cm column.

Comparative Data

Efficient Endotoxin Removal with Macro-Prep High Q Resin

A panel of resins was evaluated to identify which could be used for endotoxin polishing. Of the 20 resins initially screened for the removal of endotoxin/B-(1,3)-D-glucan from a secreted yeast protein, protein B, six were selected for further evaluation. These resins were equilibrated in PBS, pH 7.4, for endotoxin polishing of 350 ml of protein B, which had been previously purified through a Q Sepharose Fast Flow Gel Column operating in flow-through mode. The samples (5 ml) were passed over each column and the eluates were collected and assayed for the amount of endotoxin in endotoxin units (EU, Figure 3).

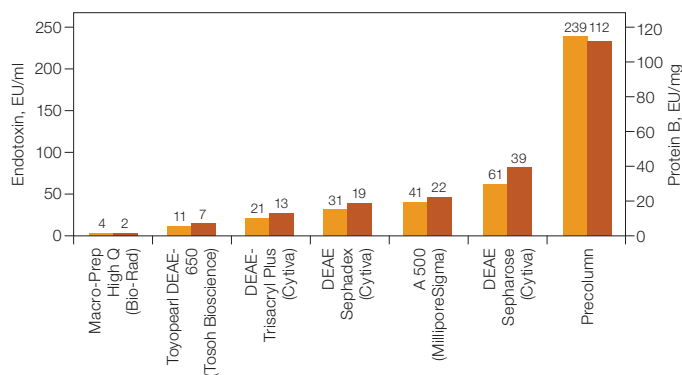


Fig. 3. Results from the evaluation of six different AEX resins for endotoxin removal from protein B. Endotoxin, EU/ml (■); protein B, EU/mg (■).

Macro-Prep High Q was chosen for the endotoxin polishing step because it resulted in the lowest amount of endotoxin per mg of protein B. The polishing step was performed on a 100 ml column equilibrated with PBS, pH 7.2. The eluate from the previous AEX step was diluted to 3.2 mg/ml and loaded at 5 ml/min. The flow-through fraction, containing protein B, was collected. Figure 4 summarizes the results of the Macro-Prep High Q AEX chromatography, which reduced the endotoxin level from 650 EU/ml to 1.29 EU/ml with a product yield of 84%.

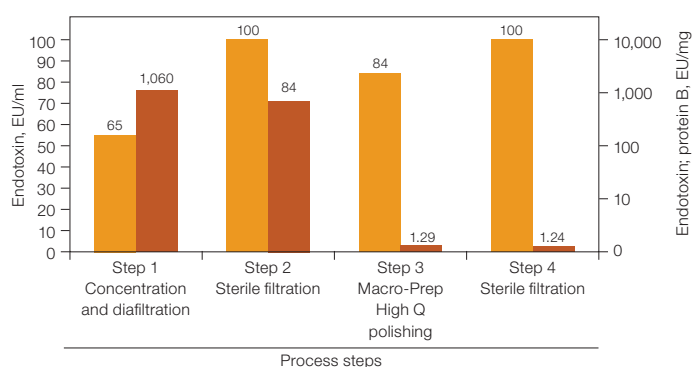


Fig. 4. Summary of results from endotoxin polishing from protein B. Step yield % (■); endotoxin (■).

Ordering Information

Catalog #	Description
Bulk Resins	
1580040	Macro-Prep High Q Support, 25 ml
1560040	Macro-Prep High Q Support, 100 ml
156-0041	Macro-Prep High Q Support, 500 ml
12003680	Macro-Prep High Q Support (in BA), 500 ml
156-0042	Macro-Prep High Q Support, 5 L
12003681	Macro-Prep High Q Support (in BA), 5 L
156-0043	Macro-Prep High Q Support, 10 L
12003682	Macro-Prep High Q Support (in BA), 10 L
1560043-3HT	Macro-Prep High Q-3HT Support, 10 L
1560040-3HT	Macro-Prep High Q-3HT Support, 100 ml
1560041-3HT	Macro-Prep High Q-3HT Support, 500 ml
1560042-3HT	Macro-Prep High Q-3HT Support, 5 L
1560043-3HT	Macro-Prep High Q-3HT Support, 10 L
1530021	Macro-Prep 25 Q Support, 50 ml
153-0024	Macro-Prep 25 Q Support, 5 L
BA, 2% benzyl alcohol.	
Prepacked Screening Tools	
12009311	EconoFit Anionic IEX Kit, includes 1 ml each of Macro-Prep High Q, Macro-Prep High Q-3HT, UNOsphere Q, Nuvia Q, Macro-Prep DEAE, and Nuvia HP-Q Resins
12009275	EconoFit Macro-Prep High Q Column, 1 x 1 ml, 7 x 25 mm
12009268	EconoFit Macro-Prep High Q Column, 1 x 5 ml, 16 x 25 mm
12009267	EconoFit Macro-Prep High Q Columns, 5 x 1 ml, 7 x 25 mm
12009269	EconoFit Macro-Prep High Q Columns, 5 x 5 ml, 16 x 25 mm

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Macro-Prep DEAE Resin



ANION
EXCHANGE

RESIN FUNCTIONS

- Weak anion exchanger with excellent flow properties for resolving complex biological mixtures at high flow rates
- Mechanically and chemically stable resins over a wide range of operating conditions

RESIN CHARACTERISTICS

- Resin consists of rigid polymethacrylate beads
- Resin exhibits minimal shrinkage and swelling in response to variations in pH or ionic strength of buffers, enabling its effective use in low- and medium-pressure chromatography

SUPPORTING INFORMATION AND APPLICATION EXAMPLES

- Macro-Prep DEAE Support for Chromatography of Biomolecules, bulletin 1942
- Purification of IgG, bulletin 2524
- Instruction manual, document LIT271

Bead Properties

Property	Description
Type of ion exchanger	Weak anion
Functional group	$-\text{N}^+(\text{C}_2\text{H}_5)_2$
Base matrix composition	Polymethacrylate
Mean particle size*	65 μm
Total ionic capacity	100–250 $\mu\text{eq/ml}$
Dynamic binding capacity	≥ 30 mg BSA/ml
Recommended linear flow rate	50–300 cm/hr
pH stability**	1–10
Shipping solution	20% ethanol or 2% benzyl alcohol
Regeneration	1–2 M NaCl
Sanitization	Acetic acid or phosphoric acid solution, pH ≥ 1.5
Storage conditions	20% ethanol, 1 M acetic acid, or 50 mM phosphoric acid
Storage temperature	Room temperature
Chemical stability	1% SDS, 8 M guanidine HCl, 100% ethanol
Shelf life	5 years

* Mean particle size after derivatization. Underivatized bead size is 50 μm .

** The use of basic reagents >pH 10 should be evaluated for each application.

Binding Capacity

Macro-Prep DEAE Resin has high dynamic binding capacity even at high flow rates (Figure 1).

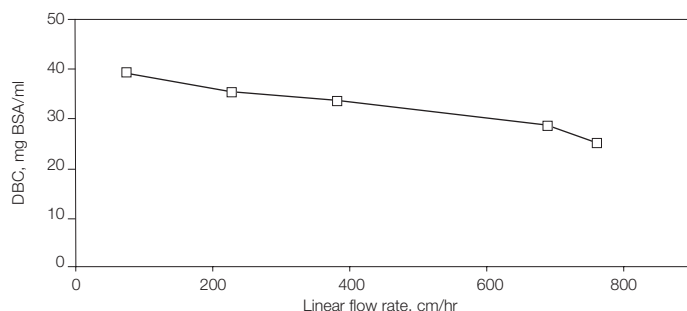


Fig. 1. The effect of flow rate on the DBC of the Macro-Prep DEAE Resin.

A sample containing 3 mg/ml BSA was separated on a 1 x 10 cm (3 ml) Macro-Prep DEAE Column. Sample was loaded onto the column in 10 mM Tris-HCl, pH 7.6 (buffer A), and was eluted in 1 M NaCl in buffer A.

Good Pressure/Flow Performance

Figure 2 shows the linear relationship between flow rate and backpressure on the Macro-Prep Resin.

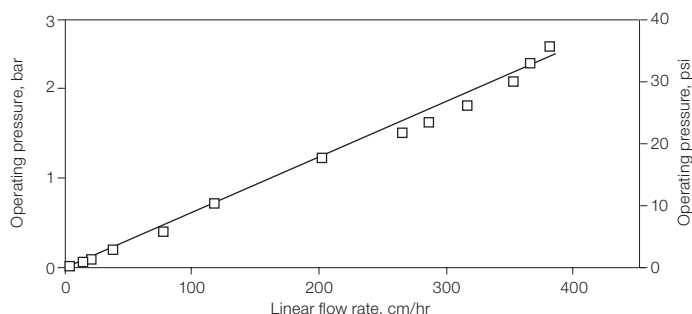


Fig. 2. Operating pressure of 50 μm Macro-Prep Resin in a 14 x 17 cm column.

Comparative Data

The particle size distribution of Macro-Prep DEAE Resin is narrow, with a derivatized mean particle size of 65 μm (Figure 3). The porosity of the bead is matched to exact specifications through a precise polymerization process. Regular pores with a narrow size distribution ensure consistent, reliable behavior. The resin exhibits consistently high resolution at high flow rates (Figure 4).

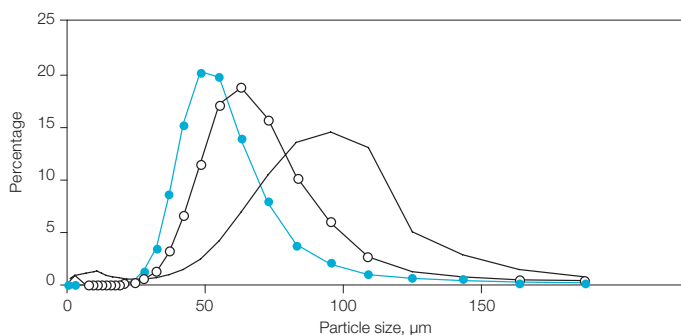


Fig. 3. Particle size distribution of the Macro-Prep DEAE Resin and two other commercially available DEAE resins. Macro-Prep DEAE Resin (—●—); TSK DEAE 650M (—○—); DEAE Sepharose Fast Flow (—).

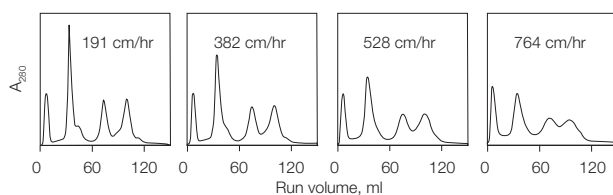


Fig. 4. The effect of flow rate on separation and peak symmetry. A 5 ml sample containing 3.65 mg/ml each myoglobin, conalbumin, BSA, and soybean trypsin inhibitor was loaded onto a column containing 6.56 ml Macro-Prep DEAE Resin. The sample was loaded in 50 mM Tris-HCl, pH 7.6 (buffer A), and eluted in a gradient of 0–1 M NaCl in buffer A as follows: buffer A for 15 ml, 0–0.35 M NaCl for 85 ml, 0.35–0.65 M NaCl for 15 ml, 0.65–1 M NaCl for 15 ml.

In addition, this weak AEX resin shows higher retention times and better separation properties than other commercially available resins (Figure 5). The resin is also able to efficiently handle high sample loads (Figure 6).

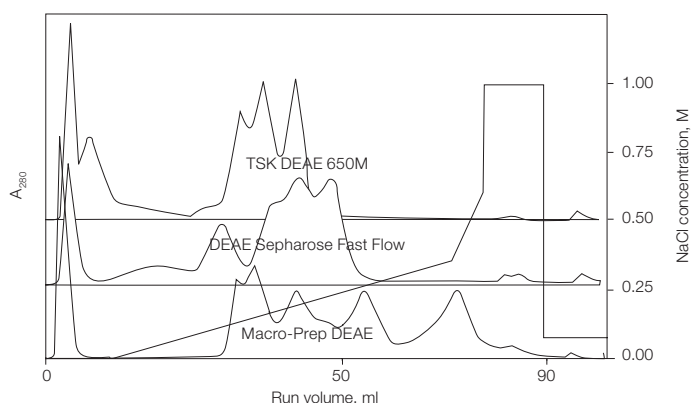


Fig. 5. Sample separation with the Macro-Prep DEAE Resin and two other commercially available DEAE resins. A 100 μ l sample containing 9 mg/ml each of myoglobin, conalbumin, ovalbumin, BSA, and soybean trypsin inhibitor was loaded onto a 0.5 cm ID column containing 2 ml of resin. Sample was loaded onto the column in 50 mM Tris-HCl, pH 7.6 (buffer A) and was eluted with 1 M NaCl in buffer A. The flow rate was 153 cm/hr (0.5 ml/min). Scales offset for comparison.

Ordering Information

Catalog #	Description
Bulk Resins	
1580020	Macro-Prep DEAE Support , 25 ml
1560020	Macro-Prep DEAE Support , 100 ml
156-0021	Macro-Prep DEAE Support , 500 ml
12003675	Macro-Prep DEAE Support (in BA) , 500 ml
156-0022	Macro-Prep DEAE Support , 5 L
12003676	Macro-Prep DEAE Support (in BA) , 5 L
156-0023	Macro-Prep DEAE Support , 10 L
12003677	Macro-Prep DEAE Support (in BA) , 10 L
BA, 2% benzyl alcohol.	

Prepacked Screening Tools

12009311	EconoFit Anionic IEX Kit , includes 1 ml each of Macro-Prep High Q, Macro-Prep High Q-3HT, UNOsphere Q, Nuvia Q, Macro-Prep DEAE, and Nuvia HP-Q Resins
12009274	EconoFit Macro-Prep DEAE Column , 1 x 1 ml, 7 x 25 mm
12009265	EconoFit Macro-Prep DEAE Column , 1 x 5 ml, 16 x 25 mm
12009264	EconoFit Macro-Prep DEAE Columns , 5 x 1 ml, 7 x 25 mm
12009266	EconoFit Macro-Prep DEAE Columns , 5 x 5 ml, 16 x 25 mm

Screen this resin for your application today.



Visit [bio-rad.com/ResinSample](https://www.bio-rad.com/ResinSample) to request a sample.

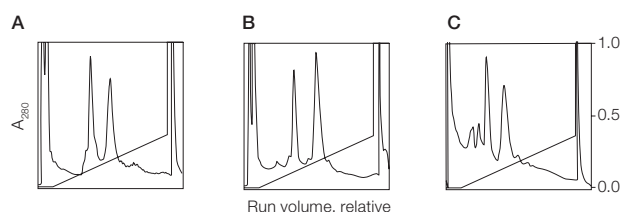


Fig. 6. Scalability of separations on Macro-Prep DEAE Resin. Separation of a yeast enzyme concentrate (Sigma) under various conditions of scale. **A**, a 50 mg sample was separated on a 7.4 ml column (10 mm ID) using a flow rate of 78 cm/hr (1 ml/min) in an analytical flow cell; **B**, a 500 mg sample was separated on a 64 ml column (25 mm ID) using a flow rate of 78 cm/hr (6.4 ml/min) in an analytical flow cell; **C**, a 1 g sample was separated on a 64 ml column (25 mm ID) using a flow rate of 305 cm/hr (25 ml/min) in a preparative flow cell. For all separations, sample was loaded in 50 mM Tris-HCl, pH 8.3, washed with 1 CV of the same buffer, and eluted in a linear gradient of 0–0.35 M NaCl in the same buffer for 22.5 CV.

Purification Workflows

Application	Capture Step Resins	Intermediate Step Resins	Polish Step Resins/Media
Equine IgG purification	Macro-Prep DEAE	—	CHT Ceramic Hydroxyapatite Type I

AG 1, AG MP-1M, AG 2, and AG 4 Resins



ANION
EXCHANGE

RESIN FUNCTIONS

- Separation of low molecular weight compounds such as inorganic ions, organic acids, nucleic acids, or carbohydrates
- Removal of organic and inorganic impurities
- AG 1-X2: sorption, exchange, and separation of low molecular weight inorganic anions and in applications such as cyclic nucleotide assays and fractionation of organic acids
- AG 1-X8: sorption and fractionation of relatively high molecular weight substances such as peptides, ribo- and deoxyribonucleotides, and uranium
- AG 2: separation of sugars, sugar alcohols, and glycosides using a step gradient and borate buffers without isomerizing some sugars
- AG 4: exchange anions such as organic acids using weak interactions, allowing easy elution and regeneration, as well as absorb mineral acids such as hydrochloric acid and perchloric acid, to yield a neutralized solution

RESIN CHARACTERISTICS

- AG 1 and AG 2 Resins are strongly basic anion exchangers with quaternary ammonium functional groups attached to the styrene-divinylbenzene copolymer lattice
- AG 1 Resin is slightly more basic and slightly more resistant to oxidation than AG 2 Resin due to differences in the structure of the quaternary functional group
- AG MP-1M Resin is the macroporous equivalent of AG 1 Resin, with an effective surface area that is approximately 23 square meters per dry gram, 20% porosity
- AG 4 Resin is a weakly basic anion exchanger with a hydrophilic acrylic matrix with a tertiary amino functional group
- Available in 4 narrow mesh ranges (20–50, 50–100, 100–200, 200–400), providing high resolution and reproducibility
- Convert to several ionic forms easily

- Analytical grade resin has been exhaustively sized, purified, and converted to make it suitable for accurate, reproducible analytical techniques
- Biotechnology grade resin is an analytical grade resin that has been certified to contain less than 100 microorganisms per gram of resin

SUPPORTING INFORMATION AND APPLICATION EXAMPLES

- Protein and peptide purification applications, bulletin 1801
- AG Resins flier, bulletin 6904
- AG 1, AG MP-1, and AG 2 Strong Anion Exchange Resin Instruction Manual, document LIT212
- AG 3, AG 4, and Bio-Rex 5 Anion Exchange Resin Instruction Manual, document LIT207

Bead Properties

Property	AG 1			AG MP-1M	AG 2	AG 4
Type of ion exchanger	Strong anion					Weak anion
Functional group	R-CH ₂ N ⁺ (CH ₃) ₃				R-CH ₂ N ⁺ (CH ₃) ₂ C ₂ H ₄ OH	R-CH ₂ N ⁺ (CH ₃) ₂
Base matrix composition	Styrene-divinylbenzene					Acrylic matrix
Particle size, μm	75–180	63–150	45–106	38–75	300–1,180	75–150
	106–250	106–250	106–180	75–150		
	180–500	180–425	180–428			
			300–1,180			
Crosslinkage, %	2	4	8	Macroporous	8	4
Approximate MW exclusion, Da	2,700	1,400	1,000	>1,000,000	1,000	1,400
Minimum capacity, meq/ml	0.6	1.0	1.2	1	1.2	0.8
Nominal density, mg/g	0.65	0.7	0.75	0.7	0.75	0.7
Ionic form	Chloride, hydroxide, acetate	Chloride	Chloride, hydroxide, acetate, formate	Chloride	Chloride	Free base
pH stability	0–14					
Regeneration	Equilibrate in correct ionic form; IEX followed by neutralization					
Sanitization	1 M NaOH, 70% ethanol. Autoclave except OH form; OH form sanitization not required					
Storage temperature	Ambient temperatures					
Chemical stability	Acid, base, organic solvents; see document LIT212					Various solvents (alcohols, hydrocarbons); see document LIT207
Shelf life	5 years					

Good Pressure/Flow Performance

The flow rate in a chromatography column increases with increasing particle size. However, the attainable resolution increases with decreasing particle size and narrower size distribution ranges. Particle size is given in either mesh size or micron size as shown in Table 1. The larger the mesh size number, the smaller the particle size.

Large mesh resin (20–50 and 50–100 mesh) is used primarily for preparative applications and batch operations where the resin and sample are slurried together. Medium mesh resin (100–200) may be used in batch as well as column applications. Medium mesh is an ideal general purpose particle size for use in analytical- and preparative-scale column chromatography. Fine mesh resin (200–400) is used for high-resolution analytical separations.

Table 1. Mesh and equivalent micron diameters.

Mesh (U.S. standard)	Micron Diameter (1 µm = 0.001 mm)
16	1,180
20	850
40	425
50	300
80	180
100	150
140	106
200	75
270	53
325	45
400	38

Technical Description

AG 1 Resins that have lower crosslinkages (for example, AG 1-X2) are useful for higher molecular weight molecules such as peptides, nucleotides, and uranium complexes. Resins with higher crosslinkages (for example, AG 1-X8) are useful for low molecular weight inorganic anions and for applications such as cyclic nucleotide assays and fractionation of organic acids.

AG MP-1M Resin is the macroporous equivalent of AG 1 Resin. Its effective surface area is approximately 23 square meters per dry gram, 20% porosity.

AG 2 Resins are similar to AG 1 Resins but are less basic and less resistant to oxidation. AG 2 Resins are capable of separating sugars, sugar alcohols, and glucocones. The isomerization of some sugars, which occurs with AG 1 Resins, does not occur with AG 2 Resins. Table 2 shows the approximate molecular weight exclusion limits in water for resins of various crosslinkages.

AG 1 and AG 2 Resins are best used for the separation of low molecular weight (<3,000 Da) compounds, including inorganic ions, organic acids, nucleic acids, carbohydrates, and metal ions. They can be used for nitrate removal, purification of sugar compounds, and pharmaceutical and fermentation applications.

Table 2. Approximate molecular weight exclusion limits for ion exchange resins in water.

Crosslinking, %	Approximate MW Exclusion Limit for Globular Molecules, Da
2	2,700
4	1,400
8	1,000
10	800
12	400

Compared to the AG 1 and AG 2 styrene-divinylbenzene matrix, which is hydrophobic, the AG 4 is softer, and therefore sensitive to crushing from excess flow rates or very large column sizes. This material is hydrophilic and is suitable for use with high molecular weight organic compounds, proteins, and carbohydrates.

Purification Workflows

Application	Capture Step Resins	Intermediate Step Resins	Polish Step Resins/Media
Impurity removal	AG 1-X8	Nuvia cPrime	CHT Type II

Ordering Information

Catalog #	Ionic Form	Mesh Size	Nominal Capacity, meq/ml	Bead Diameter, μm	Nominal Density, g/ml	Package Size, kg
Bulk Resins						
AG 1-X2 Resin, Analytical Grade						
140-1231	Chloride	50–100	0.6	180–500	0.65	0.5
140-1241	Chloride	100–200	0.6	106–250	0.65	0.5
140-1251	Chloride	200–400	0.6	75–180	0.65	0.5
140-1253	Acetate	200–400	0.6	75–180	0.65	0.5
140-1255	Acetate	200–400	0.6	75–180	0.65	10
140-1256	Acetate	200–400	0.6	75–180	0.65	1
AG 1-X2 Resin, Biotechnology Grade						
1431255	Hydroxide	200–400	0.6	75–180	0.65	0.1
AG 1-X4 Resin, Analytical Grade						
140-1331	Chloride	50–100	1.0	180–425	0.70	0.5
140-1341	Chloride	100–200	1.0	106–250	0.70	0.5
140-1342	Chloride	100–200	1.0	106–250	0.70	10
140-1351	Chloride	200–400	1.0	63–150	0.70	0.5
AG 1-X4 Resin, Biotechnology Grade						
1431345	Hydroxide	100–200	1.0	63–150	0.70	0.1
AG 1-X8 Resin, Analytical Grade						
140-1421	Chloride	20–50	1.2	300–1,180	0.75	0.5
140-1422	Hydroxide	20–50	1.2	300–1,180	0.75	0.5
140-1424	Hydroxide	20–50	1.2	300–1,180	0.75	10
140-1425	Chloride	20–50	1.2	300–1,180	0.75	1
140-1426	Chloride	20–50	1.2	300–1,180	0.75	10
140-1431	Chloride	50–100	1.2	180–425	0.75	0.5
140-1432	Chloride	50–100	1.2	180–425	0.75	10
140-1441	Chloride	100–200	1.2	106–180	0.75	0.5
140-1443	Acetate	100–200	1.2	106–180	0.75	0.5
140-1444	Formate	100–200	1.2	106–180	0.75	0.5
140-1445	Chloride	100–200	1.2	106–180	0.75	10
140-1446	Acetate	100–200	1.2	106–180	0.75	10
140-1447	Acetate	20–50	1.2	106–180	0.75	10
140-1451	Chloride	200–400	1.2	45–106	0.75	0.5
140-1453	Acetate	200–400	1.2	45–106	0.75	0.5
140-1454	Formate	200–400	1.2	45–106	0.75	0.5
140-1456	Chloride	200–400	1.2	45–106	0.75	10
AG 1-X8 Resin, Biotechnology Grade						
1432445	Hydroxide	100–200	1.2	106–180	0.75	0.1
1432446	Hydroxide	200–400	1.2	45–106	0.75	0.1
AG MP-1M Resin, Analytical Grade						
141-1831	Chloride	50–100	1.0	150–300	0.70	0.5
141-1841	Chloride	100–200	1.0	75–150	0.70	0.5
141-1842	Chloride	100–200	1.0	75–150	0.70	10
141-1851	Chloride	200–400	1.0	38–75	0.70	0.5
141-1852	Chloride	200–400	1.0	38–75	0.70	10
141-1853	Nitrate	200–400	1.0	38–75	0.70	0.5
AG 2-X8 Resin, Analytical Grade						
140-2421	Chloride	20–50	1.2	300–1,180	0.75	0.5
140-2423	Chloride	20–50	1.2	300–1,180	0.75	10
AG 4-X4 Resin, Analytical Grade						
140-4341	Free base	100–200	0.8	75–150	0.70	0.5

Screen this resin for your application today.



Visit [bio-rad.com/AGSample](https://www.bio-rad.com/AGSample) to request a sample.

ION EXCHANGE

MIXED-MODE / MULTIMODAL

AFFINITY

HYDROPHOBIC INTERACTION

SIZE EXCLUSION / GEL FILTRATION

Bio-Rex 5 Resin



ANION
EXCHANGE

RESIN FUNCTIONS

- Exchange anions such as iodide, bromide, and chloride with weak interactions
- Remove organic acids from sugars

RESIN CHARACTERISTICS

- Weak anion exchanger that contains predominantly tertiary, and some quaternary, amines
- Exhaustively sized, purified, and converted to make them suitable for accurate, reproducible analytical techniques
- Available in 2 mesh sizes (100–200 and 200–400) and both are analytical grade purity

SUPPORTING INFORMATION AND APPLICATION EXAMPLES

- Bio-Rex Resins flier, bulletin 6907
- AG 3, AG 4, and Bio-Rex 5 Anion Exchange Resin Instruction Manual, document LIT207

Bead Properties

Property	Description
Type of ion exchanger	Weak anion
Functional group	$\text{R-N}^+(\text{CH}_3)_2\text{C}_2\text{H}_4\text{OH}$
Base matrix composition	Styrene-divinylbenzene
Particle size	45–75 and 75–150 μm
Crosslinkage	4%
Approximate MW exclusion	1,400 Da
Operating pH stability	<7
pH stability	0–14
Minimum capacity	1.1 meq/ml
Nominal density	0.7 mg/g
Ionic form	Chloride

Ordering Information

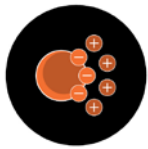
Catalog #	Description
Bulk Resins	
140-7841	Bio-Rex 5 Anion Exchange Resin , analytical grade, 100–200 mesh, chloride form, 500 g
140-7843	Bio-Rex 5 Anion Exchange Resin , analytical grade, 100–200 mesh, chloride form, 2 kg
140-7851	Bio-Rex 5 Anion Exchange Resin , analytical grade, 200–400 mesh, chloride form, 500 g

Screen this resin for your application today.



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Nuvia S Resin



CATION
EXCHANGE

RESIN FUNCTIONS

- Best-in-class DBC at high flow rates leading to faster cycle time for downstream processing during capture and polish purifications of antibodies, viruses, and proteins
- High binding performance over a wide operational window (pH and salt) for flexible DOE
- Reduced space requirements and cycle time can significantly improve productivity by contributing to smaller column footprint, decreased capital cost, and less buffer consumption

RESIN CHARACTERISTICS

- Hydrophilic and rigid polyacrylamido-based resin
- Designed with large pore diameter and optimized surface chemistry for binding of large molecules at high flow rates
- Mechanically and chemically stable resin with wide range of operating conditions

SUPPORTING INFORMATION AND APPLICATION EXAMPLES

- mAb purification, bulletin 5984
- IgM purification, bulletin 6966
- Biosimilar purification, bulletins 7038 and 7130
- Aggregate removal, bulletin 5984
- Purification strategy for clinical-grade monoclonal antibody, bulletin 6241
- Instruction manual, document 10018215

Bead Properties

Property	Description
Type of ion exchanger	Strong cation
Functional group	$-\text{SO}_3^-$
Base matrix composition	Polyacrylamide
Mean particle size	85 μm
Total ionic capacity	90–150 $\mu\text{eq/ml}$
Dynamic binding capacity*	>110 mg/ml at 300 cm/hr
Recommended linear flow rate	50–300 cm/hr
Compression factor (settled bed volume/packed bed volume)	1.15–1.20
pH stability	Short term: 2–14 Long term: 4–13
Shipping solution	20% ethanol + 0.1 M NaCl or 2% benzyl alcohol
Regeneration	1–2 M NaCl
Sanitization	0.5–1.0 M NaOH
Storage conditions	20% ethanol or 0.1 M NaOH
Storage temperature	Room temperature
Chemical stability	1.0 M NaOH (20°C), up to 1 week 0.1 M NaOH (20°C), up to 5 years
Shelf life	5 years

* 10% BT capacity determined with 4.5 mg/ml hlgG in 40 mM Na acetate + 30 mM NaCl, pH 5.0.

High Binding Capacity

With optimized bead size distribution and mechanical properties, Nuvia S consistently delivers very high binding capacity at fast linear velocities while maintaining low column pressure (Figure 1). The wide operating window allows users to operate at increasingly higher flow rates without compromising binding capacity.

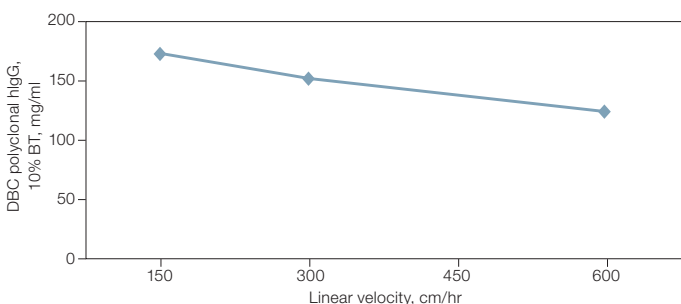


Fig. 1. Dynamic binding capacity vs. flow velocity. A 1.1 cm diameter column was packed to a 20 cm bed height with Nuvia S Resin. Polyclonal hlgG (4.5 mg/ml) in 40 mM NaOAc and 30 mM NaCl, pH 5.0, was loaded until 10% BT was observed.

Nuvia S is designed with proprietary surface extender technology and tuned for optimized charge, flow, and binding kinetics. The unique resin design gives Nuvia S best-in-class dynamic binding characteristics (Figure 2).

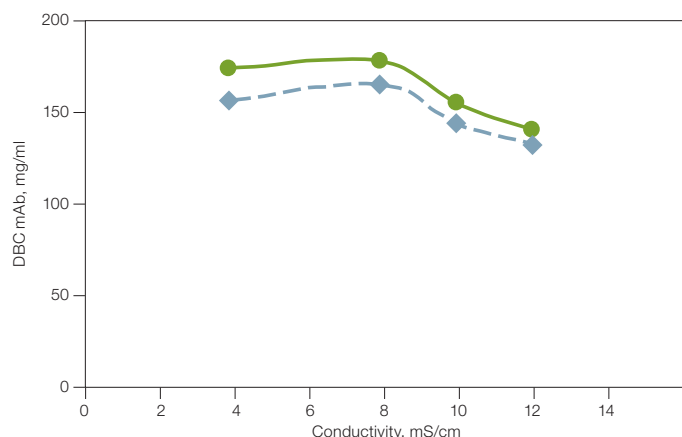


Fig. 2. Binding of a mAb by Nuvia S Resin. Column size, 1.1 x 10 cm. The sample mAb (4.5–5.0 mg/ml) was loaded onto the column in 20 mM NaOAc, pH 5.0 + NaCl (adjusted for conductivity) until 5% (◆) and 10% (●) BT capacity were observed.

Good Pressure/Flow Performance

Figure 3 shows the effect of flow rate and load viscosity on Nuvia S Resin backpressure.

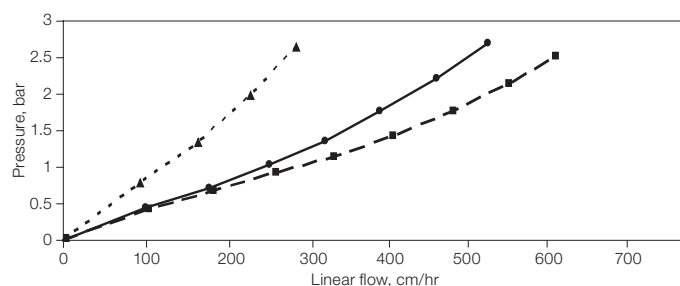


Fig. 3. Nuvia S Resin pressure/flow performance. A 20 cm diameter x 20 cm bed height column was packed to a compression factor of 1.17. Water (—■—); 10% glycerol (—●—); 30% glycerol (---▲---).

Comparative Data

High Productivity

The binding capacity of polyclonal hlgG to Nuvia S was compared to that of other commercially available CEX resins (Figure 4).

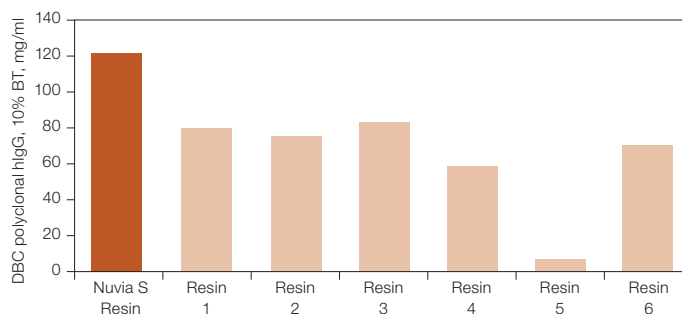


Fig. 4. Binding of polyclonal hlgG by Nuvia S Resin. Comparison of Nuvia S and other commercially available CEX resins. Column size, 0.7 x 5.5 cm. The sample was loaded onto the column in 40 mM NaOAc, pH 5.0 + 30 mM NaCl, washed, and then eluted with 40 mM NaOAc, pH 5.0 + 1 M NaCl.

Nuvia S exhibits efficient binding for hIgG in the linear velocity range of 300–600 cm/hr, significantly outperforming all commercial CEX resins tested (Figure 5). The hIgG binding capacity on Nuvia S was found to be 114 mg/ml even at a high flow rate of 600 cm/hr. Such performance allows Nuvia S to deliver the speed and throughput needed in process manufacturing of mAbs. For initial capture, operation at high flow rates offers the additional advantage of minimizing the target antibody's exposure to proteases present in the cell culture feedstream.

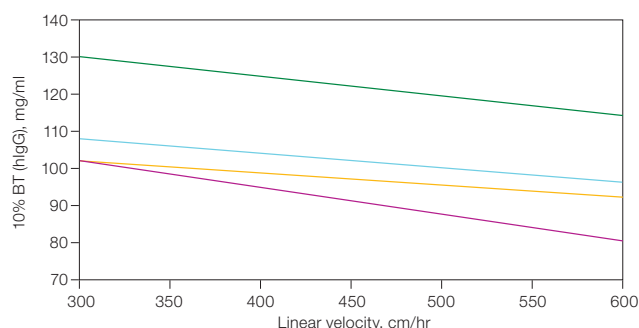


Fig. 5. Binding of hIgG by Nuvia S Resin at various linear velocities. Comparison of Nuvia S with commercially available CEX resins. Nuvia S (—); CEX 1 (—); CEX 2 (—); CEX 3 (—). Column size, 1.1 x 9.5 ± 0.3 cm.

Nonaffinity Capture

Tissue culture fluid (TCF) containing mAb G was purified on Nuvia S. HCP and DNA clearance (Table 1) were sufficient to replace affinity chromatography in a three-step process.

Table 1. Impurity clearance on Nuvia S.

	IgG, mg/ml	DNA, ng/ml	HCP, ng/ml
TCF	1.42	176	40,028
Euate	11.87	211	480
Clearance	ND	1.7 logs	2.8 logs

An 8 ml column (0.5 x 10.0 cm) was packed with Nuvia S. The 6x diluted TCF sample was washed in 20 mM sodium acetate, pH 5.0 (buffer A), and eluted in a gradient of buffer A + 1 M NaCl, pH 5.0, at 200 cm/hr.

Purification Workflows

Application	Capture Step Resins	Intermediate Step Resins	Polish Step Resins/Media
mAb	Nuvia S	Nuvia Q	Nuvia cPrime
Whey protein	One step with Nuvia S		

Ordering Information

Catalog #	Description
Bulk Resins	
1560311	Nuvia S Media, 25 ml
12003704	Nuvia S Media (in BA), 25 ml
1560313	Nuvia S Media, 100 ml
12003705	Nuvia S Media (in BA), 100 ml
156-0315	Nuvia S Media, 500 ml
12003706	Nuvia S Media (in BA), 500 ml
12018126	Nuvia S Media, 5 L
156-0317	Nuvia S Media, 10 L
12003707	Nuvia S Media (in BA), 10 L
BA, 2% benzyl alcohol.	

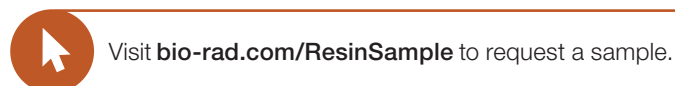
Prepacked Screening Tools

12009310	EconoFit Cationic IEX Kit, includes 1 ml each of Macro-Prep High S, UNOsphere S, Nuvia S, Nuvia HR-S, and Macro-Prep CM Resins
12009291	EconoFit Nuvia S Column, 1 x 1 ml, 7 x 25 mm
732-4701	Foresight Nuvia S Plates, 2 x 96-well, 20 µl
732-4801	Foresight Nuvia S RoboColumn Unit, 200 µl
732-4802	Foresight Nuvia S RoboColumn Unit, 600 µl
732-4720	Foresight Nuvia S Column, 1 x 1 ml
732-4740	Foresight Nuvia S Column, 1 x 5 ml

Case Studies

- Biosimilar purification, page 124
- mAb purification, page 131
- Protein purification, page 173

Screen this resin for your application today.



Nuvia HR-S Resin



CATION
EXCHANGE

RESIN FUNCTIONS

- Exceptional selectivity and resolution during separation of closely related biomolecules and charge variants
- High dynamic binding capacity leads to high recovery and good productivity during intermediate and polish purifications, thereby reducing cost of goods and increasing process economy

RESIN CHARACTERISTICS

- Hydrophilic and rigid polyacrylamide-based resin
- Designed with large pore diameter and optimized surface chemistry for binding of large molecules at high flow rates
- Mechanically and chemically stable resin with wide range of operating conditions

SUPPORTING INFORMATION AND APPLICATION EXAMPLES

- Product information sheet, bulletin 6448
- Minimize on-column aggregates, bulletin 7000
- Aggregate removal during mAb purification, bulletin 6439
- Instruction manual, document 10033316

Bead Properties

Property	Description
Type of ion exchanger	Strong cation
Functional group	-SO ₃ ⁻
Base matrix composition	Polyacrylamide
Mean particle size	50 µm
Total ionic capacity	100–180 µeq/ml
Dynamic binding capacity*	>70 mg/ml at 300 cm/hr
Recommended linear flow rate	50–200 cm/hr
Compression factor (settled bed volume/packed bed volume)	1.20–1.30
pH stability	Short term: 2–14 Long term: 4–13
Shipping solution	20% ethanol + 0.1 M NaCl or 2% benzyl alcohol
Regeneration	1–2 M NaCl
Sanitization	0.5–1.0 M NaOH
Storage conditions	20% ethanol or 0.1 M NaOH
Storage temperature	Room temperature
Chemical stability	1.0 M NaOH (20°C), up to 5 weeks 0.1 M NaOH (20°C), up to 5 years
Shelf life	5 years

* 10% BT capacity determined with 5.0 mg/ml hlgG in 20 mM Na acetate, pH 5.0.

High Binding Capacity

Nuvia HR-S exhibits high binding capacity over a wide operating pH range. The binding capacity of the test protein lactoferrin decreases as the salt concentration increases and optimal binding capacity is seen at lower pH conditions (Figure 1).

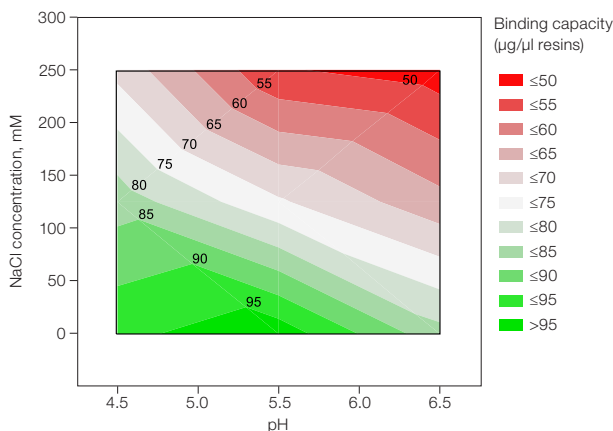


Fig. 1. DOE of lactoferrin binding to Nuvia HR-S at varying pH and NaCl concentrations. Experiments were performed in mini spin columns packed with 400 µl of resin and the dry weight was used to determine the absolute mg of the packed bed. The loading buffer was 20 mM citrate + a variable NaCl concentration. After washing, bound lactoferrin was eluted with buffer containing 1 M NaCl and the binding capacity was calculated based on the amount of lactoferrin eluted divided by the volume of the resin.

Good Pressure/Flow Performance

Figure 2 shows a linear relationship between pressure and flow up to 200 cm/hr for Nuvia HR-S.

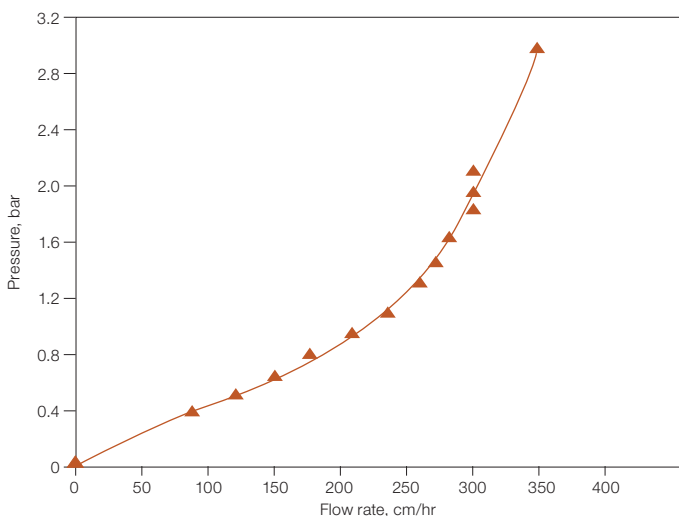


Fig. 2. Nuvia HR-S Resin pressure/flow performance. A 20 cm diameter x 20 cm bed height column was packed to a compression factor of 1.25.

High Resolution between Monomer and Aggregates

Nuvia HR-S delivers the resolution performance that bioprocess scientists need to separate HMW impurities like aggregates from monomers during mAb purification. An example of this high-resolution separation is shown in Figure 3, where a mixture of monomer and aggregate was separated on Nuvia HR-S Resin. Figure 4 shows HPLC-SEC progressive pools collected from the separation of the chromatogram in Figure 3. The data show that even with relatively nonstringent pooling, aggregate levels are significantly reduced. From a starting aggregate concentration of 8.9%, the final eluate pool contained 0.11% residual aggregate with an overall monomer recovery of 82%. Based on the data from Figure 3, an optimized gradient was used for further analysis, as shown in Figure 5.

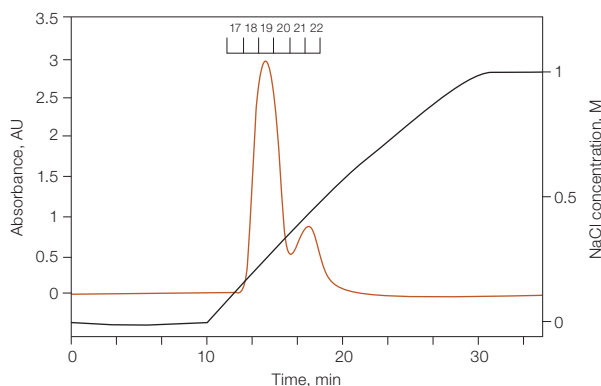


Fig. 3. Separation of aggregates from monomer by Nuvia HR-S before optimization. Absorbance (—); NaCl concentration (—).

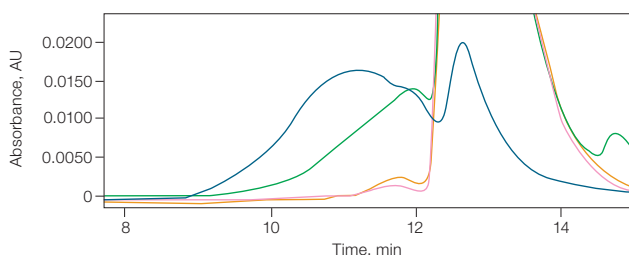
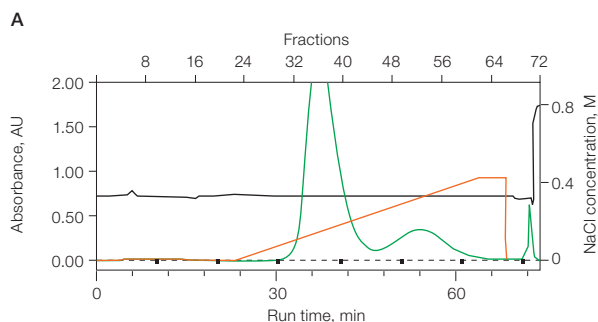


Fig. 4. SEC profiles. Load (—); selected pools (Fr 17+18+19+20) (—); (Fr 17+18+19) (—); aggregated pool (Fr 21+22) (—).



B

Chromatography Condition	Specification
Column	7 x 56 mm Nuvia HR-S Resin packed bed column
Load	20 ml aggregate and monomer mixture at a loading level of 46 mg/ml
Flow rate	150 cm/hr
Equilibration	40 mM Na acetate, pH 5.0
Gradient elution	40 mM Na acetate, 0–0.4 M NaCl, pH 5.0

Fig. 5. Monoclonal antibody separation on Nuvia HR-S Resin. A, chromatogram of the purification of monomer from aggregate; B, chromatography conditions. A_{280} (—); pH (—); gradient elution (—).

Comparative Data

Superior Aggregate Content and Monomer Recovery Using Nuvia HR-S

Nuvia HR-S Resin was compared to a commercially available small particle size CEX resin (Resin 1) to determine the aggregate content and monomer recovery as fractions were progressively pooled during elution. Nuvia HR-S delivered a final aggregate content of <0.3% and a recovery of >80%, while the recovery using agarose-based Resin 1 was <70% for the same final aggregate content (Figure 6).

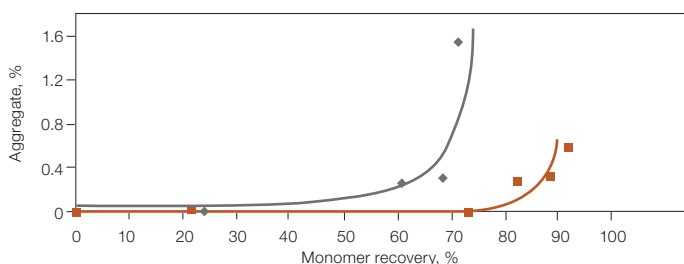


Fig. 6. Performance of Nuvia HR-S (■) vs. Resin 1 (◆).

Minimal On-Column Aggregate Generation

Six commercial resins including Nuvia HR-S were tested for the formation of on-column aggregates during mAb purification. Nuvia HR-S produced the smallest amount of on-column aggregates as shown in Figure 7.

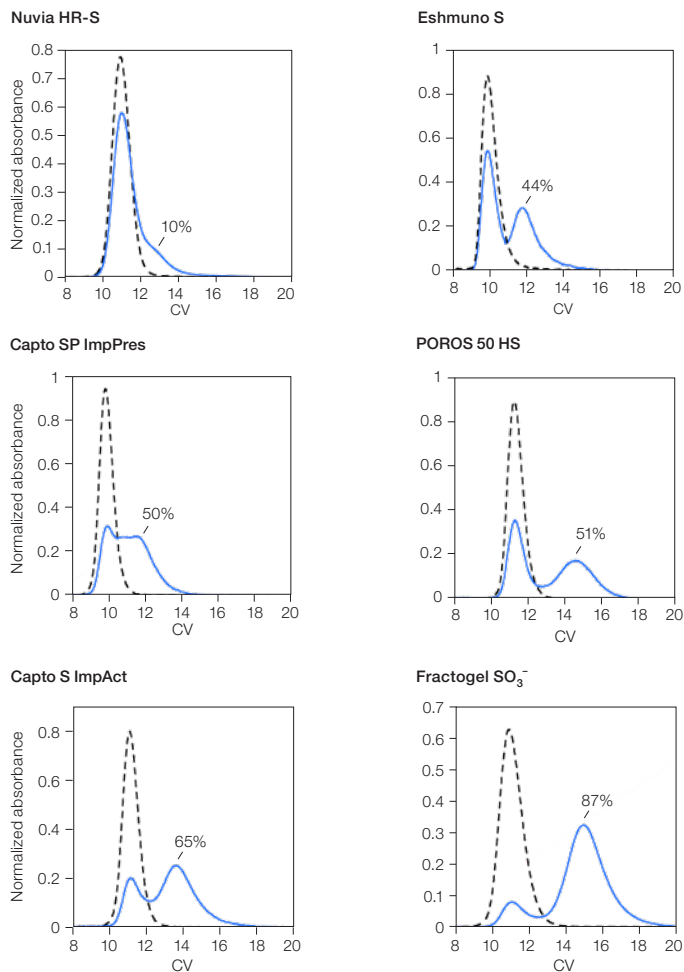


Fig. 7. On-column aggregates. 0 min hold time (---); 1,000 min hold time (—); percentage of on-column aggregate shown in plots. For further details, refer to Bio-Rad bulletin 7000.

Purification Workflows

Application	Capture Step Resins	Intermediate Step Resins	Polish Step Resins/Media
mAb	Nuvia S	Nuvia HR-S	CHT Ceramic Hydroxyapatite
mAb	Nuvia S	Nuvia HR-S	Nuvia cPrime

Ordering Information

Catalog # Description

Bulk Resins

1560511	Nuvia HR-S Media, 25 ml
12003712	Nuvia HR-S Media (in BA), 25 ml
1560513	Nuvia HR-S Media, 100 ml
12003713	Nuvia HR-S Media (in BA), 100 ml
156-0515	Nuvia HR-S Media, 500 ml
12003714	Nuvia HR-S Media (in BA), 500 ml
12018125	Nuvia HR-S Media, 5 L
156-0517	Nuvia HR-S Media, 10 L
12003715	Nuvia HR-S Media (in BA), 10 L

BA, 2% benzyl alcohol.

Prepacked Screening Tools

12009310	EconoFit Cationic IEX Kit, includes 1 ml each of Macro-Prep High S, UNOsphere S, Nuvia S, Nuvia HR-S, and Macro-Prep CM Resins
12009284	EconoFit Nuvia HR-S Column, 1 x 1 ml, 7 x 25 mm
732-4707	Foresight Nuvia HR-S Plates, 2 x 96-well, 20 µl
732-4831	Foresight Nuvia HR-S RoboColumn Unit, 200 µl
732-4832	Foresight Nuvia HR-S RoboColumn Unit, 600 µl
732-4723	Foresight Nuvia HR-S Column, 1 ml
732-4743	Foresight Nuvia HR-S Column, 5 ml

Case Studies

- Charge variant separation, page 134

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UNOsphere S Resin



CATION
EXCHANGE

RESIN FUNCTIONS

- Delivers high productivity through high binding capacity and low backpressure for the separation of proteins, nucleic acids, plasmids, viruses, and other biomolecules
- Has the ability to self-generate an on-column pH gradient, which gives it resolving power not attainable by other S-type resins
- Provides optimal balance between resolution and binding capability

RESIN CHARACTERISTICS

- Hydrophilic and rigid polyacrylamide-based resin
- Large pore diameter, large surface area, and optimized functionalities

SUPPORTING INFORMATION AND APPLICATION EXAMPLES

- Product information sheet, bulletin 2669
- Guidelines for Using UNOsphere S Cation Exchange Support, bulletin 5515
- Enhanced Purification of Similarly Charged Proteins with Self-Generated On-Column Resin pH Gradients, bulletin 7026
- Development of an Efficient Manufacturing Process for Adenovirus, bulletin 6719
- Purification of Murine IgG1 Using UNOsphere S and CHT Ceramic Hydroxyapatite Chromatography, bulletin 2780
- Purification of Transgenic Antibody from Corn Seed Using UNOsphere S and CHT Ceramic Hydroxyapatite Supports, bulletin 2774
- Instruction manual, document 4110109

ION EXCHANGE

MIXED-MODE / MULTIMODAL

AFFINITY

HYDROPHOBIC INTERACTION

SIZE EXCLUSION / GEL FILTRATION

Bead Properties

Property	Description
Type of ion exchanger	Strong cation
Functional group	$-\text{SO}_3^-$
Base matrix composition	Polyacrylamide
Mean particle size	80 μm
Total ionic capacity	219–319 $\mu\text{eq/ml}$
Dynamic binding capacity*	60 mg IgG/ml at 150 cm/hr 30 mg IgG/ml at 1,200 cm/hr
Recommended linear flow rate	50–300 cm/hr
Compression factor (settled bed volume/packed bed volume)	1.15–1.20
pH stability	1–14
Shipping solution	20% ethanol + 0.1 M NaCl or 2% benzyl alcohol
Regeneration	1–2 M NaCl
Sanitization	0.5–1.0 M NaOH
Storage conditions	20% ethanol or 0.1 M NaOH
Storage temperature	Room temperature
Chemical stability	1.0 M NaOH (20°C), up to 12 weeks 0.1 M NaOH (20°C), up to 5 years 1.0 M HCl (20°C), up to 200 hr
Shelf life	5 years

* 10% BT capacity determined with 4.5 mg/ml hlgG in 1.1 x 10 cm column.

Binding Capacity

The mean particle size of UNOsphere S Resin is 80 μm , which generates a low backpressure. The highly macroporous nature of UNOsphere S Resins provides high binding capacities that range from 40–60 mg IgG/ml resin in the linear velocity range of 150–600 cm/hr (Figure 1).

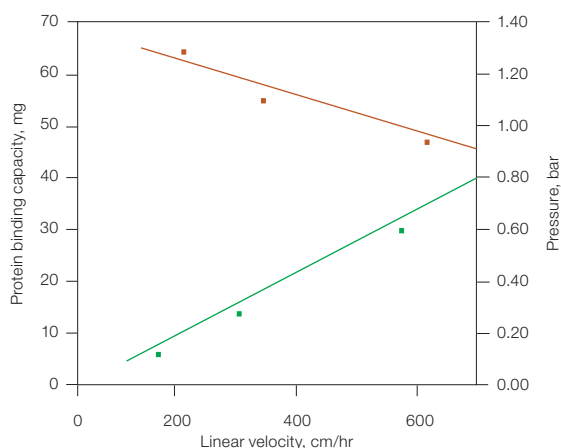


Fig. 1. Binding and backpressure properties of UNOsphere S Resin. Column size, 1.1 x 20 cm; sample, 2 mg/ml hlgG; buffer, 50 mM sodium acetate, pH 5.0. Backpressure (—); 10% BT capacity (—).

Good Pressure/Flow Performance

UNOsphere S Resins were designed to achieve the highest productivity (grams of drug per operational hour per liter of media) possible. UNOsphere Resins can be run at the highest linear velocities and loading capacities allowed by the column and chromatography system. Figure 2 illustrates the pressure/flow performance for UNOsphere S Resin.

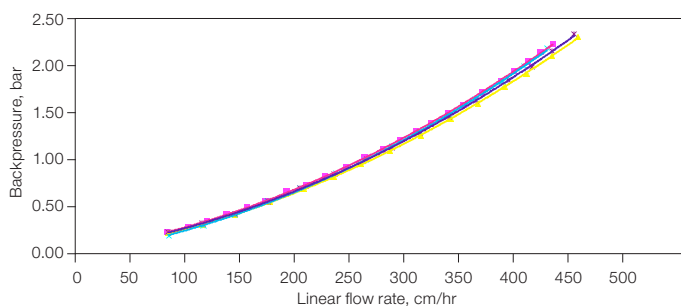


Fig. 2. Normalized pressure/flow curves for comparison of running buffers on UNOsphere S. Column size, 45 x 20 cm. 10 mM sodium acetate, 150 mM NaCl, pH 5.0 (—); 0.1 M NaCl (—); 1.0 M NaCl (—); PBS, pH 7.0 (—).

High Efficiency Capture of mAbs from Crude Feedstreams

Murine IgG1 (6.6 mg) was captured and eluted from a 2 ml UNOsphere S Column (Figure 3); assays of the load and eluate demonstrated a recovery of 97%. No antibody was detected in the flowthrough.

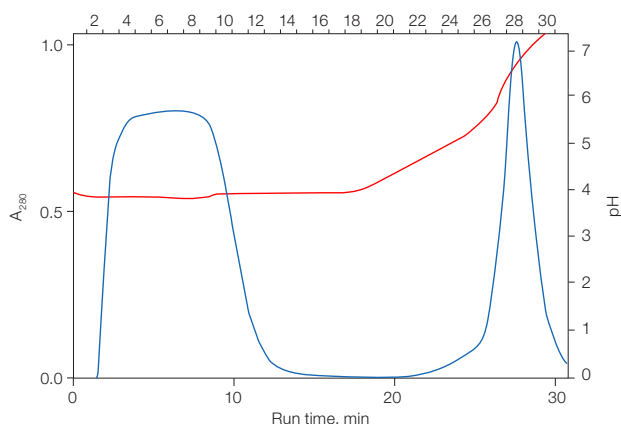


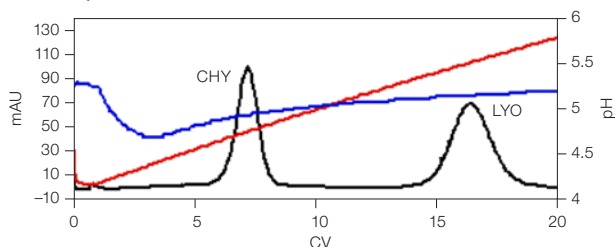
Fig. 3. Purification of murine IgG1 on a UNOsphere S Column. Column size, 0.5 x 10 cm (2 ml); sample, 15 ml (6.6 mg) of murine IgG1-conditioned medium. The sample was loaded onto the column in 20 mM citrate-phosphate buffer, pH 4.0, washed, and eluted in a linear gradient of 0–100% 20 mM citrate-phosphate, pH 8.0, in 10 CV at a flow rate of 2.0 ml/min (600 cm/hr). Each fraction was 2.0 ml. A_{280} (—); buffer pH (—).

Comparative Data

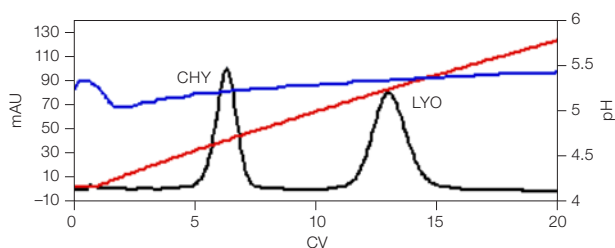
Enhanced Resolution of Proteins Due to On-Column pH Generation

Three commercial S-type resins were tested to resolve a mixture of α -chymotrypsin (pI 9.1) and lysozyme (pH 11). Although the two proteins can be resolved on all three resins, UNOsphere S, due to its efficient pH gradient induction (dip in the blue pH line), provides the best resolution between the purified protein peaks (Figure 4).

A. UNOsphere S



B. Nuvia HR-S



C. SP Sepharose Fast Flow

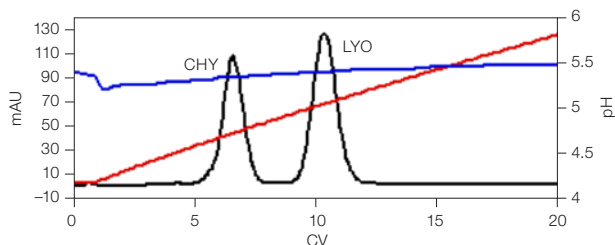


Fig. 4. Resolution of a mixture of α -chymotrypsinogen and lysozyme.
pH (—); salt gradient (—); protein (—); CHY, α -chymotrypsinogen; LYO, lysozyme.

Purification Workflows

Application	Capture Step Resins	Intermediate Step Resins	Polish Step Resins/Media
mAb	UNOsphere S	UNOsphere Q	CHT Ceramic Hydroxyapatite

Ordering Information

Catalog #	Description
Bulk Resins	
1560111	UNOsphere S Support, 25 ml
1560113	UNOsphere S Support, 100 ml
156-0115	UNOsphere S Support, 500 ml
12003702	UNOsphere S Support (in BA), 500 ml
156-0117	UNOsphere S Support, 10 L
12003703	UNOsphere S Support (in BA), 10 L
BA, 2% benzyl alcohol.	

Prepacked Screening Tools

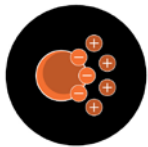
12009310	EconoFit Cationic IEX Kit, includes 1 ml each of Macro-Prep High S, UNOsphere S, Nuvia S, Nuvia HR-S, and Macro-Prep CM Resins
12009308	EconoFit UNOsphere S Column, 1 x 1 ml, 7 x 25 mm
12009305	EconoFit UNOsphere S Column, 1 x 5 ml, 16 x 25 mm
12009304	EconoFit UNOsphere S Columns, 5 x 1 ml, 7 x 25 mm
12009306	EconoFit UNOsphere S Columns, 5 x 5 ml, 16 x 25 mm
732-4710	Foresight UNOsphere S Plates, 2 x 96-well, 20 μ l
732-4813	Foresight UNOsphere S RoboColumn Unit, 200 μ l
732-4814	Foresight UNOsphere S RoboColumn Unit, 600 μ l
732-4730	Foresight UNOsphere S Column, 1 x 1 ml
732-4750	Foresight UNOsphere S Column, 1 x 5 ml

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Macro-Prep High S Resin



CATION
EXCHANGE

RESIN FUNCTIONS

- Strong cation exchanger with a unique selectivity leading to high yield and purity for basic and neutral proteins and peptides
- Mechanically and chemically stable resins specifically designed to achieve excellent resolution at very high flow rates

RESIN CHARACTERISTICS

- Resin consists of rigid polymethacrylate beads
- Resin exhibits minimal shrinkage and swelling in response to variations in pH or ionic strength of buffers, enabling their effective use in low- and medium-pressure chromatography

SUPPORTING INFORMATION AND APPLICATION EXAMPLES

- Product information sheet, bulletin 5643
- Virus purification, bulletin 1985
- mAb purification, bulletin 2079
- Instruction manual, document LIT271

Bead Properties

Property	Description
Type of ion exchanger	Strong cation
Functional group	$-\text{SO}_3^-$
Base matrix composition	Polymethacrylate
Mean particle size*	60 μm
Total ionic capacity	120–200 $\mu\text{eq/ml}$
Dynamic binding capacity	≥ 49 mg hlgG/ml
Recommended linear flow rate	50–300 cm/hr
pH stability**	1–10
Shipping solution	20% ethanol
Regeneration	1–2 M NaCl
Sanitization	Acetic acid or phosphoric acid solution, pH ≥ 1.5
Storage conditions	20% ethanol, 1 M acetic acid, or 50 mM phosphoric acid
Storage temperature	Room temperature
Chemical stability	1% SDS, 8 M guanidine HCl, 100% ethanol
Shelf life	5 years

* Mean particle size after derivatization. Underivatized bead size is 50 μm .

**The use of basic reagents >pH 10 should be evaluated for each application.

Good Pressure/Flow Performance

Figure 1 shows the linear relationship between pressure and flow up to ~400 cm/hr.

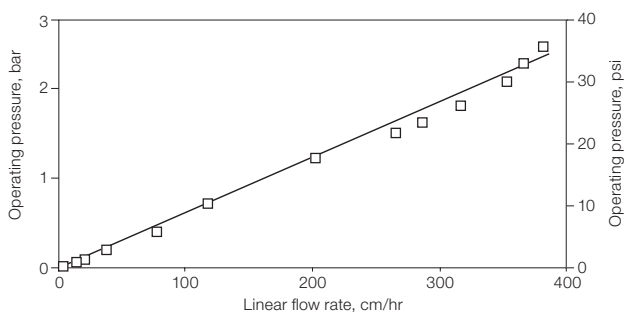


Fig. 1. Operating pressure of 50 μm Macro-Prep Resin in a 14 x 17 cm column.

Superior Performance

Macro-Prep High S Resin is an excellent choice for rapid purification. This methacrylate copolymer bead provides high-resolution separations at very high flow rates (Figure 2).

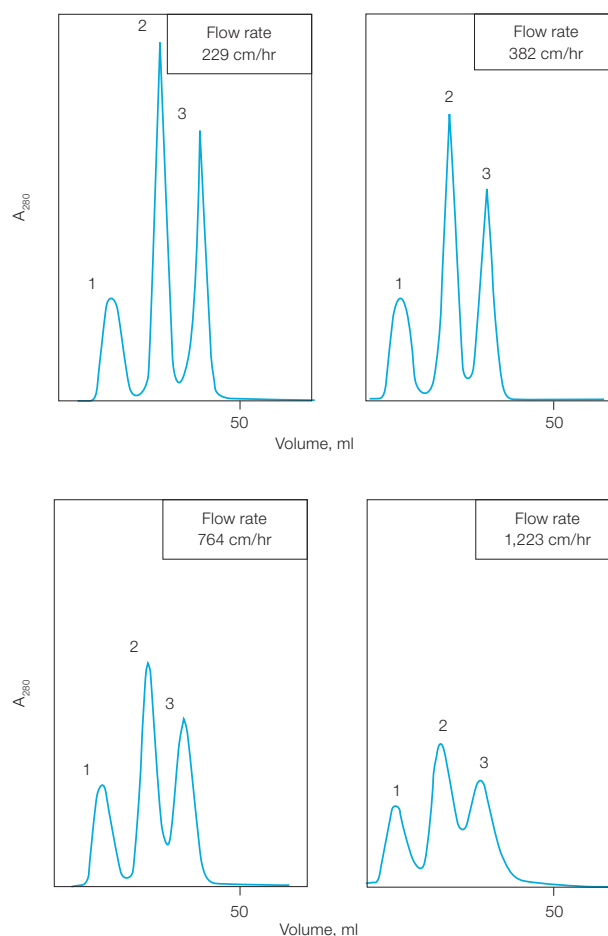
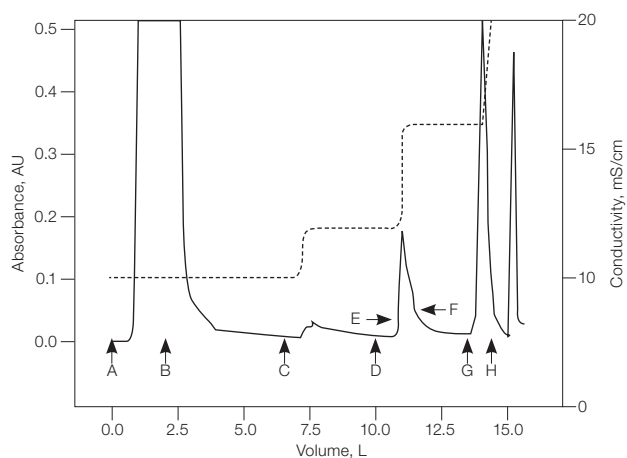


Fig. 2. Effect of flow rate on separation and peak symmetry. A 5 ml sample (1.4 mg/ml) of myoglobin (peak 1), ribonuclease A (peak 2), and cytochrome c (peak 3) was run on a 1 x 13 cm (8.7 ml) column at each of the indicated flow rates. The buffer was 20 mM potassium phosphate, pH 8.0, with a gradient of 0–1 M KCl over 75 ml.

Human monoclonal IgM was purified from bioreactor supernatants in a two-step procedure involving CEX and AEX chromatography. CEX chromatography resulted in 75% purity and a yield of 76% (Figure 3).



Resin: 2,000 ml Macro-Prep High S
Column: Vantage-S (9 x 50 cm), MilliporeSigma
Flow rate: 75 ml/min
Event table:

A-B	Load protein fraction from previous run
B	50 mM sodium phosphate, pH 7.2
C	50 mM sodium phosphate, 30 mM NaCl, pH 7.2
D	50 mM sodium phosphate, 100 mM NaCl, pH 7.2
E-F	Collect partially purified IgM
G	50 mM sodium phosphate, 1 M NaCl, pH 7
H	1.0 M NaOH

Fig. 3. Purification of IgM using Macro-Prep High S. A_{280} (—); conductivity (---).

Purification Workflows

Application	Capture Step Resins	Intermediate Step Resins	Polish Step Resins/Media
mAb	Macro-Prep High S	—	CHT Type I

Ordering Information

Catalog #	Description
Bulk Resins	
1580030	Macro-Prep High S Support, 25 ml
1560030	Macro-Prep High S Support, 100 ml
156-0031	Macro-Prep High S Support, 500 ml
12003670	Macro-Prep High S Support (in BA), 500 ml
156-0032	Macro-Prep High S Support, 5 L
12003671	Macro-Prep High S Support (in BA), 5 L
156-0033	Macro-Prep High S Support, 10 L
12003672	Macro-Prep High S Support (in BA), 10 L
BA, 2% benzyl alcohol.	

Prepacked Screening Tools

12009310	EconoFit Cationic IEX Kit, includes 1 ml each of Macro-Prep High S, UNOsphere S, Nuvia S, Nuvia HR-S, and Macro-Prep CM Resins
12009276	EconoFit Macro-Prep High S Column, 1 x 1 ml, 7 x 25 mm
12009271	EconoFit Macro-Prep High S Column, 1 x 5 ml, 16 x 25 mm
12009270	EconoFit Macro-Prep High S Columns, 5 x 1 ml, 7 x 25 mm
12009272	EconoFit Macro-Prep High S Columns, 5 x 5 ml, 16 x 25 mm

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Macro-Prep CM Resin



CATION
EXCHANGE

RESIN FUNCTIONS

- Weak cation exchanger providing exceptional capacity, resolution, reproducible results, and throughput
- Mechanically and chemically stable resin over a wide range of operating conditions

RESIN CHARACTERISTICS

- Resin consists of rigid polymethacrylate beads that enhance access to ionic sites and a hydrophilic matrix that reduces nonspecific binding
- Resin exhibits minimal shrinkage and swelling in response to variations in pH or ionic strength of buffers, enabling their effective use in low- and medium-pressure chromatography

SUPPORTING INFORMATION AND APPLICATION EXAMPLES

- Practical Guide: Selecting the Optimal Resins for the Process Purification of Native and Recombinant Proteins, bulletin 6810
- Instruction manual, document LIT271

Bead Properties

Property	Description
Type of ion exchanger	Weak cation
Functional group	—COO ⁻
Base matrix composition	Polymethacrylate
Mean particle size*	60 µm
Total ionic capacity	170–250 µeq/ml
Dynamic binding capacity	≥25 mg hemoglobin/ml
Recommended linear flow rate	50–300 cm/hr
pH stability**	1–10
Shipping solution	20% ethanol or 2% benzyl alcohol
Regeneration	1–2 M NaCl
Sanitization	Acetic acid or phosphoric acid solution, pH ≥1.5
Storage conditions	0.1 N NaOH
Storage temperature	Room temperature
Chemical stability	1% SDS, 8 M guanidine HCl, 100% ethanol
Shelf life	5 years

* Mean particle size after derivatization. Underivatized bead size is 50 µm.

**The use of basic reagents >pH 10 should be evaluated for each application.

Good Pressure/Flow Performance

Figure 1 shows the effect of flow rate on the Macro-Prep CM backpressure.

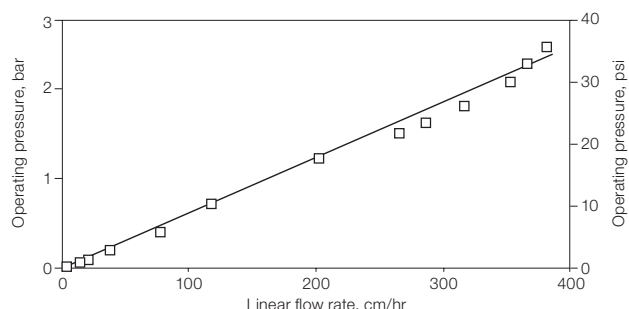


Fig. 1. Operating pressure of 50 µm Macro-Prep Resin in a 14 x 17 cm column.

Ordering Information

Catalog #	Description
Bulk Resins	
1580070	Macro-Prep CM Support , 25 ml
1560070	Macro-Prep CM Support , 100 ml
156-0071	Macro-Prep CM Support , 500 ml
12003685	Macro-Prep CM Support (in BA) , 500 ml
156-0073	Macro-Prep CM Support , 10 L
12003686	Macro-Prep CM Support (in BA) , 10 L
	BA, 2% benzyl alcohol.

Prepacked Screening Tools

12009310	EconoFit Cationic IEX Kit , includes 1 ml each of Macro-Prep High S, UNOsphere S, Nuvia S, Nuvia HR-S, and Macro-Prep CM Resins
12009273	EconoFit Macro-Prep CM Column , 1 x 1 ml, 7 x 25 mm

Publications

Application	Reference
Histone purification	Li A et al. (2010). Phosphorylation of histone H2A.X by DNA-dependent protein kinase is not affected by core histone acetylation, but it alters nucleosome stability and histone H1 binding. <i>J Biol Chem</i> 285, 17,778–17,788.
Clearance of cationic species	Vitali A et al. (1998). Purification and partial characterization of a peroxidase from plant cell cultures of <i>Cassia didymobotrya</i> and biotransformation studies. <i>Biochem J</i> 331, 513–519. Porter EM et al. (1998). Isolation of human intestinal defensins from ileal neobladder urine. <i>FEBS Lett</i> 434, 272–276.

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AG 50W and AG MP-50 Resins



CATION
EXCHANGE

RESIN FUNCTIONS

- Ultrapurification of buffers and ionic reagents
- Separation of small peptides and amino acids
- Removal of cations, and metal separations, including water softening and nitrate removal
- Purification of sugar compounds, and other pharmaceutical and fermentation applications

RESIN CHARACTERISTICS

- Strong cation exchange resins composed of sulfonic acid functional groups attached to a styrene-divinylbenzene copolymer lattice
- Lower physical resistance to shrinking and swelling, so they absorb more water and swell to a larger wet diameter than highly crosslinked resins of equivalent dry diameter
- AG 50W Resin is sized to provide high resolution and reproducibility, and converts to several ionic forms easily
- AG MP-50 is equivalent to AG 50W Resin but its effective surface area is approximately 35 square meters per dry gram, or 30–35% porosity
- Analytical grade resin has been exhaustively sized, purified, and converted to make it suitable for accurate, reproducible analytical techniques
- Biotechnology grade resin is an analytical grade resin that has been certified to contain less than 100 microorganisms per gram of resin

SUPPORTING INFORMATION AND APPLICATION EXAMPLES

- Protein and Peptide Purification Applications, bulletin 1801
- AG Resins flier, bulletin 6904
- AG 50W and AG MP-50 Cation Exchange Resins Instruction Manual, document LIT203
- Molecular Biology Grade AG 50W-X8 Cation Exchange Resin Instruction Manual, document LIT304

Bead Properties

Property	Description
Type of ion exchanger	Strong cation
Functional group	SO ₃ ⁻
Base matrix composition	Styrene-divinylbenzene
Minimum capacity	2% crosslinkage: 0.6 meq/ml 4% crosslinkage: 1.1 meq/ml 8% crosslinkage: 1.7 meq/ml 12% crosslinkage: 2.1 meq/ml Macroporous (MP): 1.5 meq/ml
Nominal density	2% crosslinkage: 0.7 g/ml 4% crosslinkage: 0.8 g/ml 8% crosslinkage: 0.8 g/ml 12% crosslinkage: 0.85 g/ml MP: 0.8 g/ml
Recommended linear flow rate based on application	Removing trace ions: 5–10 cm/min Separations with very few components: 1–3 cm/min Separations of multicomponent samples: 0.3–1.0 cm/min Using high-resolution resins with small particle size: 0.1–0.2 cm/min
Pressure vs. flow performance	Dependent on particle size, see Table 3
pH stability	0–14
Shipping solution	Wet mesh
Regeneration	Equilibrate with 5 CV 3 M HCl followed by 8 CV distilled water
Sanitization	Resins are stable in acid, base, and organic solvents and may be autoclaved
Storage conditions	Ambient temperatures
Chemical stability	Resins are stable in acid, base, and organic solvents and may be autoclaved (except the OH form)
Shelf life	Resins are stable for at least two years when stored in the original unopened container at ambient temperatures and protected from ultraviolet light

AG 50 Strong Acid CEX Resins are composed of sulfonic acid functional groups attached to a styrene-divinylbenzene copolymer lattice (Figure 1). The amount of resin crosslinking determines the bead pore size. A resin with lower crosslinkage has a more open structure, permeable to higher molecular weight substances than a highly crosslinked resin. It also has lower physical resistance to shrinking and swelling, so it absorbs more water and swells to a larger wet diameter than a highly crosslinked resin of equivalent dry diameter. Typical applications of 2% crosslinked AG 50W-X2 Resin and 4% crosslinked AG 50W-X4 Resin include separation or concentration of peptides, nucleotides, and amino acids. Resins with high percentage crosslinkage, such as 8% crosslinked AG 50W-X8 Resin, 12% crosslinked AG-50W-X12 Resin, and 16% crosslinked AG 50W-X16 Resin, are used for the separation of small peptides and amino acids, removal of cations, and metal separations. Industrial applications also include water softening, nitrate removal, purification of sugar compounds, pharmaceutical, and fermentation applications. Table 1 shows approximate molecular weight exclusions for each percentage crosslinkage.

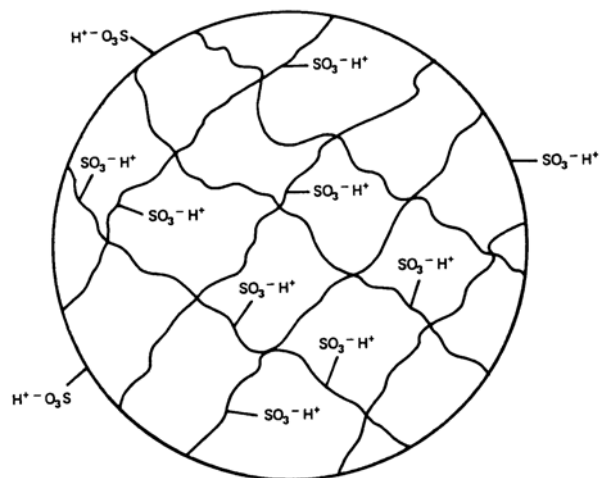


Fig. 1. AG 50 Resin functional group.

Table 1. Approximate molecular weight exclusion limits for ion exchange resins in water.

Crosslinking, %	Approximate MW Exclusion Limit for Globular Molecules, Da
2	2,700
4	1,400
8	1,000
10	800
12	400

When using IEX resins, the counterions on the resin are replaced by sample ions that have the same charge. In applications involving a CEX resin, such as AG 50 Resin, neutral molecules and anions do not interact with the resin. AG 50 Resin is available with H⁺, Na⁺, or NH₃⁺ counterions. A resin can be converted from one ionic form to another. Usually the resin is used in an ionic form with a lower selectivity for the functional group than the sample ions to be exchanged. The sample ions are then exchanged when introduced and can be eluted by introducing an ion with higher affinity for the resin or a high concentration of an ion with equivalent or lower affinity. Table 2 shows the relative selectivity of various counterions.

Table 2. Relative selectivity of various counterions.

Counterion	Relative Selectivity for AG 50W-X8 Resin	Counterion	Relative Selectivity for AG 50W-X8 Resin
H ⁺	1.0	Fe ²⁺	2.55
Li ⁺	0.85	Zn ²⁺	2.7
Na ⁺	1.5	Co ²⁺	2.8
NH ₄ ⁺	1.95	Cu ²⁺	2.9
K ⁺	2.5	Cd ²⁺	2.95
Rb ⁺	2.6	Ni ²⁺	3.0
Cs ⁺	2.7	Ca ²⁺	3.9
Cu ⁺	5.3	Sr ²⁺	4.95
Ag ⁺	7.6	Hg ²⁺	7.2
Mn ²⁺	2.35	Pb ²⁺	7.5
Mg ²⁺	2.5	Ba ²⁺	8.7

In general, the lower the selectivity of the counterion the more readily it exchanges for another ion of like charge. The order of selectivity can also be used to estimate effectiveness for different ions as eluants, with the most highly selective being the most efficient. Finally, the order of selectivity can be used to estimate the difficulty of converting the resin from one form to another. Conversion from a highly selected to a less highly selected form requires an excess of the new ion.

Good Pressure/Flow Performance

The flow rate in a chromatography column increases with increasing particle size. However, the attainable resolution increases with decreasing particle size and narrower size distribution ranges. Particle size is given either in mesh or micron size. The larger the mesh size number, the smaller the particle size (Table 3). Large mesh resin (20–50 and 50–100 mesh) is used primarily for preparative applications and batch operations where the resin and sample are slurried together. Medium mesh resin (100–200 mesh) may be used in batch as well as column applications. Medium mesh is an ideal general purpose particle size for use in analytical and preparative-scale column chromatography. Fine mesh resin (200–400) is used for high-resolution analytical separations.

Table 3. Mesh and equivalent micron diameters.

Mesh (U.S. standard)	Micron Diameter (1 μm = 0.001 mm)
16	1,180
20	850
40	425
50	300
80	180
100	150
140	106
200	75
270	53
325	45
400	38

Ordering Information

Catalog #	Ionic Form	Mesh Size	Nominal Capacity, meq/ml	Bead Diameter, μm	Nominal Density, g/ml	Package Size, kg
Bulk Resins						
AG 50W-X2 Resin, Analytical Grade						
142-1231	Hydrogen	50–100	0.6	300–1,180	0.70	0.5
142-1232	Hydrogen	50–100	0.6	300–1,180	0.70	10
142-1241	Hydrogen	100–200	0.6	106–300	0.70	0.5
142-1242	Hydrogen	100–200	0.6	106–300	0.70	10
142-1251	Hydrogen	200–400	0.6	75–180	0.70	0.5
AG 50W-X2 Resin, Biotechnology Grade						
1435241	Hydrogen	100–200	0.6	106–300	0.70	0.1
AG 50W-X4 Resin, Analytical Grade						
142-1331	Hydrogen	50–100	1.1	180–425	0.80	0.5
142-1341	Hydrogen	100–200	1.1	106–250	0.80	0.5
142-1351	Hydrogen	200–400	1.1	75–150	0.80	0.5
AG 50W-X4 Resin, Biotechnology Grade						
1435341	Hydrogen	200–400	1.1	75–150	0.80	0.1
AG 50W-X8 Resin, Analytical Grade						
142-1421	Hydrogen	20–50	1.7	300–1,180	0.80	0.5
142-1423	Hydrogen	20–50	1.7	300–1,180	0.80	10
142-1431	Hydrogen	50–100	1.7	180–425	0.80	0.5
142-1432	Hydrogen	50–100	1.7	180–425	0.80	10
142-1441	Hydrogen	100–200	1.7	106–250	0.80	0.5
142-1442	Hydrogen	100–200	1.7	106–250	0.80	10
142-1451	Hydrogen	200–400	1.7	63–150	0.80	0.5
142-1452	Hydrogen	200–400	1.7	63–150	0.80	10
AG 50W-X8 Resin, Molecular Biology Grade						
143-1442	Sodium	100–200	1.7	106–250	0.8	1
AG 50W-X8 Resin, Biotechnology Grade						
1435441	Hydrogen	100–200	1.7	106–250	0.80	0.1
1435451	Hydrogen	200–400	1.7	63–150	0.80	0.1
AG 50W-X12 Resin, Analytical Grade						
142-1254	Hydrogen	200–400	2.1	53–106	0.85	1
142-1641	Hydrogen	100–200	2.1	106–250	0.85	0.5
142-1651	Hydrogen	200–400	2.1	53–106	0.85	0.5
AG MP-50 Resin, Analytical Grade						
143-0832	Hydrogen	50–100	1.5	150–300	0.80	1
143-0841	Hydrogen	100–200	1.5	75–150	0.80	0.5

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Bio-Rex 70 Resin



CATION
EXCHANGE

RESIN FUNCTIONS

- Purification and fraction of peptides, proteins, antibodies, and other cationic molecules

RESIN CHARACTERISTICS

- Weak cation exchanger that contains carboxylic acid exchange groups on a hydrophilic macroreticular acrylic polymer lattice
- Available in 4 mesh sizes (20–50, 50–100, 100–200, and 200–400)
- Analytical grade resin has been exhaustively sized, purified, and converted to make it suitable for accurate, reproducible analytical techniques
- Biotechnology grade resin is an analytical grade resin that has been certified to contain less than 100 microorganisms per gram of resin

SUPPORTING INFORMATION AND APPLICATION EXAMPLES

- Bio-Rex Resins flier, bulletin 6907
- Bio-Rex Weakly Acidic Cation Exchange Resin Instruction Manual, document LIT206

Bead Properties

Property	Description
Type of ion exchanger	Weak cation
Functional group	–COO [–]
Base matrix composition	Macroreticular acrylic polymer lattice
Particle size	45–75, 75–150, 150–300, and 300–1,180 µm
Mesh	20–50, 50–100, 100–200, and 200–400
Pore size range	700–4,000 Å
pH stability	5–14
Na ⁺ capacity	0.5 meq/ml
H ⁺ capacity	2.4 meq/ml
Nominal density	0.6 mg/g
Ionic form	Sodium

Ordering Information

Catalog #	Description
142-5822	Bio-Rex 70 Cation Exchange Resin , analytical grade, 20–50 mesh, sodium form, 500 g
142-5832	Bio-Rex 70 Cation Exchange Resin , analytical grade, 50–100 mesh, sodium form, 500 g
142-5833	Bio-Rex 70 Cation Exchange Resin , analytical grade, 50–100 mesh, sodium form, 10 kg
142-5842	Bio-Rex 70 Cation Exchange Resin , analytical grade, 100–200 mesh, sodium form, 500 g
142-5843	Bio-Rex 70 Cation Exchange Resin , analytical grade, 100–200 mesh, sodium form, 10 kg
142-5852	Bio-Rex 70 Cation Exchange Resin , analytical grade, 200–400 mesh, sodium form, 500 g
1435832	Bio-Rex 70 Cation Exchange Resin , biotechnology grade, 50–100 mesh, sodium form, 100 g
1435852	Bio-Rex 70 Cation Exchange Resin , biotechnology grade, 200–400 mesh, sodium form, 100 g
142-5853	Bio-Rex 70 Cation Exchange Resin , biotechnology grade, 200–400 mesh, sodium form, 10 kg

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Chelex 100 Resin



CATION
EXCHANGE

RESIN FUNCTIONS

- Scavenges metal contaminants without altering the concentration of nonmetal ions

RESIN CHARACTERISTICS

- Styrene-divinylbenzene copolymer containing paired iminodiacetate ions, which act as chelating groups in binding polyvalent metal ions
- Strong selectivity for heavy metal ions
- Strong attraction for transition metals in high-salt solutions
- Extensively purified
- Sized to provide high resolution and reproducibility

SUPPORTING INFORMATION AND APPLICATION EXAMPLES

- Chelex 100 Resin flier, bulletin 6908
- Chelex 100 and Chelex 20 Chelating Ion Exchange Resin Instruction Manual, document LIT200
- Chelex 100 Iron Form Resin for Glyphosate Clean-Up Instruction Sheet, document LIT237

Bead Properties

Property	Description
Type of ion exchanger	Strong cation
Functional group	$\text{R-CH}_2\text{N}(\text{CH}_2\text{COO}^-)_2$
Base matrix composition	Styrene-divinylbenzene
Particle size	75–150, 150–300, and 300–1,180 μm
Total ionic capacity	0.4 meq/ml (defined as $\text{Cu}(\text{NH}_3)_4^{2+}$ uptake)
Mesh	50–100, 100–200, and 200–400
Recommended linear flow rate	≥ 20 cm/min (separations of strong cations from weak cations) <4 cm/min (separations of similar species)
pH stability	2–14
Shipping solution	Dry powder
Regeneration	Two-step process: 1. Convert to the hydrogen form using acid 2. Convert to the desired ionic form using the hydroxide of the cation desired
Sanitization	0.5–1.0 N NaOH
Storage conditions	Store in salt form such as sodium or ammonium
Chemical stability	1 M acid, 1 M base, organic solvents, 8 M urea, reducing agents
Shelf life	5 years

The chelating Chelex Resin is efficiently regenerated in dilute acid and operates in basic, neutral, and weakly acidic solutions of pH 4 or higher. At very low pH, the resin acts as an anion exchanger. Figure 1 shows the zwitterionic forms of the Chelex Resin as a function of pH.

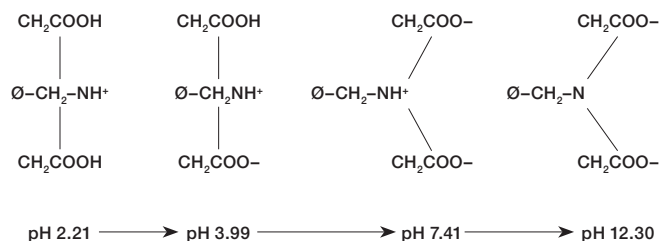


Fig. 1. Change in structure of Chelex Resin ligand with increasing pH.

Applications

Chelex 100 Resins can be used in many processes, including analysis of trace metals in natural waters, reagents, biochemicals, and physiological fluids; removal of trace metals from reagents, biochemicals, physiological fluids, culture media, soils, and enzyme systems; recovery of metals from process streams; and chromatography of closely related metals. Chelex 100 Resin is usually used in a batch process.

Trace Metal Removal

Chelex Resin offers a rapid method for thoroughly removing trace metal contaminants that could have an effect on biological fluids or biological systems under study. A unique IEX resin that is more selective for multivalent metals than the standard CEX resins, Chelex Resin will scavenge multivalent metal ion contaminants without altering the concentration of nonmetallic ions. In most cases, neither column nor batch treatment with Chelex 100 Resin has any effect on protein concentration or enzyme activity. Where low protein recovery is a problem, the protein can be dialyzed against a buffer containing Chelex 100 Resin.

Glyphosate Concentration and Cleanup

Chelex 100 Resin may be used to concentrate glyphosate (N-(phosphonomethyl)glycine) when the resin is in its iron form. When environmental water or an aqueous extract of crops, such as soybeans, grapes, cabbage, or alfalfa, is applied to Chelex 100 Resin (100–200 mesh, iron form), glyphosate forms a complex with the iron form of the resin. The glyphosate may subsequently be eluted with 6 M HCl and applied to AG 1-X8 Resin (200–400 mesh, chloride form) for AEX cleanup. The glyphosate, along with its breakdown product, aminomethyl phosphonic acid, may then be quantified with an Aminex Column. For more information, see document LIT237.

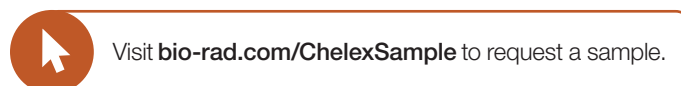
Metal Analysis

Trace metals can be concentrated by adsorption to Chelex Resin. For recommended use, see document LIT200.

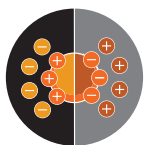
Ordering Information

Catalog #	Description
Bulk Resins	
142-2822	Chelex 100 Resin , analytical grade, 50–100 mesh, sodium form, 500 g
142-2832	Chelex 100 Resin , analytical grade, 100–200 mesh, sodium form, 500 g
1422825	Chelex 100 Resin , analytical grade, 100–200 mesh, iron form, 100 g
142-2842	Chelex 100 Resin , analytical grade, 200–400 mesh, sodium form, 500 g
1432832	Chelex 100 Resin , biotechnology grade, 100–200 mesh, sodium form, 100 g
142-2823	Chelex 100 Resin , analytical grade, 50–100 mesh, sodium form, 10 kg
142-2834	Chelex 100 Resin , analytical grade, 100–200 mesh, sodium form, 10 kg
142-2826	Chelex 100 Resin , analytical grade, 100–200 mesh, iron form, 10 kg
142-2827	Chelex 100 Resin , analytical grade, 100–200 mesh, iron form, 1 kg
142-2843	Chelex 100 Resin , analytical grade, 200–400 mesh, sodium form, 10 kg

Screen this resin for your application today.



AG 11 A8 Resin



ION
EXCHANGE

RESIN FUNCTIONS

- Chromatographic desalting
- Separation of acids from nonionic molecules
- Separation of salts from organic materials by absorbing both anions and cations in equivalent amounts while allowing the organic compounds to pass through

RESIN CHARACTERISTICS

- Each resin bead is a molecular mixture of a cation and anion exchanger
- Resin is composed of a styrene-divinylbenzene crosslinked, rigid polymer lattice with attached quaternary ammonium groups (strongly basic anion exchange groups), which contains a web of trapped, linear, relatively flexible acrylic polymer with carboxyl groups (weakly acidic cation exchange groups)
- Made by polymerization of acrylic acid moieties inside the AG 1-X8 Resin (see page 30), which produces a spherical resin bead containing paired anion and cation exchange sites
- Analytical grade resin has been exhaustively sized, purified, and converted to make it suitable for accurate, reproducible analytical techniques
- Biotechnology grade resin is an analytical grade resin that has been certified to contain less than 100 microorganisms per gram of resin

Bead Properties

Property	Description
Type of ion exchanger	Mixed-bed strong anion and weak cation
Functional group (ligand)	$-\text{N}^+(\text{CH}_3)_3$ and $-\text{COO}^-$
Counterions	OH^- and H^+
Base matrix composition	Styrene-divinylbenzene
Particle size	180–425 μm
Mesh	40–80
Crosslinkage	8%
Approximate MW exclusion	1,000 Da
Minimum capacity	$-\text{N}^+(\text{CH}_3)_3$: ≥ 1.2 meq/ml $-\text{COO}^-$: ≥ 1.1 meq/ml
Nominal density	0.70 g/ml
Ionic form	Self-adsorbed
pH stability	4–10
Storage temperature	Ambient temperatures
Shelf life	5 years

Ordering Information

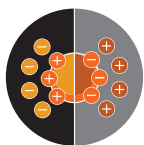
Catalog #	Description
142-7834	AG 11 A8 Ion Retardation Resin , analytical grade, 500 g
1437834	AG 11 A8 Ion Retardation Resin , biotechnology grade, 100 g

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AG 501-X8 and Bio-Rex MSZ 501(D) Resins



ION
EXCHANGE

RESIN FUNCTIONS

- Water deionization and softening
- Removal of nonionic substances such as urea, acrylamide, formamide, and glyoxal
- Removal of radioactive substances from nuclear power plant wastewater and polishing of condensate

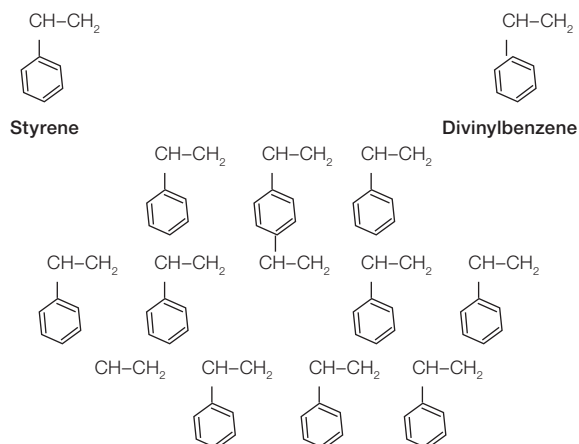
RESIN CHARACTERISTICS

- Mixed-bed ion exchangers with a styrene-divinylbenzene copolymer matrix (see Figure 1)
- AG 501-X8 contains an equal amount of AG 50W-X8 strong cation exchange resin (H^+ form) (page 51), and AG 1-X8, a strong anion exchange resin (OH^- form) (page 30)
- Bio-Rex MSZ 501(D) is similar to AG 501-X8 but with a more uniform bead size that minimizes pressure drops, increases resin capacity, and allows shorter regeneration times
- Available with a blue dye (D) visual indicator irreversibly bound to the anion exchange resin, which turns from blue to gold when the exchange capacity is exhausted
- Analytical grade resin has been exhaustively sized, purified, and converted to make it suitable for accurate, reproducible analytical techniques
- Biotechnology grade resin is an analytical grade resin that has been certified to contain less than 100 microorganisms per gram of resin
- Molecular biology grade resin is certified to contain no endo- or exonucleases and no ligase inhibitors

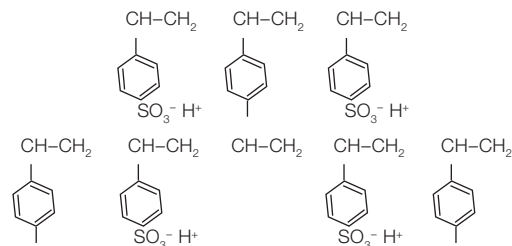
SUPPORTING INFORMATION AND APPLICATION EXAMPLES

- AG Resins flier, bulletin 6904
- AG 501-X8 and Bio-Rex MSZ 501(D) Mixed Bed Resin Instruction Manual, document LIT205

A. Crosslinked styrene-divinylbenzene copolymer



B. AG 50W crosslinked strongly acidic CEX resin



C. AG 1 crosslinked strongly basic AEX resin

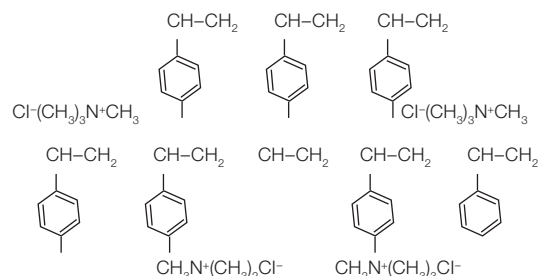


Fig. 1. Molecular structures of styrene-divinylbenzene crosslinked copolymers with attached groups for AG 50W and AG 1 Resins.

Bead Properties

Property	AG 501-X8 and AG 501-X8(D)	Bio-Rex MSZ 501(D)
Type of ion exchanger	Mixed-bed strong anion and strong cation	
Functional group (ligand)	$-N^+(CH_3)_3$ and $-SO_3^-$	
Counterions	OH^- and H^+	
Base matrix composition	Styrene-divinylbenzene	
Particle size	300–1,180 μm	600 $\pm$ 100 μm
Mesh	20–50	
Crosslinkage	8%	8 and 10%
Approximate MW exclusion	1,000 Da	
Minimum capacity	~1.5 meq/ml	
Nominal density	0.75 g/ml	
Recommended linear flow rate	Removing trace ions: 5–10 cm/min Separations with very few components: 1–3 cm/min Separations of multicomponent samples: 0.3–1.0 cm/min Using high-resolution resins with small particle size: 1.0–2.0 cm/min	
pH stability	0–14	
Shipping solution	Wet mesh	
Regeneration	See Regeneration section	
Sanitization	Resins are stable in acid, base, and organic solvents and may be autoclaved, except the OH form	
Storage conditions	Ambient temperatures	
Chemical stability	Acid, base, organic solvent	
Shelf life	Mixed-bed ion exchange resins are stable for 2 years when stored at 21°C and protected from exposure to ultraviolet light. The shelf life may be extended by storing the resin at 4°C	
Available grades	Analytical, biotechnology, molecular biology	Analytical

Note: Descriptions listed under AG 501-X8 apply to all resins in the table unless otherwise indicated.

Applications

AG 501-X8 and Bio-Rex MSZ 501(D) Resins can be used in either batch or column method, although the batch technique is much more common. Table 1 provides guidelines for column deionization for a range of volumes.

Table 1. Guidelines for column deionization.

Column Size, cm (diameter x length)	Weight of Resin, g	Deionized Water, L (approximate)*
0.5 x 20	2.6	1.5
1.0 x 50	27.0	6.7
1.5 x 100	118.0	29.0
2.5 x 100	325.0	80.0
5.0 x 100	1,300.00	325.0
15.0 x 120	13.6	3,500.0

* Based on using 100 ppm NaCl. This volume will vary depending on initial water quality.

Regeneration

Mixed-bed resins used in laboratory-scale applications are not normally regenerated because of the difficulty in separating the mixed anion and cation resins, the large volumes of regenerants required for the anion resin, and difficulty in accurately remixing chemically equivalent resins. See document LIT205 for the regeneration protocol, which is cost effective for large-scale applications.

AG 501-X8(D) Resin has a blue dye irreversibly bound to the anion exchange resin, which turns from blue to gold when the exchange capacity is exhausted, indicating regeneration is necessary.

Ordering Information

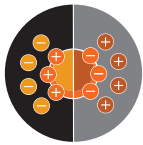
Catalog #	Description	Mesh Size	Diameter, µm	Nominal Density, g/ml	Package Size, kg
Bulk Resins					
142-6424	AG 501-X8 Mixed Bed Resin , analytical grade	20–50	300–1,180	0.75	0.5
142-6427	AG 501-X8 Mixed Bed Resin , analytical grade	20–50	300–1,180	0.75	10
1437424	AG 501-X8 Mixed Bed Resin , biotechnology grade	20–50	300–1,180	0.75	0.1
143-7428	AG 501-X8 Mixed Bed Resin , biotechnology grade	20–50	300–1,180	0.75	10
1436424	AG 501-X8 Mixed Bed Resin , molecular biology grade	20–50	300–1,180	0.75	0.1
143-6426	AG 501-X8 Mixed Bed Resin , molecular biology grade	20–50	300–1,180	0.75	10
142-6425	AG 501-X8(D) Mixed Bed Resin , analytical grade	20–50	300–1,180	0.75	0.5
1437424	AG 501-X8 Mixed Bed Resin , biotechnology grade	20–50	300–1,180	0.75	0.1
1437425	AG 501-X8(D) Mixed Bed Resin , biotechnology grade	20–50	300–1,180	0.75	0.1
143-7428	AG 501-X8 Mixed Bed Resin , biotechnology grade	20–50	300–1,180	0.75	10
1436425	AG 501-X8(D) Mixed Bed Resin , molecular biology grade	20–50	300–1,180	0.75	0.1
143-6427	AG 501-X8(D) Mixed Bed Resin , molecular biology grade	20–50	300–1,180	0.75	10
142-7425	Bio-Rex MSZ 501(D) Mixed Bed Resin	25–35	500–700	0.75	0.5
142-7426	Bio-Rex MSZ 501(D) Mixed Bed Resin	25–35	500–700	0.75	10

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Bio-Rex RG 501-X8 Resin



ION
EXCHANGE

RESIN FUNCTIONS

- Cleanup of power plant deionization systems
- Large-scale cleanup of metals from wastewater

RESIN CHARACTERISTICS

- Uniform bead size, which minimizes pressure drops and increases resin capacity; used for desalting water and other nonelectrolytes
- The resin contains a dye that changes from blue-green to gold when its exchange capacity is exhausted

SUPPORTING INFORMATION AND APPLICATION EXAMPLES

- Bio-Rex Resins flier, bulletin 6907
- AG 501-X8 and Bio-Rex MSZ 501(D) Mixed Bed Resin Instruction Manual, document LIT205

Ordering Information

Catalog #	Description
444-9999	Bio-Rex RG 501-X8 Resin , 500 g, technical grade ion exchange resin, H ⁺ /OH ⁻ form, 20–50 mesh, for water deionization/purification and General Electric stator systems
444-9998	Bio-Rex RG 501-X8 Resin , 1 ft (~20 kg), technical grade mixed-bed resin, H ⁺ /OH ⁻ form, 20–50 mesh, for water deionization/purification and General Electric stator systems

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Chromatography Resins and Media by Technique: **Mixed-Mode / Multimodal**



ION EXCHANGE

MIXED-MODE / MULTIMODAL

AFFINITY

HYDROPHOBIC INTERACTION

SIZE EXCLUSION / GEL FILTRATION

Mixed-Mode / Multimodal Chromatography Resins and Media



MIXED-MODE



HI CATION
EXCHANGE



HI ANION
EXCHANGE



CHT CERAMIC
HYDROXYAPATITE

Mixed-mode (multimodal) chromatography depends on more than one interaction principle occurring between stationary phase and solute. Both the interactions lead to retention of the solutes, which distinguishes this technique from conventional single-mode chromatography. The multiple interactions between stationary phase and solutes can lead to unique selectivity and facilitate separation of closely related proteins and contaminants.

Selectivity during binding, washing, and elution can be achieved by manipulating buffer components, conductivity, and/or pH. Modulation of each interaction type must consider buffer effects on both the resin structure and protein. For example, the charge on the (typically weak) IEX groups on the solid support can be changed from neutral to positive (for amino-based ligands) or negative (for carboxyl-based ligands) by varying the pH of the buffer. In addition, the charge on those amino acids, which specifically bind to the IEX functionalities, can be similarly changed. In a case where the charge on the protein and ligand are opposite, binding is enhanced. When the ligand and protein share the same charge, elution is enhanced. Similarly, in a situation where both ligand and the protein binding site are neutral, hydrophobic interactions are enhanced. Generally, when the two entities are of opposite charge, hydrophobic interactions near the charged amino acids are promoted. The opposite is true when both sites have the same charge. Varying the salt concentration generally has opposite effects on ionic and hydrophobic interactions. Increasing the salt concentration reduces ionic interactions but promotes hydrophobic interactions; this is the basis for the relatively constant binding capacity of mixed-mode resins over a fairly broad conductivity range. Finally, adding a variety of mobile-phase modifiers can also affect the interactions with Nuvia bead-based resins (Table 1).

Table 1. Effect of modifiers on interactions between biomolecules and Nuvia Resins.

Modifier	Modulated Interaction
Arginine	Electrostatic ↓
	Hydrophobic ↓
	H-bonding ↓
Guanidine HCl	Electrostatic ↓
	Hydrophobic ↓
	H-bonding ↓
Urea	Hydrophobic ↓
	H-bonding ↓
Propylene glycol	Electrostatic ↑
	Hydrophobic ↓
Glycine	Electrostatic ↓
	Hydrophobic ↑

↓ interaction is weakened.

↑ interaction is strengthened.

ADVANTAGES

Mixed-mode interactions have numerous advantages beyond simply being able to discriminate between product and impurities. They include:

- Improved yield and activity via advantageous use of charge-charge repulsion between ligand and protein
- Improved design space for protein binding
- Minimal feed manipulation prior to binding
- Mild condition to preserve activity

Bio-Rad offers three different families of mixed-mode chromatography resins — Nuvia Resins, CHT Ceramic Hydroxyapatite Media, and Bio-Gel Crystalline Hydroxyapatite Media.

Table 2 lists the various interaction principles of these mixed-mode resins.

Table 2. Chromatography mode interaction principles.

Mode	Principle	Nuvia cPrime Mixed-Mode Resin	Ceramic and Crystalline Hydroxyapatites	Nuvia aPrime 4A Mixed-Mode Resin
CEX	Electrostatic binding	X	X	
AEX	Electrostatic binding			X
HIC	Hydrophobic complex formation	X		X
C-sites	Calcium chelate formation		X	
Hydrogen bonding	Hydrogen bonding	X		
Phosphate-specific interactions			X	

NUVIA MIXED-MODE RESINS

The Nuvia bead is built on a polymeric base matrix that delivers low backpressure at high flow rates. Nuvia Mixed-Mode Resins are designed with a mixed-mode ligand that provides a unique balance between hydrophobic and charged characteristics. Both Nuvia Mixed-Mode Resins have high salt tolerance and provide a wide design space for effective purification of established therapeutic proteins as well as the increasingly diverse new constructs that are in development, many of which lack an affinity handle.

CHT CERAMIC HYDROXYAPATITE MEDIA

CHT Ceramic Hydroxyapatite ($\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$) is a spherical, macroporous form of hydroxyapatite, a biodegradable, naturally occurring mineral form of calcium apatite. It has been sintered at high temperatures to modify it from a nanocrystalline to ceramic form. Unlike most other chromatography adsorbents, CHT is both the ligand and the support matrix.

BIO-GEL HYDROXYAPATITE HT AND HTP MEDIA

Bio-Gel crystalline hydroxyapatite is chemically and thermally stable, is compatible with a wide range of aqueous and organic solvents, and can be sanitized in NaOH. It has a pH tolerance of >5.5 and is autoclavable. Three types of resins are available:

- Bio-Gel HT Gel — ideal for the purification of proteins, nucleic acids, and other macromolecules, Bio-Gel HT is prepared by the method of Tiselius (1956). It is suspended in 10 mM sodium phosphate buffer containing 0.02% NaN_3 and should be stored at 4°C
- Bio-Gel HTP Gel — this is the material described by Tiselius, which has been dried by a unique process developed by Bio-Rad. It may be stored without refrigeration. When suspended in buffer, it has characteristics similar to Bio-Gel HT Gel; 1 g hydrates to 2–3 ml
- DNA grade Bio-Gel HTP Gel — a dry powder with a smaller particle size than HTP gel, this gel significantly increases its capacity and enhances its selectivity for double-stranded DNA. Due to its small particle size, it is recommended for batch mode chromatography or for use with very short columns

Bio-Rad's complete range of mixed-mode media offerings are shown in Table 3.

Table 3. Bio-Rad mixed-mode media offerings.

Resin	Interaction Type
Nuvia cPrime Resin	Aromatic hydrophobic weak cation exchange resin (HIC-CEX)
Nuvia aPrime 4A Resin	Aromatic hydrophobic weak anion exchange resin (HIC-AEX)
Ceramic Hydroxyapatite Media CHT Type I, Type II, and XT	Cation exchange calcium affinity resin
Bio-Gel HT and HTP Media	Cation exchange calcium affinity resin

REFERENCE

Tiselius A (1956). Recent advances in chromatography and electrophoresis of proteins and polypeptides. Rend Ist Sup Sanit 19 (suppl), 215–230. In Italian.

Nuvia cPrime Resin



HI CATION
EXCHANGE

RESIN FUNCTIONS

- Simultaneous high purity and recovery under mild conditions during bind-elute or flow-through purification, improves process productivity and process economy
- Effective for initial capture and polish applications
- Salt tolerant, which enables capture purification without feedstream dilution, leading to increased time and cost efficiency

RESIN CHARACTERISTICS

- Designed with a mixed-mode ligand that provides a unique balance between hydrophobic and charged characteristics and also provides an opportunity for hydrogen-bonding interactions
- Wide design space; can operate across a wide range of salt concentrations and pH
- Straightforward method development
- Fast mass transfer allows low residence time without loss of DBC

SUPPORTING INFORMATION AND APPLICATION EXAMPLES

- Product information sheet, bulletin 6242
- A Purification Strategy for Clinical-Grade Monoclonal Antibody Using Hydrophobic Cation Exchange Chromatography, bulletin 6241
- Development of an Efficient Manufacturing Process for Adenovirus, bulletin 6719
- Single-Step Purification of Raw Starch Digesting α -Amylase Using Nuvia cPrime Hydrophobic Cation Exchange Resin, bulletin 6992
- Improved Process Economics of HUMIRA Biosimilar Purification with Ion Exchange and Mixed-Mode Resins, bulletin 7130
- Purification of Recombinant Proteins on Nuvia cPrime Hydrophobic Cation Exchange Media: A Simple Approach to Method Development, bulletin 6418
- Using Design of Experiments (DOE) with Nuvia cPrime to Elucidate Capture Conditions of Recombinant Proteins, bulletin 7128
- Instruction manual, document 10023853

Bead Properties

Property	Description
Function	Mixed-mode, hydrophobic interaction and cation exchange
Functional group	Aromatic hydrophobic weak cation exchanger
Base matrix composition	Polyacrylamide
Mean particle size	70 μm
Ionic capacity	123 $\pm$ 20 $\mu\text{eq/ml}$
Dynamic binding capacity	>40 mg hlgG/ml (at 10% BT) at 300 cm/hr >60 mg lactoferrin/ml at 300 cm/hr
Recommended linear flow rate	50–600 cm/hr
Compression factor (settled bed volume/packed bed volume)	1.15–1.20
pH stability	Short term: 3–14 Long term: 4–13
Shipping solution	20% ethanol + 30 mM Na_2SO_3 , or 2% benzyl alcohol
Regeneration	1 N NaOH
Sanitization	1 N NaOH
Storage conditions	0.1 N NaOH
Chemical stability	25% acetic acid, 8 M urea, 6 M guanidine HCl, 6 M potassium thiocyanate, 3 M sodium chloride, 1% Triton X-100, 2% SDS, 70% ethanol, 30% isopropyl alcohol
Shelf life	5 years

High Binding Capacity

Nuvia cPrime is designed for versatile capture and high recovery at high flow rates (Figure 1) and across a wide range of salt concentrations and pH (Figure 2). These properties, summarized in Table 1, may allow for direct loading without the need for dilution. Integrating a Nuvia cPrime step into a multicolumn process is operationally simple.

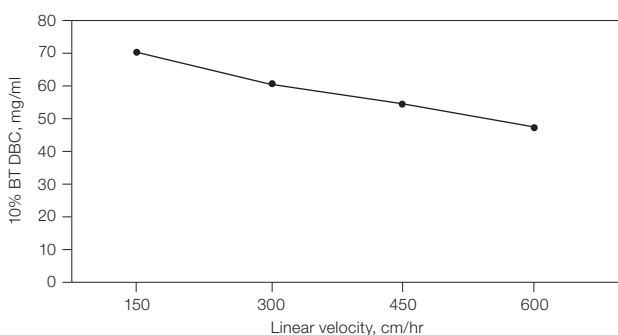


Fig. 1. DBC vs. linear velocity of Nuvia cPrime Resin. A 1.1 x 9.6 cm column was loaded with 5.25 mg/ml lactoferrin in 20 mM NaOAc + 150 mM NaCl, pH 4.5, until 10% BT was observed.

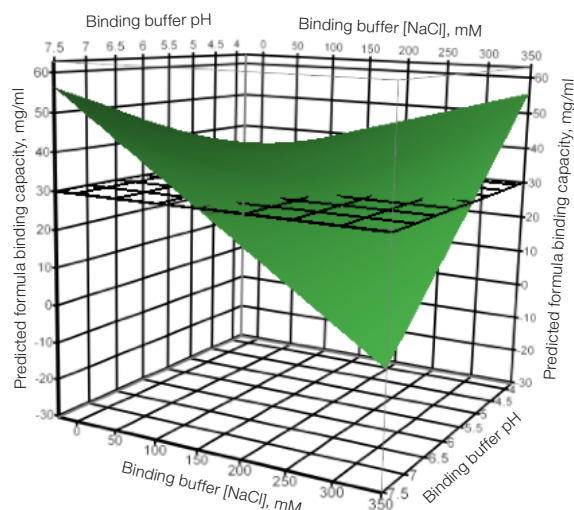


Fig. 2. Predicted binding capacity of Nuvia cPrime for mAb2 at varying pH and NaCl concentrations.

Table 1. mAb X binding capacity and recovery as a function of Nuvia cPrime flow rate.

Flow Rate, cm/hr	DBC, 10% BT, mAb X, mg/ml	Recovery, %
150	40	88
200	33	85
250	30	80

Good Pressure/Flow Performance

Nuvia cPrime is built on a porous polymeric base matrix that delivers low backpressure at high flow rates (Figure 3). It is also chemically and mechanically stable. Fast mass transfer dynamics ensure efficient chromatography at high flow, making Nuvia cPrime Resin an operationally superior choice for commercial-scale applications.

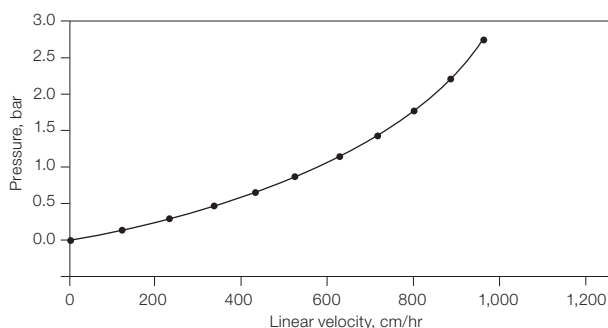


Fig. 3. Nuvia cPrime displays low backpressure at high flow rates. Flow performance of Nuvia cPrime Resin in a Bio-Rad InPlace Column. A 20 x 20 cm column with 1.17 axial compression was used.

Performance Data

The balance between hydrophobicity, charged interaction, and the highly hydrophilic base matrix of Nuvia cPrime empowers method developers with new ways to directly exploit various modes of interaction. This allows purification of challenging or sensitive proteins or separation of closely related protein species, such as isoforms and variants from posttranslational modifications and product aggregates and fragments.

Partially purified mAb1 from a Nuvia Q polishing step can be loaded onto a Nuvia cPrime Column following a simple pH adjustment to positively charge mAb1, without extensive dilution or buffer exchange for conductivity reduction. A high concentration of salts, such as $(\text{NH}_4)_2\text{SO}_4$, was not needed for promoting the binding of target protein either. Such behavior makes Nuvia cPrime distinctive from conventional hydrophobic interaction resins, which require high salt levels for effective protein binding. Under the selected condition, only the desired full-length mAb1 was retained by the column (Figure 4A). A 25 kD L-chain fragment of this antibody appeared in the column flow-through fractions, as visualized on SDS-PAGE (Figure 4B) and identified by mass spectrometry (data not shown) and HPLC-SEC (Figure 4C).

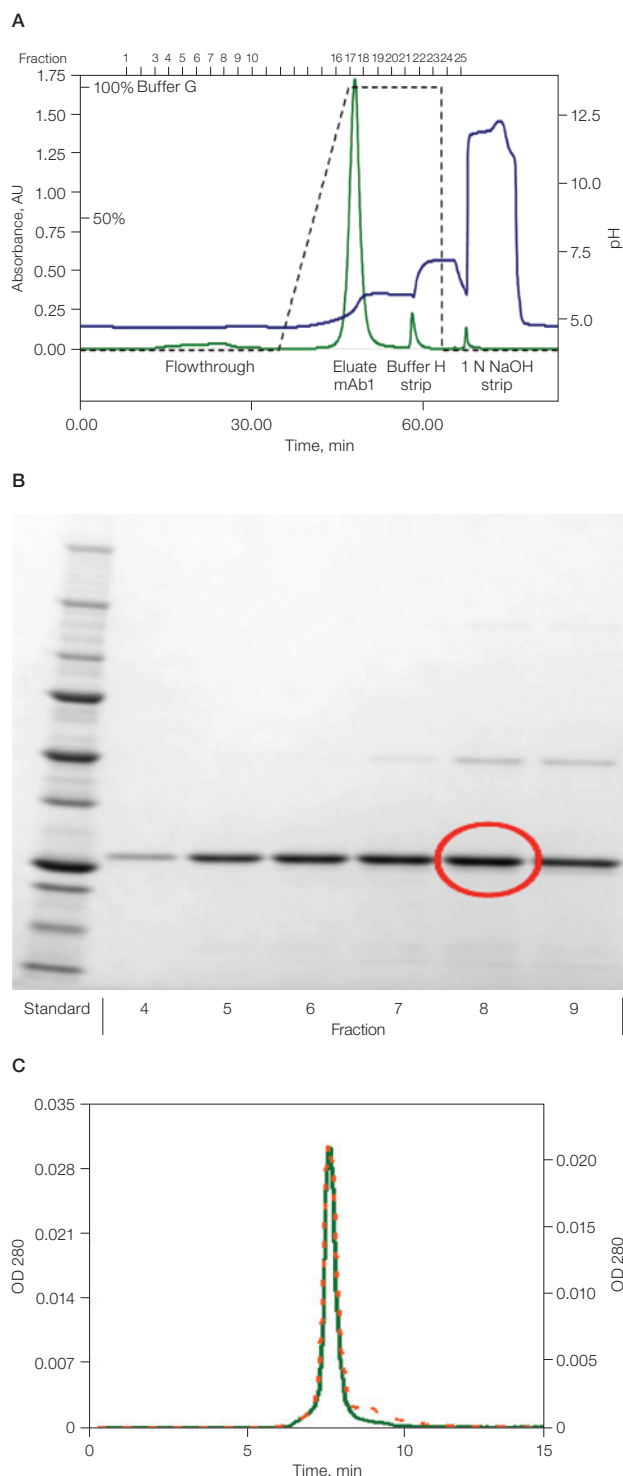


Fig. 4. Final polishing purification of mAb1 using Nuvia cPrime. **A**, chromatogram of mAb1 purification on a Nuvia cPrime Column. OD 280 (—); pH (—); percentage of buffer G (---). mAb1 eluted from the column (fractions 17–19) was collected and subjected to purity analysis. **B**, SDS-PAGE analysis of column flowthrough (fractions 4–9); Precision Plus Protein Unstained Standards were used. A major protein impurity was excised from the gel (circled in red) and analyzed by MALDI-TOF mass spectrometry. **C**, HPLC-SEC comparison of pooled Nuvia Q fractions (---) and pooled Nuvia cPrime fractions (—).

Process Workflow

The mixed-mode nature of the Nuvia cPrime ligand and its associated range of interactions allows for a directed and intuitive approach to method development and process

optimization (Figure 5). Alternatively, a simple DOE exercise will quickly guide developers to optimum loading, wash, and elution conditions afforded by the resin's large design space.

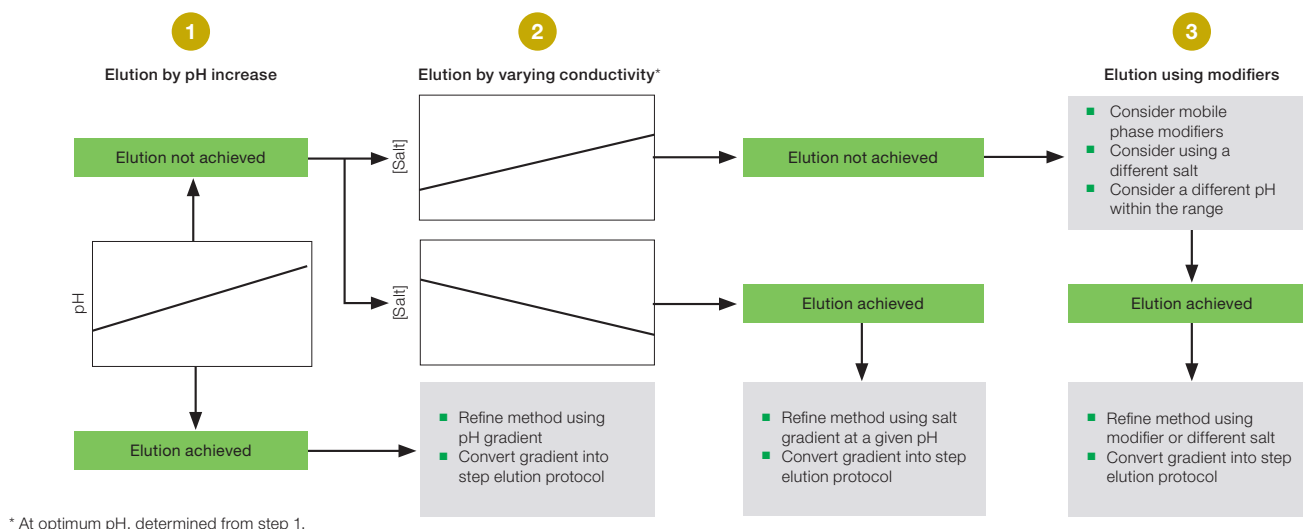


Fig. 5. Recommended approach to method development.

Purification Workflows

Application	Capture Step Resins	Intermediate Step Resins	Polish Step Resins/Media
mAb	Nuvia S	Nuvia Q	Nuvia cPrime
Enzyme	Nuvia cPrime	Nuvia Q	CHT Ceramic Hydroxyapatite
Protein	Nuvia IMAC	Nuvia cPrime	CHT Ceramic Hydroxyapatite
Adenovirus	Nuvia cPrime	—	Nuvia Q

Ordering Information

Catalog #	Description
Bulk Resins	
1563401	Nuvia cPrime Media, 25 ml
12003716	Nuvia cPrime Media (in BA), 25 ml
1563402	Nuvia cPrime Media, 100 ml
12003717	Nuvia cPrime Media (in BA), 100 ml
156-3403	Nuvia cPrime Media, 500 ml
12003718	Nuvia cPrime Media (in BA), 500 ml
156-3404	Nuvia cPrime Media, 1 L
12003719	Nuvia cPrime Media (in BA), 1 L
156-3405	Nuvia cPrime Media, 5 L
12003720	Nuvia cPrime Media (in BA), 5 L
156-3406	Nuvia cPrime Media, 10 L
12003721	Nuvia cPrime Media (in BA), 10 L

BA, 2% benzyl alcohol.

Catalog #	Description
Prepacked Screening Tools	
12009315	EconoFit Aggregate Clearance Kit , includes 1 ml each of Nuvia cPrime, Nuvia aPrime 4A, Nuvia HR-S, CHT XT Media
12009314	EconoFit Large Biomolecule Kit , includes 1 ml each of Nuvia Q; Nuvia HP-Q; Nuvia cPrime; Nuvia aPrime 4A; CHT II, 40 µm; and CHT XT Media
12009313	EconoFit mAb Kit , includes 1 ml each of Nuvia S, Nuvia Q, Nuvia cPrime, Nuvia aPrime 4A, Nuvia HR-S, and CHT XT Media
12009281	EconoFit Nuvia cPrime Column , 1 x 1 ml, 7 x 25 mm
732-4705	Foresight Nuvia cPrime Plates , 2 x 96-well, 20 µl
732-4807	Foresight Nuvia cPrime RoboColumn Unit , 200 µl
732-4808	Foresight Nuvia cPrime RoboColumn Unit , 600 µl
732-4722	Foresight Nuvia cPrime Column , 1 ml
732-4742	Foresight Nuvia cPrime Column , 5 ml

Case Studies

- mAb purification, page 131
- Adenovirus purification, page 151

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Nuvia aPrime 4A Resin



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RESIN FUNCTIONS

- Simultaneous high purity and recovery under mild conditions during bind-elute or flow-through purification
- Condenses multiple steps into one; improves process productivity and process economy
- Salt tolerant, which enables minimum feedstream manipulation, leading to better time and cost efficiency

RESIN CHARACTERISTICS

- Ligand density and hydrophobicity are designed to facilitate selective and reversible binding of target molecules for maximal purity and recovery
- Wide design space; can operate across a wide range of salt concentrations and pH
- Straightforward method development

SUPPORTING INFORMATION AND APPLICATION EXAMPLES

- Product information sheet, bulletin 7142
- Effects of Buffer Composition on Protein Purification with a Hydrophobic Anion Exchange Resin, Nuvia aPrime 4A, bulletin 7207
- Purification of Antibodies with Nuvia aPrime 4A Hydrophobic Anion Exchange Resin, bulletin 7193
- Purification of a Recombinant Bacterial DyP-Type Peroxidase with a Hydrophobic Anion Exchange Resin, bulletin 7182
- Efficient Purification of a Recombinant Protein from *Escherichia coli* Fermentate with Nuvia aPrime 4A Resin, bulletin 7302
- Maximize Monomer Purity and Recovery during mAb Purification flier, bulletin 7135
- Get the Best Monomer Recovery during Protein Purification flier, bulletin 7136
- Get the Best HCP and Aggregate Clearance during mAb Purification flier, bulletin 7310
- Virus Reduction and Impurity Removal Potential of Nuvia aPrime 4A Resin, bulletin 3254
- Instruction manual, document 10000098359

Go to [bio-rad.com/Library](https://www.bio-rad.com/Library) to access these webinars and videos:

- AEX Mixed-Mode Chromatography Resin for High Selectivity and Recovery video
- Mixed-Mode Chromatography — Optimizing Target Purity and Recovery with Buffer Additives
- A New Mixed-Mode Media for Biomolecule Purification
- Two-Step Purification of Low-Expressing Recombinant Exoprotein A Using Mixed-Mode Chromatography
- A Mixed-Mode Approach — Alternative to a Three-Step mAb Purification Process
- What You Need to Know About Packing Resins at Process Scale
- Mixed-Mode Chromatography Resins for Biomolecule Purification
- Integrating Viral Clearance into Your Process Development — A Case Study Using a Mixed-Mode Chromatography Resin video
- Assessing Viral Clearance in Early Phase Process Development

Bead Properties

Property	Description
Function	Mixed-mode, hydrophobic interaction and anion exchange
Functional group	Aromatic hydrophobic strong anion exchanger
Base matrix composition	Polyacrylamide
Mean particle size	50 µm
Ionic capacity	80–120 µeq/ml
Dynamic binding capacity (DBC)*	≥50 mg/ml at 300 cm/hr
Recommended linear flow rate	50–300 cm/hr
Pressure/flow performance**	<3 bar at flow rate of 300 cm/hr
Compression factor	1.1–1.25
pH stability***	Short term: 2–14
Shipping solution	20% ethanol + 1 M NaCl
Regeneration	1 M NaOH
Sanitization	1 M NaOH
Storage conditions	20% ethanol
Storage temperature	Room temperature
Chemical stability†	1 M NaOH, 1 M HCl, 25% acetic acid, 8 M urea, 6 M guanidine HCl, 3 M NaCl, 1% Triton X-100, 20–70% ethanol, 30% isopropanol
Shelf life††	5 years

* 10% breakthrough capacity determined with 1.0 mg/ml of an acidic monoclonal antibody (mAb; pI ~6.9) in 20 mM NaPO₄, pH 7.8.

** 20 x 20 cm packed bed (1.25 compression factor).

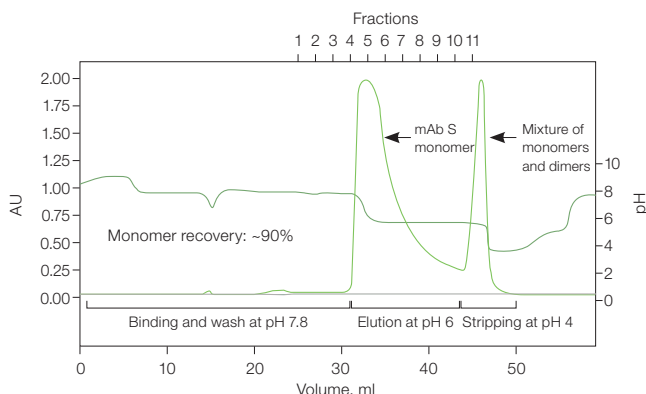
*** No significant change in ligand density after 200 hr contact at 22°C.

† No significant change in ligand density after 1 week contact at 22°C.

†† Stored at room temperature in 20% ethanol + 1 M NaCl.

Simultaneous High Purity and Recovery

Nuvia aPrime 4A Resin's ability to clear high molecular weight species from an acidic mAb (pI ~6.9) feedstream was tested. At pH 7.8, Nuvia aPrime 4A interacts with the mAb through both electrostatic and hydrophobic interactions. The mAb monomer was eluted at the mild condition of ~pH 6 with an overall recovery of ~90%. The high molecular weight impurities were retained until the resin was stripped with a pH 4 buffer (Figure 1).



Sample	Monomer, %	Dimer, %	Tetramer, %
Load	72.4	21.7	5.9
Eluate	100	ND	ND

Fig. 1. Purification of mAb S (pI ~6.9) on Nuvia aPrime 4A Resin. A solution containing ~10 mg of target mAb in 20 mM NaPO₄, pH 7.8, was injected onto a 1 ml column and eluted by a buffer at pH 6. Clear separation of the high molecular weight impurities from the monomeric fraction was observed. AU (—); pH (—). AU, absorbance units; ND, not done.

Straightforward Method Development

The mixed-mode nature of the Nuvia aPrime 4A ligand facilitates straightforward method development and process optimization. A simple DOE protocol can also be used by method developers to optimize their process conditions. The ease of method development for purification of a basic protein with a pI of ~8.45 was tested with the goal to find the optimal condition for the highest monomer recovery and purity. A fractional factorial screening design with two center points was performed. Binding capacity was low under all tested conditions (Figure 2A), suggesting that flow-through purification was more appropriate for this protein. Figures 2B and 2C show the high molecular weight impurity clearance and monomer recovery in response to purification buffer composition.

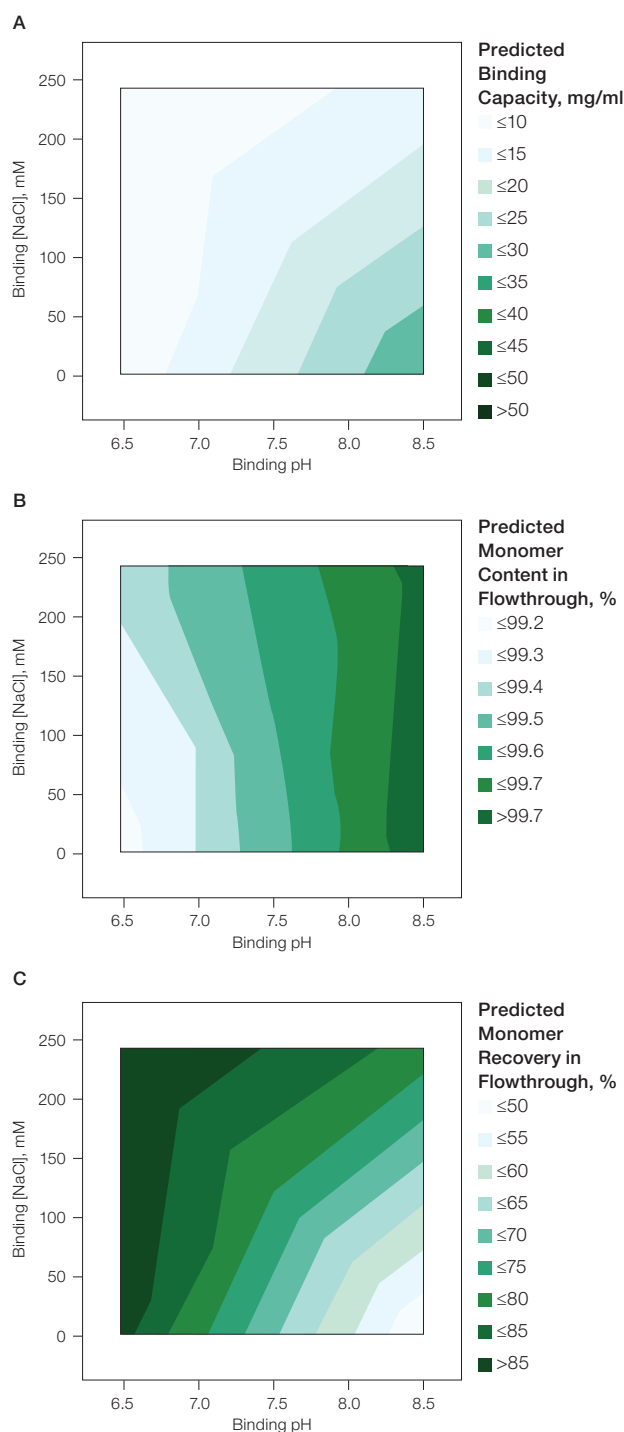


Fig. 2. Basic protein (pI ~8.45) purification using Nuvia aPrime 4A Resin. **A**, effect of binding buffer pH and NaCl on the binding of the basic protein by Nuvia aPrime 4A; **B**, effect of binding buffer pH and NaCl on monomer content in the flow-through fraction; **C**, effect of binding buffer pH and NaCl on monomer recovery in the flow-through fraction.

Good Pressure/Flow Performance

Nuvia aPrime 4A Resin is designed with a bead size optimized to achieve fast mass transfer dynamics and process-scale purification of biomolecules at high flow rates without being limited by column pressure. The column pressure remains below 3 bar up to a linear velocity of 350 cm/hr (Figure 3).

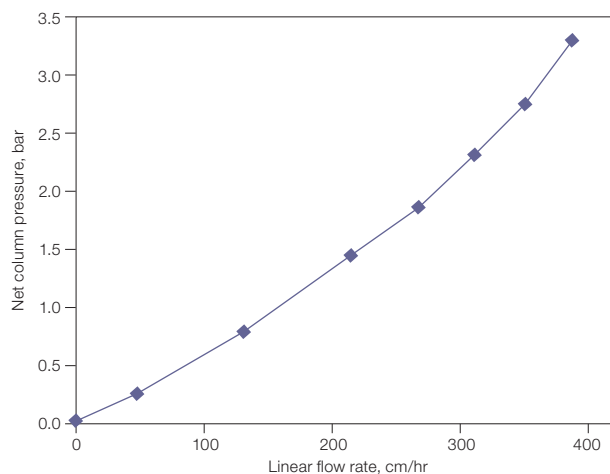


Fig. 3. Pressure/flow performance of Nuvia aPrime 4A Resin. Nuvia aPrime 4A Resin slurry prepared in 1x PBS, pH 7.5, was packed into a 20 x 20 cm column by axial compression at 300 cm/hr with a compression factor of 1.25.

Robust Performance and Recovery

Nuvia aPrime 4A is built on a mechanically stable base bead. The chemical stability of Nuvia aPrime 4A allows the resin to perform consistently with minimal changes to dynamic binding capacity or recovery even after multiple exposures to NaOH (Figure 4).

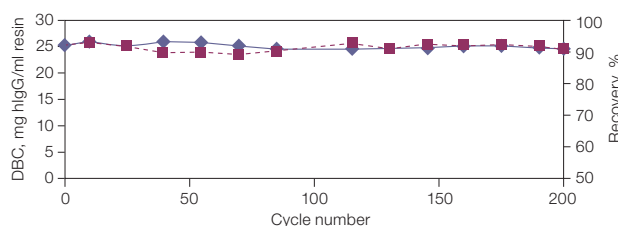
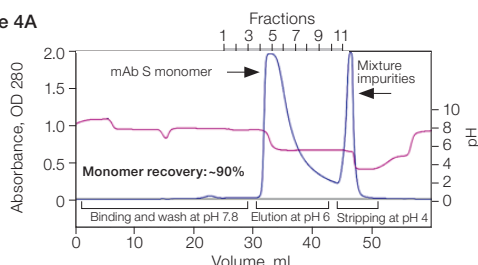


Fig. 4. Stability, reusability, and recovery with Nuvia aPrime 4A Resin. Polyclonal hlgG (1.0 mg/ml) in equilibration buffer (50 mM Na₂PO₄, pH 8.0) was loaded onto a 1 ml Bio-Scale Mini Column (0.56 x 4 cm) packed with Nuvia aPrime 4A. The column was operated at 150 cm/hr. hlgG was eluted with 100 mM sodium citrate, pH 3.0. The column was cleaned in place for 30 min with 1 M NaOH. The 10% DBC and recovery of hlgG at linear velocity of 150 cm/hr was determined after every ten cycles. DBC (—◆—); recovery (---■---). DBC, dynamic binding capacity; hlgG, human IgG.

Competitive Data

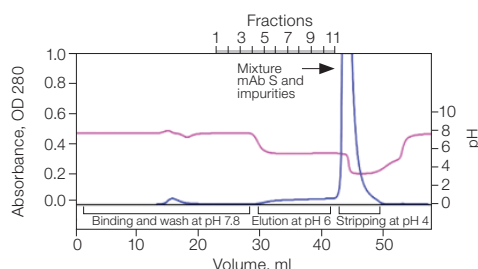
The efficiency of three commercial mixed-mode resins for the purification of mAb S (approximate pI of 6.9) from a UNOsphere SUPrA eluate containing about 25% aggregated mAb was tested. Only Nuvia aPrime 4A Resin produced a mAb S elution at a mild pH of 6.0 with a monomer recovery of about 90% (Figure 5A). The two other resins showed elution of the antibody only at pH 4.0 and with significant contamination from dimers and aggregates (Figures 5B and 5C).

A. Nuvia aPrime 4A



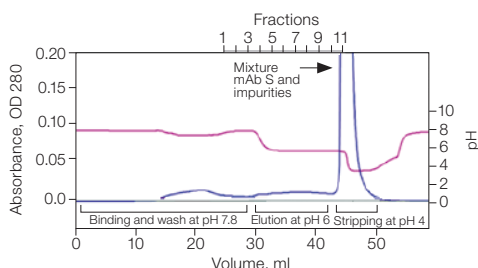
Sample	Monomer, %	Dimer, %	Tetramer, %
Load	72.4	21.7	5.9
Eluate	100	ND	ND

B. Resin 1



Sample	Monomer, %	Dimer, %	Tetramer, %
Load	72.4	21.7	5.9
Eluate	Target not eluted at pH 6		

C. Resin 2

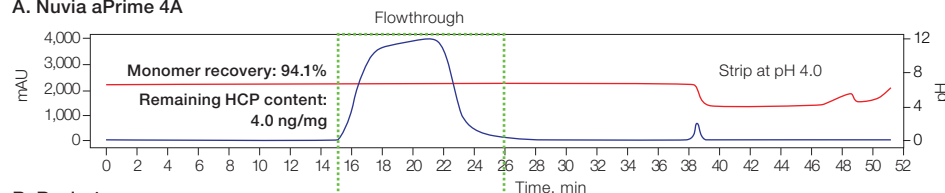


Sample	Monomer, %	Dimer, %	Tetramer, %
Load	72.4	21.7	5.9
Eluate	Target not eluted at pH 6		

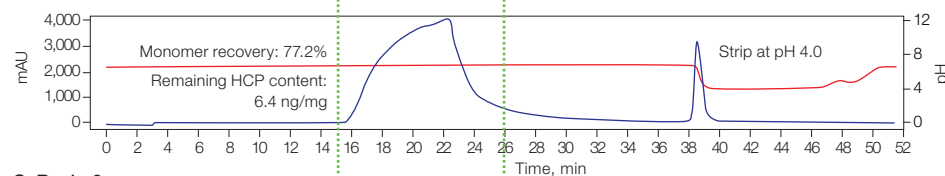
Fig. 5. Elutions of mAb S at pH 6.0. **A**, with Nuvia aPrime 4A, monomer recovery reaches 90%. **B** and **C**, with two commercially available resins, mAb S is not eluted at pH 6.0. Bound mAb S is stripped from the column at pH 4.0 in a mixture with dimers and aggregates. OD 280 (—); pH (—).

A DOE study showed that Nuvia aPrime 4A offers better monomer recovery than other commercially available resins for the purification of a basic protein (pI ~8.4) at the optimal FT operation pH of 6.5 (Figure 6). Nuvia aPrime 4A Resin's superior performance is the result of its optimized pore structures and surface properties, which offer efficient mass transfer and minimal nonspecific interaction between biomolecules and this chromatography media.

A. Nuvia aPrime 4A



B. Resin 1



C. Resin 2

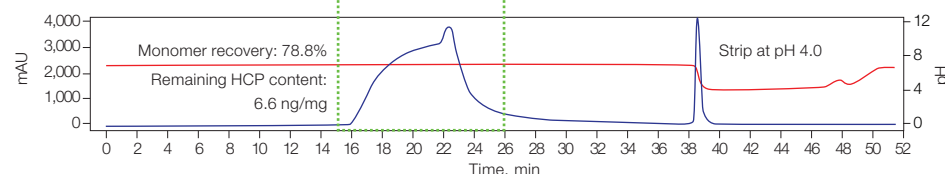


Fig. 6. Chromatograms from the respective 1 ml columns with monomer recovery and remaining HCP content indicated.

FT conditions: 99% monomer content in sample load, 150–200 ng/mg starting HCP content, 20 mM NaPi, 100 mM NaCl, pH 6.5. OD 280 (—); pH (—).

Process Workflow

The schematic in Figure 7 outlines a general method development rationale. In most cases, conducting a few simple designs of experiments to identify optimal binding and elution conditions will yield an effective, robust, and scalable method.

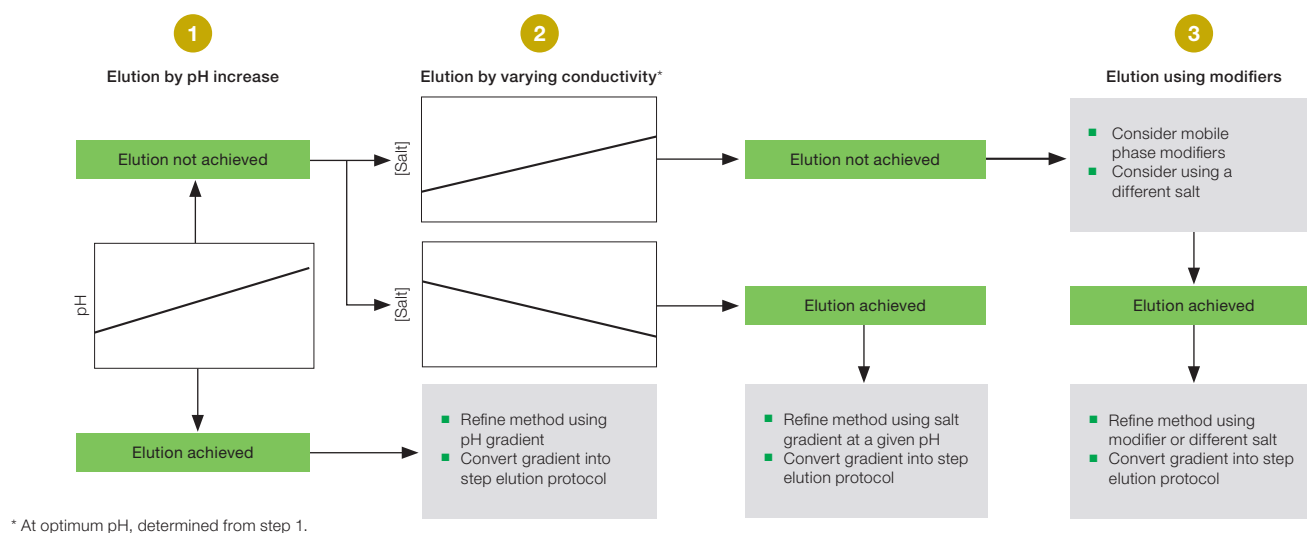


Fig. 7. Recommended approach to method development.

Purification Workflow

Application	Capture Step Resins	Intermediate Step Resins	Polish Step Resins/Media
Recombinant protein	Nuvia aPrime 4A	—	CHT Ceramic Hydroxyapatite
Enzyme	Nuvia IMAC	—	Nuvia aPrime 4A

Ordering Information

Catalog # Description

Bulk Resins

12007397	Nuvia aPrime 4A Media , 25 ml
12007396	Nuvia aPrime 4A Media , 100 ml
12007379	Nuvia aPrime 4A Media , 500 ml
12007380	Nuvia aPrime 4A Media , 5 L
12007391	Nuvia aPrime 4A Media , 10 L

Large volumes and special packaging for industrial applications are available upon request.

Prepacked Screening Tools

12009315	EconoFit Aggregate Clearance Kit , includes 1 ml each of Nuvia cPrime, Nuvia aPrime 4A, Nuvia HR-S, CHT XT Media
12009314	EconoFit Large Biomolecule Kit , includes 1 ml each of Nuvia Q; Nuvia HP-Q; Nuvia cPrime; Nuvia aPrime 4A; CHT II, 40 µm; and CHT XT Media
12009313	EconoFit mAb Kit , includes 1 ml each of Nuvia S, Nuvia Q, Nuvia cPrime, Nuvia aPrime 4A, Nuvia HR-S, and CHT XT Media
12009280	EconoFit Nuvia aPrime 4A Column , 1 x 1 ml, 7 x 25 mm
12007411	Foresight Nuvia aPrime 4A Plates , 2 x 96-well, 20 µl
12007394	Foresight Nuvia aPrime 4A RoboColumn Unit , 200 µl
12007395	Foresight Nuvia aPrime 4A RoboColumn Unit , 600 µl
12007392	Foresight Nuvia aPrime 4A Column , 1 ml
12007393	Foresight Nuvia aPrime 4A Column , 5 ml

Case Studies

- DyP purification, page 166

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MIXED-MODE / MULTIMODAL

AFFINITY

HYDROPHOBIC INTERACTION

SIZE EXCLUSION / GEL FILTRATION

CHT Ceramic Hydroxyapatite Media



CHT CERAMIC
HYDROXYAPATITE

RESIN FUNCTIONS

- Unique separation properties with unparalleled selectivity and resolution that can often separate proteins shown to be homogeneous by electrophoretic and other chromatographic techniques
- Single-step removal of multiple impurities leads to increased efficiencies in process productivity and economy for downstream processing
- Removal of high molecular weight impurities such as aggregates and dimers, often to below detectable levels
- Diverse binding capabilities of CHT Media for a variety of biomolecules, including monoclonal antibodies and viruses, allow its use throughout the downstream process

RESIN CHARACTERISTICS

- Unique separation properties are attributed to its unique composition, $\text{Ca}_{10}(\text{PO}_4)_6\text{OH}_2$, and its multiple interactions with biomolecules. Biomolecules can interact with CHT through metal affinity interactions via calcium atoms (C sites) and/or cation exchange interactions via phosphate groups (P sites)
- Different particle size and pore size options available to fit the application needs

SUPPORTING INFORMATION AND APPLICATION EXAMPLES

- CHT Type I and Type II product information sheet, bulletin 5667
- CHT XT product information sheet, bulletin 7080
- Foresight Pro Column Product Information Sheet, bulletin 3291
- How CHT Ceramic Hydroxyapatite Works, bulletin 5709
- How Does CHT Ceramic Hydroxyapatite Media Work poster, bulletin 6902
- Good Practices for CHT Ceramic Hydroxyapatite Media Usage poster, bulletin 6932
- Recommendations for CHT Ceramic Hydroxyapatite Column Packing poster, bulletin 6965
- Packing and Slurry-in-Place Procedures for CHT Ceramic Hydroxyapatite Using a Bio-Rad InPlace Process-Scale Chromatography Column, bulletin 5739
- Process-Scale Packing Procedures for CHT Ceramic Hydroxyapatite in Open Column Systems: Best Practices, bulletin 5831

continues

- High-Throughput Calcium Analysis of Hydroxyapatite Column Effluent, bulletin 6278
 - EDTA Complexometric Titration of Hydroxyapatite Column Effluent, bulletin 6067
 - Removing the Aggregates of an Acidic Monoclonal Antibody with CHT Ceramic Hydroxyapatite XT Media, bulletin 7109
 - Improve mAb Purification Cost Efficiency with the Right Resin flier, bulletin 7099
 - Maximize Monomer Recovery during mAb Purification flier, bulletin 7100
 - Development of a Non-Affinity Based Purification Platform for Neutral/Basic IgMs, bulletin 6966
 - Purification of Murine IgG₁ Using UNOsphere S and CHT Ceramic Hydroxyapatite Chromatography, bulletin 2780
 - High-Throughput Calcium Analysis of Hydroxyapatite Column Effluent, bulletin 6278
 - Purification of Transgenic Antibody from Corn Seed Using UNOsphere S and CHT Ceramic Hydroxyapatite Supports, bulletin 2774
 - Chimeric IgG Monoclonal Antibody Purification: A Comparative Study Using CHT Ceramic Hydroxyapatite and CFT Ceramic Fluoroapatite Chromatographic Media, bulletin 5853
 - Purification of Horse IgG_T Using Macro-Prep DEAE and CHT Ceramic Hydroxyapatite Type I Supports, bulletin 2524
 - Separation of Fab and Fc Fragments from Monoclonal Antibody Papain Digest on Ceramic Hydroxyapatite and Ceramic Fluoroapatite poster, bulletin 5913
 - Purification of an Acidic IgM with a Strong Anion Exchange Resin Tailored for Large Biomolecule Purification, bulletin 7127
 - Achieve High-Level Impurity Clearance during mAb Purification with a Single Media flier, bulletin 3225
 - Monoclonal Antibody Purification: Polish Purification Resins product information sheet, bulletin 6946
 - Single-Step Influenza and Dengue Virus Purification with Mixed-Mode CHT Ceramic Hydroxyapatite XT Media, bulletin 7115
 - Mammalian Virus Purification Using Ceramic Hydroxyapatite, bulletin 6549
 - Purification of Recombinant Human Alpha1-Antitrypsin from Transgenic Milk Using CHT Ceramic Hydroxyapatite, bulletin 2535
 - Purification of a Calcium-Binding Protein Using CHT Ceramic Hydroxyapatite, bulletin 2575
 - Purification of an Acidic Enzyme Using Ceramic Hydroxyapatite Chromatography: Effective Removal of Acidic Impurities, bulletin 7129
 - Plasmid Purification Using CHT Ceramic Hydroxyapatite Support, bulletin 2731
 - Plasmid DNA Adsorption Behavior on CHT Ceramic Hydroxyapatite Type II Support, bulletin 2946
 - Mixed-Mode Chromatography for mAb S Aggregate Removal: Comparison of CHT Ceramic Hydroxyapatite, Capto adhere, and Capto adhere ImpRes, bulletin 6749
 - Removal of Aggregate from an IgG₄ Product Using CHT Ceramic Hydroxyapatite Resin, bulletin 2940
 - Maximize Monomer Recovery during mAb Purification flier, bulletin 7100
 - Instruction manual, document 6086
- Go to **[bio-rad.com/Library](https://www.bio-rad.com/Library)** to access these webinars and videos:
- What You Need to Know About Packing Resins at Process Scale
 - Assessing Viral Clearance in Early Phase Process Development
 - Mixed-Mode Chromatography Resins for Biomolecule Purification

continues

- Virus Purification Strategies and Considerations
- CHT Column Packing Is Easier Than You Think
- CHT Prepacked Process-Scale Columns
- What You Need to Know About Packing Resins at Process Scale
- A Mixed-Mode Approach — Alternative to a Three-Step mAb Purification Process
- Integrating Viral Clearance into Your Process Development — A Case Study Using a Mixed-Mode Chromatography Resin

Types

There are three types of CHT available, all with the $\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$ backbone:

- CHT Ceramic Hydroxyapatite Types I and II are available in two particle sizes, 40 and 80 μm .* CHT Ceramic Hydroxyapatite XT is available in the 40 μm particle size
- CHT Type I has the highest overall protein binding capacity among the three varieties. It also has the best capacity for acidic proteins in general
- CHT XT is similar in application usage to CHT Type I and is a more stable matrix, providing significantly longer usable column lifetime at process scale
- CHT Type II, with its large pore size, is the most effective for purification of large biomolecules such as IgM and viruses

* CHT Ceramic Hydroxyapatite Types I and II are also available in a 20 μm particle size, though we do not recommend using this size in process-scale columns.

Bead Properties

Property	CHT Type I	CHT Type II	CHT XT
Function	Mixed-mode, cation (phosphate), and affinity (calcium)	Mixed-mode, cation (phosphate), and affinity (calcium)	Mixed-mode, cation (phosphate), and affinity (calcium)
Functional group	Ca^{2+} , PO_4^{3-} , OH^-	Ca^{2+} , PO_4^{3-} , OH^-	Ca^{2+} , PO_4^{3-} , OH^-
Mean particle size*	40, 80 μm	40, 80 μm	40 μm
Dynamic binding capacity	≥ 25 mg lysozyme/g CHT** 25–60 mg IgG/ml CHT*	≥ 12.5 mg lysozyme/g CHT** 15–25 mg IgG/ml CHT*	≥ 60 mg/ml mAb G*** (pl 9.2) 17–25 mg lysozyme/g†
Recommended linear flow rate	50–300 cm/hr	50–300 cm/hr	50–300 cm/hr
Tap-settled density (g/ml tap-settled bed)	0.63 g/ml	0.63 g/ml	0.67 g/ml††
Compression factor)	Incompressible	Incompressible	Incompressible
pH stability	6.5–14	6.5–14	6.5–14
Shipping solution	Dry	Dry	Dry
Regeneration	400 mM sodium phosphate, pH ≥ 7.5 0.4–1 M potassium phosphate, pH ≥ 7.5 1 M trisodium phosphate, pH 11–12	400 mM sodium phosphate, pH ≥ 7.5 0.4–1 M potassium phosphate, pH ≥ 7.5 1 M trisodium phosphate, pH 11–12	400 mM sodium phosphate, pH ≥ 7.5 0.4–1 M potassium phosphate, pH ≥ 7.5 1 M trisodium phosphate, pH 11–12
Sanitization	1 N NaOH	1 N NaOH	1 N NaOH
Autoclavability (bulk)	121°C, 20 min in phosphate buffered saline, pH 7	121°C, 20 min in phosphate buffered saline, pH 7	121°C, 20 min in phosphate buffered saline, pH 7
Recommended column storage	0.1 N NaOH	0.1 N NaOH	0.1 N NaOH
Chemical compatibility	Stable in all commonly used organic solvents Stable in all commonly used buffer components, except for chelating agents (citrate, EDTA, etc.) Stable in solutions at pH 6.5 or greater	Stable in all commonly used organic solvents Stable in all commonly used buffer components, except for chelating agents (citrate, EDTA, etc.) Stable in solutions at pH 6.5 or greater	Stable in all commonly used organic solvents Stable in all commonly used buffer components, except for chelating agents (citrate, EDTA, etc.) Stable in solutions at pH 6.5 or greater
Shelf life (dry, unused material)	5 years	5 years	5 years

* 40 μm particles, 300 cm/hr, 5 mM sodium phosphate, 25 mM NaCl, pH 6.5.

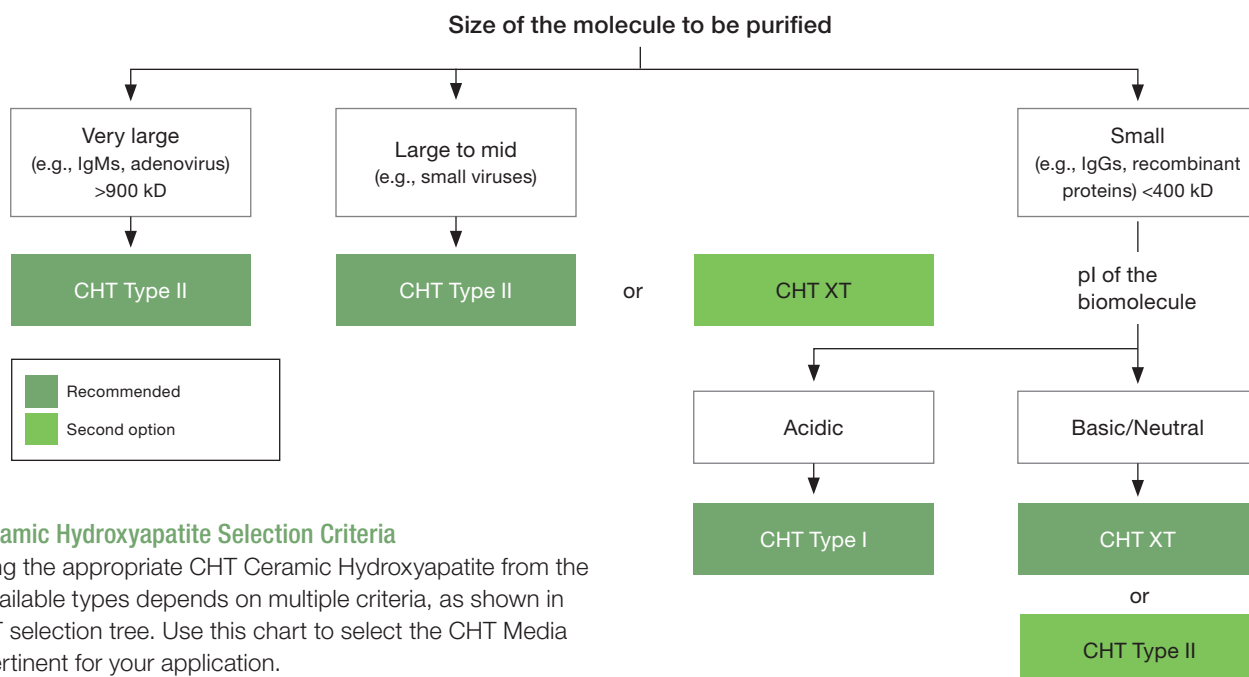
** 1 ml/min (~475–500 cm/hr).

*** At 100 cm/hr with 5 mM sodium phosphate, 25 mM sodium chloride, pH 7, in a 0.5 x 9.5 cm column.

† At 500 cm/hr with 10 mM sodium phosphate, pH 6.8.

†† Value from preliminary data.

Note: A small amount (up to 5 mM) of sodium phosphate should be added to all unbuffered solutions as a counterion.



CHT Ceramic Hydroxyapatite Selection Criteria

Choosing the appropriate CHT Ceramic Hydroxyapatite from the three available types depends on multiple criteria, as shown in the CHT selection tree. Use this chart to select the CHT Media most pertinent for your application.

Binding Capacity

Hydroxyapatite contains two types of binding sites: positively charged calcium and negatively charged phosphate groups. These sites are distributed throughout the crystal structure of the matrix. Solute species dominantly interact through CEX via the phosphate groups and/or through metal affinity via the calcium atoms.

CEX occurs when protein amino groups interact ionically with the negatively charged phosphates. The amino groups are similarly repelled by the calcium sites. Binding depends on the combined effects of these interactions. These IEX interactions can be disrupted by adding neutral salts, such as sodium chloride, or buffering species, such as phosphate, to the mobile phase. CEX interactions also weaken with increasing pH. Hence, the addition of salt or phosphate or an increase in pH can be used to weaken the interaction. Studies with model proteins have demonstrated that AEX, which might be expected from interactions of negatively charged surface residues with calcium, is negligible.

Calcium affinity occurs via interactions with carboxyl clusters and/or phosphoryl groups on proteins or other molecules (for example, nucleic acids); these groups are simultaneously repelled by the negative charge of the CHT phosphate groups. The affinity interaction is 15–60 times stronger than ionic interactions alone and, like classical metal-affinity interactions, is not affected by increasing ionic strength using typical elution ions (for example, chloride). Species binding through calcium affinity may adsorb more strongly as the ionic strength increases due to ionic shielding of the charge repulsion from the CHT phosphate sites. Metal affinity interactions can be dissociated by phosphate in the mobile phase.

Figure 1 illustrates CEX metal affinity to carboxyl and phosphoryl groups.

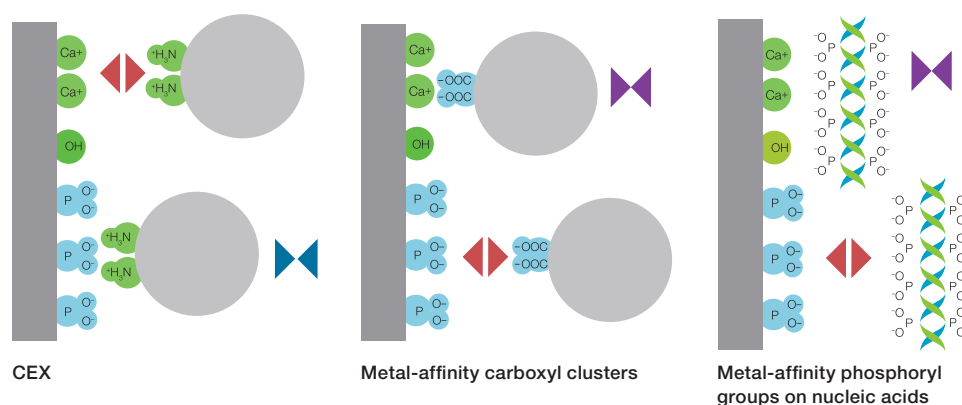


Fig. 1. Schematic representation of CHT binding mechanism. Biomolecule (●); metal affinity (●); electrostatic repulsion (↔); electrostatic attraction (↔).

Acidic proteins such as albumin bind chiefly by metal-affinity interactions. Sodium chloride at 1.0 M reduces retention time by approximately 10% in the presence of phosphate gradients, indicating a minor contribution by CEX. To elute acidic proteins, phosphate buffers are usually required.

Basic proteins such as IgG bind chiefly by CEX interactions and may be selectively eluted with either phosphate or salts.

Good Pressure/Flow Performance

CHT is a rigid sphere and typically exhibits low backpressure at high flow rates relative to its particle size (Figure 2). Figure 3 shows the effect that the different CHT particle sizes have on the separation of proteins.

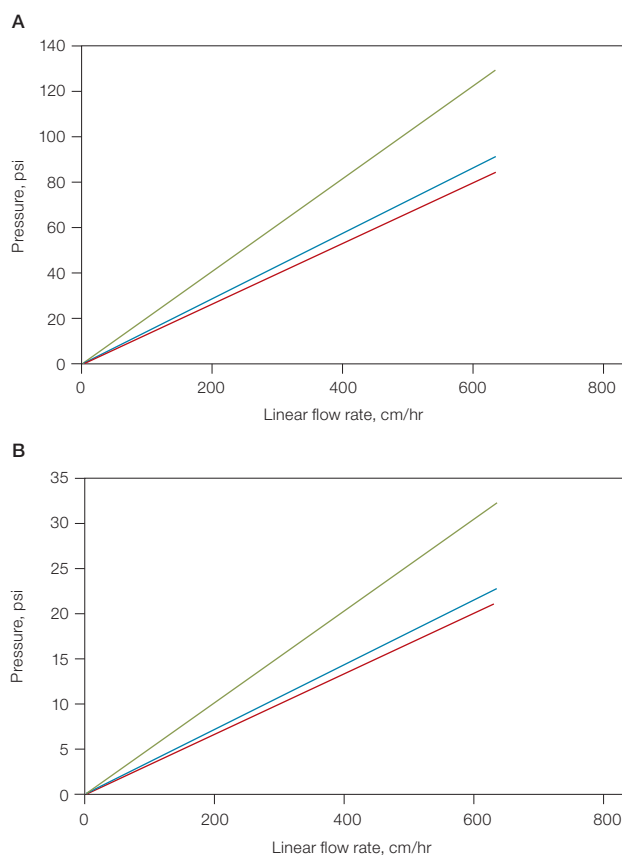


Fig. 2. Estimated pressure for 40 µm CHT (A) and 80 µm CHT (B). Media packed to 20.0 cm bed height vs. 1.0 M NaOH (1.80 cP), 0.5 M sodium phosphate buffer, pH 6.8 (1.27 cP), and PBS (1.18 cP). 1.18 cP (—); 1.27 cP (—); 1.80 cP (—). cP, centipoise.

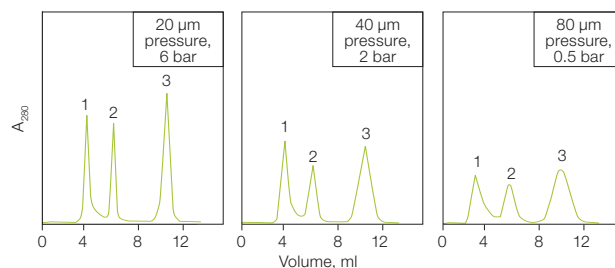
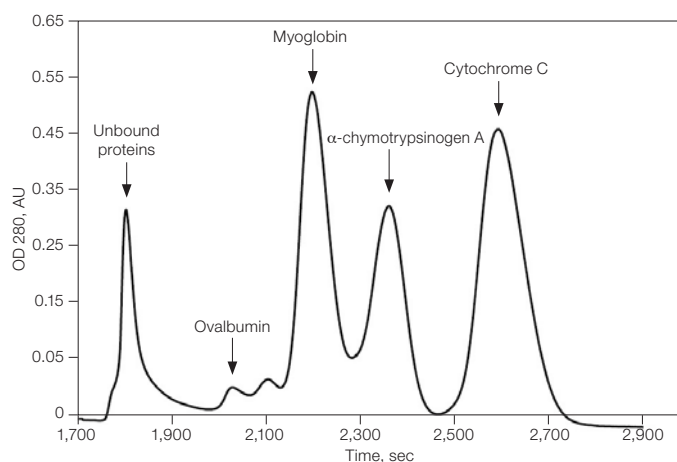


Fig. 3. Effect of particle size on separation of proteins. A 10 µl sample of 10 mg/ml BSA (peak 1), 1.3 mg/ml lysozyme (peak 2), and 5 mg/ml cytochrome c (peak 3) was run on each 4 x 100 mm column packed with the indicated particle size of CHT Ceramic Hydroxyapatite at a flow rate of 478 cm/hr. The elution buffer was a linear gradient of 1–400 mM sodium phosphate, pH 6.8, over 15 min.

Performance Data

CHT has unique separation properties with unmatched selectivity and resolution, including protein separations (Figures 4 and 5) and clearance (Figures 6 and 7; Tables 1 and 2).



Flow rate: 1.57 ml/min
 Pre-equilibrium: 400 mM sodium phosphate, pH 6.8, 4 CV
 Loading buffer: 5 mM sodium phosphate, pH 6.8, 15 CV
 Post-load wash: 5 mM sodium phosphate, pH 6.8, 1 CV
 Elution: Linear gradient elution 0–75% 400 mM sodium phosphate, pH 6.8, over 15 CV
 Stripping: 400 mM sodium phosphate, pH 6.8, 3 CV
 Sanitization: 1 M sodium hydroxide, 3 CV

Fig. 4. Separation of protein standards. A mixture of the following protein samples in 10 ml of 5 mM sodium phosphate, pH 6.8, was used for purification: 120 mg of ovalbumin, 75 mg of myoglobin, 60 mg of α-chymotrypsinogen A, 75 mg of cytochrome C. One hundred microliters of this mixture was loaded onto a 0.7 x 5.4 cm column with a packed bed volume of 2 ml of CHT XT.

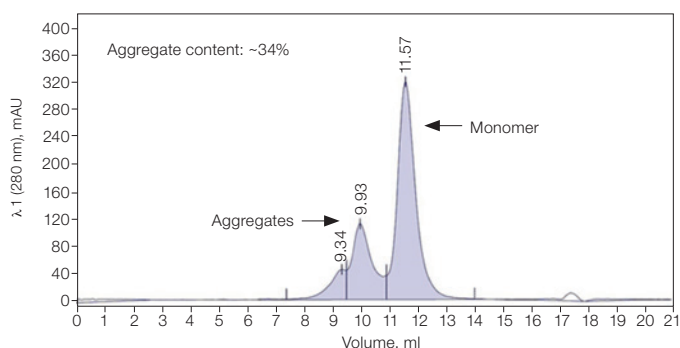
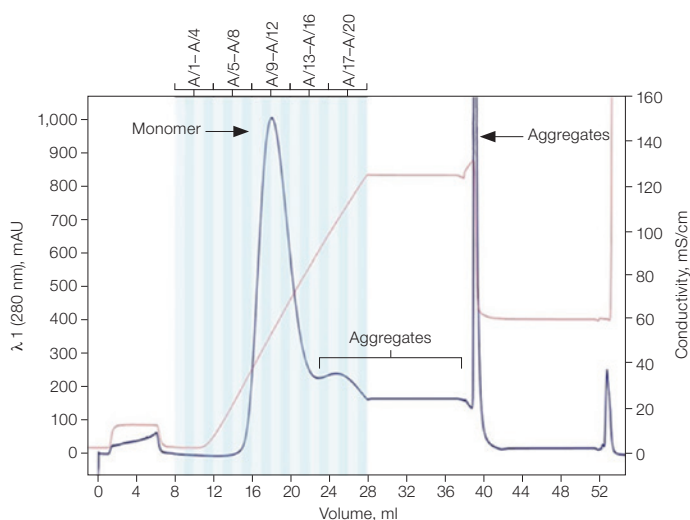


Fig. 5. SEC analysis of mAb S eluted from UNOsphere SUPra. SEC reveals the level of mAb S aggregates in the eluate from UNOsphere SUPra.



Equilibration: 10 CV of 10 mM NaPO₄, pH 7, 300 cm/hr
 Load: 5 ml of CHT load (containing 8.5 mg mAb S at a concentration of 1.7 mg/ml)
 Wash: 3 CV of 10 mM NaPO₄, pH 7
 Elution: 20 CV of linear gradient: 0 to 100% of 10 mM NaPO₄, 1 M NaCl, pH 7
 Strip: 4 CV of 400 mM NaPO₄, pH 7
 Sanitization: 3 CV of 1 N NaOH

Fig. 6. Separation of mAb S monomers from aggregates on a CHT XT Column. The blue vertical bands represent collected column fractions. Fractions 10–14 were pooled. OD 280 (—); conductivity (—).

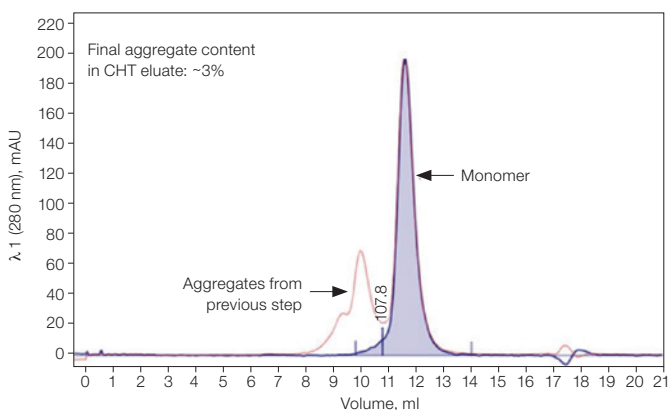


Fig. 7. SEC analysis of mAb S eluted from CHT XT. SEC showing the level of mAb S aggregates in the eluate from CHT XT. Load (—); eluate (—).

Table 1. Reduction of mAb S aggregate contamination by CHT XT.

Sample	Monomer Content, %
CHT XT load	66.0
CHT XT eluate	96.9

Table 2. Impurity clearance by CHT XT.

Impurity	Load	Pool
HCP, ppm	2,635	229
dsDNA, ppm	112	3
Leached rPA, ppm	11	1
Endotoxin (LAL Endotoxin Assay), EU/mg	11	1
Endotoxin (EndoLISA Assay), EU/mg	422	16

Comparative Data

The efficacy of three commercial mixed-mode resins for the purification of mAb S (pI ~6.9) in a Protein A eluate containing about 75% monomeric and 25% aggregated mAb were tested. Each media reduced the aggregate levels to ≤0.5% (Table 3). However, CHT Ceramic Hydroxyapatite Type I, 40 μm offered significantly higher monomer recovery in the smallest eluate volume (Figure 8 and Table 3).

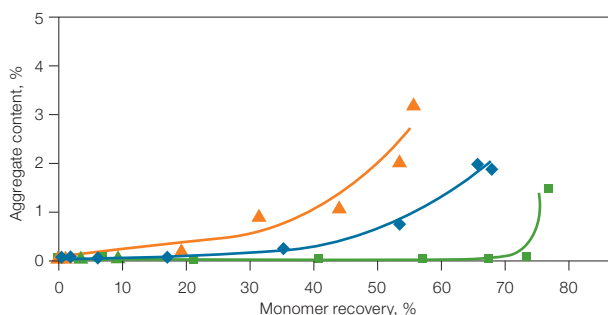


Fig. 8. A bind-and-elute strategy was employed for the polish purification of mAb S with the three media. Salt and pH gradients were optimized from DOE studies for the separation of aggregate from monomer. CHT Column, 1 ml, 0–1,000 mM sodium chloride gradient (■); Capto adhere Column, 1 ml, pH 8–5 gradient (◆); Capto adhere ImpRes Column, 1 ml, pH 8–5 gradient (▲).

Table 3. Comparison of CHT, Capto adhere, and Capto adhere ImpRes mAb S Aggregate clearance and monomer recovery.

Media	mAb S Monomer Content, %	mAb S Monomer Recovery, %	Eluate Volume, CV
CHT*	99.5	82.7	5
Capto adhere**	99.5	48.8	14
Capto adhere ImpRes***	99.5	61.7	14

* Step elution with 10 mM sodium phosphate and 550 mM sodium chloride, pH 7.0.

** Step elution with 50 mM sodium acetate, pH 5.5.

*** Step elution with 50 mM sodium acetate, pH 5.4.

Purification Workflows

Application	Capture Step Resins	Intermediate Step Resins	Polish Step Resins/Media
mAb	Nuvia S	Nuvia Q	CHT Ceramic Hydroxyapatite
IgM	Nuvia HP-Q	—	CHT Ceramic Hydroxyapatite
IgM	Nuvia cPrime	—	CHT Ceramic Hydroxyapatite

Ordering Information

Catalog # Description

Bulk Resins

CHT Ceramic Hydroxyapatite, Type I

1584000	CHT Ceramic Hydroxyapatite, 40 µm, Type I, 10 g
1570040	CHT Ceramic Hydroxyapatite, 40 µm, Type I, 100 g
157-0041	CHT Ceramic Hydroxyapatite, 40 µm, Type I, 1 kg
157-0045	CHT Ceramic Hydroxyapatite, 40 µm, Type I, 5 kg
1588000	CHT Ceramic Hydroxyapatite, 80 µm, Type I, 10 g
1570080	CHT Ceramic Hydroxyapatite, 80 µm, Type I, 100 g
157-0081	CHT Ceramic Hydroxyapatite, 80 µm, Type I, 1 kg
157-0085	CHT Ceramic Hydroxyapatite, 80 µm, Type I, 5 kg

CHT Ceramic Hydroxyapatite, Type II

1584200	CHT Ceramic Hydroxyapatite, 40 µm, Type II, 10 g
1574000	CHT Ceramic Hydroxyapatite, 40 µm, Type II, 100 g
157-4100	CHT Ceramic Hydroxyapatite, 40 µm, Type II, 1 kg
157-4500	CHT Ceramic Hydroxyapatite, 40 µm, Type II, 5 kg
1588200	CHT Ceramic Hydroxyapatite, 80 µm, Type II, 10 g
1578000	CHT Ceramic Hydroxyapatite, 80 µm, Type II, 100 g
157-8100	CHT Ceramic Hydroxyapatite, 80 µm, Type II, 1 kg
157-8500	CHT Ceramic Hydroxyapatite, 80 µm, Type II, 5 kg

CHT Ceramic Hydroxyapatite XT Media

12002457	CHT Ceramic Hydroxyapatite XT, 40 µm, 10 g
12002454	CHT Ceramic Hydroxyapatite XT, 40 µm, 100 g
12002456	CHT Ceramic Hydroxyapatite XT, 40 µm, 1 kg
12002455	CHT Ceramic Hydroxyapatite XT, 40 µm, 5 kg

Prepacked Screening Tools

Foresight Plates*

732-4716	Foresight CHT Type I Plates, 40 µm, 20 µl
732-4718	Foresight CHT Type II Plates, 40 µm, 20 µl
12003151	Foresight CHT XT Plates, 40 µm, 20 µl

Foresight RoboColumn Units**

732-4822	Foresight CHT Type I RoboColumn Unit, 40 µm, 200 µl
732-4823	Foresight CHT Type I RoboColumn Unit, 40 µm, 600 µl
732-4825	Foresight CHT Type II RoboColumn Unit, 40 µm, 200 µl
732-4826	Foresight CHT Type II RoboColumn Unit, 40 µm, 600 µl
12003152	Foresight CHT XT RoboColumn Unit, 40 µm, 200 µl
12003148	Foresight CHT XT RoboColumn Unit, 40 µm, 600 µl

Foresight Columns

732-4735	Foresight CHT Type I Column, 40 µm, 1 ml
732-4755	Foresight CHT Type I Column, 40 µm, 5 ml
732-4736	Foresight CHT Type II Column, 40 µm, 1 ml
732-4756	Foresight CHT Type II Column, 40 µm, 5 ml
12003150	Foresight CHT XT Column, 40 µm, 1 ml
12003149	Foresight CHT XT Column, 40 µm, 5 ml

EconoFit Columns

12009255	EconoFit CHT Type I Column, 40 µm, 1 x 1 ml, 7 x 25 mm
12009253	EconoFit CHT Type I Column, 40 µm, 1 x 5 ml, 16 x 25 mm
12009254	EconoFit CHT Type I Columns, 40 µm, 5 x 5 ml, 16 x 25 mm
12009256	EconoFit CHT Type I Column, 80 µm, 1 x 1 ml, 7 x 25 mm
12009259	EconoFit CHT Type II Column, 40 µm, 1 x 1 ml, 7 x 25 mm
12009257	EconoFit CHT Type II Column, 40 µm, 1 x 5 ml, 16 x 25 mm
12009258	EconoFit CHT Type II Columns, 40 µm, 5 x 5 ml, 16 x 25 mm
12009260	EconoFit CHT Type II Column, 80 µm, 1 x 1 ml, 7 x 25 mm
12009261	EconoFit CHT XT Column, 1 x 1 ml, 7 x 25 mm

* 2 x 96-well plates.

** Package size: one row of eight columns.

Catalog #

Description

EconoFit Kits

12009315	EconoFit Aggregate Clearance Kit , includes 1 ml each of Nuvia cPrime, Nuvia aPrime 4A, Nuvia HR-S, CHT XT Media
12009311	EconoFit CHT Kit , includes 1 ml each of CHT I, 40 µm; CHT II, 40 µm; MPC I, 40 µm; and CFT II, 40 µm Media
12009314	EconoFit Large Biomolecule Kit , includes 1 ml each of Nuvia Q; Nuvia HP-Q; Nuvia cPrime; Nuvia aPrime 4A; CHT II, 40 µm; and CHT XT Media
12009313	EconoFit mAb Kit , includes 1 ml each of Nuvia S, Nuvia Q, Nuvia cPrime, Nuvia aPrime 4A, Nuvia HR-S, and CHT XT Media

Process Prepacked Columns

Foresight Pro Columns, CHT Type I, 40 µm

12014659	Foresight Pro Column , CHT Type I, 40 µm, 0.2 L, 5 cm ID x 10 cm length
12014660	Foresight Pro Column , CHT Type I, 40 µm, 0.4 L, 5 cm ID x 20 cm length
12014671	Foresight Pro Column , CHT Type I, 40 µm, 0.5 L, 8 cm ID x 10 cm length
12014672	Foresight Pro Column , CHT Type I, 40 µm, 1.0 L, 8 cm ID x 20 cm length
12014673	Foresight Pro Column , CHT Type I, 40 µm, 0.8 L, 10 cm ID x 10 cm length
12014674	Foresight Pro Column , CHT Type I, 40 µm, 1.5 L, 10 cm ID x 20 cm length
12014675	Foresight Pro Column , CHT Type I, 40 µm, 1.3 L, 13 cm ID x 10 cm length
12014676	Foresight Pro Column , CHT Type I, 40 µm, 2.7 L, 13 cm ID x 20 cm length
12014677	Foresight Pro Column , CHT Type I, 40 µm, 3.1 L, 20 cm ID x 10 cm length
12014678	Foresight Pro Column , CHT Type I, 40 µm, 6.3 L, 20 cm ID x 20 cm length
12014679	Foresight Pro Column , CHT Type I, 40 µm, 4.5 L, 24 cm ID x 10 cm length
12014680	Foresight Pro Column , CHT Type I, 40 µm, 9.0 L, 24 cm ID x 20 cm length
12014681	Foresight Pro Column , CHT Type I, 40 µm, 8.5 L, 33 cm ID x 10 cm length
12014682	Foresight Pro Column , CHT Type I, 40 µm, 17 L, 33 cm ID x 20 cm length

Foresight Pro Columns, CHT Type II, 40 µm

12014697	Foresight Pro Column , CHT Type II, 40 µm, 0.2 L, 5 cm ID x 10 cm length
12014698	Foresight Pro Column , CHT Type II, 40 µm, 0.4 L, 5 cm ID x 20 cm length
12014699	Foresight Pro Column , CHT Type II, 40 µm, 0.5 L, 8 cm ID x 10 cm length
12014700	Foresight Pro Column , CHT Type II, 40 µm, 1.0 L, 8 cm ID x 20 cm length
12014701	Foresight Pro Column , CHT Type II, 40 µm, 0.8 L, 10 cm ID x 10 cm length
12014702	Foresight Pro Column , CHT Type II, 40 µm, 1.5 L, 10 cm ID x 20 cm length
12014703	Foresight Pro Column , CHT Type II, 40 µm, 1.3 L, 13 cm ID x 10 cm length
12014704	Foresight Pro Column , CHT Type II, 40 µm, 2.7 L, 13 cm ID x 20 cm length
12014705	Foresight Pro Column , CHT Type II, 40 µm, 3.1 L, 20 cm ID x 10 cm length
12014706	Foresight Pro Column , CHT Type II, 40 µm, 6.3 L, 20 cm ID x 20 cm length
12014707	Foresight Pro Column , CHT Type II, 40 µm, 4.5 L, 24 cm ID x 10 cm length
12014708	Foresight Pro Column , CHT Type II, 40 µm, 9.0 L, 24 cm ID x 20 cm length
12014709	Foresight Pro Column , CHT Type II, 40 µm, 8.5 L, 33 cm ID x 10 cm length
12014710	Foresight Pro Column , CHT Type II, 40 µm, 17 L, 33 cm ID x 20 cm length

continues

Catalog #	Description
Foresight Pro Columns, CHT XT, 40 µm	
12014725	Foresight Pro Column , CHT XT, 40 µm, 0.2 L, 5 cm ID x 10 cm length
12014726	Foresight Pro Column , CHT XT, 40 µm, 0.4 L, 5 cm ID x 20 cm length
12014727	Foresight Pro Column , CHT XT, 40 µm, 0.5 L, 8 cm ID x 10 cm length
12014728	Foresight Pro Column , CHT XT, 40 µm, 1.0 L, 8 cm ID x 20 cm length
12014729	Foresight Pro Column , CHT XT, 40 µm, 0.8 L, 10 cm ID x 10 cm length
12014730	Foresight Pro Column , CHT XT, 40 µm, 1.5 L, 10 cm ID x 20 cm length
12014731	Foresight Pro Column , CHT XT, 40 µm, 1.3 L, 13 cm ID x 10 cm length
12014732	Foresight Pro Column , CHT XT, 40 µm, 2.7 L, 13 cm ID x 20 cm length
12014733	Foresight Pro Column , CHT XT, 40 µm, 3.1 L, 20 cm ID x 10 cm length
12014734	Foresight Pro Column , CHT XT, 40 µm, 6.3 L, 20 cm ID x 20 cm length
12014735	Foresight Pro Column , CHT XT, 40 µm, 4.5 L, 24 cm ID x 10 cm length
12014736	Foresight Pro Column , CHT XT, 40 µm, 9.0 L, 24 cm ID x 20 cm length
12014737	Foresight Pro Column , CHT XT, 40 µm, 8.5 L, 33 cm ID x 10 cm length
12014738	Foresight Pro Column , CHT XT, 40 µm, 17 L, 33 cm ID x 20 cm length

Case Studies

- Aggregate removal during mAb purification, pages 143 and 146
- Virus purification, page 161

Screen this resin for your application today.



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Bio-Gel Hydroxyapatite HT and HTP Media



CRYSTALLINE
HYDROXYAPATITE

RESIN FUNCTIONS

- Preparative fractionation of monoclonal antibodies, proteins, enzymes, nucleic acids, and other macromolecules

RESIN CHARACTERISTICS

- High capacity for nucleic acids and proteins
- Low nonspecific adsorption
- Chemical and thermal stability, including the ability to be autoclaved
- Available as a suspension or as a dry powder
- DNA grade Bio-Gel HTP Gel has a smaller particle size than HTP Gel, significantly increasing its capacity and enhancing its selectivity for double-stranded DNA

SUPPORTING INFORMATION AND APPLICATION EXAMPLES

- Bio-Gel HT, Bio-Gel HTP, and DNA Grade Bio-Gel HTP Hydroxyapatite Instruction Manual, document LIT217

Bead Properties

Property	Bio-Gel HTP Gel	Bio-Gel HTP Gel, DNA Grade	Bio-Gel HT Gel
Function	Mixed-mode	Cation (phosphate)	Affinity (calcium)
Functional group	Ca ²⁺	PO ₄ ³⁻	OH ⁻
Particle size	10–100 µm		
Flow rate, cm/hr*	>35	>5	>25
BSA adsorbed, mg/dry gram**	10	10	10
Calf thymus DNA adsorbed, mg/dry gram	>500	>800	>500
Hydrated volume, ml/g	—	2–3	2–3
Storage	Dry form at room temperature; hydrated at 4°C		4°C

* Flow rate determined in a 1.5 x 10 cm column with 40 cm H₂O hydrostatic pressure.

** Batchwise uptake.

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Ordering Information

Catalog #	Description
Hydrated Resins (hydroxyapatite gel suspended in sodium phosphate buffer)	
1300150	Bio-Gel HT Hydroxyapatite , 250 ml
130-0151	Bio-Gel HT Hydroxyapatite , 500 ml
130-0152	Bio-Gel HT Hydroxyapatite , 5 L
Dry Resins	
1300420	Bio-Gel HTP Hydroxyapatite , 100 g, hydroxyapatite chromatography media, powder, 1 g hydrates to 2–3 ml of gel
130-0421	Bio-Gel HTP Hydroxyapatite , 1 kg, hydroxyapatite chromatography media, powder, 1 g hydrates to 2–3 ml of gel
130-0425	Bio-Gel HTP Hydroxyapatite , 5 kg, hydroxyapatite chromatography media, powder, 1 g hydrates to 2–3 ml of gel
130-0426	Bio-Gel HTP Hydroxyapatite , 10 kg, hydroxyapatite chromatography media, powder, 1 g hydrates to 2–3 ml of gel
1300520	Bio-Gel HTP Hydroxyapatite , 100 g, DNA grade hydroxyapatite chromatography media, powder, smaller particle size than HTP gel with enhanced selectivity for dsDNA
130-0521	Bio-Gel HTP Hydroxyapatite , 500 g, DNA grade hydroxyapatite chromatography media, powder, smaller particle size than HTP gel with enhanced selectivity for dsDNA
130-0522	Bio-Gel HTP Hydroxyapatite , 5 kg, DNA grade hydroxyapatite chromatography media, powder, smaller particle size than HTP gel with enhanced selectivity for dsDNA

Chromatography Resins and Media by Technique: **Affinity**



ION EXCHANGE

MIXED-MODE / MULTIMODAL

AFFINITY

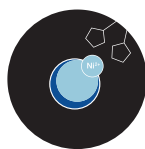
HYDROPHOBIC INTERACTION

SIZE EXCLUSION / GEL FILTRATION

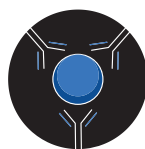
Affinity Chromatography Resins



AFFINITY



IMAC



ANTIBODY



AFFI-GEL

Affinity chromatography relies on the ability of ligands to bind to specific well-defined sites on proteins. The most commonly used affinity ligand is Protein A, which binds to many subclasses of immunoglobulin G (IgG). Binding is typically performed directly from fermentation or other processes and high yields and purities can be expected.

Another commonly used class of affinity ligands is that employed in immobilized metal affinity chromatography (IMAC). IMAC has consistently been one of the most popular methods for purifying recombinant proteins. There are three primary distinguishing differences between available IMAC resins: type of metal, type of ligand, and type of bead. The choice of metal ion to be immobilized on the IMAC ligand depends on the application. While trivalent cations such as Al^{3+} , Ga^{3+} , and Fe^{3+} , or the tetravalent Zr^{4+} , are preferred for capture of phosphoproteins and phosphopeptides, divalent Cu^{2+} , Ni^{2+} , Zn^{2+} , and Co^{2+} ions are used for purification of histidine-tagged proteins and proteins with other presentations of surface histidine and/or cysteine. The ligands are designed to specifically bind transition-state metals. Iminodiacetic acid (IDA) and nitrilotriacetic acid (NTA) are the two most common ligands used in IMAC resins. IMAC resins are best known as specific ligands for histidine-tagged proteins; however, a polyhistidine sequence is not necessary for good binding. The final distinguishing characteristic of the resin is the bead or base matrix. Bio-Rad uses proprietary technology to make polyacrylamide-based UNOsphere beads. These beads combine excellent mass transfer properties with a rigid, chemically resilient structure. See Figure 1 for an affinity model.

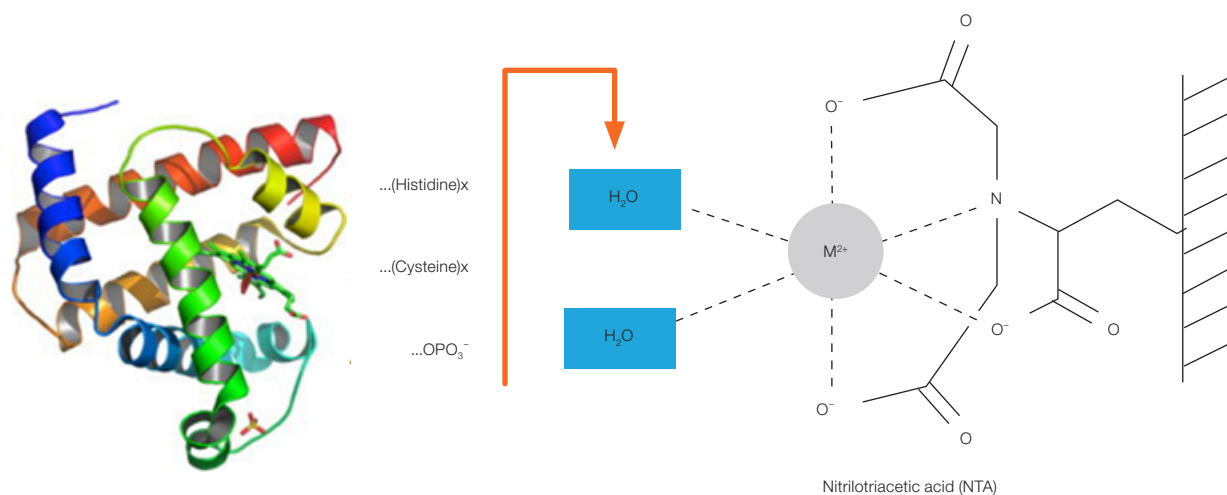


Fig. 1. Affinity model of histidine, cysteine, and phosphate groups to a metal (M) ion chelated by an NTA group. M²⁺ = Ni²⁺, Cu²⁺, Zn²⁺, Ca²⁺, Fe³⁺.

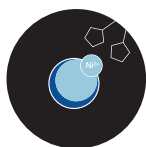
NUVIA IMAC RESIN

Based on UNOsphere beads, Bio-Rad offers an IMAC resin, Nuvia IMAC. This resin is made with the NTA ligand for superior binding capacity. Nuvia IMAC can be used for the capture purification of histidine-tagged proteins as its binding affinity is high enough that proteins with noncontiguous histidine sequences are eliminated during a wash step. Nuvia IMAC can also be employed in intermediate polishing steps in orthogonal approaches that could also include IEX and HIC resins. Nuvia IMAC Resin is designed for easy scalability. It is available in small to bulk sizes to transition from laboratory- to pilot- to manufacturing-scale purifications.

AFFI-GEL RESINS

Ready-to-use resins, including Protein A resins, and activated resins are available for affinity purification of affinity-tagged proteins, monoclonal antibodies, and other biomolecules. Base matrices are available with ligands already coupled for convenience. These ligands have unique specificity and can be used to capture specific molecules. An activated medium for spontaneous ligand immobilization lets you connect any ligand, such as an antibody, enzyme, or antigen, to an affinity matrix of your choice. This flexibility allows researchers to customize their purification process based on the specific needs of their target molecule. Additionally, the availability of activated media simplifies the overall purification workflow. These resins are used in various applications, ranging from drug discovery to production.

Nuvia IMAC Resin



IMAC

RESIN FUNCTIONS

- Immobilized metal affinity resin providing an ultra-high DBC over a wide range of flow rates
- Polymeric bead allowing wide range of flow rates without high pressure issues
- Highly selective binding of recombinant histidine-tagged proteins

RESIN CHARACTERISTICS

- A metal chelate affinity chromatography resin that contains nitrilotriacetic acid (NTA) as the chelating ligand for di- or trivalent metal ions
- Inert hydrophilic surface of the base bead and the chemical structure of NTA
- Optimized particle size and high porosity allow operation at increasingly higher flow rates without compromising binding capacity
- Stable in a broad range of chemicals and compatible with all reagents used for histidine-tagged protein purification
- Also available as uncharged Nuvia NTA support to charge the resin with metal ligand of your choice

SUPPORTING INFORMATION AND APPLICATION EXAMPLES

- Product information sheet, bulletin 6859
- A Nuvia IMAC Resin–Based Purification Workflow for the Production of Highly Pure Recombinant Proteins, bulletin 6964
- Reference Methods for Packing Columns with Nuvia IMAC Resin: From Lab-Scale to Process-Scale, bulletin 7028
- Purification of SARS-CoV-2 Spike RBD Using Nuvia IMAC Resin, bulletin 3603
- Instruction manual, document 10044307

Bead Properties

Property	Description
Function	Metal affinity
Functional group	Nitrilotriacetic acid (NTA)
Base matrix composition	Polyacrylamide
Mean particle size	50 µm
Ligand density	≥18 µmol/ml
Ni content	≥100 µmol/g, Ni-charged
Nitrilotriacetic acid	≥18 µmol/ml, uncharged
Dynamic binding capacity*	>40 mg/ml at 300 cm/hr
Recommended linear flow rate	50–300 cm/hr
Compression factor (settled bed volume/packed bed volume)	1.15–1.20
pH stability	2–14
Shipping solution	20% ethanol or 2% benzyl alcohol
Regeneration	50 mM EDTA, pH 8.0 (stripping) 1 M NaOH (CIP/SIP) 0.1 M Ni ₂ SO ₄ (recharging)
Sanitization	1 M NaOH
Storage conditions	20% ethanol
Storage temperature	Room temperature
Chemical compatibility**	Reducing agents (BME, TCEP, DTT) Denaturing agents (urea, guanidine HCl) Detergents (Triton X-100, NP-40, CHAPS, CHAPSO) Additives (glycerol)
Chemical stability***	1 M NaOH, 48 hr 0.01 M HCl, 1 week
Shelf life	5 years

* 10% breakthrough capacity determined with 1.2 mg/ml of a 40 kD histidine-tagged protein in 50 mM sodium phosphate, 5 mM imidazole, 300 mM NaCl (pH 7.5).

** Recharging is not required. Refer to the instruction manual for a complete list.

*** Recharging is required. Refer to the instruction manual for a complete list.

High Binding Capacity at Fast Flow Rates

The Nuvia IMAC Resin's superior mechanical strength and optimized pore surface area ensures high dynamic binding capacities even at the fast flow rates needed for process production efficiency (Figure 1). This ensures it delivers the productivity needed in downstream manufacturing.

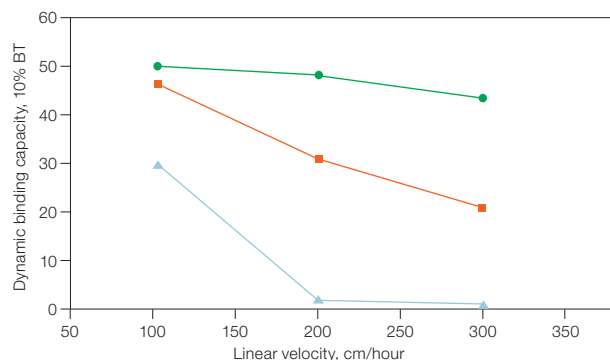


Fig. 1. Dynamic binding capacity vs. flow velocity of Nuvia IMAC Resin.

Comparison of DBCs between Nuvia IMAC and two other commercially available IMACs. The resins were packed into 1 ml columns (0.5 x 5 cm). Histidine-tagged green fluorescent protein (GFP) (1.2 mg/ml) in 50 mM sodium phosphate, 5 mM imidazole, and 300 mM NaCl, pH 7.5, was loaded onto each column until 10% breakthrough (BT) was observed. Nuvia IMAC (●); agarose IMAC resin A (■); agarose IMAC resin B (▲).

Good Pressure/Flow Performance

The column pressure remains below 2 bar at a linear velocity of 300 cm/hr (Figure 2).

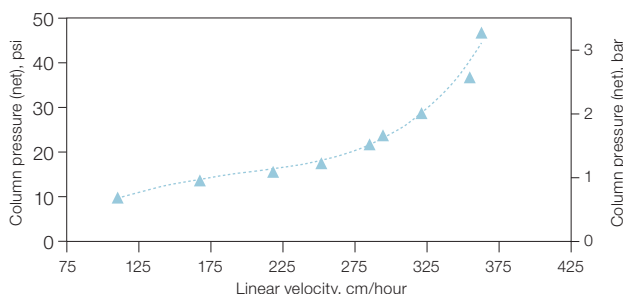


Fig. 2. Pressure/flow performance of Nuvia IMAC Resin. Uncharged Nuvia IMAC Resin slurry prepared in water was packed into a 20 x 20 cm column by axial compression with a compression factor of 1.2.

Excellent Recovery

Nuvia IMAC has an inert and hydrophilic base bead surface that minimizes nonspecific interaction with the beads due to charge or hydrophobicity. The pore structure of the beads is also optimized to allow efficient mass transfer under dynamic conditions. These features contribute to the excellent recovery of histidine-tagged proteins with Nuvia IMAC (Figure 3; target protein recovery is 96.8%).

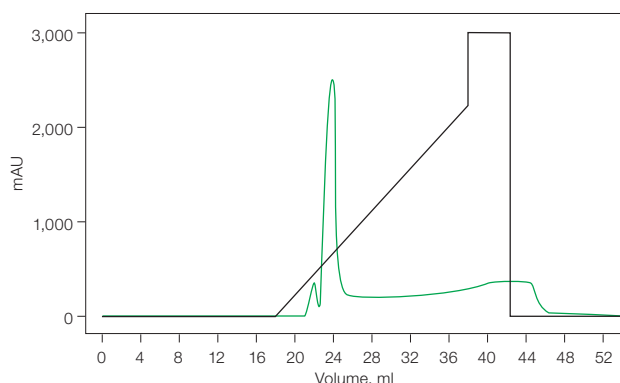


Fig. 3. Recovery of a histidine-tagged protein with Nuvia IMAC Resin. Ten milligrams of a 45 kD histidine-tagged protein were loaded onto a 1 ml column of Nuvia IMAC (0.46 x 5 cm) at a flow rate of 300 cm/hr. The equilibration and post-loading wash was performed with buffer A (50 mM sodium phosphate, 300 mM NaCl, pH 7.5). The elution was performed in a 20 CV linear gradient with 0–75% buffer B (50 mM sodium phosphate, 1 M imidazole, 300 mM NaCl, pH 7.5) and then held at 100% buffer B for 5 CV. Recovery of target protein was 96.8%. A₂₈₀ (—); %B (—).

Reproducible Performance and Reusability

Nuvia IMAC Resin shows no significant loss in its DBC over more than 100 cycles of repeated use (Figure 4). It is produced by a validated manufacturing process to ensure batch-to-batch reproducibility.

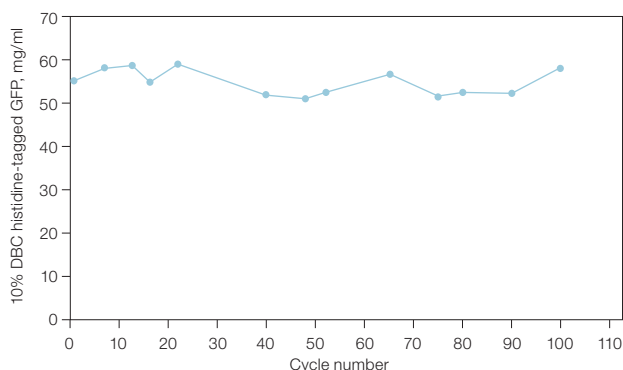


Fig. 4. Reusability of Nuvia IMAC Resin. Histidine-tagged GFP was loaded onto a 0.5 x 5.1 cm column packed with Nuvia IMAC with a compression factor of 1.25. The column was run at 300 cm/hr. The binding buffer was 50 mM sodium phosphate, 5 mM imidazole, and 300 mM NaCl, pH 7.5, and the elution buffer contained 250 mM imidazole. Solutions for column regeneration are listed in the Bead Properties table.

Purification of Histidine-Tagged Recombinant Protein

Nuvia IMAC is highly selective for histidine-tagged proteins. Figure 5 shows the purification of a histidine-tagged GFP from a crude *Escherichia coli* lysate. The various fractions shown on the chromatogram were analyzed by SDS-PAGE (Figure 6). The purity of the purified bands was estimated by densitometry to be >90%.

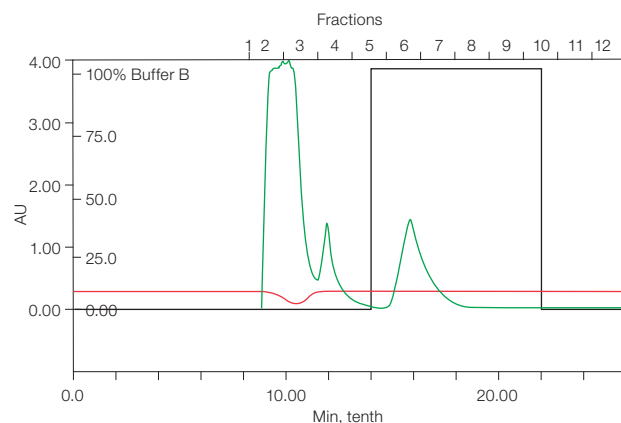


Fig. 5. Purification of histidine-tagged GFP on a Nuvia IMAC Column. *E. coli* extract (2 ml) was loaded on a 0.46 x 5 cm column in 50 mM sodium phosphate, 10 mM imidazole, and 300 mM NaCl, pH 7.5 (buffer A). The target protein was eluted with 10 column volumes (CV) 50 mM sodium phosphate, 125 mM imidazole, and 300 mM NaCl, pH 7.5 (buffer B). A₂₈₀ (—); conductivity (—); %B (—).

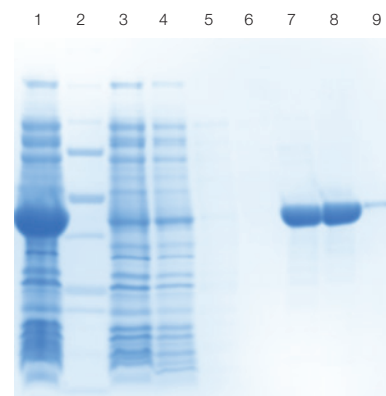


Fig. 6. Analysis of histidine-tagged GFP purified on a Nuvia IMAC Column. Fractions collected from the chromatography run shown in Figure 3 were analyzed by SDS-PAGE. 5 µl of each of the fractions were loaded onto the gel. Lane 1, *E. coli* lysate; lane 2, Precision Plus Protein Unstained Standards; lanes 3–9, fractions 3–9.

Purification Workflows

Application	Capture Step Resins	Intermediate Step Resins	Polish Step Resins/Media
Protein	Nuvia IMAC	Nuvia cPrime	CHT Ceramic Hydroxyapatite
Enzyme	Nuvia IMAC	—	Nuvia aPrime 4A

Ordering Information

Catalog #	Description
Bulk Resins*	
7800800	Nuvia IMAC Ni-Charged Resin, 25 ml
7800801	Nuvia IMAC Ni-Charged Resin, 100 ml
780-0802	Nuvia IMAC Ni-Charged Resin, 500 ml
12002782	Nuvia IMAC Ni-Charged Resin, 10 L
12003233	Nuvia IMAC Ni-Charged Resin, 5 L
12018135	Nuvia IMAC Uncharged Resin, 25 ml
12018162	Nuvia IMAC Uncharged Resin, 500 ml
12004039	Nuvia IMAC Uncharged Resin, 10 L
12004040	Nuvia IMAC Uncharged Resin, 5 L
Prepacked Screening Tools	
12004035	Foresight Nuvia IMAC Ni-Charged Plates, 2 x 96-well, 20 µl
12004051**	Foresight Nuvia IMAC Ni-Charged RoboColumn Unit, 200 µl
12004052**	Foresight Nuvia IMAC Ni-Charged RoboColumn Unit, 600 µl
12004038	Foresight Nuvia IMAC Ni-Charged Column, 1 ml
12004037	Foresight Nuvia IMAC Ni-Charged Column, 5 ml
12009289***	EconoFit Nuvia IMAC Uncharged Column, 1 x 1 ml, 7 x 25 mm

* Larger quantities available upon request.

** Package size: one row of eight columns.

*** Maximum pressure: 45 psi.

Case Studies

- Recombinant protein purification, page 170

Screen this resin for your application today.



Visit [bio-rad.com/ResinSample](https://www.bio-rad.com/ResinSample) to request a sample.

ION EXCHANGE

MIXED-MODE / MULTIMODAL

AFFINITY

HYDROPHOBIC INTERACTION

SIZE EXCLUSION / GEL FILTRATION

Affi-Gel Protein A and Affi-Prep Protein A Resins



AFFI-GEL AND
AFFI-PREP

RESIN FUNCTIONS

- Purification of immunoglobulins (IgG)
- Selective removal of IgG from serum
- Adsorption of immune complexes for antigen purification

RESIN CHARACTERISTICS

- Protein A ligand
- Able to produce high-purity IgGs
- High affinity for mammalian IgG
- High capacities for mouse IgG1 as well as other subclasses with MAPS optimized buffer

SUPPORTING INFORMATION AND APPLICATION EXAMPLES

- Affi-Gel Protein A Agarose for IgG Purification instruction sheet, document LIT379
- Affi-Gel Protein A MAPS II Kit Instruction Manual, document LIT88
- Affi-Prep Protein A Matrix Instruction Manual, document LIT230

Bead Properties

Property	Affi-Gel Protein A	Affi-Prep Protein A
Function	Affinity Protein A	Affinity Protein A
Functional group	Protein A	Protein A
Base matrix composition	Agarose	Polymethacrylate
Particle size range or mean	90–180 µm	50 µm
Binding capacity (protein)	1 ml of Affi-Gel Protein A agarose will bind 6–8 mg/ml mouse IgG1 I	Mouse IgG1 monoclonal antibody: 8–10 mg/ml Loading capacities: IgG2a: 13–15 mg/ml IgG2b: 13–15 mg/ml IgG3: 8–10 mg/ml Human IgG polyclonal antibody: 16–23 mg/ml Rat, sheep, bovine, equine, goat, rabbit, dog, and porcine IgG: 9–16 mg/ml IgM: 5–7 mg/ml*
Recommended linear flow rate	15–25 cm/hr	100–600 cm/hr
pH stability	2–10	2–14
Shipping solution	0.05 M sodium phosphate, pH 7.5, with 0.05% NaN ₃	0.05 M sodium phosphate, pH 7.5, containing 0.05% sodium azide
Chemical stability	Excellent resistance to denaturing agents such as urea, chaotropic salts such as guanidine HCl or potassium thiocyanate, and acid and base (pH 2–10)	pH 2–14, 1 N NaOH, 8 M urea, 50% methanol, 6 M guanidine HCl
Shelf life	1 year	1 year

* Approximately 50% of all IgMs will bind to Protein A.

Ordering Information

Catalog #	Description
Bulk Resins	
1536153	Affi-Gel Protein A Gel , 5 ml
1536154	Affi-Gel Protein A Media , 50 ml
1536159	Affi-Gel Protein A MAPS II Kit , includes 5 ml Affi-Gel Protein A Gel; binding, elution, and regeneration buffers; and 1 x 10 cm Econo-Column™ Chromatography Column
1560006	Affi-Prep Protein A Support , 5 ml
1560005	Affi-Prep Protein A Support , 25 ml
1536160	Affi-Gel Protein A MAPS II Buffers Kit , includes binding, elution, and regeneration buffers
1536161	Protein A MAPS II Binding Buffer , 5 L
1536162	Protein A MAPS II Elution Buffer , 5 L

Screen this resin for your application today.

ION EXCHANGE

MIXED-MODE / MULTIMODAL

AFFINITY

HYDROPHOBIC INTERACTION

SIZE EXCLUSION / GEL FILTRATION

Affi-Gel Blue Resin



AFFI-GEL

RESIN FUNCTIONS

- Albumin removal and enzyme purification

RESIN CHARACTERISTICS

- Affinity resin with a blue dye that functions as an ionic, hydrophobic, aromatic, or sterically active binding site

SUPPORTING INFORMATION AND APPLICATION EXAMPLES

- Product information sheet, bulletin 1107
- Instruction manual, document LIT590

Bead Properties

Property	Description
Function	Affinity
Functional group	Cibracon Blue F3GA
Base matrix composition	Agarose
Particle size	150–300 µm (50–100 mesh) 75–150 µm (100–200 mesh)
Ligand density	≥1.9 mg/ml
Binding capacity	>11 mg/ml
Maximum operating pressure	15 psi
Recommended linear flow rate	15–25 cm/hr
pH stability	4–10
Shipping solution	0.05% sodium azide in water
Chemical compatibility	Compatible in alcohols
Sanitization	0.1 N NaOH
Storage conditions	0.02% sodium azide, 4°C
Shelf life	3 years

Applications

- Albumin removal
- Enzyme purification

Ordering Information

Catalog #	Description
Bulk Resins	
1537301	Affi-Gel Blue Gel , 50–100 mesh, 100 ml
1537302	Affi-Gel Blue Gel , 100–200 mesh, 100 ml
153-7303	Affi-Gel Blue Gel , 100–200 mesh, 1 L
12009234	EconoFit Affi-Gel Blue Column , 1 x 5 ml
12009235	EconoFit Affi-Gel Blue Columns , 5 x 5 ml

Screen this resin for your application today.

CM Affi-Gel Blue Resin



AFFI-GEL

RESIN FUNCTIONS

- Initial purification of serum proteins to remove albumin and serum proteases

RESIN CHARACTERISTICS

- Affinity and cation exchange resin
- Blue dye functions as an ionic, hydrophobic, aromatic, or sterically active binding site

SUPPORTING INFORMATION AND APPLICATION EXAMPLES

- Instruction manual, document LIT308

Bead Properties

Property	Description
Function	Affinity and cation exchange
Functional group	Cibracon Blue F3GA and carboxymethyl
Base matrix composition	Agarose gel
Particle size	75–150 μm (100–200 mesh)
Ligand density	≥ 1.7 mg dye/ml resin; ~ 20 meq/ml carboxyl groups
Binding capacity	0.17–0.5 ml serum/ml gel
Maximum operating pressure	15 psi
Recommended linear flow rate	15–25 cm/hr
pH stability	2–11
Shipping solution	0.02% sodium azide
Chemical compatibility	Compatible in alcohols
Sanitization	0.1 N NaOH
Storage conditions	0.02% sodium azide, 4°C
Shelf life	3 years

Applications

- Serum protein purification

Ordering Information

Catalog #	Description
Bulk Resins	
1537304	CM Affi-Gel Blue Gel, 100 ml
153-7305	CM Affi-Gel Blue Gel, 1 L

Screen this resin for your application today.

DEAE Affi-Gel Blue Resin



AFFI-GEL

RESIN FUNCTIONS

- Purification of serum albumin, as it has greater affinity for this target than other ion exchange resins

RESIN CHARACTERISTICS

- Affinity and anion exchange resin
- Blue dye functions as an ionic, hydrophobic, aromatic, or sterically active binding site

SUPPORTING INFORMATION AND APPLICATION EXAMPLES

- Instruction manual, document 4006049

Bead Properties

Property	Description
Function	Affinity and anion exchange
Functional group	Cibracon Blue F3GA and diethylaminoethyl
Base matrix composition	Agarose gel
Particle size	150–300 µm (50–100 mesh)
Ligand density	≥1.7 mg dye/ml resin; ~20 meq/ml diethylaminoethyl groups
Binding capacity	0.2–1 ml of serum/ml gel
Maximum operating pressure	15 psi
Recommended linear flow rate	15–25 cm/hr
pH stability	2–11
Shipping solution	0.02% sodium azide
Chemical compatibility	Compatible in alcohols
Sanitization	0.1 N NaOH
Storage conditions	0.02% sodium azide, 4°C
Shelf life	1 year

Applications

- Serum protein purification

Ordering Information

Catalog # Description

Bulk Resins

1537307 DEAE Affi-Gel Blue Gel, 100 ml
153-7306 DEAE Affi-Gel Blue Gel, 1 L
153-7308 DEAE Affi-Gel Blue Gel, 5 L

Resin Screening Tools

12009262 EconoFit DEAE Affi-Gel Blue Column, 1 x 5 ml
12009263 EconoFit DEAE Affi-Gel Blue Columns, 5 x 5 ml

Screen this resin for your application today.

Affi-Gel Boronate Resin



AFFI-GEL

RESIN FUNCTIONS

- Efficient separation of nucleotides, nucleosides, catecholamines, and sugars

RESIN CHARACTERISTICS

- Affinity boronate resin
- High binding capacity for molecules containing cis-diol

SUPPORTING INFORMATION AND APPLICATION EXAMPLES

- Affi-Gel Boronate Affinity Gel to Separate Ribonucleotides, Ribonucleosides, Sugars, Catecholamines, and Coenzymes, bulletin 1066

Bead Properties

Property	Description
Function	Affinity
Functional group	Phenyl boronate
Base matrix composition	Polyacrylamide gel
Particle size	45–90 μm
Ligand density	1.05 $\pm$ 0.15 meq/g
Binding capacity	130 μmol sorbitol/ml gel
Maximum operating pressure	15 psi
Recommended linear flow rate	15–25 cm/hr
Storage conditions	4°C under neutral pH conditions, up to 2 months
Shelf life	2 years as dry material

Applications

- Adenylate cyclase assay
- Isolation of catecholamines from urine
- Modified nucleosides in urine
- Separation of mono- and oligonucleotides
- Sugars
- Assay of ribonucleotide reductase

Ordering Information

Catalog #	Description
Bulk Resins	
1536103	Affi-Gel Boronate Media , 5 g
1536104	Affi-Gel Boronate Media , 50 g

Screen this resin for your application today.

Affi-Gel 10 Resin



AFFI-GEL

RESIN FUNCTIONS

- Efficient coupling of neutral or basic proteins with pI from 6.5 to 11

RESIN CHARACTERISTICS

- Activated immunoaffinity supports
- Derivatized crosslinked agarose gel bead support
- Able to couple ligands spontaneously in aqueous or nonaqueous/anhydrous conditions
- High-efficiency coupling for all ligands with primary amine

SUPPORTING INFORMATION AND APPLICATION EXAMPLES

- Affi-Gel 10 and 15 Activated Affinity Media, bulletin 1085
- Instruction manual, document LIT156

Bead Properties

Property	Description
Function	Activated affinity
Functional group	N-hydroxysuccinimide esters of a derivatized crosslinked agarose gel
Base matrix composition	Copolymer of acrylamide and N,N'-methylenebisacrylamide
Particle size	75–300 μm (50–200 mesh)
Ligand density	15–20 $\mu\text{moles/ml}$
Binding capacity (protein)	25 mg/ml of gel
Maximum operating pressure	15 psi
Recommended linear flow rate	15–25 cm/hr
pH stability	Coupled gel: 2–11
Shipping solution	100% isopropanol
Chemical compatibility	Many organic solvents, e.g., alcohols, dimethylsulfoxide (DMSO), dioxane, formamide without free amines, or mixtures of these solvents
Chemical stability	During coupling, avoid primary amine- or sulfhydryl-containing compounds (e.g., Tris, glycine, or β -mercaptoethanol), as these will inactivate the N-hydroxysuccinamide groups
Storage conditions	Uncoupled Affi-Gel 10 Resin: store in a freezer (-10°C) to retain 80% of original activity for at least 1 year. Store at -70°C to further extend shelf life. Add preservatives as appropriate after coupling
Shelf life	2 years

Ordering Information

Catalog #	Description
Bulk Resins	
1536099	Affi-Gel 10 Media , 25 ml
1536046	Affi-Gel 10 Media , 4 x 25 ml
153-1000	Affi-Gel 10 Media , 1 L
1536098	Affi-Gel 10/15 Combination , 2 x 25 ml Affi-Gel 10 Gel and 2 x 25 ml Affi-Gel 15 Gel

Screen this resin for your application today.

Affi-Gel 102 Resin



AFFI-GEL

RESIN FUNCTIONS

- Immobilization of ligands that contain primary or terminal carboxyl groups

RESIN CHARACTERISTICS

- Activated immunoaffinity supports
- Amino-terminal crosslinked agarose gel, with a 6-atom hydrophilic arm
- High-efficiency coupling spontaneously in aqueous or nonaqueous/anhydrous conditions
- Offers flexible alternative chemistries and economy

SUPPORTING INFORMATION AND APPLICATION EXAMPLES

- Instruction manual, document LIT329

Bead Properties

Property	Description
Function	Activated affinity
Functional group	Use with 1-ethyl-3-(3-dimethylaminopropyl) carbodiimide hydrochloride (EDAC) carbodiimide coupling reagent, which immobilizes ligands that contain primary or terminal carboxyl groups
Base matrix composition	Crosslinked agarose gel, with a 6-atom hydrophilic arm
Particle size	75–300 µm (50–200 mesh)
Ligand density	12–20 EDAC groups/ml resin
Maximum operating pressure	15 psi
Shipping solution	0.02% sodium azide
Chemical compatibility	Many organic solvents, e.g., alcohols, dimethylsulfoxide (DMSO), isopropanol, ethanol
Shelf life	3 years

Ordering Information

Catalog #	Description
Bulk Resins	
1532401	Affi-Gel 102 Gel , 50 ml
1532402	Affi-Gel 102 Gel , 1 L
1530990	EDAC , 5 g, 1-ethyl-3-(3-dimethylaminopropyl) carbodiimide hydrochloride (EDAC or EDC) carbodiimide coupling reagent

Screen this resin for your application today.

Chromatography Resins and Media by Technique: **Hydrophobic Interaction**



ION EXCHANGE

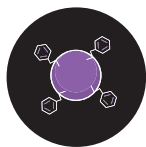
MIXED-MODE / MULTIMODAL

AFFINITY

HYDROPHOBIC INTERACTION

SIZE EXCLUSION / GEL FILTRATION

Hydrophobic Interaction Chromatography Resins



HYDROPHOBIC
INTERACTION

Hydrophobic interaction chromatography (HIC) involves the separation of protein molecules based on a differential interaction of these molecules with hydrophobic sites on the surface of a solid support. In the separation process, hydrophobic patches on the protein surface interact with hydrophobic molecules immobilized on a hydrophilic solid phase surface. Hydrophobic interactions are driven by the extrusion of a monomolecular layer of ordered water molecules covering two adjacent hydrophobic surfaces into less ordered bulk with a concomitant increase in entropy. This entropy-driven interaction between nonpolar groups in water is the basis for HIC.

The hydrophobic interactions on the protein surface are due primarily to amino acids containing aromatic (tyrosine, tryptophan) or alkyl (leucine, valine, isoleucine) side chains. Binding and elution of proteins from HIC resins is the opposite with IEX ligands. HIC interactions are strengthened by increasing salt concentration, often using ammonium sulfate. Desorption is achieved by lowering the salt concentration and/or adding mobile phase modifiers such as alcohols or surfactants. There are two basic types of HIC resins. The first contains aromatic ligands such as a phenyl group. The second contains alkyl ligands and can either have a straight chain or be branched. The interactions between a given protein and HIC resins are not necessarily predictable, and it is best to try members of both resin classes.

HIC resins are often used because they provide an orthogonal purification technique to IEX or affinity chromatography. In addition, many proteins, when they denature or are incorrectly folded, expose hydrophobic amino acids that are otherwise not accessible. Therefore, HIC resins can be used to remove these species from a process feedstream. HIC is often used to remove aggregated target proteins during a purification process. Since HIC requires high salt concentrations for binding, this step is frequently employed following ammonium sulfate precipitation or elution from IEX resins. HIC is typically the second or third step in a purification workflow. HIC resins are rarely used in the capture step due to their relatively low binding capacity and because large fermentation volumes would require unacceptably large amounts of salt for the binding step.

MACRO-PREP RESINS

Bio-Rad offers two HIC resins based on the Macro-Prep support:

- Macro-Prep Methyl HIC Resin — weakly hydrophobic, for purifying compounds that have strongly hydrophobic regions
- Macro-Prep t-Butyl HIC Resin — mildly hydrophobic, for purifying compounds that have few or weakly hydrophobic regions

NUVIA MIXED-MODE RESINS

Bio-Rad offers two mixed-mode resins with HIC functionalities: Nuvia cPrime and Nuvia aPrime 4A. Details on these two resins are available on pages 70 and 74, respectively, in Hydrophobic Ion Exchange Resins in the Mixed-Mode / Multimodal Chromatography Resins and Media section.

BIO-BEADS RESINS

Bio-Beads SM-2 Resins are nonpolar macroporous polymeric beads with high surface area for adsorbing organics with molecular weights less than 2,000 daltons. They are often used in aqueous solution and with solvents, including alcohols, petroleum ether, diethyl ether, and hexane, or with solvent mixtures, without expansion or contraction of beads.

Macro-Prep Methyl and Macro-Prep t-Butyl Resins



HYDROPHOBIC
INTERACTION

RESIN FUNCTIONS

- Particularly suited for HIC operations requiring high throughput and high recovery of target
- Designed specifically for intermediate purification steps that remove host-cell contaminants from partially purified targets

RESIN CHARACTERISTICS

- Resin consists of rigid polymethacrylate beads with a unique surface chemistry, which contributes to high productivity
- These resins are chemically and thermally stable, so they can be autoclaved and can withstand treatment in acid, chaotropic agents, or detergents while retaining high protein-binding capacities

SUPPORTING INFORMATION AND APPLICATION EXAMPLES

- Product information sheet, bulletin 1841
- Instruction manual, document LIT486

Bead Properties

Property	Macro-Prep Methyl	Macro-Prep t-Butyl
Function	Weak hydrophobic interaction	Mild hydrophobic interaction
Functional group	Methyl	-COO ⁻ and t-butyl
Base matrix composition	Polymethacrylate	Polymethacrylate
Mean particle size	60 µm	60 µm
Total ionic capacity	<2 µeq/ml	Approximately 120 µeq/ml
Dynamic binding capacity	15 mg HSA/ml	25 mg HSA/ml
Recommended linear flow rate	50–300 cm/hr	50–300 cm/hr
pH stability*	1–10	1–10
Shipping solution	20% ethanol or 2% benzyl alcohol	20% ethanol or 2% benzyl alcohol
Regeneration	70% ethanol	70% ethanol
Sanitization	Acetic acid or phosphoric acid solution, pH ≥1.5	Acetic acid or phosphoric acid solution, pH ≥1.5
Storage conditions	20% ethanol, 1 M acetic acid, or 50 mM phosphoric acid	20% ethanol, 1 M acetic acid, or 50 mM phosphoric acid
Storage temperature	Room temperature	Room temperature
Chemical stability	8 M urea, 6 M guanidine HCl, 100% ethanol	8 M urea, 6 M guanidine HCl, 100% ethanol
Shelf life	5 years	5 years

* The use of basic reagents >pH 10 should be evaluated for each application.

Macro-Prep HIC Resin Forms

- Macro-Prep Methyl HIC Resin — weakly hydrophobic, for purifying compounds that have strongly hydrophobic regions
- Macro-Prep t-Butyl HIC Resin — mildly hydrophobic, for purifying compounds that have few or weakly hydrophobic regions

Good Pressure/Flow Performance

The rigidity of Macro-Prep HIC Resins allows high flow rates without bed compression (Figure 1).

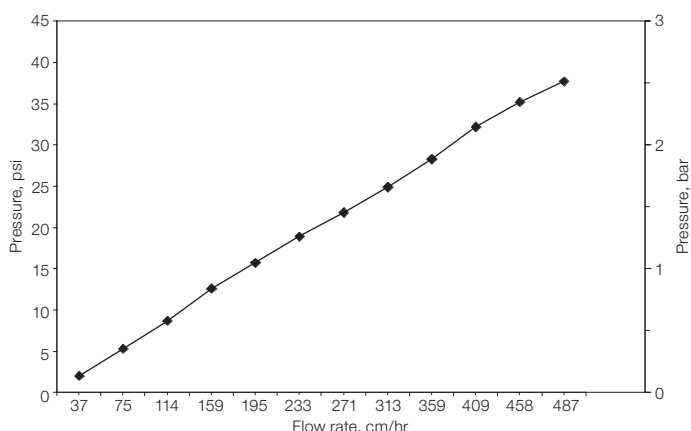


Fig. 1. Pressure vs. flow for Macro-Prep Resins. Resin bed dimensions were 14 cm diameter by 25 cm length.

Protein Purification Data

Macro-Prep Methyl HIC Resin is ideal for purification of proteins with strongly hydrophobic regions. Macro-Prep t-Butyl HIC Resin is ideal for purification of proteins with few or weakly hydrophobic regions. Macro-Prep t-Butyl HIC Resin can operate on an interaction mechanism based on both hydrophobicity and charge. The t-butyl groups are mildly hydrophobic. Depending on the pH of the loading and elution buffers, the matrix carboxyl groups can be exploited to ionically bind or repel target molecules or contaminants in combination with

hydrophobic interactions. This is an ideal strategy that minimizes the product loss commonly experienced with HIC resin due to denaturation of proteins upon exposure to hydrophobic surfaces. The two different ligands provide alternative selectivities for easier optimization of separation (Figure 2 and Table 1). These methacrylate copolymer beads provide high resolution at very high flow rates.

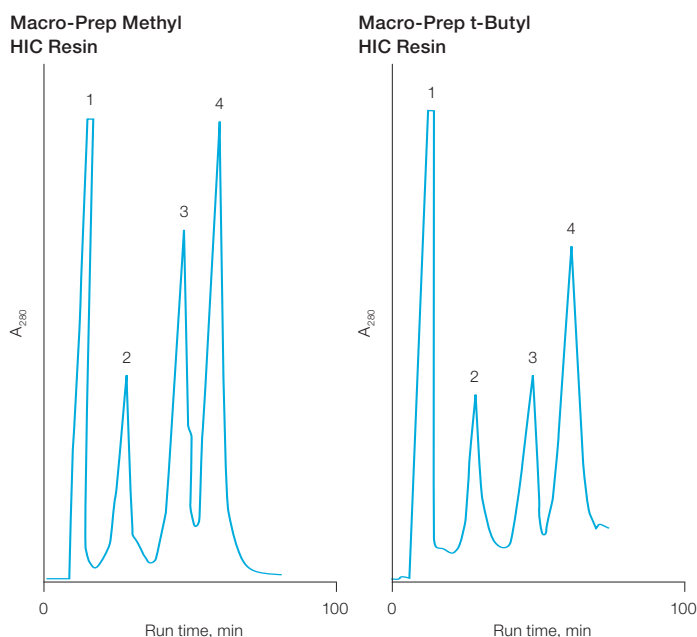


Fig. 2. Sample separation of proteins on Macro-Prep Methyl and t-Butyl HIC Resins. A 50 µl sample (0.6 mg total protein) of cytochrome c (peak 1), ovalbumin (peak 2), α-amylase (peak 3), and ferritin (peak 4) was run on a 1 x 10 cm (4.5 ml) column at a linear flow rate of 38 cm/hr. The sample was eluted with 0.1 M Na₂PO₄, pH 7.0, 1.85 M (NH₄)₂SO₄ for 10 min, followed by a gradient from 1.85 to 0 M (NH₄)₂SO₄ over 90 min. The retention times for each protein are shown in Table 1 for comparison.

Table 1. Comparison of retention times using Macro-Prep Methyl and t-Butyl ligands.

Peak	Protein	Retention Times, min	
		Methyl	t-Butyl
1	Cytochrome c	7.30	10.41
2	Ovalbumin	32.25	40.22
3	α -amylase	62.81	71.50
4	Ferritin	81.49	90.90

Macro-Prep Methyl and t-Butyl Resins can be sanitized quickly and efficiently in 0.15% peracetic acid.

Ordering Information

Catalog # Description

Macro-Prep Methyl HIC Bulk Resins

1580080 Macro-Prep Methyl HIC Support, 25 ml
 1560080 Macro-Prep Methyl HIC Support, 100 ml
 156-0081 Macro-Prep Methyl HIC Support, 500 ml
 12003689 Macro-Prep Methyl HIC Support (in BA), 500 ml
 156-0082 Macro-Prep Methyl HIC Support, 5 L
 12003690 Macro-Prep Methyl HIC Support (in BA), 5 L
 156-0083 Macro-Prep Methyl HIC Support, 10 L
 12003691 Macro-Prep Methyl HIC Support (in BA), 10 L

Macro-Prep t-Butyl HIC Bulk Resins

1580090 Macro-Prep t-Butyl HIC Support, 25 ml
 1560090 Macro-Prep t-Butyl HIC Support, 100 ml
 156-0091 Macro-Prep t-Butyl HIC Support, 500 ml
 12003694 Macro-Prep T-Butyl HIC Support (in BA), 500 ml
 156-0093 Macro-Prep t-Butyl HIC Support, 10 L
 12003695 Macro-Prep T-Butyl HIC Support (in BA), 10 L

BA, 2% benzyl alcohol.

Prepacked Screening Tools

12009277 EconoFit Macro-Prep Methyl HIC Column, 1 x 1 ml, 7 x 25 mm
 12009278 EconoFit Macro-Prep t-Butyl HIC Column, 1 x 1 ml, 7 x 25 mm

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Bio-Beads SM-2 Resin



HYDROPHOBIC
INTERACTION

RESIN FUNCTIONS

- Adsorption of nonpolar substances or surfactants from aqueous solutions
- Removal of detergents such as Triton X-100
- Removal of organics such as polyaromatic hydrocarbons from water
- Cleanup of drugs from plasma and urine
- Cleanup of biological metabolites and pesticides
- Extraction of dyes and mycotoxins from food products

RESIN CHARACTERISTICS

- Nonpolar polystyrene adsorbent
- Neutral, highly crosslinked macroporous polymeric microspheres
- High surface area for adsorbing organics with MW <2,000 Da from aqueous solutions
- Uniform pore sizes
- May be used in aqueous solution and with solvents, including alcohols, petroleum ether, diethyl ether, and hexane, or with solvent mixtures, without expansion or contraction of the beads
- Analytical grade resin has been exhaustively sized, purified, and converted to make it suitable for accurate, reproducible analytical techniques
- Biotechnology grade resin is an analytical grade resin that has been certified to contain less than 100 microorganisms per gram of resin

SUPPORTING INFORMATION AND APPLICATION EXAMPLES

- Bio-Beads Resins flier, bulletin 6905
- Bio-Beads SM-2 Hydrophobic Adsorbents Application Bibliography, bulletin 1461
- Bio-Beads SM Hydrophobic and Polar Interaction Adsorbents Instruction Manual, document LIT195

Bead Properties

Property	Description
Function	Hydrophobic nonpolar polystyrene adsorbent
Base matrix composition	Styrene-divinylbenzene
Dipole moment	0.3 debye
Particle size	300–1,180 µm
Average pore diameter	90 Å
Capacity (Triton X-100 detergent)	0.07 g/g
Density	1.02 g/ml
Shelf life	5 years at room temperature

Compounds Concentrated or Separated by Bio-Beads SM-2 Adsorbents

Trace organics	Morphine
PAH compounds	Methaqualone
Hydrocarbons and PCBs	Sulfas
Aminocarb insecticides	Free rhodamine
Carbamate insecticides	Prostaglandins
Ethyl and methyl parathion	Steroids
Metals	Bile acids
Carboxylic acids	Hormones
Phenolic acids	Purines and pyrimidines
Flavonoids	Acid dyes
Mycotoxins	Naringin and limonin
Proline and hydroxyproline	Detergents

Ordering Information

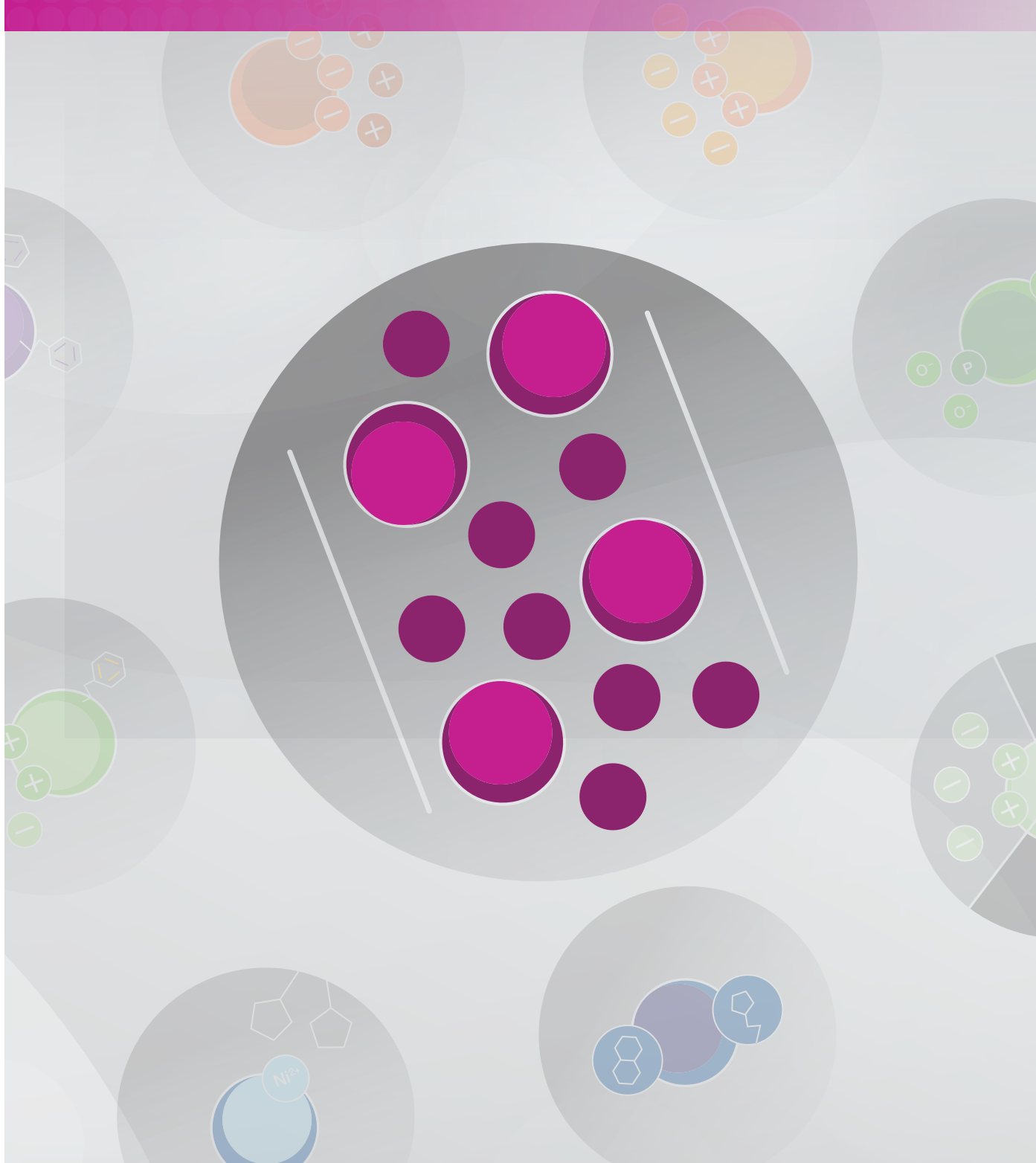
Catalog #	Description
Bulk Resins	
1523920	Bio-Beads SM-2 Adsorbents , 20–50 mesh, 100 g
1528920	Bio-Beads SM-2 Adsorbents , biotechnology grade, 20–50 mesh, 25 g
152-3922	Bio-Beads SM-2 Adsorbents , 20–50 mesh, 1 kg
152-3923	Bio-Beads SM-2 Adsorbents , 20–50 mesh, 10 kg

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Chromatography Resins and Media by Technique: **Size Exclusion / Gel Filtration**



ION EXCHANGE

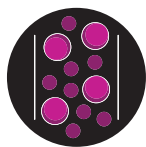
MIXED-MODE / MULTIMODAL

AFFINITY

HYDROPHOBIC INTERACTION

SIZE EXCLUSION / GEL FILTRATION

Size Exclusion / Gel Filtration Chromatography Resins



SIZE EXCLUSION /
GEL FILTRATION

Bio-Rad size exclusion chromatography resins are engineered for gentle, high-resolution separation of sensitive compounds in aqueous or organic solutions. Our size exclusion chromatography resins are available in various matrices and mesh sizes to accommodate a wide range of applications.

BIO-GEL A RESINS

Bio-Gel A Resins, ideal for purification of antibodies and aggregates, consist of agarose beads in which the pore size is controlled by the percentage of agarose in the gel. They are compatible with all commonly used buffers and can be used with high-salt buffers without significantly changing the bed volume. Bio-Gel A Resins may be used at pH 4–13 and at temperatures 2–30°C. The fractionation range is <10,000–5,000,000 daltons.

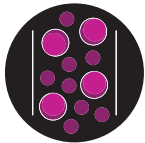
BIO-GEL P RESINS

Bio-Gel P Polyacrylamide Resins, for high-resolution gel filtration, are prepared by copolymerization of acrylamide and N,N'-methylene-bis-acrylamide. Bio-Gel P Resins are available in several particle size ranges with molecular weight exclusion limits ranging from 100 to 100,000 daltons. These resins provide efficient, gentle gel filtration of sensitive compounds. Bio-Gel P Resins are autoclavable at pH 5.5–6.5 and operate over a pH range of 2–10 at room temperature. Flow rate and resolution increase with increasing temperature in the range of 4–80°C.

BIO-BEADS SX RESINS

Bio-Beads S-X Resins are neutral, porous styrene-divinylbenzene beads for SEC of lipophilic polymers and other solutes requiring organic eluants for solubility. Exclusion limits range from 400 to 14,000 daltons.

Bio-Gel A Resins



SIZE EXCLUSION /
GEL FILTRATION

RESIN FUNCTIONS

- Ideal for purification of antibodies and aggregates
- Bio-Gel A-1.5m Gel: fractionation range <10,000–1,500,000 Da
- Bio-Gel A-5m Gel: fractionation range 10,000–5,000,000 Da

RESIN CHARACTERISTICS

- Agarose beads
- Pore size is controlled by percentage of agarose in the gel
- Excellent shrinking and swelling properties
- Compatible with high-salt buffers without bed volume change

SUPPORTING INFORMATION AND APPLICATION EXAMPLES

- Bio-Gel Resins flier, bulletin 6906
- Extracellular Vesicle Isolation by Size Exclusion Chromatography, bulletin 3540
- Instruction manual, document 4006139

ION EXCHANGE

MIXED-MODE / MULTIMODAL

AFFINITY

HYDROPHOBIC INTERACTION

SIZE EXCLUSION / GEL FILTRATION

Bead Properties

Property	Description
Function	Size exclusion
Base matrix composition	Beaded agarose gel
Particle size	Coarse: 150–300 µm (50–100 mesh) Medium: 75–150 µm (100–200 mesh) Fine: 38–75 µm (200–400 mesh)
Maximum operating pressure	15 psi
Fractionation range/ exclusion limit	Bio-Gel A-1.5m: <10,000–1,500,000 Da Bio-Gel A-5m: 10,000–5,000,000 Da
Hydrated bed size	Coarse: 150–300 µm Medium: 75–150 µm Fine: 38–75 µm
Recommended linear flow rate	Coarse: 20–25 cm/hr Medium: 15–20 cm/hr Fine: 7–13 cm/hr
pH stability	4–9; not stable in solutions with pH >7 for more than 12 hours
Temperature range*	4–30°C
Shipping solution	Fully hydrated in water, with 0.05% sodium azide added as a preservative
Cleaning	Urea or guanidine HCl
Chemical compatibility	Commonly used buffers, 20% ethanol
Storage temperature*	4°C to room temperature
Storage conditions	0.02% sodium azide; for long-term storage, maintain pH neutrality
Shelf life	5 years

* Above this temperature range the gel softens, and freezing causes irreversible collapse of the gel structure and alteration of the exclusion limit.

Size Exclusion Performance

Size exclusion chromatography (SEC) is a method used to separate molecules based on their size. Bio-Gel A consists of spherical beads with specific size distribution pores. Small molecules diffuse into the pores, while large molecules are eluted in the column's void volume. The resolution depends on factors such as particle size, pore size, flow rate, column length, and sample volume.

Applications

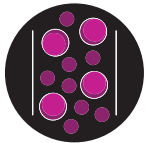
- Purification of antibodies and aggregates
- Isolation of immune complexes
- DNA fractionation
- Chromatin isolation and purification
- Extracellular vesicle isolation

Ordering Information

Catalog #	Description
Bulk Resins	
151-0450	Bio-Gel A-1.5m Gel , fine, 500 ml
151-0430	Bio-Gel A-1.5m Gel , coarse, 500 ml
151-0440	Bio-Gel A-1.5m Gel , medium, 500 ml
151-0450	Bio-Gel A-1.5m Gel , fine, 500 ml
151-0451	Bio-Gel A-1.5m Gel , fine, 10 L
151-0731	Bio-Gel A-5m Gel , coarse, 10 L
151-0741	Bio-Gel A-5m Gel , medium, 10 L
151-0751	Bio-Gel A-5m Gel , fine, 1 L
151-0752	Bio-Gel A-5m Gel , fine, 10 L

Screen this resin for your application today.

Bio-Gel P Resins



SIZE EXCLUSION /
GEL FILTRATION

RESIN FUNCTIONS

- Efficient, gentle gel filtration of sensitive compounds
- Ideal for purification of antibodies and aggregates
- High-resolution separation ranging from 100 to 100,000 daltons

RESIN CHARACTERISTICS

- Size exclusion media
- Polyacrylamide gel
- Extremely hydrophilic and essentially nonionic
- Does not support microbial growth or leach carbohydrates
- Excellent shrinking and swelling properties
- Compatible with high-salt buffers without bed volume change

SUPPORTING INFORMATION AND APPLICATION EXAMPLES

- Bio-Gel Resins flier, bulletin 6906
- Desalting with Bio-Gel P-6DG Desalting Gel, bulletin 2068
- Instruction manual, document LIT174

ION EXCHANGE

MIXED-MODE / MULTIMODAL

AFFINITY

HYDROPHOBIC INTERACTION

SIZE EXCLUSION / GEL FILTRATION

Bead Properties

All Bio-Gel P Resins

Property	Description
Function	Size exclusion
Base matrix composition	Copolymer of acrylamide and N,N'-methylenebisacrylamide
Maximum operating pressure	15 psi
pH stability	2–10 (short term) at room temperature
Shipping solution	Shipped as dry powder
Cleaning	Urea or guanidine HCl
Sanitization	Autoclavable at pH 5.5–6.5 at 120°C for 15–30 min. When autoclaving, Bio-Gel P Gels may swell 4–25 times the original volume; swelling increases with increasing pore size
Chemical compatibility	Compatible with dilute organic acids, 8 M urea, chaotropic agents, detergents, and miscible organic solvents. Alcohol, up to 20% v/v, can be used to enhance the solubility of nucleotides, peptides, and tannins without altering the exclusion properties of the gels. Formamide may also be used at full strength since Bio-Gel P Gels are completely swollen in this solvent
Storage temperature	4°C to room temperature
Storage conditions	Dry, at room temperature. Once hydrated, store in distilled water or aqueous buffers at 4°C with 0.02% sodium azide
Shelf life	5 years

Specific Bio-Gel P Resins

Resins	Bead Size	Fractionation Range/ Exclusion Limit, Da	Hydrated Bed Volume/ Mass Dry Gel, ml/g*	Particle Size, µm	Recommended Linear Flow Rate, cm/hr
Bio-Gel P-2 Gel	Fine	100–1,800	3	45–90	5–10
Bio-Gel P-2 Gel	Extra fine	100–1,800	3	<45	<10
Bio-Gel P-4 Gel	Medium	800–4,000	4	90–180	15–20
Bio-Gel P-4 Gel	Fine	800–4,000	4	45–90	10–15
Bio-Gel P-4 Gel	Extra fine	800–4,000	4	<45	<10
Bio-Gel P-6 Gel	Medium	1,000–6,000	6.5	90–180	15–20
Bio-Gel P-6 Gel	Fine	1,000–6,000	6.5	45–90	10–15
Bio-Gel P-6 Gel	Extra fine	1,000–6,000	6.5	<45	<10
Bio-Gel P-6DG Gel	Desalting grade	1,000–6,000	6.5	80–140	15–20
Bio-Gel P-10 Gel	Medium	1,500–20,000	7.5	90–180	15–20
Bio-Gel P-10 Gel	Fine	1,500–20,000	7.5	45–90	10–15
Bio-Gel P-30 Gel	Medium	2,500–40,000	9	90–180	7–13
Bio-Gel P-30 Gel	Fine	2,500–40,000	9	45–90	6–11
Bio-Gel P-60 Gel	Medium	3,000–60,000	11	90–180	4–6
Bio-Gel P-60 Gel	Fine	3,000–60,000	11	45–90	3–5
Bio-Gel P-100 Gel	Medium	5,000–100,000	12	90–180	4–6
Bio-Gel P-100 Gel	Fine	5,000–100,000	12	45–90	3–5

* In aqueous buffers, 1 gram hydrates to the volume listed.

Desalting Applications

Bio-Gel P-6DG Gel is ideal for desalting applications. The chromatograms in Figures 1 and 2 illustrate the impact on desalting and dilution by providing a visual representation of how different sample volumes can affect the recovery and purity of the material. The sample volume should not exceed 25% of the gel volume (Figure 1). When minimum sample dilution is the most important factor, the largest sample size that gives adequate desalting is recommended. Figure 2 shows chromatograms with varying sample-to-gel ratios.

The impact of desalting on the recovery and purity of the material can be observed in the chromatograms. Higher flow rates during desalting can result in faster separation of molecules, but may also lead to lower resolution. Additionally, dilution can further affect the recovery and purity of the material, as it can alter the concentration and composition of the sample being analyzed.

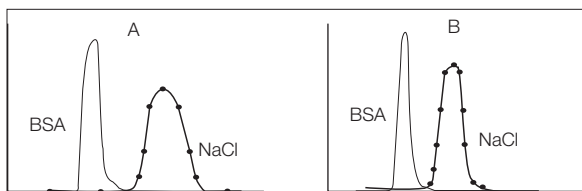


Fig. 1. Desalting of BSA on Bio-Gel P-6DG Gel. A, 1.5 ml sample, containing 1.25 mg/ml BSA in 2 M NaCl, was run on a 1.0 x 10 cm Econo-Column Chromatography Column, containing 6 ml Bio-Gel P-6DG, at a flow rate of 39 ml/hr (gravity flow); B, 10 ml sample, containing 2 mg/ml BSA in 2 M NaCl, was run on a 2.5 x 10 cm Econo-Column Chromatography Column, containing 40 ml Bio-Gel P-6DG, at a flow rate of 120 ml/hr (gravity flow).

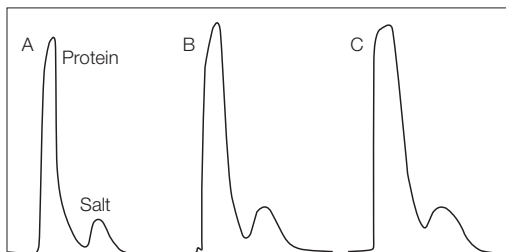


Fig. 2. Effect of sample volume on desalting. A, 1 ml sample (20% of gel volume), containing 1.3 mg/ml BSA and 2 mg/ml NaN₃, was run on a 1.0 x 10 cm Econo-Column Chromatography Column, containing 5 ml Bio-Gel P-6DG, at a flow rate of 30 ml/hr; B, 1.5 ml sample (30% of gel volume); all other conditions as in A; C, 2.0 ml sample (40% of gel volume); all other conditions as in A.

Applications

- High-efficiency desalting
- Buffer exchange
- Purification of proteins, peptides, nucleic acids, viruses, plasmids, and other macromolecules

Ordering Information

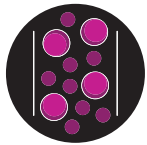
Catalog #	Description
Bulk Resins	
1504114	Bio-Gel P-2 Gel, fine, 100 g
150-4115	Bio-Gel P-2 Gel, fine, 500 g
12002617	Bio-Gel P-2 Gel, fine, 1 kg
1504118	Bio-Gel P-2 Gel, extra fine, 100 g
1504120	Bio-Gel P-4 Gel, medium, 100 g
12020840	Bio-Gel P-4 Gel, medium, 1 kg
12020799	Bio-Gel P-4 Gel, medium, 10 kg
1504124	Bio-Gel P-4 Gel, fine, 100 g
1504128	Bio-Gel P-4 Gel, extra fine, 100 g
1504130	Bio-Gel P-6 Gel, medium, 100 g
1504134	Bio-Gel P-6 Gel, fine, 100 g
150-4135	Bio-Gel P-6 Gel, fine, 10 kg
1504138	Bio-Gel P-6 Gel, extra fine, 100 g
1500738	Bio-Gel P-6DG Gel, 100 g
150-0739	Bio-Gel P-6DG Gel, 1 kg
1504140	Bio-Gel P-10 Gel, medium, 100 g
1504144	Bio-Gel P-10 Gel, fine, 100 g
150-4149	Bio-Gel P-10 Gel, fine, 10 kg
1504150	Bio-Gel P-30 Gel, medium, 100 g
150-4151	Bio-Gel P-30 Gel, medium, 10 kg
1504154	Bio-Gel P-30 Gel, fine, 100 g
12002610	Bio-Gel P-30 Gel, fine, 1 kg
150-4155	Bio-Gel P-30 Gel, fine, 10 kg
1504160	Bio-Gel P-60 Gel, medium, 100 g
1504164	Bio-Gel P-60 Gel, fine, 100 g
12002609	Bio-Gel P-60 Gel, fine, 10 kg
1504170	Bio-Gel P-100 Gel, medium, 100 g
1504174	Bio-Gel P-100 Gel, fine, 100 g
Prepacked Columns	
7322010	Econo-Pac 10DG Desalting Columns, 30
7325314	Bio-Scale Mini Bio-Gel P-6 Desalting Cartridges, 50 ml, 5 units
7325312	Bio-Scale Mini Bio-Gel P-6 Desalting Cartridge, 50 ml, 1 unit
7325304	Bio-Scale Mini Bio-Gel P-6 Desalting Cartridges, 10 ml, 5 units
12009238	EconoFit Bio-Gel P-6 Desalting Column, 1 x 5 ml
12009239	EconoFit Bio-Gel P-6 Desalting Columns, 5 x 5 ml

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Bio-Beads S-X Resins



SIZE EXCLUSION /
GEL FILTRATION

RESIN FUNCTIONS

- Size exclusion media for lipophilic polymers and solutes that require organic eluents
- Exclusion limit range from 400 to 14,000 daltons

RESIN CHARACTERISTICS

- Pore dimensions and exclusion limits are influenced by the eluant employed
- Used with benzene, toluene, xylene, carbon tetrachloride, and mixtures of solvents

SUPPORTING INFORMATION AND APPLICATION EXAMPLES

- Bio-Beads Resins flier, bulletin 6905
- Instruction manual, document LIT263

Bead Properties

Property	Description
Function	Size exclusion
Base matrix composition	Neutral, porous styrene-divinylbenzene beads
Mesh size/MW operating range	1% crosslinkage: 200–400/600–14,000 Da 3% crosslinkage: 200–400/up to 2,000 Da 8% crosslinkage: 200–400/up to 1,000 Da 12% crosslinkage: 200–400/up to 400 Da
Particle size	40–80 µm
Swollen bed volume	1% crosslinkage: 7.5 ml/g benzene 3% crosslinkage: 4.75 ml/g benzene 8% crosslinkage: 3.1 ml/g benzene 12% crosslinkage: 2.5 ml/g benzene
Nominal density	2% crosslinkage: 0.65 g/ml 4% crosslinkage: 0.7 g/ml 8% crosslinkage: 0.75 g/ml
Recommended linear flow rate	1% crosslinked resins: gravity flow 3% crosslinked resins: up to 5 ml/min 8–12% crosslinked resins: 600 cm/hr
pH stability	1–14
Shipping solution	Shipped as dry powder
Regeneration	Solvents: methylene chloride, toluene, or tetrahydrofuran
Sanitization	1 N NaOH; beads can be autoclaved
Chemical compatibility*	Avoid strong oxidizing reagents such as chromic acid, nitric acid, and hydrogen peroxide
Shelf life	3 years

* Explosive mixtures may form in the presence of strong oxidizing reagents.

Bio-Beads S-X Resins are composed of neutral, porous styrene-divinylbenzene copolymer beads with exclusion limits ranging from 400 to 14,000 daltons. They are ideal for fractionating and separating low molecular weight organic polymers and hydrophobic substances. The beads' pore size and exclusion limits are influenced by the eluant used, typically nonpolar, aromatic solvents. Bio-Beads S-X Resins use gel filtration for separation, requiring mobile eluants, and must be used in column mode.

As an example, a variety of lipophilic polymer substances have been successfully purified on beds of Bio-Beads S-X as shown in Figure 1.

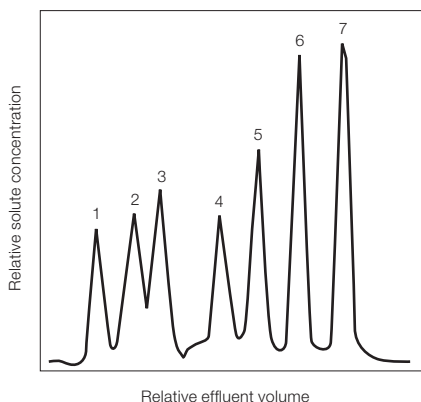


Fig. 1. Separation of triglycerides and hydrocarbons on Bio-Beads S-X1 and S-X2 Resins in benzene (two beds in series). 1, tristearin; 2, trimyristin; 3, trilaurin; 4, tricaprylin; 5, tricaproin; 6, hexadecane; 7, undecane.

Applications

- Particularly suitable for the fractionation and separation of low molecular weight organic polymers and other hydrophobic substances
- Quantification of pesticides and rodenticides
- Measurement of organic priority pollutants in sludge
- Separation of polycyclic aromatic compounds
- Assessment of tissue reaction to biomaterial
- Fractionation of halogenated environmental contaminants
- Separation of tall oil components, lipids, alkalines, long-chain fatty acids, hydrocarbons, and polystyrenes

Ordering Information

Catalog #	Description
Bulk Resins	
1522150	Bio-Beads S-X1 Support , 100 g
1522750	Bio-Beads S-X3 Support , 100 g
1523350	Bio-Beads S-X8 Support , 100 g
1523650	Bio-Beads S-X12 Support , 100 g
152-2151	Bio-Beads S-X1 Support , 1 kg
152-2753	Bio-Beads S-X3 Support , 1 kg

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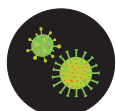
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ION EXCHANGE

MIXED-MODE / MULTIMODAL

AFFINITY

HYDROPHOBIC INTERACTION

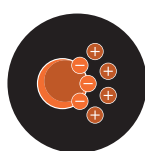
SIZE EXCLUSION / GEL FILTRATION

CASE STUDY

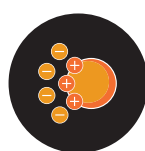
Biosimilar Monoclonal Antibody Purification Using Nuvia S, Nuvia Q, and Nuvia cPrime Resins



ANTIBODY
PURIFICATION



CATION
EXCHANGE



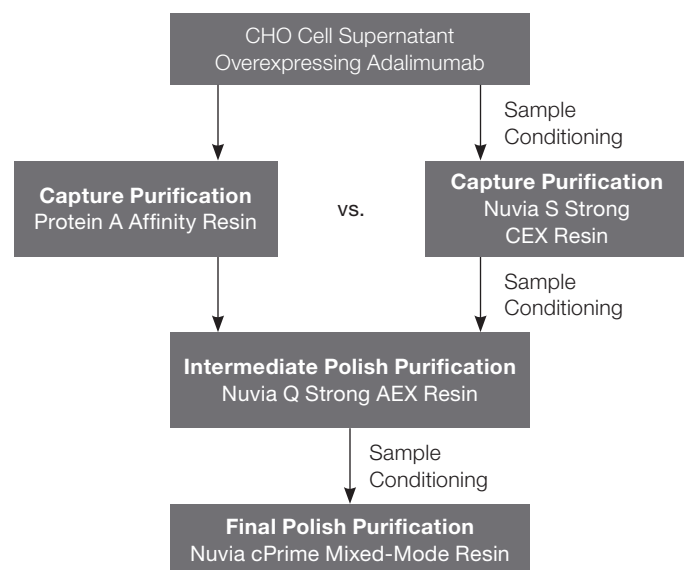
ANION
EXCHANGE



HI CATION
EXCHANGE

Biosimilars are significantly cheaper than biologics, owing to their shorter development time and fewer involved costs. Biosimilar purification costs constitute more than half of the total development costs. A significant number of these costs are due to the expense of Protein A-based media. Here, we propose an alternative but equally effective approach to biosimilar purification. With an adalimumab (HUMIRA) biosimilar, we show that using Nuvia S Cation Exchange Resin can not only help minimize purification costs, but also overcome the limitations of Protein A capture, including low binding capacity and Protein A leaching.

Protocol



Methods

Two parallel workflows starting with Nuvia S and Protein A capture resins were used for HUMIRA purification. The protocols used are shown and the details can be found in bulletin 7130.

Capture by Nuvia S CEX Resin

Chromatographic purification was conducted using the NGC Chromatography System (Bio-Rad Laboratories, Inc.). A 1 ml Foresight Nuvia S Column (Bio-Rad, #732-4720) was equilibrated with 10 column volumes (CV) of 25 mM sodium acetate (pH 4.5), 5 mM NaCl (buffer A). The CHO cell supernatant containing the biosimilar was adjusted to pH 4.5 with 1 M hydrochloric acid (HCl) and applied to the pre-equilibrated Nuvia S Column at a flow rate of ~120 cm/hr. The column was washed with 5 CV of buffer A followed by 5 CV of 25 mM sodium acetate (pH 4.75), 75 mM NaCl. The bound biosimilar was eluted with 10 CV of 25 mM sodium acetate (pH 5.0), 250 mM NaCl (buffer B).

Capture by Protein A Affinity Resin

MabSelect SuRe Protein A Resin (Cytiva, #17-5438-01) was packed in a 1 ml column according to the manufacturer's instructions and the biosimilar was loaded onto it. The bound biosimilar was eluted with 10 CV of 100 mM sodium acetate (pH 3.5), 100 mM NaCl.

Intermediate Polishing by Nuvia Q AEX Resin

The eluate from the capture step was concentrated and buffer-exchanged into 25 mM HEPES (pH 8.25), 5 mM NaCl (buffer C) using an Amicon Ultra-4 Centrifugal Filter Unit (50 kD MW cutoff) (EMD Millipore, #UFC805024). This sample was applied on a 1 ml Foresight Nuvia Q Column (Bio-Rad, #732-4721) equilibrated with 10 CV of buffer C. The flow-through fractions were collected for further purification.

Final Polishing by Nuvia cPrime Mixed-Mode Resin

The pooled flow-through fractions from the previous step were concentrated, buffer-exchanged with buffer B, and applied to a Foresight Nuvia cPrime Column (Bio-Rad, #732-4722) pre-equilibrated with 10 CV of the same buffer. The biosimilar was eluted with 5 CV of 25 mM Tris (pH 8.0), 250 mM NaCl. The column was then stripped with 10 CV of 25 mM Tris (pH 8.0), 800 mM NaCl to remove all bound impurities.

Capture with Nuvia S Resin

Column: 1 ml Foresight Nuvia S Column

Step	Buffer	CV	Flow Rate, cm/hr
Equilibration	25 mM sodium acetate (pH 4.5), 5 mM NaCl (buffer A)	10	120
Sample	CHO cell supernatant containing the biosimilar was adjusted to pH 4.5 with 1 M HCl and applied to the pre-equilibrated Nuvia S Column	—	120
Wash	Buffer A Followed by 25 mM sodium acetate (pH 4.75), 75 mM NaCl	5 5	120
Elution	25 mM sodium acetate (pH 5.0), 250 mM NaCl (buffer B)	10	120

Capture by Protein A Affinity Resin

MabSelect SuRe Protein A Resin was packed in a 1 ml column according to the manufacturer's instructions and the biosimilar was loaded onto it. The bound biosimilar was eluted with 10 CV of 100 mM sodium acetate (pH 3.5), 100 mM NaCl.

Intermediate Polishing by Nuvia Q AEX Resin

Column: 1 ml Foresight Nuvia Q Column

Step	Buffer	CV	Flow Rate, cm/hr
Equilibration	25 mM HEPES (pH 8.25), 5 mM NaCl (buffer C)	10	120
Sample	The eluate from the capture step was concentrated and buffer-exchanged into buffer C using an Amicon Ultra-4 Centrifugal Filter Unit (50 kD MW cutoff)	—	120

The flow-through fractions were collected for further purification.

Final Polishing by Nuvia cPrime Mixed-Mode Resin

Column: 1 ml Foresight Nuvia cPrime Column

Step	Buffer	CV	Flow Rate, cm/hr
Equilibration	Buffer B	10	120
Sample	The pooled flow-through fractions from the previous step were concentrated, buffer-exchanged with buffer B, and applied to a Foresight Nuvia cPrime Column	—	120
Wash	Buffer B	10	120
Elution	25 mM Tris (pH 8.0), 250 mM NaCl	5	120
Strip	25 mM Tris (pH 8.0), 800 mM NaCl	10	120

Results

Both the Protein A and Nuvia S purification workflows in 96-well plates showed the same impurity clearance levels after the polish step (Table 1). Nuvia S Resin has a higher DBC and lower price than Protein A and hence improves process economy of this workflow (Table 2). Purification using 1 ml Foresight Columns showed the same results for the Nuvia A workflow (Figures 1 and 2, Table 3).

ELISA and cell-based assays showed that the purified mAb was functionally active (Figure 3).

Table 1. Comparison of HCP clearance on 96-well filter plates between Nuvia S Resin and a Protein A resin.

	HCP, ppm Nuvia S Capture	HCP, ppm Protein A Capture
CHO cell supernatant	72,000	90,300
Capture purification	6,800	980
Nuvia Q intermediate polish purification	250	20
Nuvia cPrime final polish purification	5	20

Table 2. Comparison of cost and binding capacity between Nuvia S Resin and a Protein A resin.

	Nuvia S Resin	MabSelect SuRe Protein A Resin
Technique	Strong cation exchange chromatography	Affinity chromatography
Binding capacity	≥110 mg/ml	~35 mg/ml
Cost (25 ml)*	\$110	\$2,791

* Pricing mentioned is valid for 2018.

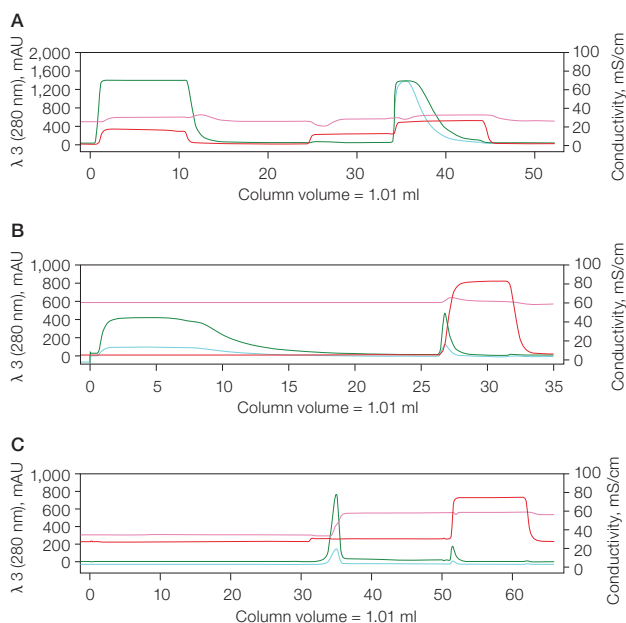


Fig. 1. Chromatograms from purification of the biosimilar. A, capture purification with Nuvia S Resin. The biosimilar obtained from fractions 36 to 44 was subjected to further purification and purity analysis. **B,** intermediate polish purification with Nuvia Q Resin. The column flow-through fractions from 1 to 12 were subjected to further purification and purity analysis. **C,** final polish purification on Nuvia cPrime Resin. The biosimilar eluted from column fractions 33–37 was collected and subjected to purity analysis. λ 2 (255 nm) (—); λ 3 (280 nm) (—); conductivity (—); pH (—).

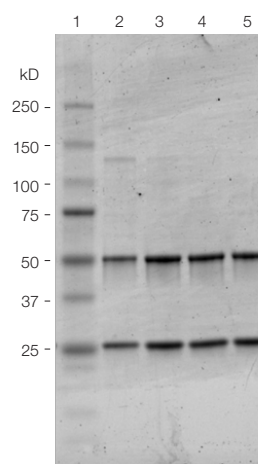


Fig. 2. SDS-PAGE analysis of the chromatographic fractions following purification on Foresight Columns. Lane 1, Precision Plus Protein Unstained Protein Standards; lane 2, CHO supernatant; lane 3, Nuvia S sample; lane 4, Nuvia Q sample; lane 5, Nuvia cPrime sample.

Table 3. HCP and hcDNA clearance with 1 ml Foresight Columns.

Sample	HCP, ppm	Overall Reduction, %	hcDNA, pg/μl	Overall Reduction, %
CHO supernatant (crude)	170,000	—	26,000	—
Nuvia S Cation Exchange (capture)	10,300	93.91	300	98.86
Nuvia Q Anion Exchange (intermediate polish)	120	99.93	0	100
Nuvia cPrime Mixed-Mode (final polish)	6	99.99	—	—

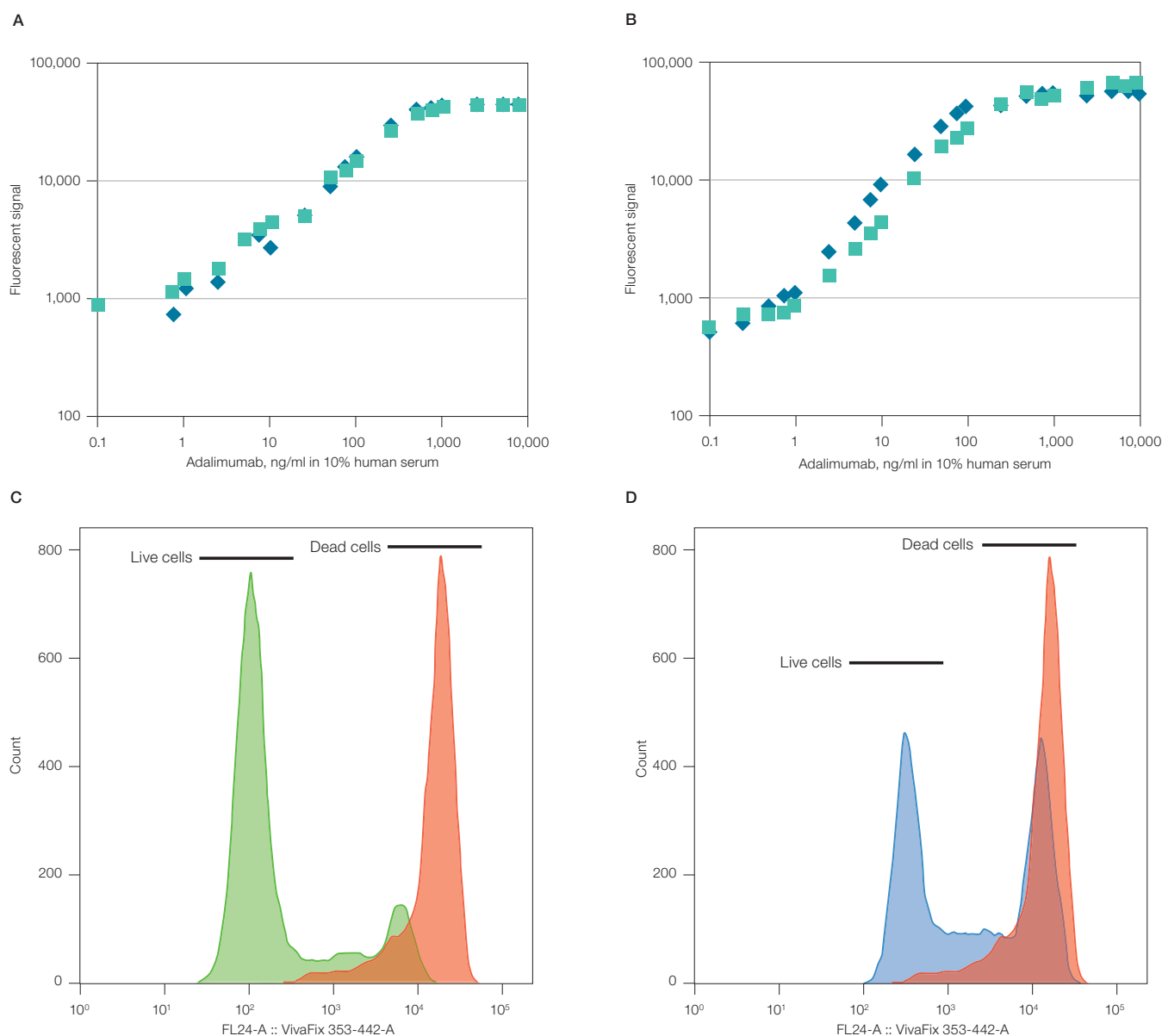


Fig. 3. Functional testing of purified adalimumab. **A**, pharmacokinetic antigen capture ELISA; comparison of capture of TNF- α by the adalimumab standard (◆) and the purified adalimumab (■) in the 96-well plate. **B**, pharmacokinetic bridging ELISA; comparison of anti-adalimumab binding by the adalimumab standard (◆) and the purified adalimumab (■) in the 96-well plate. Both adalimumabs show similar binding trends to TNF- α and the anti-adalimumab antibody. **C**, comparison of cells treated with TNF- α only (■) and cells treated with TNF- α plus adalimumab purified from lysate (PFL) (■). **D**, comparison of cells treated with TNF- α only (■) vs. cells treated with TNF- α and control anti-TNF- α antibody (purchased purified adalimumab) (■). Both adalimumab biosimilars show a protective effect on the cells after treatment with TNF- α .

Summary

We show that an IEX and mixed-mode resin-based workflow can deliver a highly pure and active adalimumab biosimilar, equal to that achieved using a purification workflow involving Protein A. Use of the Nuvia S Cation Exchanger for capture led to better process economics, not only by reducing cost but also by enhancing the process speed, a result of the significantly higher binding capacity of the resin.

Get complete details of this case study in Improved Process Economics of HUMIRA Biosimilar Purification with Ion Exchange and Mixed-Mode Resins, bulletin 7130.



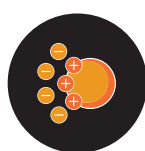
Visit [bio-rad.com/ResinSample](https://www.bio-rad.com/ResinSample) to request a sample of the resins used in this workflow.

CASE STUDY

IgM Purification Using Nuvia HP-Q Resin and CHT Ceramic Hydroxyapatite Media



ANTIBODY
PURIFICATION



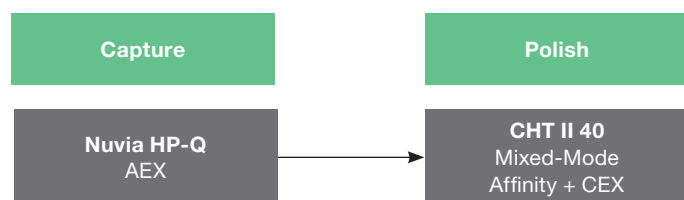
ANION
EXCHANGE



CHT CERAMIC
HYDROXYAPATITE

IgM antibodies are one of the fastest growing groups of diagnostic and therapeutic candidates. The large size of IgMs makes their purification very challenging. Bio-Rad has developed a strong AEX resin, Nuvia HP-Q, to overcome the multiple issues faced when purifying large biomolecules. This AEX-based purification platform helps overcome the limitations of traditional ion exchange resins for IgM purification and provides a successful and scalable platform strategy. The purified IgM is suitable for diagnostic applications and potentially for pharmaceutical applications.

Protocol



Method

HEK 293 cells expressing acidic IgM (pI ~6.7) were capture purified on Nuvia HP-Q and polish purified with CHT II 40 at the following conditions.

Capture with Nuvia HP-Q Resin

Column: 1 ml Foresight Nuvia HP-Q Column

Step	Buffer	CV	Flow Rate, cm/hr
Equilibration	20 mM Na ₂ HPO ₄ /NaH ₂ PO ₄ , pH 7.5 (buffer A)	10	120
Sample	400 ml cell culture supernatant adjusted to pH 7.5 with 800 ml equilibration buffer	–	120
Wash	Buffer A	20	120
Elution	Linear gradient from 0 to 20% buffer B (1 M NaCl, 20 mM Na ₂ HPO ₄ /NaH ₂ PO ₄ , pH 7.5)	10	120

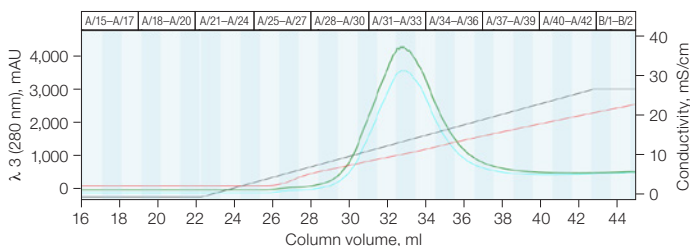
Polish with CHT Type II

Column: 1 ml Foresight CHT II 40 Column

Step	Buffer	CV	Flow Rate, cm/hr
Equilibration	10 CV of 1 M NaCl, 20 mM Na ₂ HPO ₄ /NaH ₂ PO ₄ , pH 7.0	10	120
Sample	Nuvia HP-Q elution pool, supplemented with NaCl, with pH adjusted to 7.0 using diluted H ₃ PO ₄	–	120
Wash	1 M NaCl, 20 mM Na ₂ HPO ₄ /NaH ₂ PO ₄ , pH 7.0	10	120
Elution	Linear gradient from 0 to 100% buffer D (1 M NaCl, 400 mM Na ₂ HPO ₄ /NaH ₂ PO ₄ , pH 7.0)	20	120

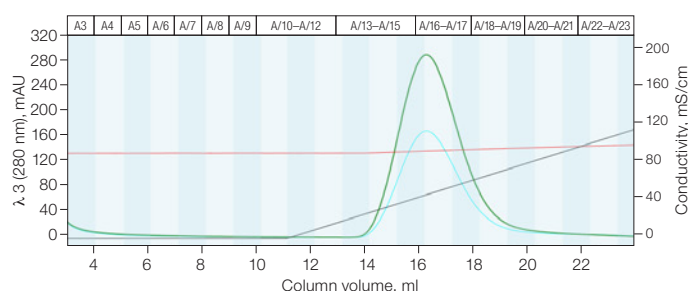
Results

The acidic IgM was purified on Nuvia HP-Q and CHT II-40 (Figures 1 and 2). The CHT II 40 eluate showed a single band of IgM in the nonreducing gel and two bands for the heavy and the light chains, respectively, on the reducing gel (Figure 3). The purity was further confirmed by SEC analysis. Contaminant peaks were seen on the SEC plot with the Nuvia HP-Q eluate (Figure 4A) but only a single sharp peak for IgM was seen on the SEC plot with the CHT II 40 eluate (Figure 4B). Analysis of the contaminant levels in each of the samples showed a reduction in the DNA and host cell protein content after Nuvia HP-Q purification. Polish purification with CHT II 40 resulted in minimizing these contaminants further (Table 1).



Column: Foresight Nuvia HP-Q, 1 ml
 Equilibration: 10 CV of 20 mM Na₂HPO₄/NaH₂PO₄, pH 7.5 (buffer A)
 Sample: 400 ml cell culture supernatant adjusted to pH 7.5 with 800 ml equilibration buffer
 Flow rate: 120 cm/hr
 Wash: 20 CV buffer A
 Elution: 10 CV linear gradient from 0 to 20% buffer B
 Buffer A: 20 mM Na₂HPO₄/NaH₂PO₄, pH 7.5
 Buffer B: 1 M NaCl, 20 mM Na₂HPO₄/NaH₂PO₄, pH 7.5

Fig. 1. Capture purification of IgM on Nuvia HP-Q Resin. Chromatography run of IgM on Nuvia HP-Q with a 0–200 mM NaCl linear gradient. 260 nm (■); 280 nm (■); conductivity (■); %B (■).



Column: Foresight CHT II 40, 1 ml
 Equilibration: 10 CV of 1 M NaCl, 20 mM Na₂HPO₄/NaH₂PO₄, pH 7.0
 Sample: Nuvia HP-Q elution pool, supplemented with NaCl, with pH adjusted to 7.0 using diluted H₃PO₄
 Flow rate: 120 cm/hr
 Wash: 10 CV buffer C
 Elution: 20 CV linear gradient from 0 to 100% buffer D
 Buffer C: 1 M NaCl, 20 mM Na₂HPO₄/NaH₂PO₄, pH 7.0
 Buffer D: 1 M NaCl, 400 mM Na₂HPO₄/NaH₂PO₄, pH 7.0

Fig. 2. Polish purification of IgM on CHT II 40. Chromatography run of the previously eluted IgM on CHT II 40 with a 20–400 mM phosphate gradient in the presence of 1 M NaCl. 260 nm (■); 280 nm (■); conductivity (■); %B (■).

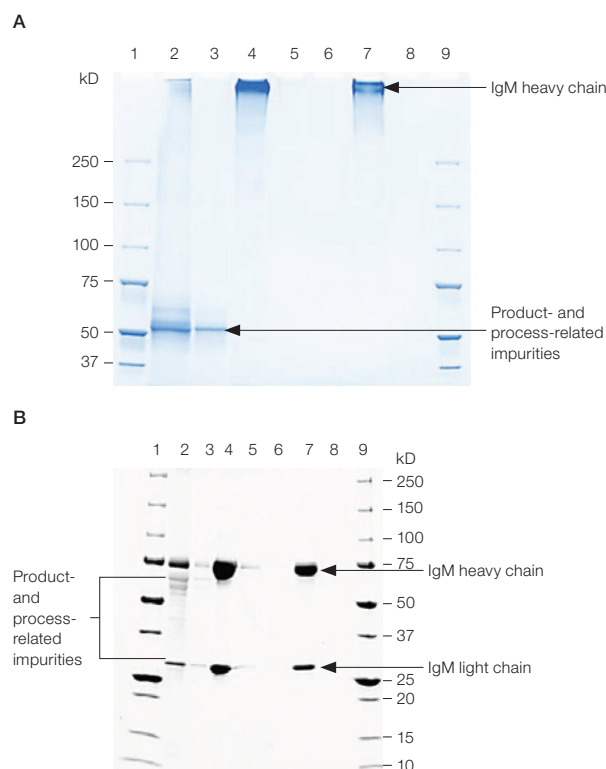


Fig. 3. Reduced SDS-PAGE analysis of IgM purified from the Nuvia HP-Q and CHT II 40 Columns. A, nonreducing Criterion XT Gel; B, reducing Criterion TGX Gel. A and B, lane 1, protein molecular weight (MW) marker; lane 2, culture supernatant; lane 3, Nuvia HP-Q flowthrough; lane 4, pooled Nuvia HP-Q eluate; lane 5, CHT II 40 flowthrough; lane 6, CHT II 40 wash; lane 7, pooled CHT II 40 eluate; lane 8, DNA peak from the CHT II 40 plot; lane 9, protein MW marker.

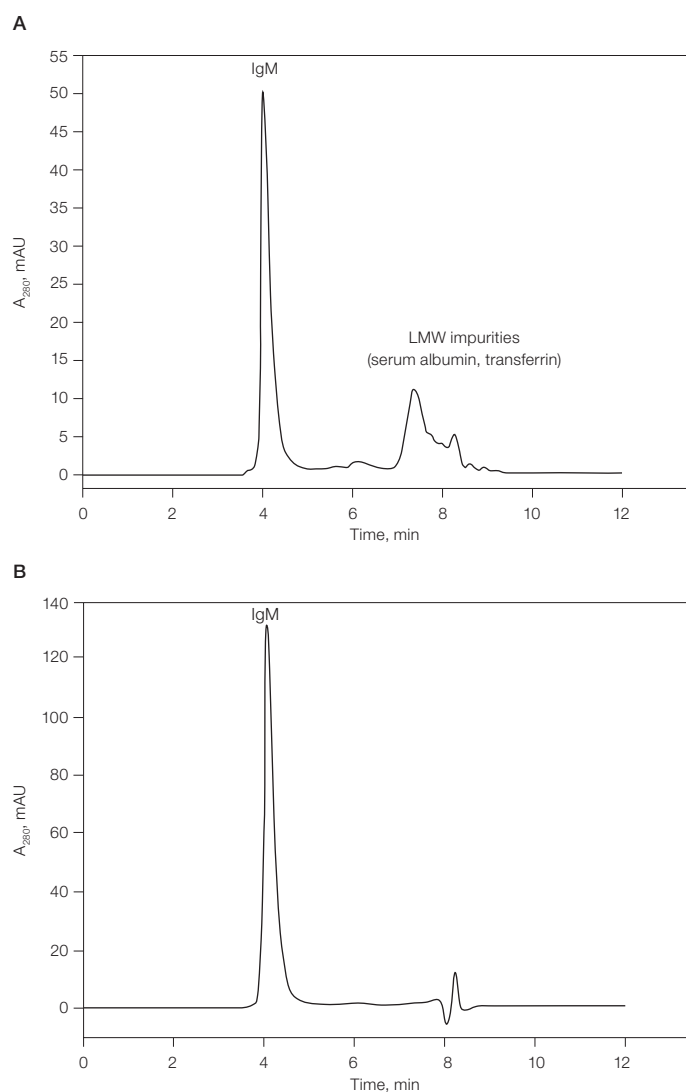


Fig. 4. SEC analysis of the Nuvia HP-Q and CHT II 40 eluates. Analysis was performed on a Dionex UltiMate 3000 System using a MAbPac SEC-1 Column. A sample volume of 3 μ l Nuvia HP-Q eluate (**A**) or 10 μ l CHT II 40 eluate (**B**) was injected. Analytical separation was achieved in isocratic mode using 50 mM Na_3PO_4 , 200 mM arginine, 300 mM NaCl, pH 6.8, as running buffer at 0.8 ml/min for 12 min. LMW, low molecular weight.

Table 1. Final purification data.

Sample	Purity, %*	Recovery, %*	DNA, pg/ μ l**	HCP, ppm***
Culture supernatant	20	—	2,332	>200,000
Nuvia HP-Q EP	63	83	1,187	21,455
CHT II 40 EP	>99	77	Not detectable	109

EP, elution pool; HCP, host cell protein.

* Determined by HPLC-SEC.

** DNA content determined by Quant-iT PicoGreen dsDNA Assay Kit.

*** HCP content determined by HEK 293 HCP ELISA Kit.

Summary

This nonaffinity-based platform approach using Nuvia HP-Q Anion Exchange Resin produces a >99% pure IgM, which meets the purification requirements for diagnostic IgM manufacturing. Using this state-of-the-art resin with high capacity and selectivity for large biomolecules also helps overcome the affinity-based challenges of low binding and/or recovery during IgM purification.

Get complete details of this case study in Purification of an Acidic IgM with a Strong Anion Exchange Resin Tailored for Large Biomolecule Purification, bulletin 7127.

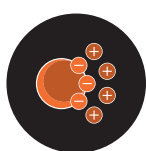
Visit [bio-rad.com/ResinSample](https://www.bio-rad.com/ResinSample) to request a sample of the resins used in this workflow.

CASE STUDY

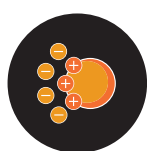
Clinical-Grade Monoclonal Antibody Purification Using Nuvia S, Nuvia Q, and Nuvia cPrime Resins



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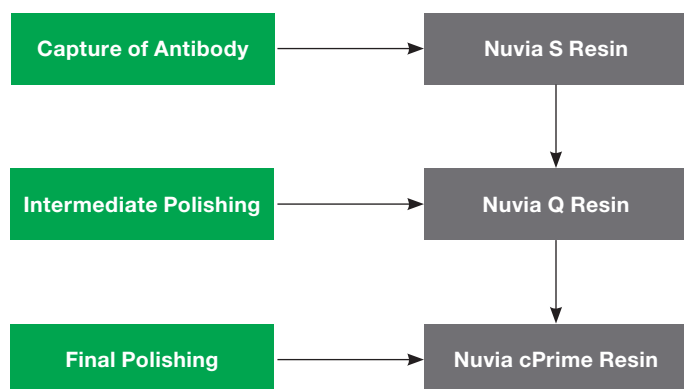
ANION
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mAbs are an immensely important class of therapeutic proteins. The elevated levels of process- and/or product-related impurities resulting from prolonged fermentation and substantially higher cell density in the expression culture have introduced numerous challenges in their purification. Chromatographic resins with high capacity, selectivity, and improved chromatographic and operational performance offer the latest productivity tools to address downstream process challenges. We have employed two next-generation ultra-high capacity ion exchangers, Nuvia S and Nuvia Q Resins, and Nuvia cPrime Hydrophobic CEX Resin, to effectively purify a mAb from CHO cell culture harvest. Our results demonstrate that this three-step nonaffinity workflow can effectively deliver highly purified mAbs with minimal feed conditioning.

Protocol



Method

Capture on Nuvia S Resin

Column: 1 ml, 0.56 x 4 cm

Step	Buffer	CV	Flow Rate, cm/hr
Equilibration	20 mM sodium acetate, 20 mM NaCl, pH 4.7 (buffer A)	15	300
Sample	CHO cell culture supernatant diluted 1:4 with dH ₂ O; adjusted to pH 4.7 with 1 M phosphoric acid; clarified with 0.2 µm filter	–	300
Wash	Buffer A	15	300
Elution	20 mM sodium acetate, 200 mM NaCl, pH 4.9 (buffer B)	15	300
Sanitization	1 M NaOH	5	300

Eluate was adjusted to pH 7.0 with 1 M NaOH.

Intermediate Polishing on Nuvia Q Resin

Column: 1 ml, 0.56 x 4 cm

Step	Buffer	CV	Flow Rate, cm/hr
Equilibration	10 mM sodium phosphate, 10 mM NaCl, pH 7.0 (buffer A)	10	300
Sample	Eluate from Nuvia S Column adjusted to pH 7.0	10	300
Wash	Buffer A	25	300
Strip	100 mM sodium phosphate, 1.5 M NaCl, pH 7.2 (buffer B)	10	300
Sanitization	1 M NaOH	5	300

The mAb in the flowthrough was collected and the pooled fractions were adjusted to pH 5.0 with 100 mM sodium acetate, 500 mM NaCl, pH 4.5.

Final Polishing on Nuvia cPrime Resin

Column: 1 ml, 0.56 x 4 cm

Step	Buffer	CV	Flow Rate, cm/hr
Equilibration	50 mM sodium acetate, 125 mM NaCl, pH 5.0 (buffer A)	10	300
Sample	Pooled fractions from Nuvia Q Column adjusted to pH 5.0	12	300
Wash	Buffer A	15	300
Elution 1	Salt gradient formed between buffer A and 50 mM sodium phosphate, 50 mM NaCl, pH 6.2 (buffer B)	15	300
Elution 2	Buffer B	5	300
Strip	200 mM sodium phosphate, 700 mM NaCl, pH 7.5	10	300
Sanitization	1 M NaOH	5	300

Results

The negatively charged groups in Nuvia S repel the host cell DNA contaminants causing the dsDNA to be presented mostly in the flow-through fractions, as revealed by the absorbance trace at 260 nm (Figure 1), resulting in a remarkable, more than 3-log reduction of the dsDNA contamination level (Table 1).

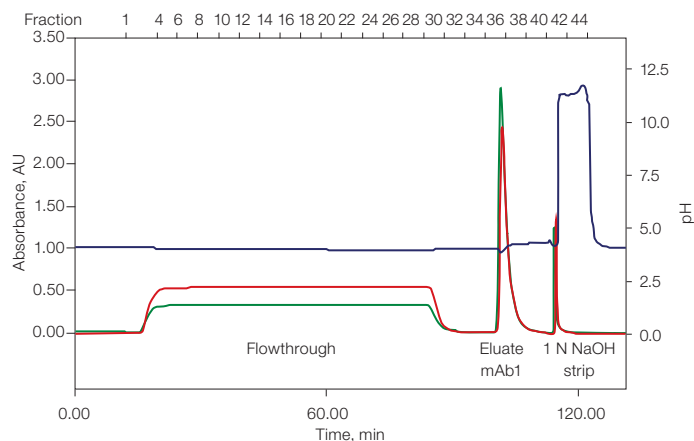


Fig. 1. Capturing mAb1 from CHO cell culture harvest using Nuvia S Resin. mAb1 eluted from the column (fractions 38–40) was subjected to further purification and purity analysis. OD 280 (—); OD 260 (—); pH (—).

Table 1. Impurity clearance.

Sample	Host Cell Proteins, ng/mg	Host Cell dsDNAs, ng/mg	Aggregate Content, %
Cell culture supernatant	6.3×10^4	9.3×10^4	Not determined
Nuvia S fraction	2.6×10^3	17	Not determined
Nuvia Q fraction	59	4.1	Not determined
Nuvia cPrime fraction	5.5	Not detected (<0.008)	<0.9

The primary concern of the Nuvia Q purification step was to maximize recovery of the mAb in the flow-through fractions while binding the highest level of contaminants to the column. We eventually conducted a flow-through purification of mAb purification at pH 7.0, slightly below the mAb's determined pI value of 7.2 (Figure 2). Both protein and DNA contaminants were dramatically reduced under this optimized condition.

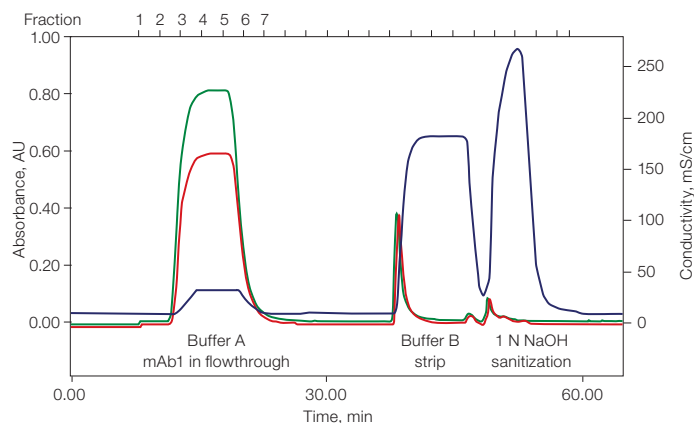


Fig. 2. Intermediate polishing purification of mAb1 in flow-through mode using Nuvia Q Resin. mAb1 in column flowthrough (fractions 3–6) was subjected to further purification and purity analysis. OD 280 (—); OD 260 (—); conductivity (—).

The partially purified mAb from the Nuvia Q polishing step could be loaded onto the Nuvia cPrime Column following a simple pH adjustment to positively charge the mAb without extensive dilution or buffer exchange for conductivity reduction. Such behavior makes Nuvia cPrime distinct from conventional hydrophobic interaction resins, which require high salt levels for effective protein binding. Under the selected condition, only the desired full-length mAb was retained by the column. The bound intact mAb was eluted by a buffer with slightly higher pH and lower conductivity, which suggests that the mAb was likely selectively bound by Nuvia cPrime via a combination of ionic and hydrophobic interactions (Figure 3). This chromatographic step was also very effective at eliminating host cell impurities (Table 1).

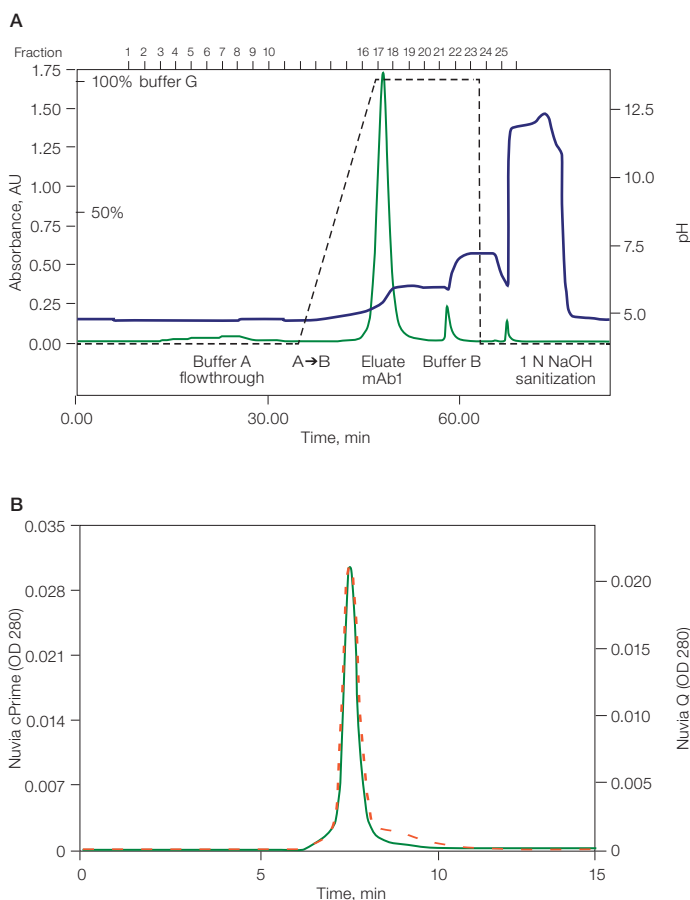


Fig. 3. Final polishing purification of mAb1 using Nuvia cPrime Resin. **A**, chromatogram of mAb1 purification on a Nuvia cPrime Column: OD 280 (—); pH (—); percentage of buffer G (---). mAb1 eluted from the column (fractions 17–19) was collected and subjected to purity analysis. **B**, HPLC-SEC comparison of pooled Nuvia Q fractions (---) and pooled Nuvia cPrime fractions (—).

The final mAb preparation was essentially nucleic acid-free because nucleic acids were unretained due to electrostatic charge repulsion between the negative hydrophilic DNA and the negatively charged hydrophobic ligand of Nuvia cPrime. The purity of the mAb after this three-step purification was assessed by HPLC-SEC analysis (Figure 4).

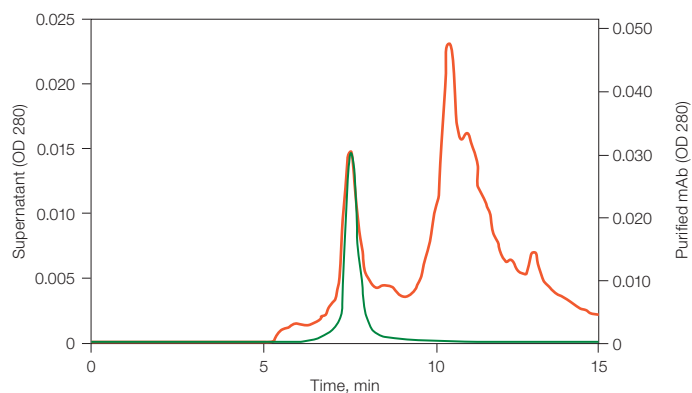


Fig. 4. Three-step purification of mAb1 using Nuvia cPrime Resin. HPLC-SEC comparison of cell culture supernatant (—) and purified mAb1 (—).

Nuvia cPrime Resin's higher affinity for the full-length mAb compared to process impurities and by-products gives this final polishing step its powerful purification benefit. It can be thought of as an orthogonal approach, combining CEX and HIC in a single step to achieve unique selectivity for a target protein molecule.

Summary

The extraordinarily high binding capacities of Nuvia S and Nuvia Q Resins make it possible to process increasing volumes of high-titer feed with existing production facilities and minimal buffer consumption, easing the pressure on capital investment and process development timelines. In the final polishing step, the orthogonal interaction modes afforded by Nuvia cPrime Resin offer unique selectivity for the full-length monomeric mAb molecules. This resin is a powerful tool for the removal of both product-related impurities and host-cell contaminants. The versatility of these resins has allowed us to arrange the order of these three chromatography steps in a sequence that requires no buffer exchange and minimal handling. This process is designed to shorten cycle time and reduce buffer consumption, thus improving overall productivity. All chromatography resins are base stable and mechanically resilient (bulletins 5987, 6129, and 6242). They can be operated at high flow rates with low backpressures, providing the productivity, robustness, and process economics demanded by today's downstream manufacture of biopharmaceuticals.

Get complete details of this case study in the Nuvia cPrime Hydrophobic Cation Exchange Resin product information sheet, bulletin 6242.



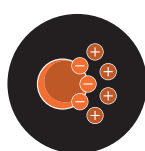
Visit [bio-rad.com/ResinSample](https://www.bio-rad.com/ResinSample) to request a sample of the resins used in this workflow.

CASE STUDY

Resolving Monoclonal Antibody Deamidation Charge Variants with Nuvia HR-S Resin



ANTIBODY
PURIFICATION



CATION
EXCHANGE

Protein deamidation at neutral or alkaline pH is of significant concern to manufacturers of biotherapeutic proteins. Deamidation at asparagine and glutamine residues has been shown to affect the immunogenicity, stability, and activity of these biotherapeutic proteins. In addition, these degradations also lead to molecular heterogeneity, which can present challenges to the manufacturing of a consistent protein product. The specific challenge is that the change in structure of the protein is minimal, as the amide group is substituted with a carboxylate. Furthermore, the overall charge of the protein changes by only -1 per deamidation, making charge-based separations such as IEX potentially difficult. The goal of the present work was to determine if a high-resolution CEX resin, Nuvia HR-S, is capable of resolving proteins differing by one or a few deamidation events.

Method

Intermediate Polishing on Nuvia HR-S Resin

Column: 1 ml, 0.5 x 5.0 cm

Step	Buffer	CV	Flow Rate, cm/hr
Equilibration	10 mM sodium acetate, pH 4.5 (buffer A)	–	300
Sample	Protein A eluate diluted 1:4 in buffer A	–	300
Wash 1	Buffer A	–	300
Wash 2	10 mM sodium citrate, 10 mM sodium phosphate, pH 5.0 (buffer B)	–	300
Elution	Gradient of buffer B to 10 mM sodium citrate, 10 mM sodium phosphate, pH 8.0	15	300

Results

An antibody with known surface asparagines and glutamines was first isolated by Protein A affinity chromatography and then further purified on Nuvia HR-S in a pH gradient. In the resulting chromatogram (Figure 1), the elution peak contains a prominent leading shoulder. The shoulder and peak were analyzed by SDS-PAGE electrophoresis and both were shown to contain antibody heavy and light chains (Figure 2). Fractions 7–13 contain the target mAb; fractions 8 and 11 were selected for further analysis by mass spectrometry (Figure 3). Analysis of the data indicated that the shoulder fragment had a higher mass by approximately 2.5 mass units. Since deamidation produces a mass change of approximately $+1$ units, this means that the light chain in the shoulder of the antibody peak resulted from two or three deamidation events.

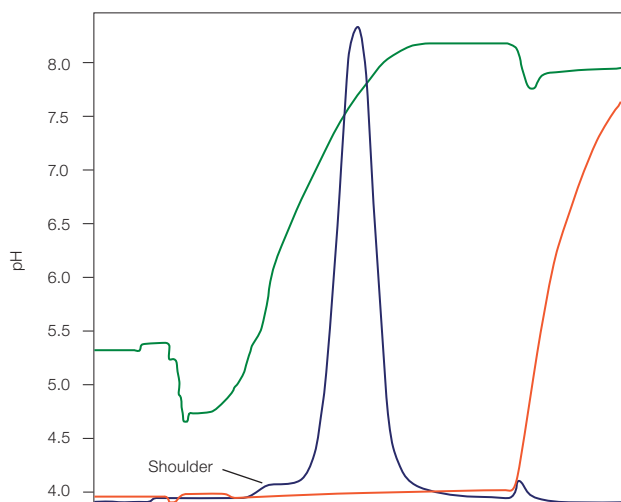


Fig. 1. Purification of the mAb on Nuvia HR-S. The distinct shoulder on the front side of the main peak is a modified mAb fragment. A_{280} (—); conductivity (—); pH (—).

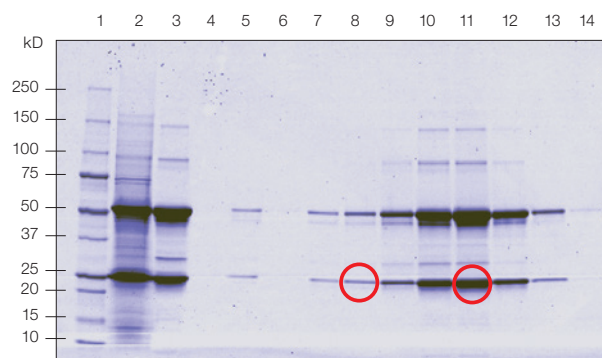


Fig. 2. SDS-PAGE analysis of chromatographic fractions. Lane 1, Precision Plus Protein Standards; 2, mAb cell culture harvest; 3, Protein A capture eluate. Lanes 4–14 show Nuvia HR-S Resin chromatography: 4–5, flowthrough 1 and 2; 6, wash; 7–14, elutions 1–8. The bands circled in red correspond to the bands used for mass spectrometry analysis in Figure 3.

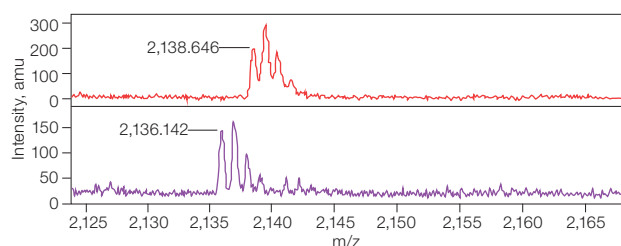


Fig. 3. Mass spectrometry analysis of the mAb. Analysis of the gel-purified proteins was carried out with an autoflex II MALDI-TOF Mass Spectrometer. Mass spectrometer analyses of the shouldered peak (top spectrum) and the main peak (lower spectrum) show a mass shift of about 2.5 amu between the two peaks, which is consistent with two or three deamidations on a single peptide. amu, atomic mass unit.

Examination of the sequence in that region revealed the presence of two asparagine and three glutamine residues, consistent with the results (Figure 4).

10	20	30	40	50	60	70
MKYLPTAA	GLLLAAQPA	HADIQNTQSP	SSLASVVGDR	VTITCRASQS	ISSTLNWYQQ	KPGKAPKLLI
80	90	100	110	120	130	140
YAASSLQSGV	PSRFSGSGSG	TDFTLTISL	QPEDFATTYC	QSYSTLLTF	GGGKVEIKR	TVAAPSVFIF
150	160	170	180	190	200	210
PPSDEQLKSG	TASVVCLLNN	FYPREVKVQW	K	VDNALQSGNSQESVTEQDSK	DSTTSLST	LTLSKADYEK
220	230	240	250	260		
HKVYACEVTH	QGLSSPVTKS	FNRGEYSARQ	STPFVCEYQG	QSSDLP		

Fig. 4. Shoulder fraction protein sequence. Examination of the shoulder fraction protein sequence confirms deamidation events on a single mAb peptide.

Summary

The present study has shown that Nuvia HR-S Resin is capable of resolving antibodies differing by no more than two or three deamidation events. The ability to resolve such closely related species can significantly aid manufacturers in producing consistent biotherapeutic products.

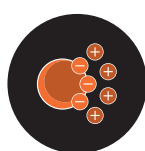
Visit [bio-rad.com/ResinSample](https://www.bio-rad.com/ResinSample) to request a sample of the resins used in this workflow.

CASE STUDY

Improving Aggregate Removal from Monoclonal Antibody Purification Using Nuvia HR-S Resin



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Ever increasing performance demands in protein purification require more selective chromatography methods to effectively remove aggregates and other impurities. At high titer levels, aggregates often present a unique purification challenge. One emerging approach to address this challenge is the use of new, smaller-particle chromatography resins that are optimized for high resolution and capacity. Such resins can be particularly productive in challenging situations and during final polishing steps (He et al. 2010). Nuvia HR-S is a high-resolution resin. It features a hydrophilic polymer matrix with an open-pore structure designed for fast and efficient mass transfer and superior flow properties at high flow rates. In this case study, the performance of Nuvia HR-S was compared to that of a high-resolution agarose-based resin.

Methods

Initial Separation of Aggregate from Monomer Using Nuvia HR-S Resin

Column: 2.1 ml, 0.7 x 5.6 cm

Step	Buffer	CV	Flow Rate, cm/hr
Equilibration	40 mM sodium acetate, pH 5 (buffer A)	3	310
Sample	Treated mAb at pH 5	9.5	310
Elution 1	Gradient of buffer A to buffer A + 1.0 M NaCl	20	310
Elution 2	Buffer A + 1.0 M NaCl	2	310
Sanitization	1 M NaOH	5	310

This method applies to Figure 1.

Comparison of Nuvia HR-S Resin with a Commercial Small Particle Size CEX Resin

Column: 2.1 ml, 0.7 x 5.6 cm

Step	Buffer	CV	Flow Rate, cm/hr
Equilibration	40 mM sodium acetate, pH 5 (buffer A)	3	150
Sample	Treated mAb at pH 5	9.5	150
Elution 1	Gradient of buffer A to buffer A + 0.4 M NaCl	20	150
Elution 2	Buffer A + 0.4 M NaCl	2	150
Sanitization	1 M NaOH	5	150

This method applies to Figure 3.

Results

Preparation of Aggregate and Monomer Test Solution

The in vitro generation of aggregates was designed to obtain a final composition of ~10% aggregates. A ratio of 2.5 volumes of monomeric antibody solution to 1 volume of aggregated antibody solution was chosen to yield a composition with the following properties:

$A_{280} = 6.3$ or 4.56 mg/ml

pH = 4.91

Conductivity = 2.6 mS/cm

Aggregate = 8.9%

A plot of log MW of six protein standards vs. their retention times yielded a straight line ($R^2 = 0.96$). Using this plot and the retention times of the aggregates, the apparent MW of the aggregate was calculated to be 386 kD.

Initial Separation of Aggregate from Monomer Using Nuvia HR-S Resin

As shown in Figure 1, there was considerable overlap of the monomer and aggregate peaks (fractions [Fr] 17–22). Depending on the fraction pooling method, monomer recovery of 66 – 81% was achieved with an aggregate content of 0.46 – 0.85% . This observation is consistent with the premise that greater recovery results in higher aggregate content. The quality of the recovered pools (Figure 2) is shown in the overlay of the enlarged front shoulders of each SEC profile. The results demonstrated the feasibility of using Nuvia HR-S to separate aggregates from the monomer and showed that the extent of residual aggregates depended on the fractions included in the pool. Additionally, the overlap of monomer and aggregates in the Nuvia HR-S elution profile suggested that a lower flow rate and a shallower gradient (0.1 – 0.4 M NaCl) could be beneficial.

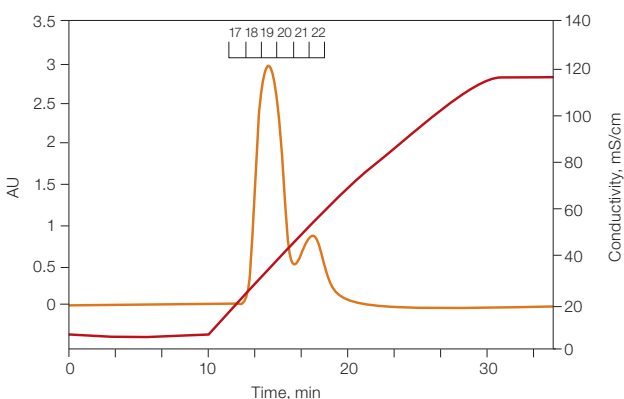


Fig. 1. Separation of aggregates from monomer by Nuvia HR-S before optimization. Absorbance (—); conductivity (—).

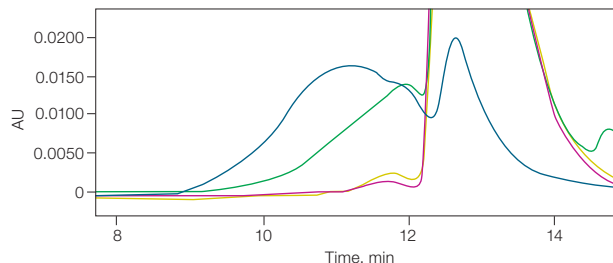


Fig. 2. SEC profiles. Load (—); selected pools (Fr 17+18+19+20) (—) and (Fr 17+18+19) (—); aggregated pool (Fr 21+22) (—).

Comparison of Nuvia HR-S with a Commercial Small Particle Size CEX Resin

A loading level of 46 mg IgG/ml resin was employed to challenge the binding capacity of Nuvia HR-S and Resin 1, and aggregate clearance by each resin was evaluated. A comparison of the elution profiles obtained from each chromatography run using the method conditions listed is shown in Figure 3. As indicated by the absorbance profile, the chromatography conditions resulted in a low yield for Resin 1 due to incomplete binding during the loading phase. In contrast, Nuvia HR-S was able to quantitatively retain the antibody before elution.

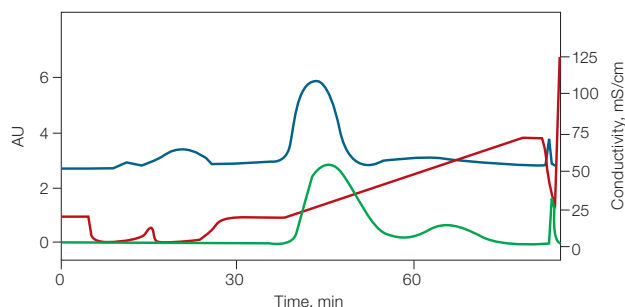


Fig. 3. Elution profiles after optimization. Nuvia HR-S (—); Resin 1 (—); conductivity (—).

Rather than subjectively determining the pertinent fractions, a strategy was designed to measure the aggregate content and the monomer recovery as fractions were progressively pooled during elution. As shown in Figure 4, both resins show an increase in aggregates as the pooling was extended to increase monomer recovery. While Nuvia HR-S was able to deliver a final aggregate content of $<0.3\%$ and a recovery of $>80\%$, Resin 1 recovered $<70\%$ of total monomer at the same target aggregate content. This is due to the lower binding capacity of Resin 1, as confirmed by the increase in its absorbance during loading (Figure 3).

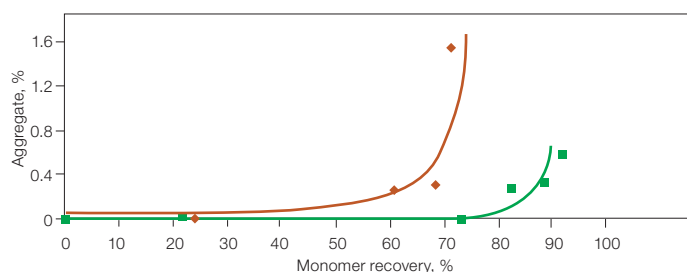


Fig. 4. Performance of Nuvia HR-S (■) vs. Resin 1 (◆).

Correlation between Monomer Recovery, Aggregate Content, and Target Conductivity

As depicted in Figure 4, aggregate recovery increases with increasing monomer recovery. In the Nuvia HR-S gradient elution, conductivity of the last fraction added to the pool was determined as the cutoff target of the pool. As shown, recovery is a function of the acceptable aggregate percentage and the target conductivity (Figure 5). Using the Fit Model function in JMP Software (SAS Institute Inc.), recovery could be correlated to those two measurements. A summary of actual and predicted data is shown in Figure 6.

The model yielded excellent correlation ($R^2 = 0.94$). This finding confirms the need to determine conductivity for elution so that both recovery and aggregate levels are within acceptable limits.

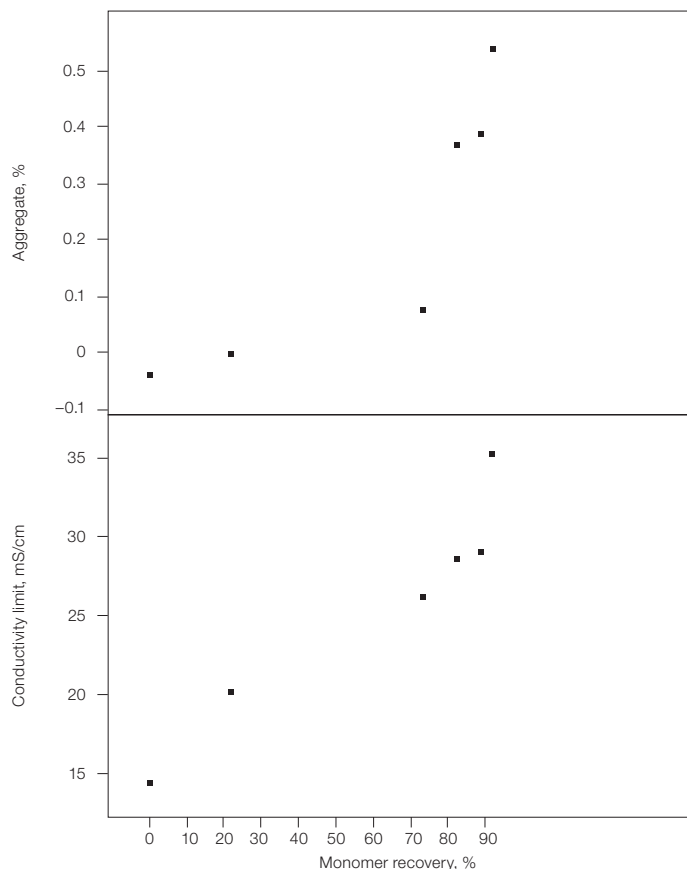
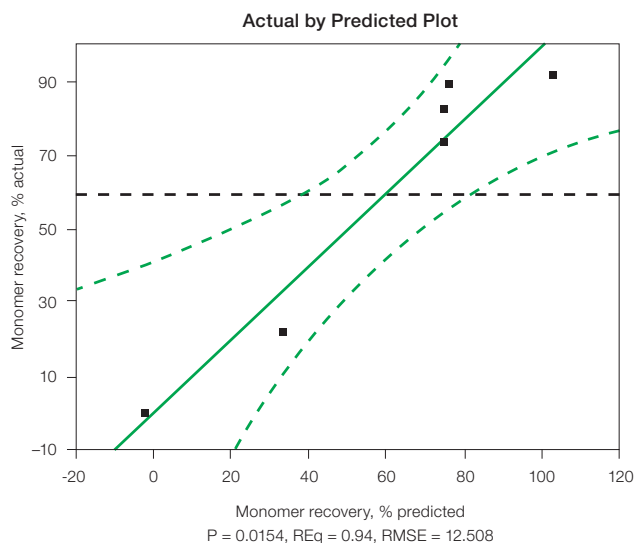


Fig. 5. Scatter plot matrix of aggregate content or conductivity vs. monomer recovery on Nuvia HR-S.



Summary of Fit

R^2	0.938098
R^2 adjusted	0.89683
Root mean square error	12.50828
Mean of response	59.78333
Observations (or sum of weights)	6

Fig. 6. Fit model of monomer recovery on Nuvia HR-S.

Summary

The data presented here demonstrate that Nuvia HR-S High-Resolution Cation Exchange Resin reduced aggregate content to <0.3% with >80% monomer recovery from a mAb feedstream containing 8.9% aggregates. In addition, Nuvia HR-S provides significantly higher recovery than a comparable agarose-based small-particle CEX resin. Finally, aggregate removal and final recovery are shown to be functions of buffer conductivity measured at the end of eluate collection.

Get references and complete details of this case study in Improving Aggregate Removal from a Monoclonal Antibody Feedstream Using High-Resolution Cation Exchange Chromatography, bulletin 6439.

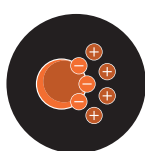
Visit [bio-rad.com/ResinSample](https://www.bio-rad.com/ResinSample) to request a sample of the resins used in this workflow.

CASE STUDY

Minimizing On-Column Aggregate Formation during Monoclonal Antibody Purification Using Nuvia HR-S Resin



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The formation of aggregates is a common occurrence during monoclonal antibody (mAb) purification and can happen for several reasons. Aggregates are undesirable due to the possibility of altered pharmacodynamics, pharmacokinetics, and/or increased immunogenicity. Generally, the formation of aggregates is shown to be product or process related. Shown here is how on-column aggregate formation is dependent on the resin type and on the resin's interactions with the loaded mAb. Among the various resins tested, Nuvia HR-S Resin, a high-resolution cation exchanger, demonstrated the lowest proportion of on-column aggregate formation with two mAbs. On-column aggregate formation can therefore be minimized by proper resin screening and subsequent optimization of resin type used for purification process development.

Method

Seven different resins were used for the case study (Table 1).

Table 1. Mean particle diameter and charge densities of all resins used.

	Fractogel SO ₃ ⁻ (M)	Eshmuno S	POROS 50 HS	POROS XS	Capto S ImpAct	Capto SP ImpRes	Nuvia S	Nuvia HR-S
Mean particle diameter, µm	74*	85*	50**	50**	50**	36–44**	85	50
Charge density, µeq/ml ^b	70–110	50–100	135	88–120	37–63	130–160	90–150	100–180

* Guo and Carta 2015.

** Values obtained from manufacturer literature.

Monoclonal Antibodies

Monoclonal antibody A was a glycosylated IgG2 antibody with a molecular mass of 150 kD and a calculated pI of 8.7 based on its amino acid sequence. Monoclonal antibody B had a molecular mass of 150 kD and a calculated pI of 7.

The following chromatography protocol was used:

Column: 0.5 x 5 cm with an actual bed volume of 1.00 ± 0.04 ml

Step	Buffer	CV	Flow Rate, cm/hr
Equilibration	Acetate buffer containing 40 mM Na ⁺	5	150
Sample	0.5 ml of 2 mg/ml mAb A in the acetate buffer	–	150
After loading, the flow was stopped and the column was held idle for different hold times ranging from 0 to 1,000 min (data shown only for the 0 and 1,000 min hold time).			
Wash	Load buffer	2	150
Elution	Load buffer followed by 20 CV gradient to 1 M NaCl in load buffer	20	150

The load buffers used in these runs were an acetate buffer containing 40 mM Na⁺ at pH 5.0 for the Fractogel Resin and acetate buffer containing 20 mM Na⁺ at pH 4.8 for all other resins. An elution flow rate of 0.3 ml/min was used with

Eshmuno S and Nuvia S whereas all other columns were operated at an elution flow rate of 1 ml/min. These conditions were selected to provide the best resolution between peaks for the different resins.

All calculations of hydrodynamic radii were done as described in Guo et al. 2016.

Results

On-Column Aggregate Formation Is Dependent on Resin Type

As shown in Figure 1, for the conditions chosen, the protein eluted in a single peak on all resins when elution was started immediately after loading and washing the column (0 minute hold time). However, different patterns of elution were seen for each of the resins when employing a 1,000 minute hold time. All resins showed a two-peak elution behavior with the first peaks appearing at the same retention time as the 0 minute hold time runs. The second elution peaks, however, showed different retention times and percentage of total eluted protein for all seven resins.

Nuvia HR-S yielded the least amount of eluted protein in the second peak (~10%) (Table 2). This result is in agreement with previously reported data showing that the “tentacle type” Fractogel displayed strong two-peak elution behavior (Guo et al. 2014, Guo and Carta 2014), whereas only one peak eluted from a macroporous CEX resin without “tentacles” under otherwise identical conditions (Guo and Carta 2015).

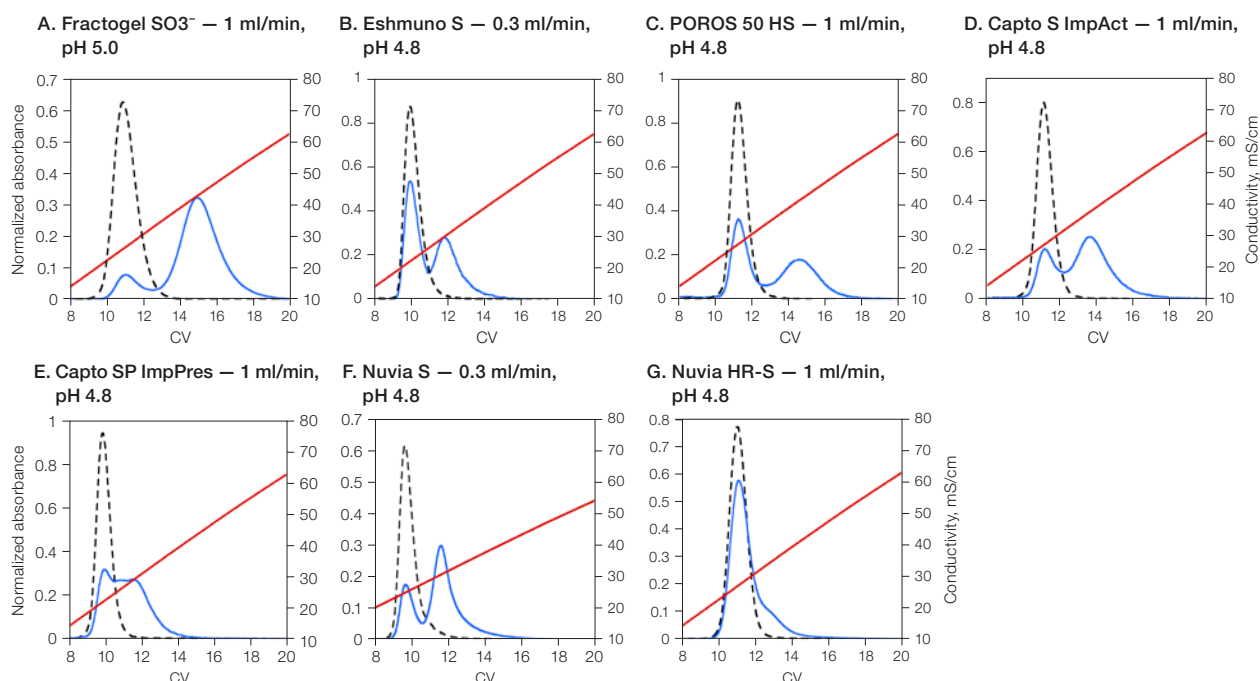


Fig. 1. Nuvia HR-S caused the least amount of on-column aggregate formation of mAb A. The load buffer for all resins from B to G was acetate buffer containing 20 mM Na⁺ at pH 4.8 and for A was acetate buffer containing 40 mM Na⁺ at pH 5.0. The elution flow rate was 1 ml/min on all resins except B and F, which had a flow rate of 0.3 ml/min. 0 min hold time, (---); 1,000 min hold time, (—); conductivity, (—). Data in A–C and F courtesy of Dr. Giorgio Carta.

Table 2. Percentage of on-column aggregate formed on different resins (protein eluting in second peak after 1,000 min hold time).

	Fractogel SO ₃ ⁻ (M)	Eshmuno S	POROS 50 HS	Capto S ImpAct	Capto SP ImpRes	Nuvia S	Nuvia HR-S
Percentage of on-column aggregation, %	87	44	51	65	50	73	10

The predominant one-peak elution profile of mAb A on Nuvia HR-S appears to be related to the resin's internal structure. The tested resins (excluding Nuvia HR-S and POROS 50 HS) contain grafted polymers designed to enhance binding capacity. In such cases, binding is not restricted to monolayer coverage, but potentially occurs throughout the grafted polymer layer (Almodóvar et al. 2011, Almodóvar et al. 2012). Therefore, these resins demonstrated a distinct two-peak elution behavior with a relatively large fraction of late-eluting protein. POROS 50 HS has a bimodal pore size distribution, which includes large “through pores,” about 500 nm in diameter, as well as small pores, about 22 nm in diameter. It is possible that proteins bind in these small pores, similar to the way they bind to a grafted polymer layer, yielding a large second elution peak. Nuvia HR-S, on the other hand, has an open pore structure without bimodal pores or grafted polymers and shows the least percentage of protein elution in the second peak (Table 2).

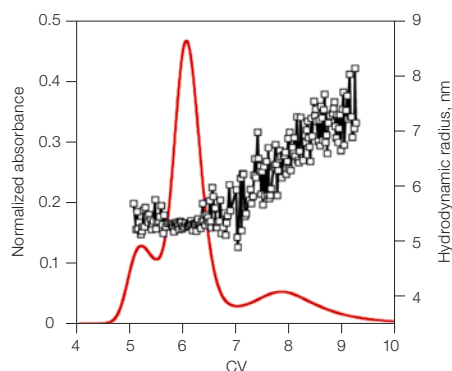
On-Column Aggregate Formation Depends on Unique mAb-Resin Structure Interactions

Due to differences in stability, different mAbs may show dissimilar behaviors on the same resin under the same conditions. To demonstrate this principle, a second monoclonal antibody, mAb B, with an approximate pI of 7 and MW of 150 kD, was studied.

Monoclonal antibody B was loaded onto Nuvia HR-S and POROS XS, and its elution profiles after zero and 30 minute hold times were studied. This antibody exhibited a two-peak elution behavior for zero hold times (Guo et al. 2016) and three-peak elution behavior for longer hold times (30 minutes) on the POROS XS (Figure 2A). In contrast, a single peak was observed even after a 30 minute hold time on Nuvia HR-S (Figure 2B). In-line hydrodynamic radius measurements confirmed that the single elution peak obtained for Nuvia HR-S consisted exclusively of monomeric species with radii of about 5.3 nm. However, with POROS XS, the hydrodynamic radii of the species increased from 5.3 nm (corresponding to the first two elution peaks) to about 7.5 nm, indicating the formation of aggregates.

To explore this behavior further, the two monomer peak fractions from the POROS XS Column were separately dialyzed against the original load buffer and then individually re-injected into either the POROS XS Column or the Nuvia HR-S Column. Re-injecting the fractions from the two peaks into the POROS XS Column resulted in the same two-peak elution behavior as the fresh mAb sample (Figure 3A). In contrast, reloading on the Nuvia HR-S Column led to the same single peak regardless of which POROS XS Resin peak-produced fraction was re-injected (Figure 3B). This confirms that the two primary mAb B elution peaks from POROS XS contained monomers.

A. POROS XS



B. Nuvia HR-S

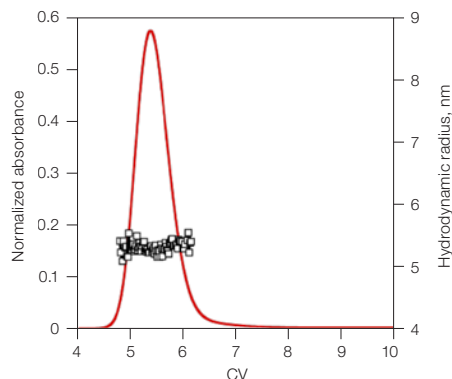
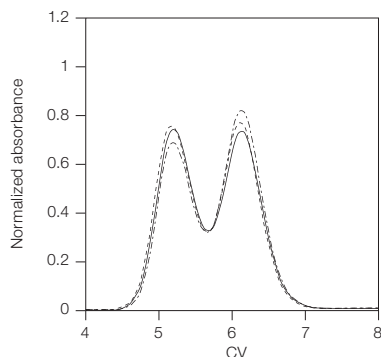


Fig. 2. Comparison of elution behavior and on-column aggregate formation of mAb B. Columns packed with POROS XS (A) and Nuvia HR-S (B) with a 30 min hold time followed by a 20 CV 0–1 M NaCl gradient in 40 mM sodium acetate at pH 5. For POROS XS, the first two peaks contain species with hydrodynamic radii of 5.3 nm, consistent with monomeric mAb B, while the third peak shows hydrodynamic radii increasing to about 7.5 nm, corresponding to a dimeric species. For Nuvia HR-S, a single peak with hydrodynamic radii of 5.3 nm indicates the presence of only monomeric species. Hydrodynamic radii (■). Data courtesy of Dr. Giorgio Carta.

The POROS XS contains two different pore sizes that bind mAb B with different strengths and at different rates, explaining the two-peak behavior. Nuvia HR-S, which has a uniform pore size, produced a single peak.

A. POROS XS



B. Nuvia HR-S

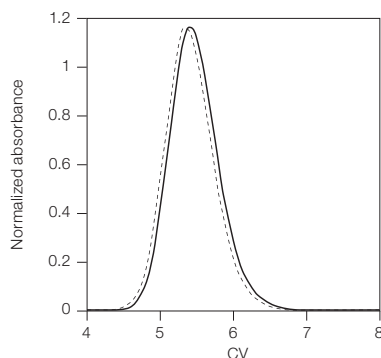


Fig. 3. Comparison of elution profiles on POROS XS and Nuvia HR-S.

Elution profiles of fresh mAb B sample were overlaid with the results obtained by re-injecting fractions from the POROS XS with 0 min hold time followed by a 20 CV 0–1 M NaCl gradient in 40 mM sodium acetate at pH 5. **A**, Results from POROS XS. **B**, Results from Nuvia HR-S. Conditions were identical for all runs. Fresh mAb sample (—); first peak fraction (---); second peak fraction (- - -). Data courtesy of Dr. Giorgio Carta.

Summary

Of all the resins tested under the mentioned conditions, Nuvia HR-S, a high-resolution CEX resin, shows minimal formation of on-column aggregates, irrespective of mAb type. Nuvia HR-S also demonstrated single-peak elution behavior after mAb B had been held on the resin for up to 30 minutes. We know this behavior is not due to its particle size because POROS XS, which has the same particle size as Nuvia HR-S, demonstrated a different elution behavior and aggregate formation. Prior studies have also shown that, when compared with another high-resolution agarose-based resin, Nuvia HR-S demonstrated comparable total aggregate removal, but at significantly higher recovery rates (Ng et al. 2014). These results underscore the importance of screening multiple resins during mAb purification process development, especially for aggregation-prone mAbs, not only to minimize the potential of on-column aggregate formation, but also to optimize recovery. Without this screening, it is difficult to predict the interaction and elution behavior of different mAbs on any given resin.

Get references and complete details of this case study in *Minimizing On-Column Monoclonal Antibody Aggregate Formation*, bulletin 7000.



Visit [bio-rad.com/ResinSample](https://www.bio-rad.com/ResinSample) to request a sample of the resins used in this workflow.

CASE STUDY

Removing the Aggregates of an Acidic Monoclonal Antibody with CHT Ceramic Hydroxyapatite XT Media



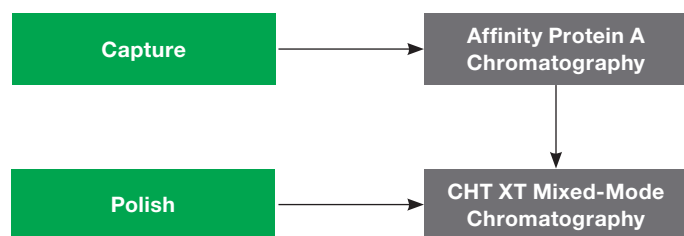
ANTIBODY
PURIFICATION



CHT CERAMIC
HYDROXYAPATITE

A major challenge in downstream purification of monoclonal antibodies (mAbs) is the formation and removal of aggregates (Vazquez-Rey and Lang 2011). The presence of aggregates can hamper the efficacy of therapeutic mAbs due to their different bioactivity, storage stability, immunogenicity, and pharmacokinetic properties relative to the monomeric versions. In many cases, antibody aggregates are the most challenging process/product-related impurities to remove. Extensive process development is necessary to obtain highly pure monomeric mAbs. CHT Ceramic Hydroxyapatite Media are known as the industry gold standard for impurity removal during monoclonal antibody (mAb) purification. The present study demonstrates the removal of aggregates from the monomers of mAb S (pI ~6.9) with CHT XT Media. mAb S was initially isolated from CHO cell culture using Protein A chromatography. The eluate from this capture step was a mixture of mAb S monomers and aggregates, with the high molecular weight species accounting for ~34% of the total mass of antibodies. Polish purification was then performed on a CHT XT Column, resulting in >96% final monomer content, thereby confirming the robustness of CHT XT in aggregate removal. In addition, excellent clearance of host cell proteins (HCPs), dsDNA, leached recombinant Protein A (rPA), and endotoxins was achieved, making CHT XT an effective solution for mAb purification.

Protocol



Method

Capture with UNOsphere SUPra Resin

Column: 0.7 x 5.5 cm

Step	Buffer	CV	Flow Rate, cm/hr
Equilibration	1x PBS, pH 7.0	5	300
Sample	10 ml CHO cell culture	–	150
Wash	1x PBS, pH 7.0	10	150
Elution	0.1 M glycine, pH 3	5	150
Stripping	0.1 M glycine, 1 M NaCl, pH 3	5	300

The eluted mAb S solution was adjusted to pH ~8.0 with 1 M Tris base and filtered through a 0.22 µm membrane. The monomer content in the eluted sample was quantified using SEC (Figure 1) as the protocol described.

The CHT load was prepared by adding 0.3 ml of 0.5 M NaPO₄ (monobasic) to 15 ml of neutralized UNOsphere SUPra eluate. The pH of this sample was adjusted to 7.0 with 1 M NaOH.

Polish with CHT XT Media

Column: 0.5 x 5.1 cm

Step	Buffer	CV	Flow Rate, cm/hr
Equilibration	10 mM NaPO ₄ , pH 7	10	300
Sample	5 ml of CHT load (containing 8.5 mg mAb S at a concentration of 1.7 mg/ml)	–	150
Wash	10 mM NaPO ₄ , pH 7	3	150
Elution	Linear gradient: 0 to 100% of 10 mM NaPO ₄ , 1 M NaCl, pH 7	20	150
Stripping	400 mM NaPO ₄ , pH 7	4	300
Sanitization	1 M NaOH	3	300

The details of the monomer content analysis with SEC and the analytical assays are described in bulletin 7109.

Aggregate Content of UNOsphere SUPra Eluate (mAb S)

About 34% mAb S aggregate content was seen in the UNOsphere SUPra eluate as shown in Figure 1. This result indicates that mAb S is pH sensitive, and the acidic condition needed for its elution from the Protein A column triggered structural changes as well as oligomerization.

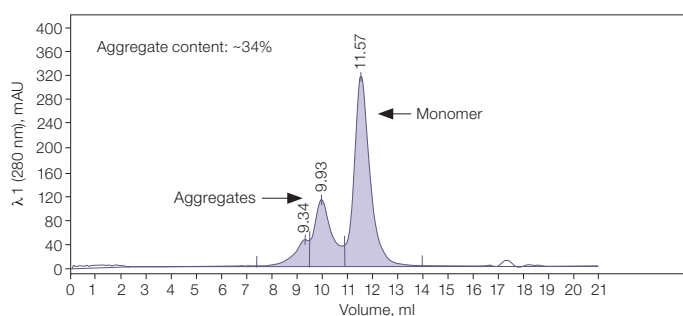


Fig. 1. SEC analysis of mAb S eluted from UNOsphere SUPra. SEC reveals the level of mAb S aggregates in the eluate from UNOsphere SUPra.

Separation of mAb S Monomers from Aggregates on CHT XT Media

The mAb S monomers were eluted as a sharp peak between fractions 10 and 14 (Figure 2), which were pooled together for monomer content analysis by SEC. The high molecular weight fractions were eluted later and seen as multiple peaks in the chromatogram.

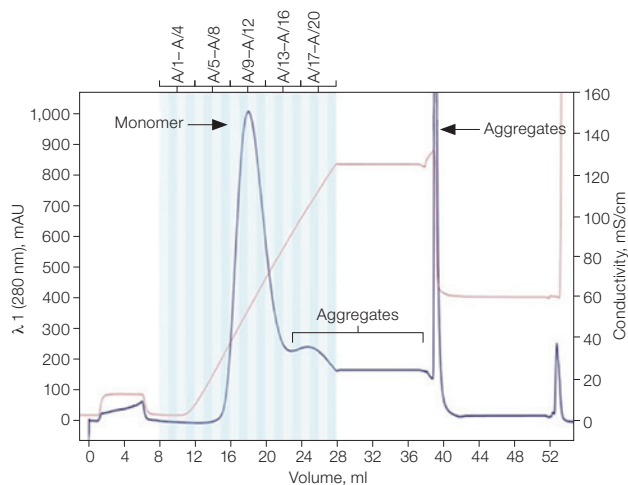


Fig. 2. Separation of mAb S monomers from aggregates on a CHT XT Column. The blue vertical bands represent collected column fractions. Fractions 10 to 14 were pooled. OD 280 (—); conductivity (—).

Aggregate Content of CHT XT Eluate (mAb S)

Significant reduction of the mAb S aggregate was achieved after purification of the Protein A eluate on CHT XT as shown in Figure 3. The final aggregate content was ~3% as opposed to the initial ~34% seen in the UNOsphere SUPra eluate. Overall purification performance of CHT XT chromatography is summarized in Table 1.

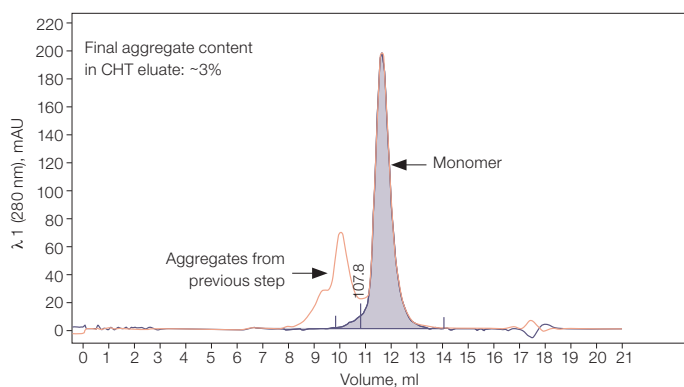


Fig. 3. SEC analysis of mAb S eluted from CHT XT. SEC showing the level of mAb S aggregates in the eluate from CHT XT. Load (—); eluate (■).

Table 1. Reduction of mAb S aggregate contamination by CHT XT.

Sample	Monomer Content, %
CHT XT load	66.0
CHT XT eluate	96.9

Clearance of HCPs, DNA, Leached Protein A, and Endotoxins by CHT XT Media

To further assess the levels of other impurities in the CHT XT eluate, various analytical assays were performed. The results are shown in Table 2. CHT XT showed multiple fold-removal of all four impurities.

Table 2. Impurity clearance by CHT XT.

Impurity	Load	Pool
HCP, ppm	2,635	229
dsDNA, ppm	112	3
Leached rPA, ppm	11	1
Endotoxin (LAL Endotoxin Assay), EU/mg	11	1
Endotoxin (EndoLISA Assay), EU/mg	422	16

Summary

This study shows that CHT XT is capable of minimizing the aggregate content during mAb purification. It can also reduce other product- and process-related impurities to significantly low levels. These data demonstrate that CHT XT is an exceptional tool for polish purification of monoclonal antibodies.

Get references and complete details of this case study in Removing the Aggregates of an Acidic Monoclonal Antibody with CHT Ceramic Hydroxyapatite XT Media, bulletin 7109.



Visit [bio-rad.com/ResinSample](https://www.bio-rad.com/ResinSample) to request a sample of the resins used in this workflow.

CASE STUDY

Comparison of Three Mixed-Mode Resins Including CHT Ceramic Hydroxyapatite Media for Aggregate Removal during Monoclonal Antibody Purification



ANTIBODY
PURIFICATION



MIXED-MODE



CHT CERAMIC
HYDROXYAPATITE

The strong acidic elution conditions needed for Protein A affinity chromatography, commonly used for the capture of monoclonal antibodies (mAbs) from clarified tissue culture fluid, can trigger structural changes and promote the oligomerization of pH-sensitive molecules. Additional column chromatography steps are necessary for the clearance of such product-related impurities. The separation of mAb monomers and aggregates has been a major challenge in downstream manufacturing, as these molecules exhibit very similar physical and chemical properties and sometimes identical chromatographic behavior. Baseline separation of these mAb species is often difficult to achieve, and extensive process development is required to balance the purity and yield of the monomeric mAbs.

The molecule of interest in this present study, mAb S, was harvested from Chinese hamster ovary (CHO) cell culture. Typically, a Protein A affinity chromatography eluate contains a mixture of mAb S monomers (~75%) and aggregates (dimers and other oligomers, ~25%). The mAb S monomers were separated from the aggregates using three mixed-mode chromatography media, CHT Ceramic Hydroxyapatite, Capto adhere, and Capto adhere ImpRes, which have all been extensively researched for their applications in mAb aggregate clearance (Gagnon 2009). The purity and recovery of monomeric mAb S purified with these chromatography media were compared. The results underscore the importance of screening multiple media to develop an efficient downstream production process with the highest possible target purity and recovery.

Methods

Affinity Chromatography Using UNOsphere SUPra Resin

Column: 1 ml, 0.56 x 4 cm

Step	Buffer	CV	Flow Rate, cm/hr
Equilibration	1x PBS, pH 7.3	10	300
Sample	CHO cell culture harvest	~20 mg in 20 ml	300
Wash	1x PBS, pH 7.3	10	300
Elution	100 mM glycine, pH 3.5	10	300

Mixed-Mode Chromatography Using CHT Ceramic Hydroxyapatite Media, Type I, 40 µm

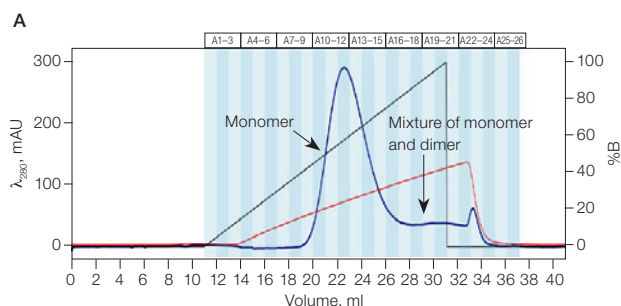
Column: 1 ml, 0.56 x 4 cm

Step	Buffer	CV	Flow Rate, cm/hr
Equilibration	10 mM sodium phosphate, pH 7.0	10	300
Sample	UNOsphere SUPra eluate (mAb) adjusted to pH 7.0	2.8 mg in 1 ml	300
Wash	10 mM sodium phosphate, pH 7.0	5	300
Elution	10 mM sodium phosphate, 550 mM NaCl, pH 7.0	15	300

This method applies to Figure 2.

Results

The mAb S aggregates were well separated from the monomer using the CHT Column with a 0–1,000 mM sodium chloride gradient. The bulk of the mAb S monomer was eluted at ~50%B, which is equivalent to ~500 mM sodium chloride (Figure 1A). Elution of the monomer was complete at ~75%B or 750 mM sodium chloride, and increasing the salt concentration resulted in leaching of the undesired mAb S oligomers from the column (Figure 1B). Step elution of mAb S was then performed using a sodium phosphate buffer containing 550 mM sodium chloride (Figure 2). Fractions A5 to A9 were pooled, and the monomer content in this eluate was determined to be 99.5% with an overall mAb S monomer recovery of 82.7%.



Column: 1 ml CHT packed in a Bio-Scale Mini Cartridge (0.56 x 4 cm)
 Sample: 2.8 mg affinity-purified mAb S in 1 ml 10 mM sodium phosphate, pH 7.0
 Flow rate: 300 cm/hr
 Equilibration: 10 mM sodium phosphate, pH 7.0, 10 CV
 Post-loading wash: 10 mM sodium phosphate, pH 7.0, 5 CV
 Elution: Buffer A: 10 mM sodium phosphate, pH 7.0
 Buffer B: 10 mM sodium phosphate, 1 M sodium chloride, pH 7.0
 A to B linear gradient, 0–100%B, 20 CV

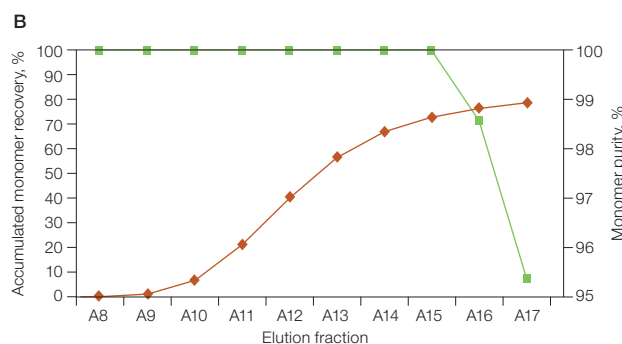


Fig. 1. Purification of mAb S on a CHT Column. A, removal of mAb S aggregates using a 0–1,000 mM sodium chloride gradient. OD 280 (—); conductivity (—); %B (—). The blue vertical bands represent where fractions were collected. B, mAb S monomer content in individual elution fractions (—) and accumulated monomer recovery in pooled elution fractions (—).

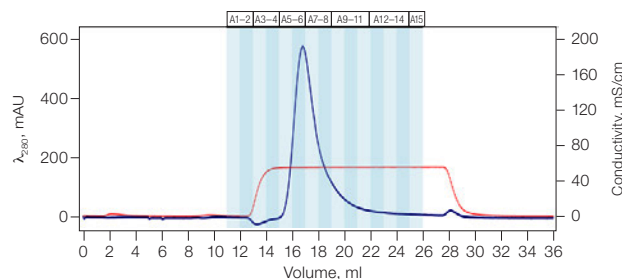


Fig. 2. Elution of mAb S monomers from a CHT Column. Elution was carried out using 10 mM sodium phosphate and 550 mM sodium chloride, pH 7.0. Fractions A5 to A9 were pooled. OD 280 (—); conductivity (—). The blue vertical bands represent where fractions were collected.

Preliminary DOE studies were performed to screen for mAb S purification conditions with Capto adhere Media. Binding was maximal with buffers containing ≤ 50 mM sodium chloride at pH > 8.0 . mAb S was only partially eluted from the column under a modest acidic condition (pH 5.0). At pH 4.0, all mAb S species were eluted from the Capto adhere Column without differentiation of monomer and higher molecular weight species. The addition of sodium chloride to the elution buffer further hampered target protein recovery, which indicates a strong hydrophobic interaction between target molecules and this hydrophobic AEX media (data not shown). A gradient of pH 8–5 was therefore employed in an effort to elute mAb S monomers selectively from the Capto adhere Column (Figure 3A). Dependency of monomer content and elution pH was observed, with the later fractions containing significantly more high molecular weight species (Figure 3B) and exhibiting a plateau of mAb S monomer recovery (Figure 3C).

To further assess the effect of the elution pH on final mAb S monomer purity and recovery, step elutions under various pH conditions were performed. The results are summarized in Table 1. The chromatography of mAb S elution at pH 5.5 is shown in Figure 4. The purity of the product was found to be very sensitive to the pH of the elution buffer. A minor pH drop, from 5.5 to 5.4, resulted in a tripling of the aggregate content in the eluate. The pooled product from a pH 5.4 step elution failed the quality control (QC) specification of $\geq 99.5\%$ monomer content. Furthermore, despite the fact that 14 CV of eluate were collected, monomer recovery was only $\sim 53\%$.

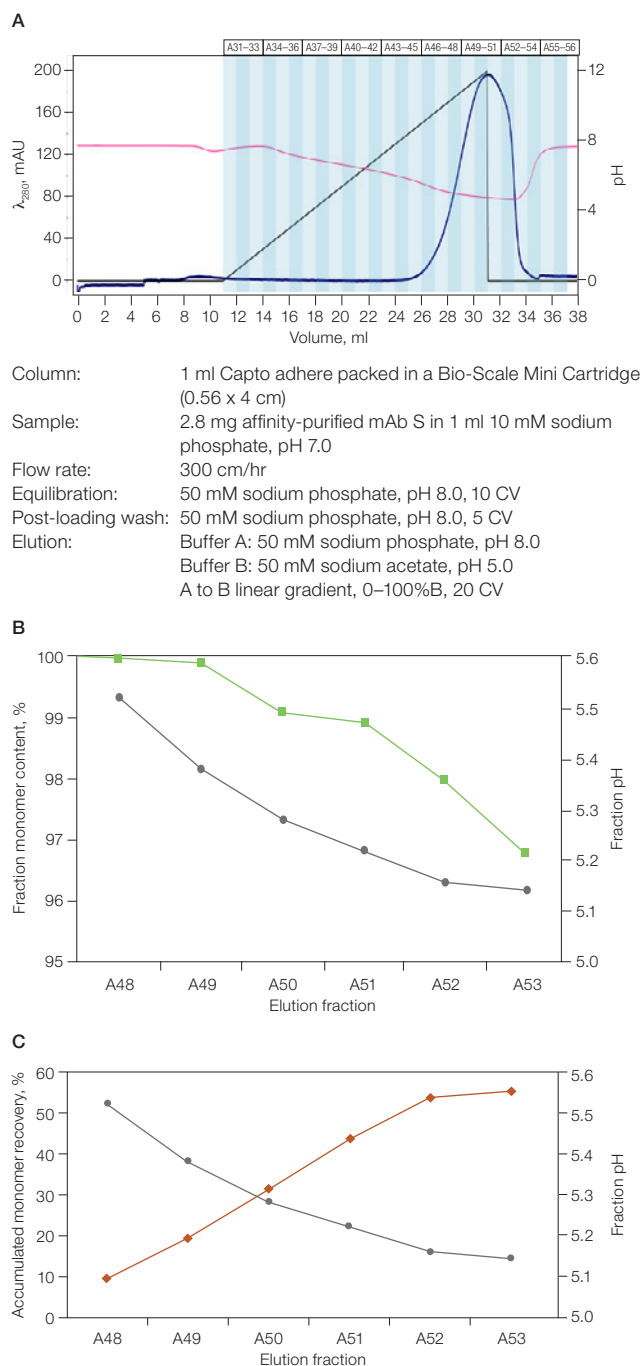
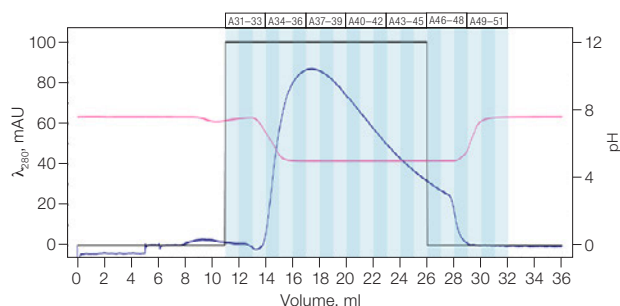


Fig. 3. Purification of mAb S on a Capto adhere Column. **A**, removal of mAb S aggregates using a pH 8–5 gradient. OD 280 (—); pH (—); %B (—). The blue vertical bands represent where fractions were collected. **B**, mAb S monomer content in individual elution fractions (—■—) and its correlation with fraction pH (—●—). **C**, accumulated mAb S monomer recovery in pooled elution fractions (—◆—) and its correlation with fraction pH (—●—).

Table 1. Pooled elution fractions from the Capto adhere Column using 50 mM sodium acetate buffer at different pH.

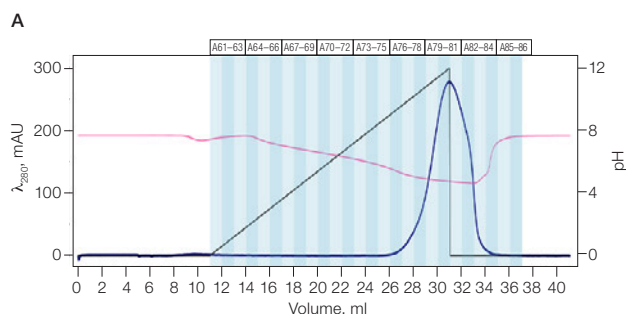
Elution pH	mAb S Monomer Content, %	mAb S Monomer Recovery, %	Eluate Volume, CV
5.2	96.9	76.9	14
5.3	98.1	64.9	14
5.4	98.8	52.5	14
5.5	99.7	48.8	14



Column: 1 ml Capto adhere packed in a Bio-Scale Mini Cartridge (0.56 x 4 cm)
Sample: 2.8 mg affinity-purified mAb S in 1 ml 50 mM sodium phosphate, pH 8.0
Flow rate: 300 cm/hr
Equilibration: 50 mM sodium phosphate, pH 8.0, 10 CV
Post-loading wash: 50 mM sodium phosphate, pH 8.0, 5 CV
Elution: 50 mM sodium acetate, pH 5.5, 15 CV

Fig. 4. Elution of mAb S monomers from a Capto adhere Column. Elution was carried out using 50 mM sodium acetate, pH 5.5. Fractions A35 to A48 were pooled. OD 280 (—); pH (—); %B (—). The blue vertical bands represent where fractions were collected.

Compared with Capto adhere, the binding of mAb S by Capto adhere ImpRes was more sensitive to the concentration of sodium chloride in the buffer. However, the elution of mAb S from this mixed-mode media seemed to be solely dependent on the pH of the buffer, as revealed by DoE screening (data not shown). Again, a pH gradient was used to resolve mAb S monomers from aggregates (Figure 5A). The pH of the elution fractions, monomer content, and monomer recovery were analyzed and the results are shown in Figures 5B and 5C.



Column: 1 ml Capto adhere ImpRes packed in a Bio-Scale Mini Cartridge (0.56 x 4 cm)
Sample: 2.8 mg affinity-purified mAb S in 1 ml 50 mM sodium phosphate, pH 8.0
Flow rate: 300 cm/hr
Equilibration: 50 mM sodium phosphate, pH 8.0, 10 CV
Post-loading wash: 50 mM sodium phosphate, pH 8.0, 5 CV
Buffer A: 50 mM sodium phosphate, pH 8.0
Buffer B: 50 mM sodium acetate, pH 5.0
A to B linear gradient, 0–100%B, 20 CV

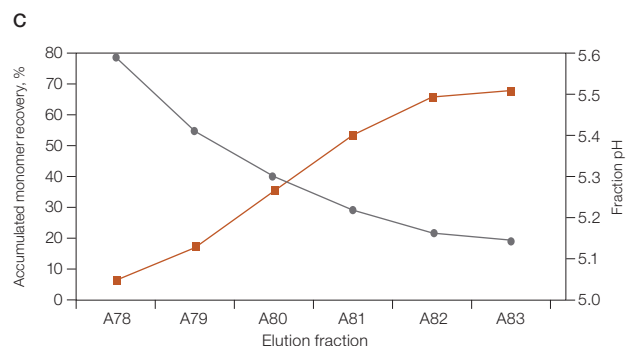
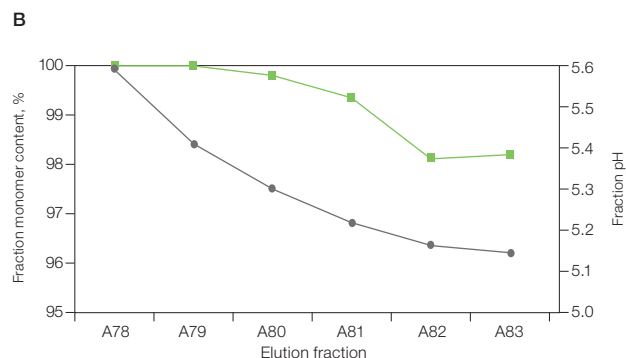
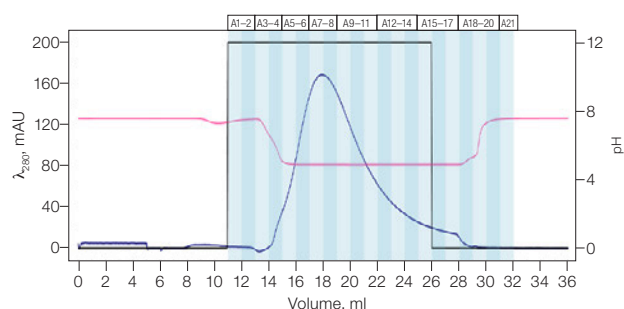


Fig. 5. Purification of mAb S on a Capto adhere ImpRes Column. A, removal of mAb S aggregates using a pH 8–5 gradient. OD 280 (—); pH (—); %B (—). The blue vertical bands represent where fractions were collected. B, mAb S monomer content in individual elution fractions (—) and its correlation with fraction pH (—). C, accumulated mAb S monomer recovery in pooled elution fractions (—) and its correlation with fraction pH (—).

Buffers at different pH were used to elute mAb S from the Capto adhere ImpRes Column in step mode (Table 2). Figure 6 depicts the chromatogram resulting from the elution of mAb S with 50 mM sodium acetate (pH 5.4). Overall, Capto adhere ImpRes offered better recovery of mAb S monomer than Capto adhere. The purity of mAb S monomer in the eluate met the QC specification of $\geq 99.5\%$ monomer content and was consistent at pH ≥ 5.4 ; however, monomer recovery was only 50–62% after 14 CV of eluate were collected under these conditions.

Table 2. Pooled elution fractions from the Capto adhere ImpRes Column with 50 mM sodium acetate buffer at different pH.

Elution pH	mAb S Monomer Content, %	mAb S Monomer Recovery, %	Eluate Volume, CV
5.3	98.6	68.3	14
5.4	99.6	61.7	14
5.5	99.6	50.5	14



Column: 1 ml Capto adhere ImpRes packed in a Bio-Scale Mini Cartridge (0.56 x 4 cm)
Sample: 2.8 mg affinity-purified mAb S in 1 ml 50 mM sodium phosphate, pH 8.0
Flow rate: 300 cm/hr
Equilibration: 50 mM sodium phosphate, pH 8.0, 10 CV
Post-loading wash: 50 mM sodium phosphate, pH 8.0, 5 CV
Elution: 50 mM sodium acetate, pH 5.4, 15 CV

Fig. 6. Elution of mAb S monomers from a Capto adhere ImpRes Column. Elution was carried out using 50 mM sodium acetate, pH 5.4. Fractions A4 to A18 were pooled. OD 280 (—); pH (—); %B (—). The blue vertical bands represent where fractions were collected.

Summary

A high level of aggregates was present in the mAb S harvest, which could have been an indication of the intrinsic hydrophobicity of mAb S or the exposure of its hydrophobic amino acid residues due to misfolding, unfolding, or denaturation. The mixture of monomers and aggregates of mAb S could not be separated by traditional cation exchangers (data not shown). Three mixed-mode media, CHT, Capto adhere, and Capto adhere ImpRes, were tested for their efficiency in clearing mAb S aggregates; their optimal performances are summarized in Table 3.

Table 3. Comparison of CHT, Capto adhere, and Capto adhere ImpRes in the clearance of mAb S aggregates.

Media	mAb S Monomer Content, %	mAb S Monomer Recovery, %	Eluate Volume, CV
CHT*	99.5	82.7	5
Capto adhere**	99.5	48.8	14
Capto adhere ImpRes***	99.5	61.7	14

* Step elution with 10 mM sodium phosphate and 550 mM sodium chloride, pH 7.0.

** Step elution with 50 mM sodium acetate, pH 5.5.

*** Step elution with 50 mM sodium acetate, pH 5.4.

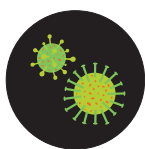
The contamination level of mAb S aggregates in the recovered mAb S monomer fractions was effectively reduced to $<0.5\%$ by all three media. However, CHT offered the best monomer recovery in the smallest eluate volume (Table 3). Since the isoelectric point (pI) of mAb S was estimated at 6.9 (data not shown), it should bind to Capto adhere-based media through strong electrostatic interaction. The difficult elution of mAb S from these two hydrophobic anion exchangers suggests that a significant hydrophobic interaction was also involved in mAb S binding to these columns. A more stringent acidic condition was necessary for protonating mAb S molecules in order to promote their dissociation from these chromatography media; however, this was carried out at the expense of selectivity between monomers and aggregates. To satisfy the QC specification of $\geq 99.5\%$ monomer in the final product, elution had to be terminated at pH 5.5 and pH 5.4 for Capto adhere and Capto adhere ImpRes, respectively. As a result, the recovery of mAb S monomer from these chromatography media was compromised. Tailing was observed and elution was incomplete even after 15 CV of elution buffer were applied.

Get references and complete details of this case study in Mixed-Mode Chromatography for mAb S Aggregate Removal: Comparison of CHT Ceramic Hydroxyapatite, Capto adhere, and Capto adhere ImpRes, bulletin 6749.

Visit [bio-rad.com/ResinSample](https://www.bio-rad.com/ResinSample) to request a sample of CHT Media.

CASE STUDY

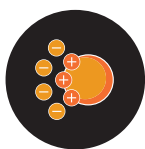
cGMP-Ready Adenovirus Purification with Nuvia cPrime and Nuvia Q Resins



VIRUS
PURIFICATION



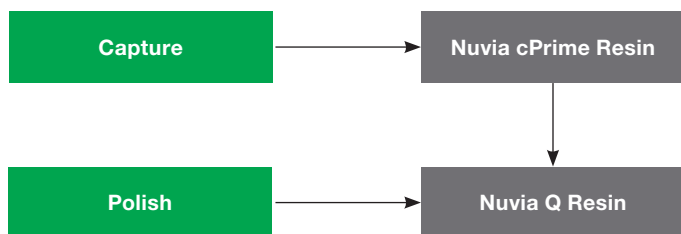
HI CATION
EXCHANGE



ANION
EXCHANGE

In this case study, we discuss the development of a two-column cGMP-ready purification process for a recombinant adenovirus after screening five Bio-Rad chromatography resins. We show that the final process yields an active, concentrated virus product with purity, HCP levels, and DNA contamination comparable to clinical-grade products. The process is readily scalable and is sufficiently simple, rapid, and efficient to be considered for the production of clinical-grade viral vectors.

Protocol



Method

Capture on Nuvia cPrime Resin

Column: 5 ml, 0.8 x 10 cm

Step	Buffer	CV	Flow Rate, cm/hr
Equilibration	25 mM histidine, pH 6.0 (buffer A)	10	120
Sample	Culture supernatant diluted 1:3 with buffer A	48	120
Wash	Buffer A	5	120
Elution	75 mM Tris, 525 mM NaCl, pH 8.5	3	120

Polishing on Nuvia Q Resin

Column: 5 ml, 0.8 x 10 cm

Step	Buffer	CV	Flow Rate, cm/hr
Equilibration	Nuvia cPrime eluate diluted 1:1 with buffer A	10	120
Sample	Nuvia cPrime eluate diluted 1:1 with buffer A	2	120
Wash 1	75 mM, 250 mM NaCl, pH 8.0	2	120
Wash 2	15 mM Tris, 440 mM NaCl, pH 8.5	3	120
Elution	75 mM Tris, 1 M NaCl, pH 7.5	5	120

Results

Process Development

Mass Capture

As shown in Table 1, three resins had potential for use in a mass capture process — Nuvia cPrime, UNOsphere Q, and Nuvia Q. Of these, Nuvia cPrime was selected because the requirements for virus binding (low salt and low pH) made it poorly suited for use after an anion exchange capture operation (where the resulting feedstream would be expected to have high ionic strength and perhaps high pH).

Developing the mass capture step using Nuvia cPrime involved replacing the buffer exchange step used in the screening study with a dilution step. Also, a nuclease digestion was introduced prior to column loading. Initial mass capture experiments were focused on reducing feedstream volumes and recovering virus. Example chromatograms, overlaid with the results of the transgene expression assay, illustrate the early progression of these experiments (Figure 1).

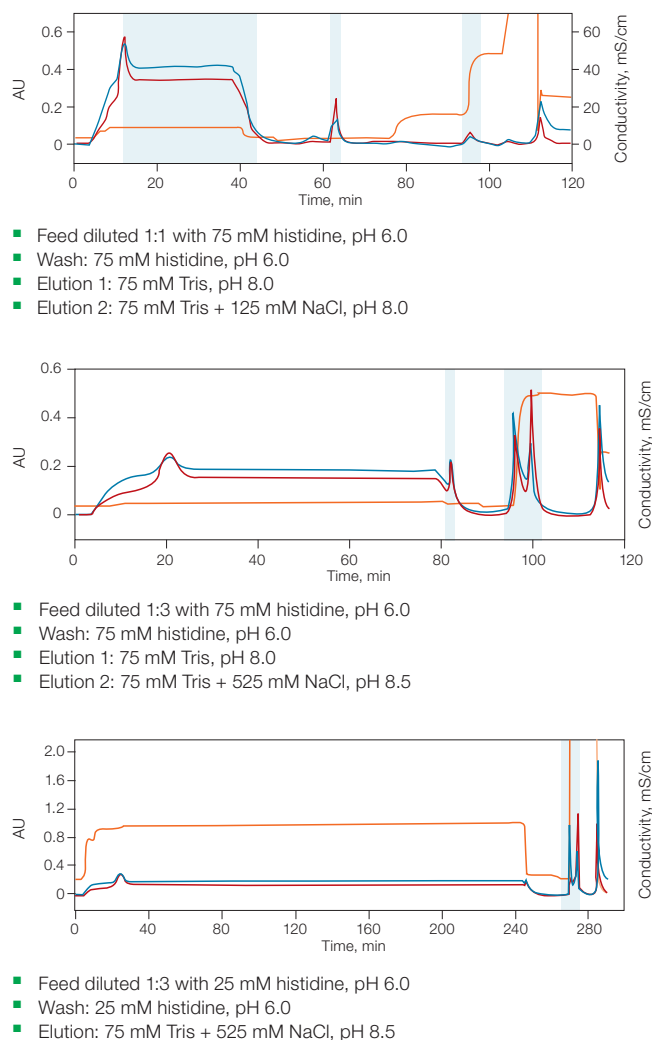


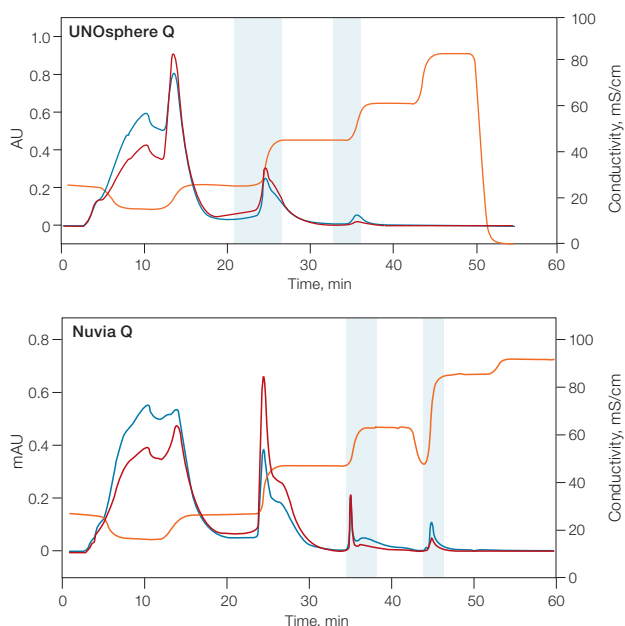
Fig. 1. Iterative development of Nuvia cPrime capture. OD 260 (—); OD 280 (—); conductivity (—). Blue shading indicates detection of significant transgene expression.

Table 1. Results from the initial resin screening and their implications.

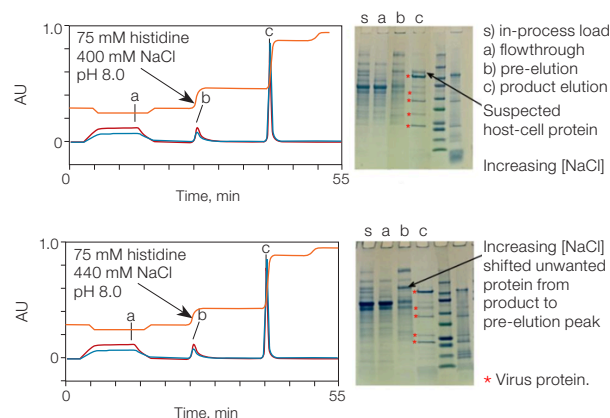
Column Type	Virus in Flowthrough/Wash	Virus in Eluate	Notes/Implications
UNOsphere S (CEX)	+++	++	Poorly suited for virus purification in both bind-and-elute and flow-through modalities
Nuvia S (CEX)	+++	++	Poorly suited for virus purification in both bind-and-elute and flow-through modalities
Nuvia cPrime (MM)	—	++++	Partial elution in 125 mM NaCl, pH 6.5; hence, dilution of crude harvest required prior to column loading
UNOsphere Q (AEX)	—	++++	Could be considered for direct mass capture
Nuvia Q (AEX)	—	++++	Could be considered for direct mass capture

AEX Chromatography

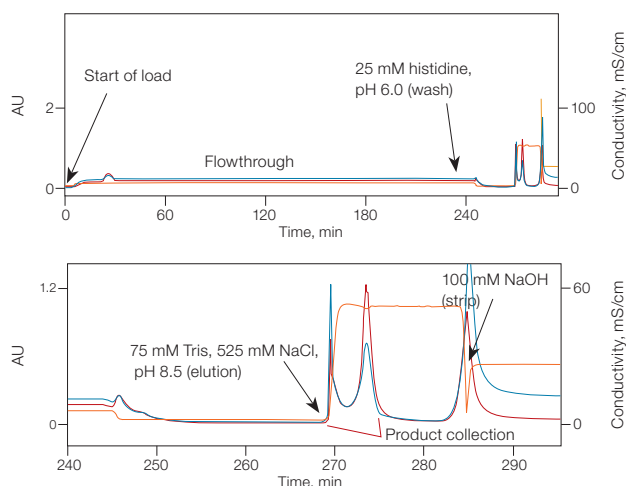
Of the two AEX resins, UNOsphere Q and Nuvia Q, Nuvia Q was selected because it could adsorb virus at higher NaCl concentrations (Figure 2). It was therefore easier to work with downstream of the Nuvia cPrime capture step, where the eluate had an NaCl concentration of approximately 500 mM going on to AEX.

**Fig. 2. Behavior of the crude harvest on the UNOsphere Q and Nuvia Q Columns.** OD 260 (—); OD 280 (—); conductivity (—). Blue shading indicates detection of significant transgene expression.

AEX experiments with Nuvia Q were focused on attaining high product purity (Figure 3). In one late-stage experiment, a suspected HCP contaminant was prevented from eluting with the final product by increasing the pre-elution NaCl concentration.

**Fig. 3. Chromatograms and gels from AEX process development experiments.** OD 260 (—); OD 280 (—); conductivity (—). * Proteins presumed to be virus capsid components.**Final Process****Mixed-Mode Chromatography**

Initial capture was accomplished using Nuvia cPrime Mixed-Mode Resin (Figure 4). This portion of the process achieved a tenfold reduction in processing volume and a significant reduction in feedstream contaminants (Figure 6, lanes 2–4).

**Fig. 4. Representative mixed-mode chromatogram.** OD 260 (—); OD 280 (—); conductivity (—).

AEX Chromatography

Final virus purification was accomplished using Nuvia Q Resin (Figure 5). This portion of the process achieved an additional twofold reduction of product volume along with a significant improvement in product purity (Figure 6, lanes 4–7). Following this operation, nonvirus proteins were no longer evident by SDS-PAGE (Figure 6, lane 7).

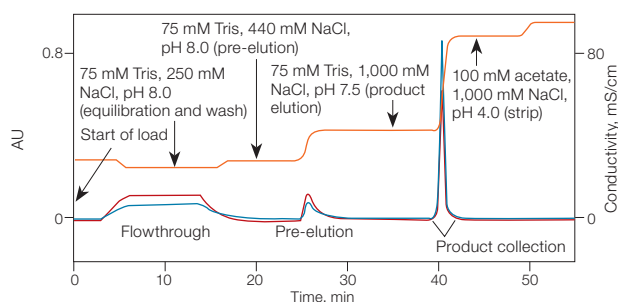


Fig. 5. Representative AEX chromatogram. OD 260 (—); OD 280 (—); conductivity (—).

Analysis of In-Process and Final Product

We used SDS-PAGE analysis to visualize the progressive reduction of contaminating proteins at each step of the purification process (Figure 6). The five most prominent viral proteins — hexon, penton, core (V), hexon (VI), and core (VII) — are readily visible in the final purified product (Figure 6, lane 7).

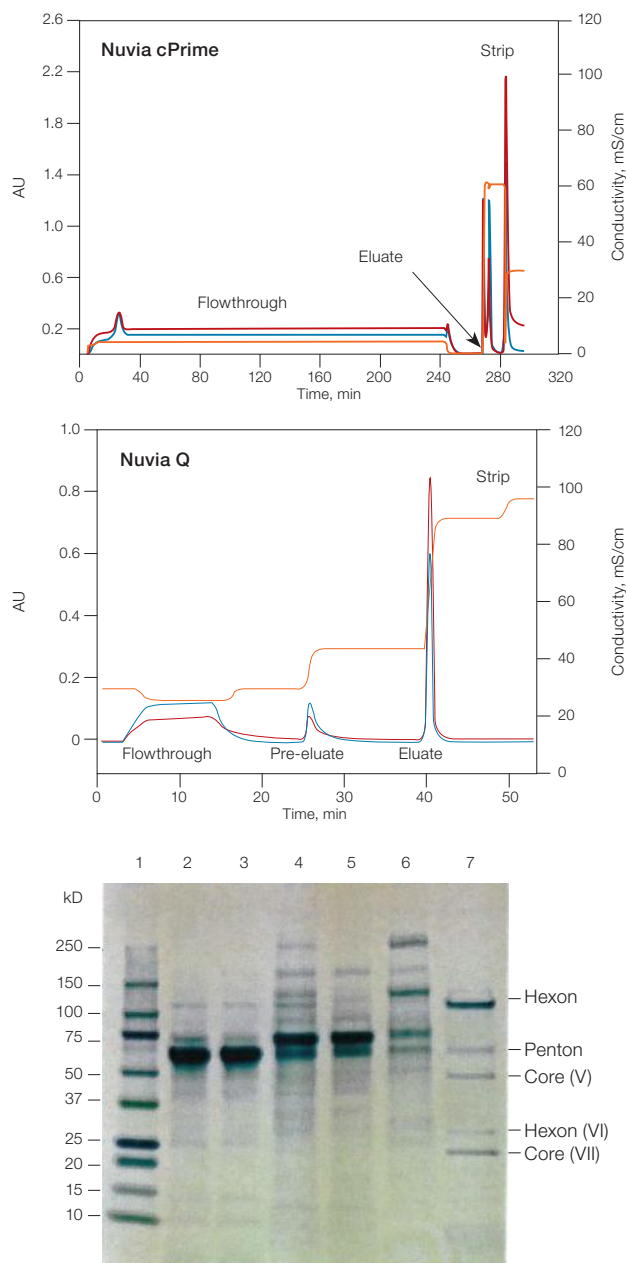


Fig. 6. SDS-PAGE of intermediates and the final product. Lane 1, MW marker; lane 2, Nuvia cPrime load; lane 3, Nuvia cPrime flowthrough; lane 4, Nuvia cPrime elution/Nuvia Q load; lane 5, Nuvia Q flowthrough; lane 6, Nuvia Q pre-elution; lane 7, Nuvia Q product. OD 260 (—); OD 280 (—); conductivity (—).

The data in Table 2 demonstrate an overall recovery of virus particles of approximately 54%, with DNA levels below detection and host cell protein at 2 ng/10¹⁰ particles. These values are well within current guidelines for clinical and perhaps commercial use.

Table 2. Viral particle recovery and impurity clearance.

Sample	Total Virus (x 10 ¹¹ particles)	Impurity Levels (ng/10 ¹⁰ particles)	
		DNA	HCP
Bulk harvest	30.6	3,144	ND
Nuclease-treated harvest	31.8	30	3,022
Nuvia cPrime eluate	18.4	ND	58
Nuvia Q eluate	16.4	<0.02	2

ND, not determined.

Summary

The final process yields an active, concentrated virus product with purity, HCP, and DNA levels comparable to clinical-grade products. While the purification method presented here was developed using the Ad5-E1+GFP model virus, it is expected to be applicable to recombinant adenoviruses in general, and to constructs derived from serotype 5 viruses in particular.

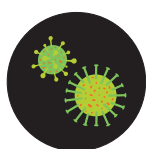
Get complete details of this case study in Development of an Efficient Manufacturing Process for Adenovirus, bulletin 6719.



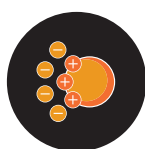
Visit bio-rad.com/ResinSample to request a sample of the resins used in this workflow.

CASE STUDY

Low Cost, Fast, cGMP Retrovirus Purification with Nuvia Q Resin



VIRUS
PURIFICATION



ANION
EXCHANGE

In this case study, we present data from the resin screening, process development, and scale-up purification of a retrovirus-like particle intended for use in human subjects. The resulting cGMP-compatible process required approximately 4 hours to purify 240 L of nuclease-treated bioreactor harvest and generated a 99% reduction in process volume and up to 65% recovery of virus particles with a final purity consistent with requirements for clinical trials. This protocol is readily scalable, fast, low cost, simple, and cGMP-compatible and can be used for the capture step in clinical-grade manufacture of retrovirus vectors.

As for any biologic intended for human use, downstream processing of retroviruses is required to remove upstream process components, such as media constituents and induction agents, and upstream process contaminants, such as host cell proteins (HCP) and DNA, while maintaining high product retention and biological activity. However, properties inherent to retroviruses and their production systems give rise to challenges not generally encountered in the downstream processing of most other biologics. These include the viruses' large size (~125 nm), their high heterogeneity with respect to surface protein and carbohydrate structure, and their susceptibility to damage by fluid shear conditions, interfacial stress, nonphysiological pH, and surfactants.

Protocol



Anion Exchange Resin Screening

Unlike cation exchangers, anion exchangers generally bind retroviruses under pH and ionic strength conditions consistent with those of most bioreactor feedstreams. This is both convenient and economical in that product purification and concentration can be accomplished concurrently. Affinity modalities were not considered because they tend to be costly from material, quality assurance, and regulatory standpoints. Several mixed-mode resins, including Capto adhere and Capto MMC (Cytiva), were screened following the manufacturer's recommendations. However, none of the mixed-mode resins tested provided satisfactory bind/elute recoveries.

Four commercial ion exchange media were screened: DEAE Sepharose Fast Flow Resin (Cytiva), Nuvia Q Resin (Bio-Rad Laboratories, Inc.), Q Sepharose Fast Flow Resin (Cytiva), and Sartobind Q Membrane Adsorber (Sartorius AG). All products were acquired from the manufacturers as preassembled or prepacked 1.0 ml devices.

Resin	Format	Manufacturer
DEAE Sepharose Fast Flow	1 ml column	Cytiva
Nuvia Q	1 ml column	Bio-Rad
Q Sepharose Fast Flow	1 ml column	Cytiva
Sartobind Q	Membrane	Sartorius

Method

Load material for resin screening was prepared by adding 840 μ l of 1.25 M sodium phosphate (NaPhos), pH 6.75, to 20 ml of freshly thawed clarified bioreactor harvest. Each device was used according to the following protocol.

Step	Buffer	CV	Flow Rate, cm/hr
Equilibration	1 M NaOH	30	340
	1 M NaCl	10	
	50 mM NaPhos, pH 7.0	50	
Loading	18 ml of material		23
Wash	50 mM NaPhos with 1.5 M NaCl, pH 7	10	23
Elution	50 mM NaPhos with 1.5 M NaCl, pH 7	10	23
Strip	50 mM NaPhos with 2.0 M NaCl, pH 7.0	10	23

Results

Quantitation of virus eluted from each medium by RT-qPCR revealed that product recovery was higher with Nuvia Q than with the other resins tested (Table 1). No significant virus was detected in any of the post-elution 2.0 M NaCl washes.

Table 1. Screening of anion exchange resins.

Anion Exchange	Resin Recovery, %
DEAE Sepharose Fast Flow	34
Nuvia Q	47
Q Sepharose Fast Flow	42
Sartobind Q	20

As shown in Figure 1, most UV-absorbing material passed through the column during the load and wash (a) and did not contain significant virus as measured by RT-qPCR (a1). During the gradient portion of the experiment, most of the absorbed material eluted between approximately 100 mM and 1,100 mM NaCl (c).

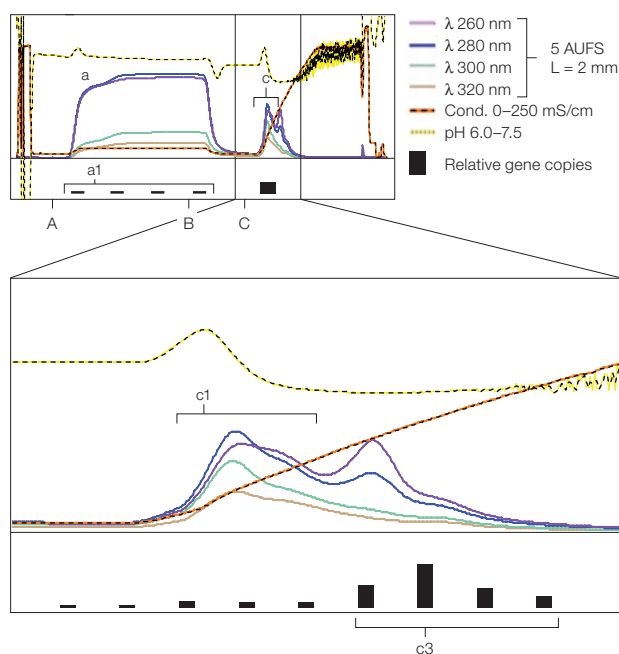


Fig. 1. Determination of approximate NaCl concentrations affecting the absorption/desorption behavior of product and contaminants.

Based on these findings, the gradient was replaced with two steps: a pre-elution wash at 440 mM NaCl and a product elution at 1,100 mM NaCl (Figure 2, C and D).

The majority of virus gene copies and virus activity (d1: black and red bars) was associated with the second peak. No significant virus was found in the flow-through and wash peaks (a1 and c1, respectively). Total product recovery was 48%.

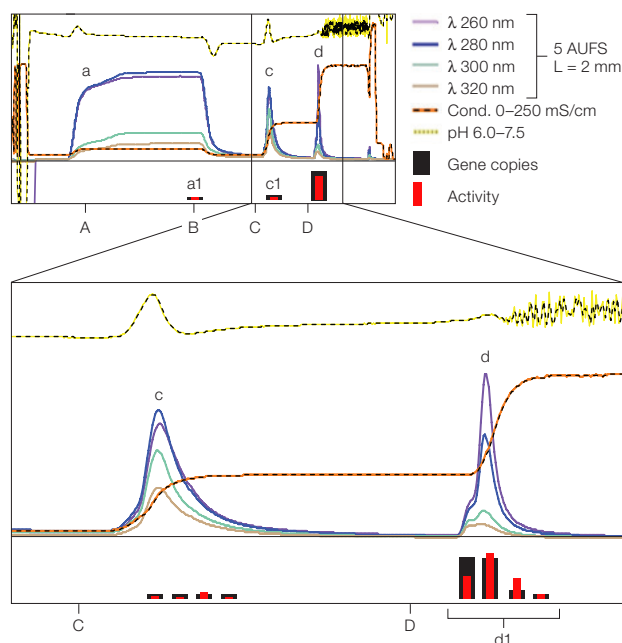


Fig. 2. Replacement of gradient elution with step elution. A single late flow-through fraction (a1) was analyzed to confirm that virus was not lost during the load and wash portions of the procedure (a). Forty-eight percent of the virus loaded, as measured by gene copy number, was recovered in the product-rich peak (d), and there was good correlation between gene copy number, a measure of virus particle concentration, and virus activity measurements (d1).

The NaCl concentration in the feedstream was increased from approximately 110 to 170 mM in order to improve column capacity by reducing the absorption of charged contaminants. In anticipation of large volume processing, NaCl and NaPhos were introduced concurrently by adding 1.0 M NaPhos, 1.2 M NaCl, pH 6.75, to the clarified harvest at a 1:19 ratio. Also, the load volume was increased from 18 to 50 ml. In the aggregate, these process changes resulted in product recovery increasing from 48 to 71%. However, the principle features of the chromatogram (Figure 3) did not change significantly as a result of these changes, and again, no significant virus was found in the flow-through and wash peaks (a1 and c1, respectively).

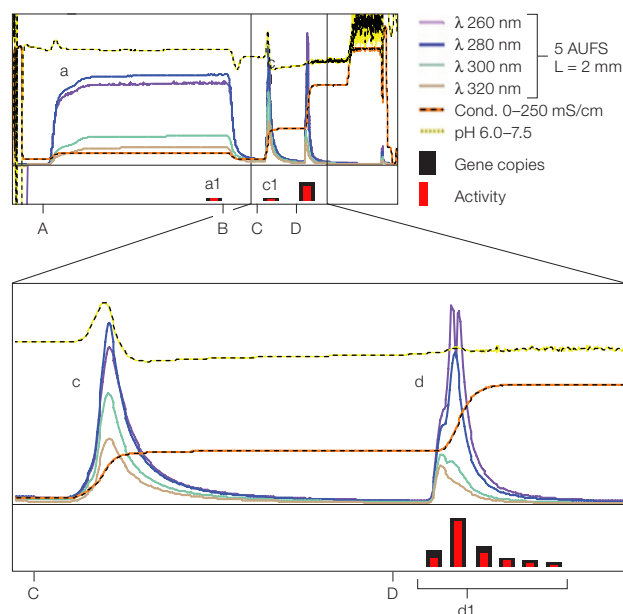


Fig. 3. Increasing the relative load volume and NaCl concentration improved product recovery. A single late flow-through fraction (a1) was analyzed to confirm that virus was not lost during the load and wash portions of the procedure (a). Seventy-one percent of the virus loaded, as measured by gene copy number, was recovered in the product-rich peak, and there remained good correlation between gene copy number, as measured by RT-qPCR (black bars), and biological activity (red bars) (d1). Load-to-CV ratio stayed 50:1

Endonuclease Digestion

Near pH 7, the presence of high molecular weight chromatin is known to cause significant amounts of both HCP and host DNA to elute from anion exchange resins. To reduce the chromatin's molecular weight, and thereby elute additional HCP and DNA during washing, the feedstream was treated with *Serratia marcescens* extracellular endonuclease prior to column chromatography. This resulted in a large reduction in the UV absorption of the product peak (Figure 4, d), without any associated reduction in product recovery, 72% as measured by RT-qPCR (d1), indicating a significant increase in product purity. This was accompanied by a corresponding increase in the size of the wash peak (c), presumably corresponding to additional HCP and DNA also being pre-eluted. No additional virus was detected in the flowthrough or washes (a1, a2, c1).

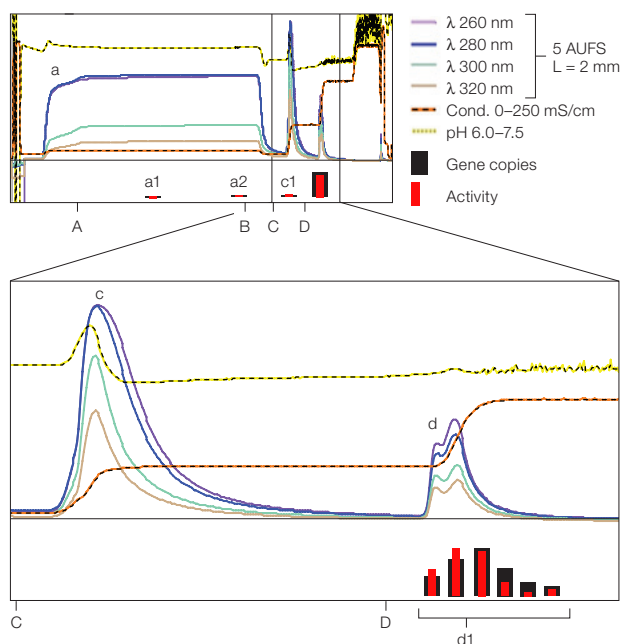


Fig. 4. Chromatogram after addition of nuclease. Compared to experiments performed without nuclease treatment, the contaminant-rich peak at the 0–440 mM transition showed increased UV absorbance and an increased A_{260}/A_{280} ratio (c), whereas the product-rich peak at the 440–1,100 mM transition exhibited a corresponding decrease in UV absorbance (d). Fractions from the first peak were pooled and analyzed to confirm that virus was not being lost during the 440 mM NaCl wash portion of the procedure (c1). Seventy-two percent of the virus loaded, as measured by gene copy number, was recovered in the product-rich peak, and there remained good correlation between gene copy number, as measured by RT-qPCR (black bars), and biological activity (red bars) (d1).

Scale-Up Studies

Process scale-up was performed as shown in Table 2.

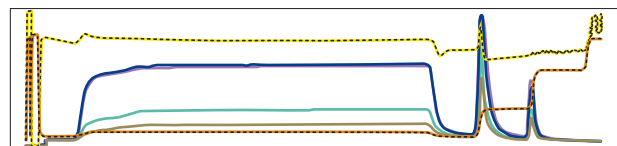
Table 2. Process parameters during downstream process development.

Parameter	A	B	C	D	E	F
Bioreactor volume, L	3	3	3	10	50	250
Load volume, L	0.018	0.050	0.5	5.3	33	240
Column diameter, cm	0.80	0.80	0.80	2.6	6.2	17.8
Column length, cm	2	2	20	20	20	20
Column volume, ml	1	1	10	105	605	4,900
Flow rate, ml/min	0.30	0.30	3.5	35	200	1,700
Relative flow rate, CV/hr	18	18	21	20	20	21
Relative load volume, CV	19	50	50	50	54	49
Linear flow rate, cm/hr	35	35	420	400	395	415
Scale factor (bioreactor)	1	1	1	3	17	83
Scale factor (chromatography)	1	3	28	295	1,800	13,000

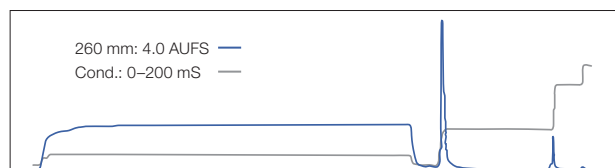
Column A, buffer system development; B, relative load volume increase and introduction of nuclease treatment; C, scale-up via increase of column length and linear flow rate; D–F, scale-up via increase of column diameter.

Representative chromatography traces were comparable across the scale-up volumes (Figure 5).

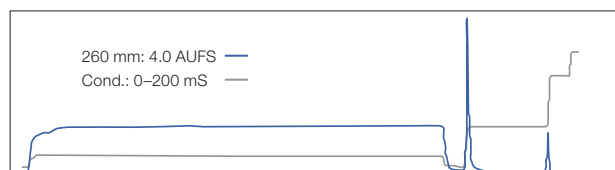
B. Bioreactor = 3 L; column = 1 ml; flow rate = 0.3 ml/min



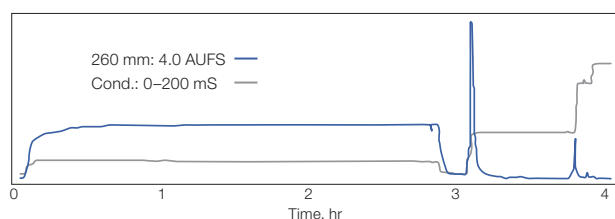
D. Bioreactor = 10 L; column = 105 ml; flow rate = 35 ml/min



E. Bioreactor = 50 L; column = 600 ml; flow rate = 200 ml/min



F. Bioreactor = 275 L; column = 5,100 ml; flow rate = 1,700 ml/min



Bioreactor/Column Scaling

	Reactor, L	Column, ml	Load, L	Flow, ml/min
B	3	1	0.06	0.3
D	10	105	6.4	35
E	50	600	36	200
F	250	5,100	275	1,700

Fig. 5. Scale-up for GMP manufacturing. Column volume was scaled from 1 to 10 ml, 105 ml, 0.6 L, and finally to 5.1 L. Corresponding flow rates were scaled from 300 μ l/min to 1.7 L/min. A load volume-to-CV ratio of approximately 50:1 was maintained throughout. Representative chromatograms at the 1 ml (B), 105 ml (D), 600 ml (E), and 5.1 L scale (F) are shown. Letter designations correspond to the columns in Table 2.

The amount of selected contaminants was analyzed in purified virus samples (Table 3).

Table 3. Analytical test results.

Assay	Per 10^{10} vp
Potency, % relative to reference	127
Residual HCP, ng	111
Residual human DNA, ng	<0.6
Residual endonuclease, ng	<0.5
Residual adenovirus E1 DNA	Not detected
Residual SV40 large T-antigen DNA	Not detected
Bioburden (cfu) <61>	<0.6
Endotoxin (EU) <85>	0.4

Final viral recovery = ~65%.

Summary

The process described here for retrovirus purification using Nuvia Q Resin constitutes an efficient single-step platform for low cost, fast, simple, and robust manufacture of retroviruses. This process is easily scalable and uses cGMP-compatible reagents.

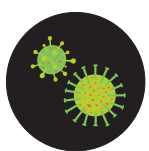
Get complete details of this case study in Low-Cost, Fast, and Scalable Downstream Purification Process Development for a Clinical-Stage Retrovirus-Like Particle, bulletin 7200.



Visit [bio-rad.com/ResinSample](https://www.bio-rad.com/ResinSample) to request a sample of Nuvia Q Resin.

CASE STUDY

Mammalian Virus Purification Using CHT Ceramic Hydroxyapatite Media



VIRUS
PURIFICATION



CHT CERAMIC
HYDROXYAPATITE

Viruses can infect mammalian cells and cause diseases such as influenza, hepatitis, yellow fever, smallpox, and AIDS. Purification of viral particles is used extensively in the study and characterization of these infectious agents. Understanding aspects of a virus, such as how it infects host cells, uses the host cells for reproduction, and evades the host immune system, aids scientists in determining how to use viruses for research and therapy. In order to study a virus, a pure, high-quality infectious population is required. Conventional techniques for mammalian virus purification, for uses such as vaccine production or biological studies, can produce material of variable quality and quantity, often with significant loss of particle infectivity.

In this case study, ceramic hydroxyapatite media is used for purification of a wide variety of mammalian viruses (Table 1). Viral activity was determined using the assays shown in Table 2. Chromatography using ceramic hydroxyapatite media is simple, easily scalable, and results in a concentrated preparation of highly active virus.

Table 1. Viral type and size.

Virus	Family	Genus	Genome	Envelope	Size, nm
Dengue	<i>Flaviviridae</i>	<i>Flavivirus</i>	ssRNA	+	50
Japanese encephalitis	<i>Flaviviridae</i>	<i>Flavivirus</i>	ssRNA	+	50
Influenza	<i>Orthomyxoviridae</i>	<i>Influenzavirus</i>	ssRNA	+	80–120
Mouse hepatitis	<i>Coronaviridae</i>	<i>Coronavirus</i>	ssRNA	+	100–150
Adenovirus	<i>Adenoviridae</i>	<i>Mastadenovirus</i>	dsDNA	–	90
Poliovirus	<i>Picornaviridae</i>	<i>Enterovirus</i>	ssRNA	–	30
Feline calicivirus	<i>Caliciviridae</i>	<i>Vesivirus</i>	ssRNA	–	30–38

Table 2. Detection methods used for viral activity.

Detection Method	Virus
Hemagglutination (HA) test	Dengue, influenza, adenovirus
Plaque assay	Japanese encephalitis
50% tissue culture infective dose (TCID ₅₀)	Poliovirus, feline calicivirus, mouse hepatitis

Methods

Purification on CHT Ceramic Hydroxyapatite Media, Type II, 40 µm

Column: 0.6 ml, 0.46 x 3.5 cm

Step	Buffer	CV	Flow Rate, cm/hr
Wash	600 mM sodium phosphate, pH 7.2 (buffer B)	8	350
Equilibration	10 mM sodium phosphate, pH 7.2 (buffer A)	16	350
Sample	Virus preparation in buffer A	16	350
Wash	Buffer A	16	350
Elution 1	Gradient of buffer A to buffer B	24	350
Elution 2	Buffer B	8	350

This method applies to Figures 1A, 2, 3, 4, 6B, and 7A.

Purification on CHT Media, Type II, 40 µm

Column: 0.6 ml, 0.46 x 3.5 cm

Step	Buffer	CV	Flow Rate, cm/hr
Wash	600 mM sodium phosphate, pH 7.2 (buffer B)	8	350
Equilibration	10 mM sodium phosphate, pH 7.2 (buffer A)	16	350
Sample	Virus preparation in buffer A	16	350
Wash	Buffer A	16	350
Elution 1	Gradient of buffer A to buffer B	24	350
Elution 2	Buffer B	8	350

This method applies to Figure 1B.

Purification on CHT Media, Type II, 40 µm

Column: 0.6 ml, 0.68 x 2.0 cm

Step	Buffer	CV	Flow Rate, cm/hr
Wash	600 mM sodium phosphate, pH 7.2 (buffer B)	8	160
Equilibration	10 mM sodium phosphate, pH 7.2 (buffer A)	16	160
Sample	Virus preparation in buffer A	16	160
Wash	Buffer A	16	160
Elution 1	Gradient of buffer A to buffer B	24	160
Elution 2	Buffer B	8	160

This method applies to Figure 5.

Purification on CHT Media Type I, 40 µm

Column: 0.6 ml, 0.46 x 3.5 cm

Step	Buffer	CV	Flow Rate, cm/hr
Wash	600 mM sodium phosphate, pH 7.2 (buffer B)	8	350
Equilibration	10 mM sodium phosphate, pH 7.2 (buffer A)	16	350
Sample	Virus preparation in buffer A	16	350
Wash	Buffer A	16	350
Elution 1	Gradient of buffer A to buffer B	24	350
Elution 2	Buffer B	8	350

This method applies to Figure 6A.

Results

Dengue Virus

Figure 1A shows the recovery of dengue virus type 2 from cell culture fluid. HA activity was recovered near the end of the gradient, separated from the bulk of A₂₈₀-absorbing material and from dsDNA (Kurosawa et al. 2012b). Figure 1B demonstrates that decreasing the flow rate by tenfold improves the sharpness of the elution peaks and, hence, separation. In both cases, recovery of HA activity was >95%. Recent studies have indicated that adsorption of dengue virus particles to the surface of CHT Type II Media is similar to their adsorption to cells (Saito et al. 2013).

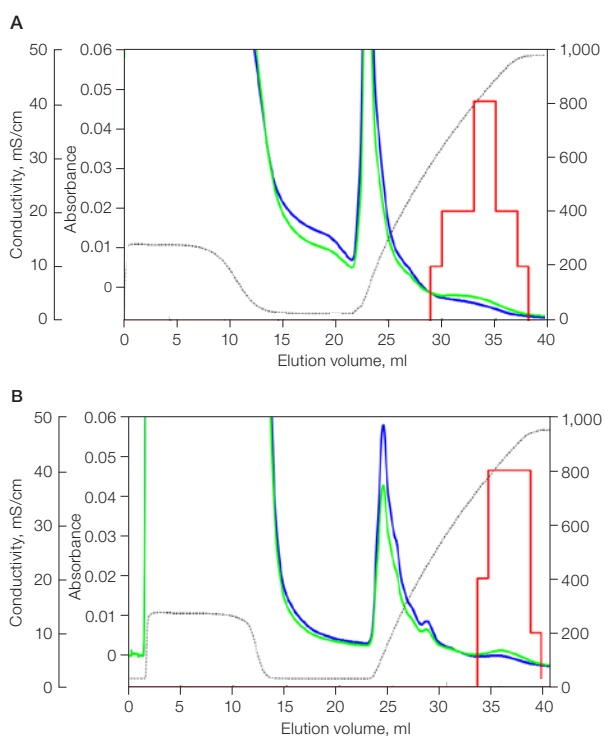


Fig. 1. Chromatograms of the separation of dengue virus type 2 by CHT Type II Media. A, flow rate at 1.0 ml/min; B, flow rate at 0.1 ml/min. UV absorbance at 260 nm (—); UV absorbance at 280 nm (—); conductivity (—); viral activity in HA test (—).

Other serotypes of dengue virus also bound to and eluted from CHT Type II Media. The approximate elution points in the sodium phosphate gradient for each serotype are shown in Table 3. Types 2 and 4 eluted at roughly the same position.

Table 3. Elution points of dengue serotypes in a sodium phosphate gradient.

Virus Serotype	Approximate Elution Point, mM
1	250
2	450
4	425

Influenza Virus

Chromatography of influenza virus A/Beijing/262/95 and A/Panama/2007/99 (Schickli et al. 2001) cultured in the presence of 0.02% and 0.20% BSA, respectively, is shown in Figure 2.

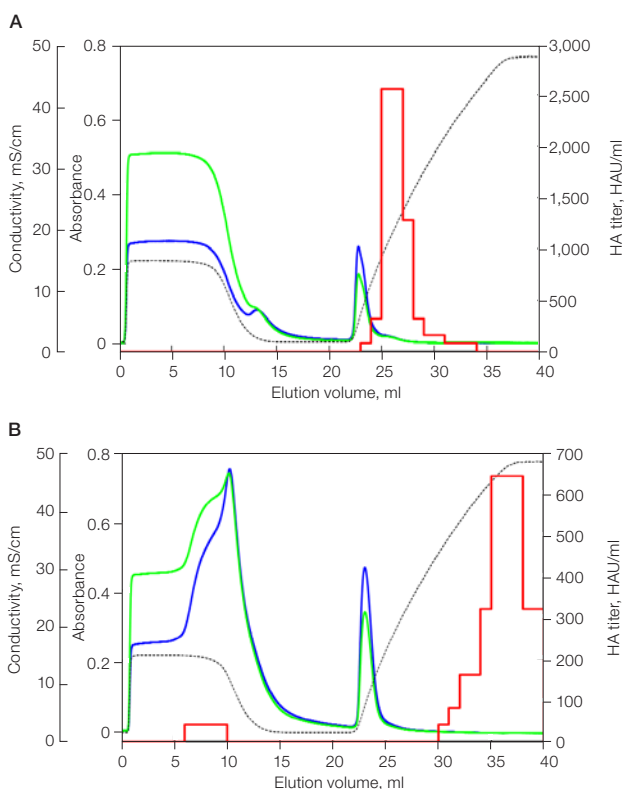


Fig. 2. Chromatography of influenza virus. A, A/Beijing/262/95; B, A/Panama/2007/99. UV absorbance at 260 nm (—); UV absorbance at 280 nm (—); conductivity (—); viral activity in HA test (—).

HA activity is separated from a small BSA peak and a significant amount of material that did not bind to the column. Recovery, as measured by the HA assay, was 98% for the A/Beijing/262/95 virus. Higher concentrations of sodium phosphate are required to elute the A/Panama/2007/99 virus. In addition, the retention time was not affected by the source (allantoic fluid vs. cell culture; data not shown).

Mouse Hepatitis Virus

Mouse hepatitis virus (MHV) is a *Coronavirus* (CoV), a genus that includes SARS-CoV. Two strains of MHV (MHV-NuU and MHV-S) (Hirano et al. 1981) were applied and bound to CHT Type II Media. Both were eluted at 26–28 minutes in the gradient (Figure 3).

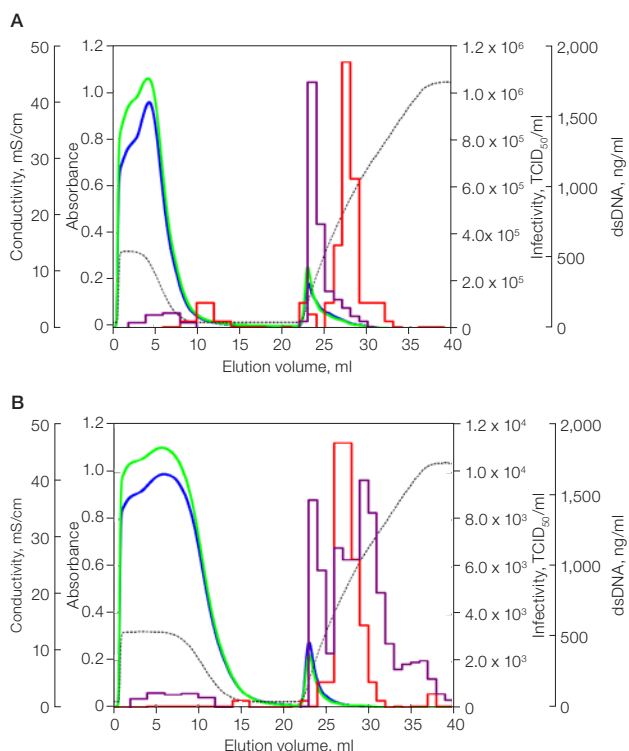


Fig. 3. Chromatography of two strains of MHV (A, MHV-NuU; B, MHV-S). UV absorbance at 260 nm (—); UV absorbance at 280 nm (—); conductivity (—); viral infectivity in TCID₅₀ (—); dsDNA (—). Culture fluid contained 10% fetal bovine serum (FBS).

Nonenveloped Viral Particles

Nonenveloped viral particles can be purified by ceramic hydroxyapatite chromatography in the same way as enveloped viruses. Adenovirus (AdV) type 27, feline calicivirus (FCV) A391 (Hirano et al. 1986), and poliovirus (PV) Sabin type 2 all adsorbed to CHT Type II Media (Figure 4), although they showed different elution times.

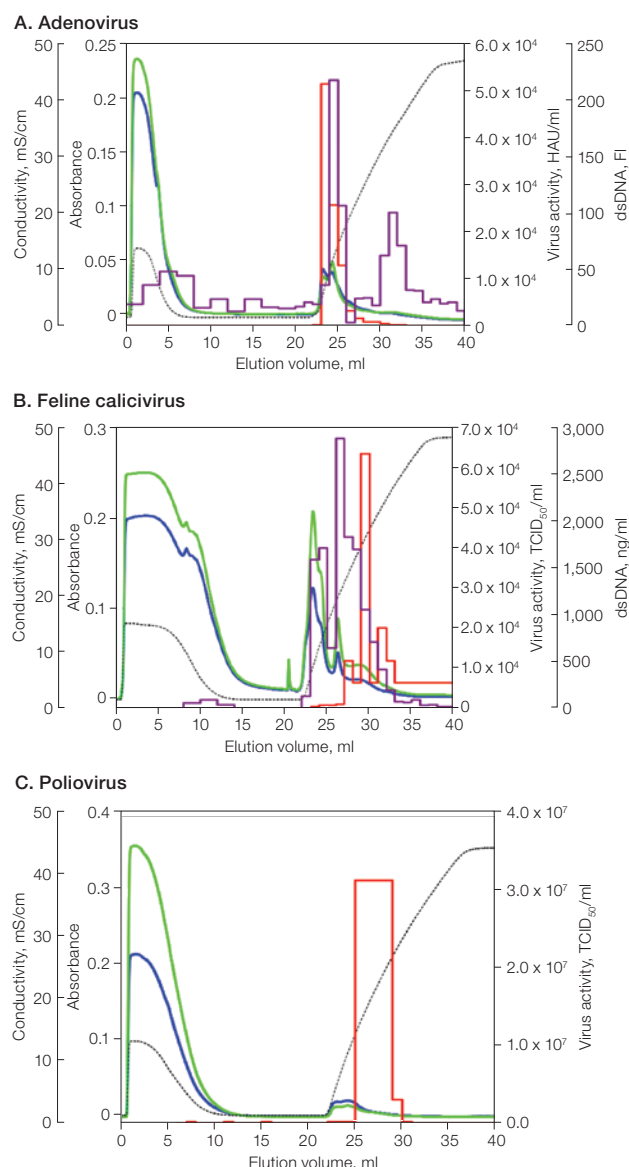


Fig. 4. Chromatograms of the separation of cell lysate (A) or culture fluid (B, C) containing nonenveloped viral particles by CHT Ceramic Hydroxyapatite Type II Media. A, AdV type 27; B, FCV A391; C, PV Sabin type 2. UV absorbance at 260 nm (—); UV absorbance at 280 nm (—); conductivity (—); viral activity (AdV in HA test, FCV and PV in TCID₅₀) (—); dsDNA (—). Cell culture fluid contained 10% FBS. FI, fluorescence intensity.

Japanese Encephalitis Virus

Japanese encephalitis virus (JEV) chromatography is shown in Figure 5 (Kurosawa et al. 2009, 2012a). Irrespective of the source or strain, the virus elutes at approximately 350 mM sodium phosphate (note that the gradient in these two cases is 10–400 mM and the column size is 6.8 x 20 mm). Again, there is good separation between protein contamination and the virus.

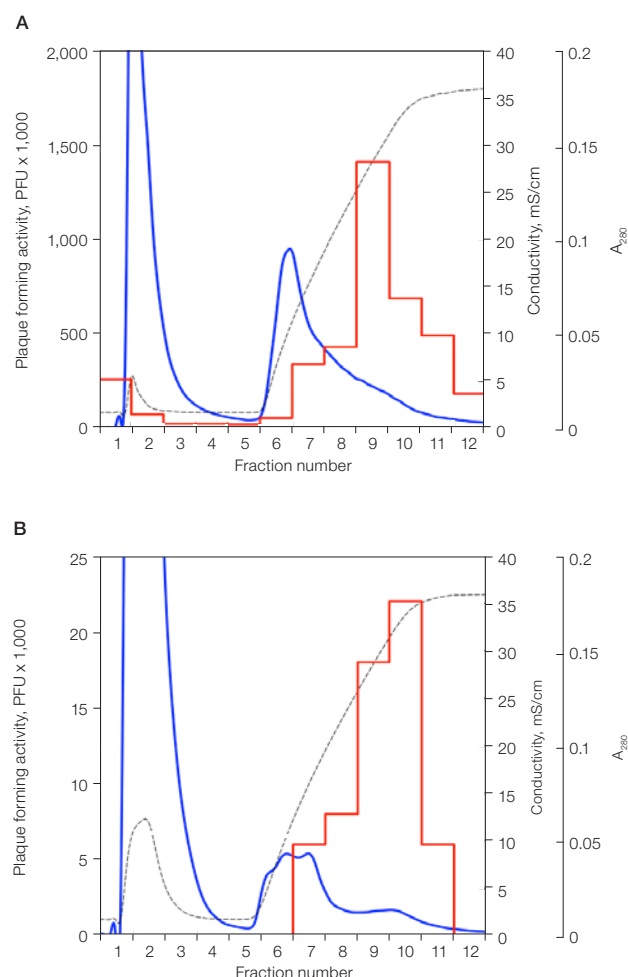


Fig. 5. JEV chromatography at pH 7.0. A, mouse brain homogenate infected with JEV JaGAR01; B, cell culture fluid of JEV Beijing. UV absorbance at 280 nm (—); conductivity (—); infectious activity in plaque assay (—). **Note:** Figure 5A is modified from Kurosawa et al. 2012a.

Effect of Hydroxyapatite Type on Separation

Figure 6 shows the separation of dengue virus type 2 from cell culture contaminants on CHT Type I and CHT Type II Media. Yields were ≥80% for both media types. Although binding and elution were achieved on both media, the separation of virus from impurities was best on CHT Type II Media. Figure 7 shows a similar study using CHT Type II Media for the purification of poliovirus, with a recovery of 88%. These results illustrate the importance of choosing the appropriate media for the separation in question.

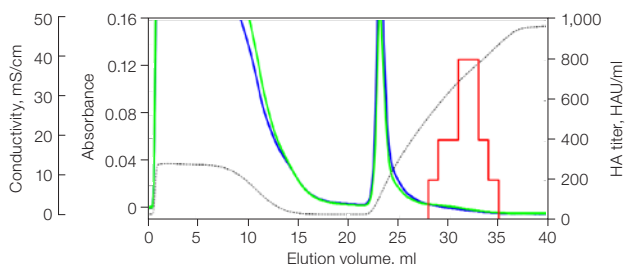
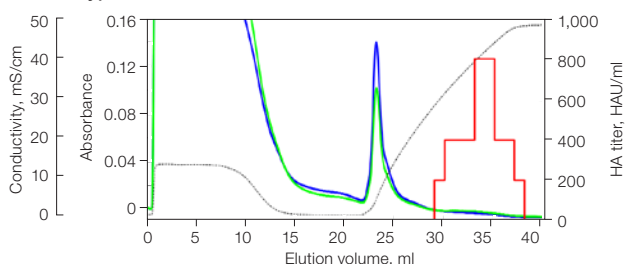
A. CHT Type I**B. CHT Type II**

Fig. 6. Chromatograms of dengue virus type 2. A, CHT Type I Media; B, CHT Type II Media. UV absorbance at 260 nm (—); UV absorbance at 280 nm (—); conductivity of elution buffer (—); viral activity in HA test (—).

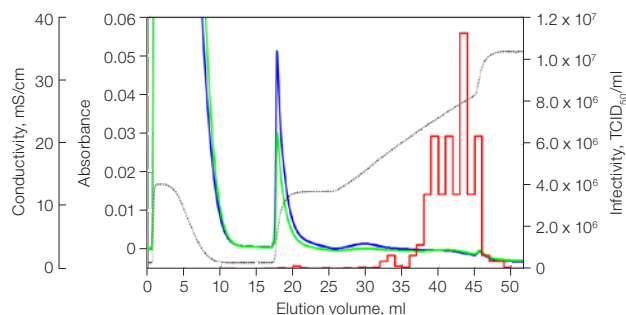


Fig. 7. CHT Type II Media chromatogram of culture fluid of polio virus at pH 6.4. UV absorbance at 260 nm (—); UV absorbance at 280 nm (—); conductivity of elution buffer (—); infectious activity in TCID₅₀ (—).
Note: the gradient in these two cases is 150–450 mM at pH 6.4 for 20 ml.

Summary

Using ceramic hydroxyapatite media provided high purity, recovery, and viral activity for seven mammalian viruses of varying size and belonging to different families. We have shown that, in at least one case, slowing the flow rate and decreasing the gradient slope allowed better purification of viral particles on CHT Type II Media, signifying the importance of determining the best settings for such factors when using apatite media. Testing different apatites is significant for determining which media type will work best for a specific virus. CHT Type II Media, because of its larger pore size, allowed better separation of the dengue virus from contaminants compared to other apatite media.

Of equal significance, the use of ceramic hydroxyapatite media is simple and provides reproducible results, allowing an alternative to the conventional methods of viral purification.

Get references and complete details of this case study in Mammalian Virus Purification Using Ceramic Hydroxyapatite, bulletin 6549.



Visit [bio-rad.com/ResinSample](https://www.bio-rad.com/ResinSample) to request a sample of the different CHT Media.

CASE STUDY

Single-Step Dye-Decolorizing Peroxidase (DyP) Purification Using Nuvia aPrime 4A Resin



PROTEIN
PURIFICATION



HI ANION
EXCHANGE

DyP (dye-decolorizing peroxidase)-type peroxidases are a family of heme-containing enzymes. Owing to their remarkable expression and robustness, they have many potential uses in a variety of biotechnology applications. Here we show purification of bacterially expressed DyPs with a hydrophobic anion exchange mixed-mode resin, Nuvia aPrime 4A, to generate a >95% pure, functionally viable sample. The workflow used in this purification is readily scalable for process production.

Method

Pseudomonas fluorescens-expressed DyP (PfDyP) was purified on Nuvia aPrime 4A Hydrophobic Anion Exchange Resin using the following protocol. Analytical assays showed the functionality of the purified enzyme.

Step	Buffer	CV	Flow Rate, cm/hr
Equilibration	50 mM K_2HPO_4/KH_2PO_4 buffer at pH 7.8 (buffer A)	10	120
Binding	4 ml cell culture supernatant adjusted to pH 7.8 with 6 ml equilibration buffer	–	120
Wash	Buffer A	20	120
Elution	Stepwise elution using 2 CV each of 10, 20, 30, 50, and 100% buffer B Buffer B: 1 M NaCl, 50 mM K_2HPO_4/KH_2PO_4 , pH 7.8	2	120

Results

Active peroxidase eluted with 1 M NaCl and showed ~95% purity with SDS-PAGE analysis (Figure 1). The purification procedure yielded ~35 mg of purified histidine-tagged PfDyP from 1 L of culture broth. The activity of the purified enzyme was shown by various assays (Figures 2–6).

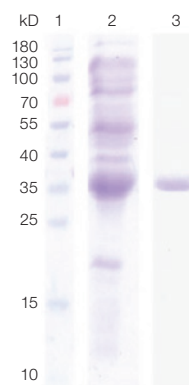


Fig. 1. SDS-PAGE analysis of the enzyme eluted from the Nuvia aPrime 4A Column. Lane 1, molecular markers; lane 2, cell-free extract; lane 3, PfDyP fraction after Nuvia aPrime 4A Mixed-Mode Resin chromatography.

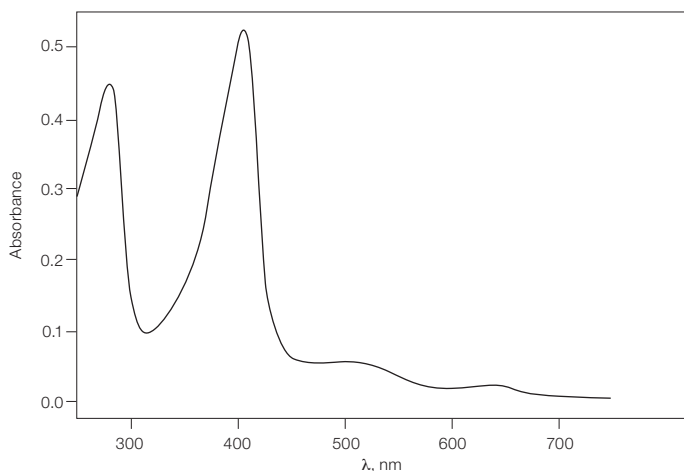


Fig. 2. UV/Vis spectra of PfDyP B2. Soret band (405 nm) is indicative of heme incorporation. An Rz value of 1.16 shows that enzyme is loaded with heme.

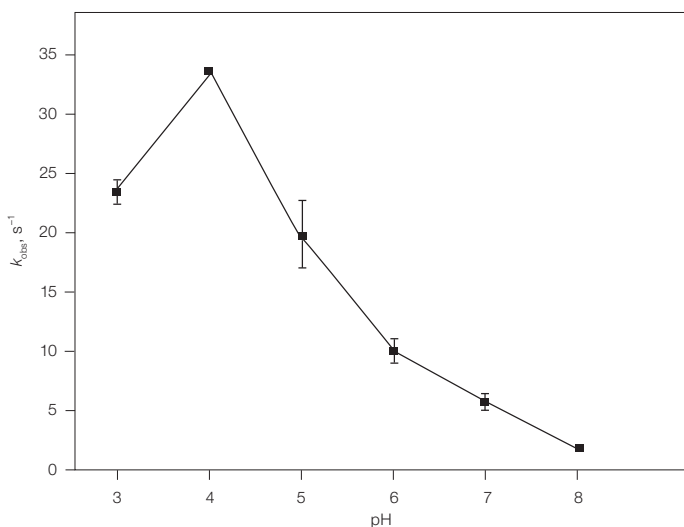


Fig. 3. Optimal pH of the purified PfDyP with the substrate ABTS.

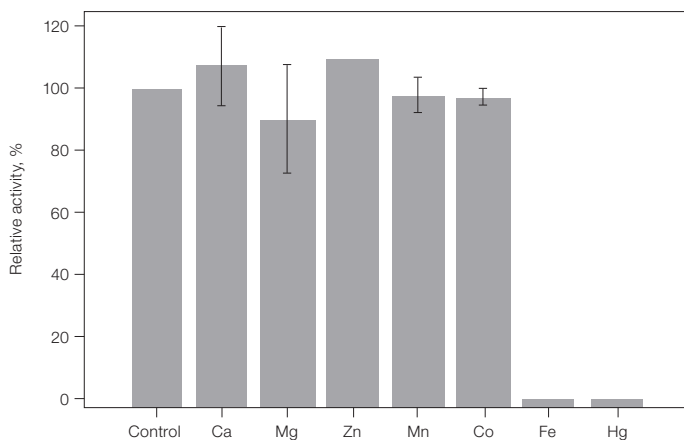


Fig. 4. Relative activity of the purified PfDyP in the presence of different cations.

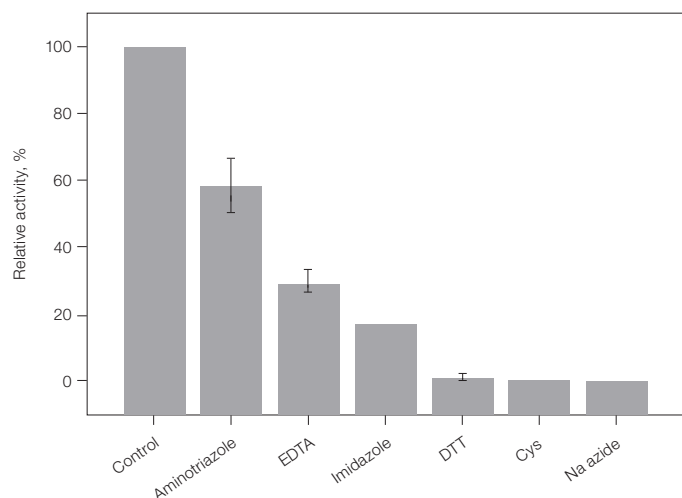


Fig. 5. Relative activity of the purified PfDyP in the presence of different reducing agents.

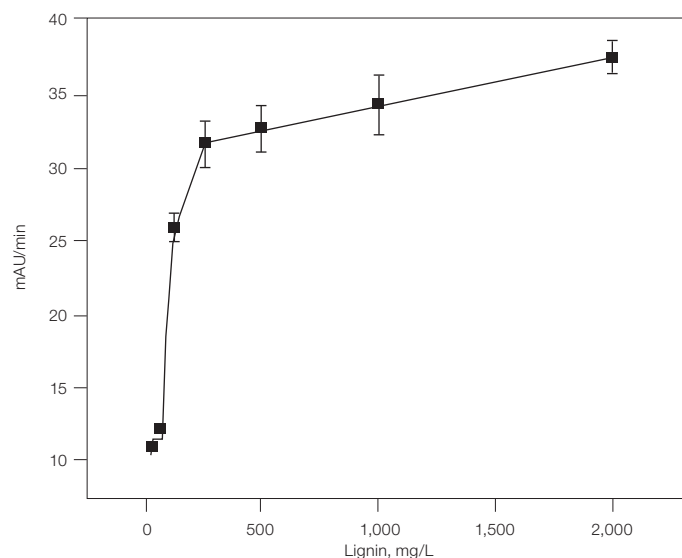


Fig. 6. Steady state activity of *P. fluorescens* DyPB with kraft lignin.

Summary

Nuvia aPrime 4A Mixed-Mode Resin can be used to purify crude samples of PfDyP to >95% purity while maintaining the stability and functionality of the active enzyme. This permits the enzyme to be used for biotechnical industry applications. The purification process is scalable and can be used for manufacturing bulk quantities of this enzyme. For targets for therapeutic purposes that require greater purification stringency, Nuvia aPrime 4A can be incorporated into a workflow with additional purification steps.

Get complete details of this case study in Purification of a Recombinant Bacterial DyP-Type Peroxidase with a Hydrophobic Anion Exchange Resin, bulletin 7182.



Visit [bio-rad.com/ResinSample](https://www.bio-rad.com/ResinSample) to request a sample of the resins used in this workflow.

CASE STUDY

Single-Step Raw Starch Digesting Alpha Amylase (RSDA) Purification Using Nuvia cPrime Resin



PROTEIN
PURIFICATION



HI CATION
EXCHANGE

Raw starch digesting α -amylase (RSDA) is an industrially important enzyme. The kinetics and mode of action of RSDA are not well understood. Further elucidation of this enzyme's mechanism of action, however, requires highly pure RSDA. Here we demonstrate a single-step purification protocol, using the mixed-mode resin Nuvia cPrime, that yields ~96% pure recombinant enzyme from *Escherichia coli*. The protocol developed here is suitable for large-scale purification of RSDA and similar enzymes.

Method

Recombinant RSDA was purified on Nuvia cPrime Hydrophobic Cation Exchange Resin using the following protocol. RSDA activity was determined by measuring the formation of reducing sugars released during starch hydrolysis in 50 mM phosphate buffer, pH 6.5, at 75°C.

Step	Buffer	CV	Flow Rate, cm/hr
Equilibration	50 mM sodium acetate, pH 5.3, 150 mM NaCl (buffer A)	20	360
Binding	20 ml of the RSDA-containing <i>E. coli</i> extract	–	360
Wash	Buffer A	20	360
Elution	30 mM Tris-HCl, pH 8.0, 0.5 M NaCl (buffer B)	10	360

Results

To identify optimal binding and elution conditions, a central composite design (CCD) or DOE study was performed with four operation parameters (A, binding pH; B, binding NaCl concentration; C, elution pH; and D, elution NaCl concentration) (data shown in bulletin 6992). The eluted enzyme showed high activity corresponding to ~96% of the total loaded enzyme activity (Table 1). SDS-PAGE analysis of the eluted sample showed a highly pure eluate fraction (Figure 1).

Table 1. Purified RSDA activity analysis.

Sample Type	Total Protein, mg	Total Activity, IU	Specific Activity, IU/mg	Yield, % Based on Total Activity
Fermentation broth	66	9,200	139	100
Eluted sample from Nuvia cPrime Column	23	8,800	382	96

These data were first published in Lončar et al. (2015) under CC BY 4.0.

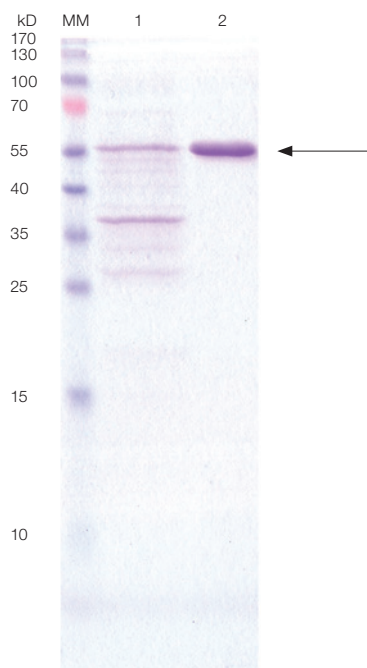


Fig. 1. RSDA purity by SDS-PAGE analysis. Lane MM: molecular markers; lane 1: cell-free extract (fermentation broth); lane 2: purified amylase. These data were first published in Lončar et al. (2015) under CC BY 4.0.

Summary

Our analysis demonstrates that Nuvia cPrime can be used to produce highly pure and active enzyme in a single step. The use of this resin overcomes the problem of compromised resin reusability caused by strong binding of fermentation broth components, such as pigments, seen with other resins. The process developed here may also be applicable to the purification of other bacterial enzymes and proteins. Nuvia cPrime Resin's multiple interaction modes, high capacity, and high selectivity can also help limit the number of steps required in a purification workflow.

Get references and complete details of this case study in Single-Step Purification of Raw Starch Digesting α -Amylase Using Nuvia cPrime Hydrophobic Cation Exchange Resin, bulletin 6992.



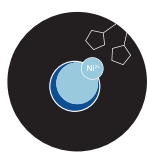
Visit [bio-rad.com/ResinSample](https://www.bio-rad.com/ResinSample) to request a sample of the resins used in this workflow.

CASE STUDY

Histidine-Tagged Protein Purification Using Nuvia IMAC and Nuvia cPrime Resins and CHT Ceramic Hydroxyapatite Media



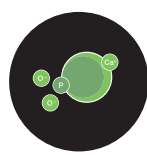
PROTEIN
PURIFICATION



IMAC



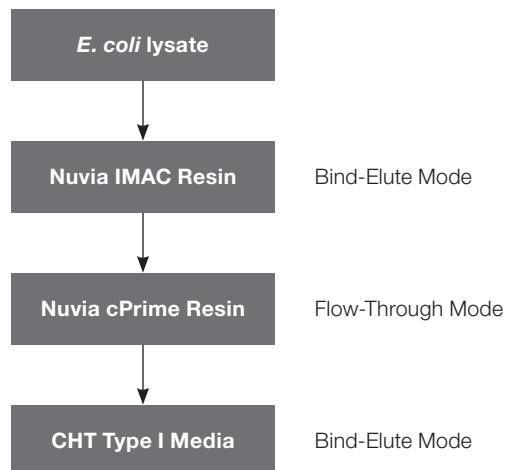
HI CATION
EXCHANGE



CHT CERAMIC
HYDROXYAPATITE

Immobilized metal affinity chromatography (IMAC) is one of the most popular tools for the preparation of recombinant proteins in a single step. As with other affinity chromatography techniques, purification with IMAC is robust and requires minimal optimization of conditions. The use of IMAC has been extended to the manufacture of protein therapeutics, vaccines, and adeno-associated virus (AAV). However, protein pharmaceuticals are held to extremely high purity standards to ensure drug efficacy and the safety of patients. In this study, we demonstrate a three column workflow for the purification of a histidine-tagged protein. Nuvia IMAC, a rigid macroporous high-capacity resin designed for process production at high flow rates, was used to capture histidine-tagged protein from *Escherichia coli* lysate. Further purification was facilitated by chromatography with two mixed-mode media, Nuvia cPrime and CHT Ceramic Hydroxyapatite.

Protocol



Method

The *E. coli*-expressed histidine-tagged recombinant protein was capture purified with Nuvia IMAC and then polish purified with Nuvia cPrime Hydrophobic Cation Exchange Resin and CHT Mixed-Mode Media under the following conditions.

Capture with Nuvia IMAC

Step	Buffer	CV	Flow Rate, cm/hr
Equilibration	50 mM sodium phosphate, 10 mM imidazole, 300 mM sodium chloride, pH 7.5 (buffer A)	10	150
Binding	<i>E. coli</i> cell lysate was centrifuged at 30,000 rpm for 5 minutes to collect the supernatant, which was then applied onto the Nuvia IMAC Column	–	150
Wash	Buffer A followed by 50 mM Tris-HCl, 5 mM sodium phosphate, pH 7.5 (buffer B)	5 each	150
Elution	50 mM Tris-HCl, 5 mM sodium phosphate, 125 mM imidazole, pH 7.5 (buffer C)	10	150

Intermediate Polish with Nuvia cPrime

Step	Buffer	CV	Flow Rate, cm/hr
Equilibration	Buffer B	10	150
Sample	The eluate from the Nuvia IMAC Column was loaded directly	–	300
Final Product	The target histidine-tagged protein was recovered in the flow-through fractions		
Strip	50 mM Tris-HCl, 5 mM sodium chloride, pH 7.5 (buffer D), followed by 50 mM sodium phosphate, 2 M sodium chloride, pH 7.5 (buffer E), followed by 1 N sodium hydroxide	5 each	120

Final Polish with CHT I-40

Step	Buffer	CV	Flow Rate, cm/hr
Equilibration	Buffer B	10	150
Sample	Partially purified flow-through sample from the Nuvia cPrime Column	–	150
Wash	50 mM sodium phosphate, 400 mM sodium chloride, pH 7.5 (buffer F)	10	150
Elution	50 mM sodium phosphate, 1.2 M sodium chloride, pH 7.5 (buffer G)	10	150
Regeneration	50 mM Tris-HCl, 5 mM sodium phosphate, 2 M sodium chloride, pH 7.5 (buffer H) followed by 1 N sodium hydroxide	2 each	150

Results

As demonstrated in Figure 1, the histidine-tagged target protein was efficiently captured from the *E. coli* lysate by chelation with Nuvia IMAC Resin and eluted with over 95% recovery and near homogeneous purity as visualized by SDS-PAGE. During the first polish purification step, Nuvia cPrime predominantly binds basic host cell proteins while leaving the acidic target protein in the flow-through fractions (Figure 2). Further purification on a CHT Column eliminated residual Ni²⁺ ions in the Nuvia cPrime Column flow-through pool, most likely through the strong interaction with the phosphoryl groups on this chromatography media (Figure 3). The summary of impurity clearance through the three-step workflow is shown in Table 1.

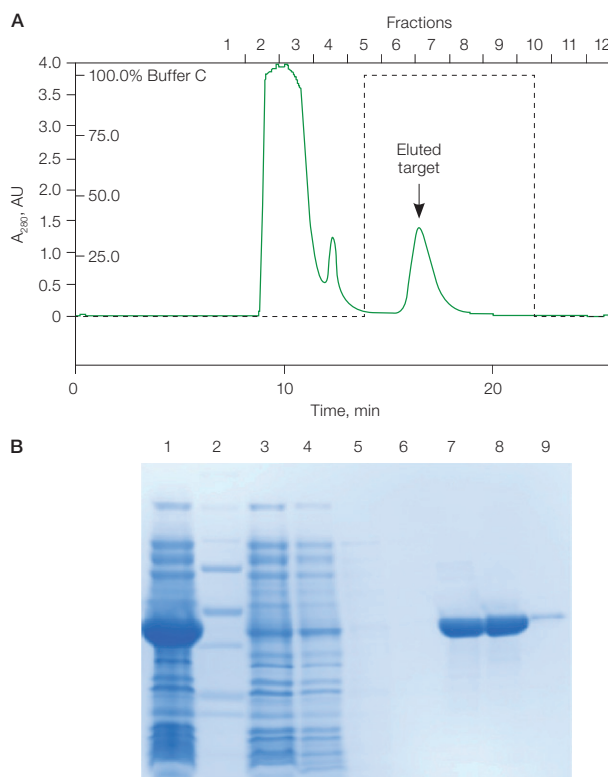


Fig. 1. Capturing histidine-tagged protein from *E. coli* cell lysate using a Nuvia IMAC Column. **A**, chromatogram of histidine-tagged protein purification on a Nuvia IMAC Column. A_{280} (—); % buffer C (---); AU, absorbance units. **B**, SDS-PAGE analysis of column fractions. Lane 1, *E. coli* cell lysate; lane 2, Precision Plus Protein Standards; lanes 3–9, column fractions 3–9. Target protein-containing fractions 7 and 8 were pooled and subjected to next purification step.

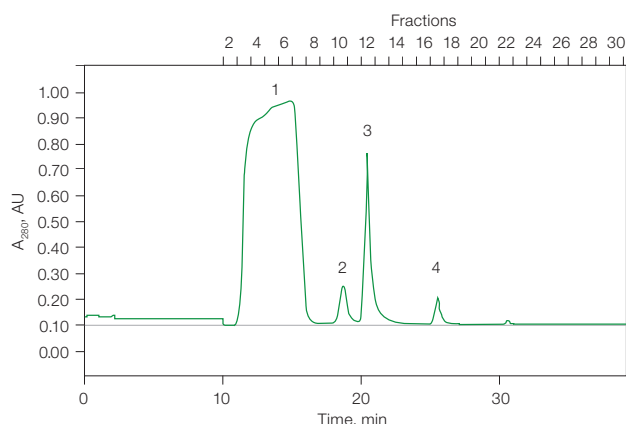


Fig. 2. Intermediate polishing purification of histidine-tagged protein in flow-through mode using a Nuvia cPrime Column. Peak 1, histidine-tagged target protein; peak 2, impurities eluted by buffer D; peak 3, impurities eluted by buffer E; peak 4, impurities stripped by 1 N NaOH. A_{280} (—); AU, absorbance units.

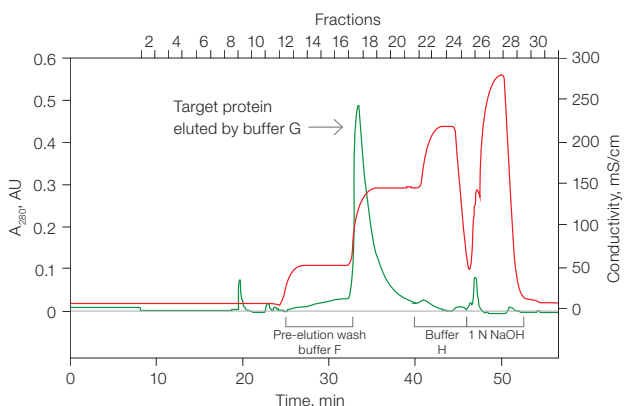


Fig. 3. Further purification of target protein using a CHT Column. Target protein eluted from the CHT Column was collected in fractions 18–22. A_{280} (—); conductivity (—); AU, absorbance units.

Summary

A three-column workflow was developed for the purification of a histidine-tagged recombinant protein from *E. coli* extract. The macroporous Ni^{2+} -charged Nuvia IMAC Resin provides efficient capture of the target protein molecules with excellent selectivity, binding capacity, and recovery. The subsequent mixed-mode chromatography steps, performed with Nuvia cPrime Resin followed by CHT Ceramic Hydroxyapatite Media, eliminate leached Ni^{2+} ions and offer further clearance of residual impurities from host cells. This workflow requires minimum feed conditioning during transitions between chromatography steps, which greatly minimizes labor and the consumption of materials. This robust purification process provides the product quality, productivity, and economics demanded by today's downstream manufacturing.

Get complete details of this case study in A Nuvia IMAC Resin–Based Purification Workflow for the Production of Highly Pure Recombinant Proteins, bulletin 6964.

Visit [bio-rad.com/ResinSample](https://www.bio-rad.com/ResinSample) to request a sample of the resins used in this workflow.

Table 1. Summary of process impurity clearance.

Samples from Purification Steps	Endotoxin, EU/mg*	Ni^{2+} , $\mu\text{g/g}^{**}$	dsDNA, ng/mg***	HCP, ng/mg†
<i>E. coli</i> lysate	1.05×10^5	—	1.34×10^4	2.87×10^5
Nuvia IMAC Column	5.13×10^3	1.400	4.63×10^2	1.11×10^4
Nuvia cPrime Column	$<3.60 \times 10^3$	0.014	2.42×10^2	6.03×10^2
CHT Column	$<1.51 \times 10^2$	Not detected	Not detected	1.46×10^2

* LAL assay, per milligram of protein.

** ICP-MS analysis, per gram of solution. Detection limit is 0.002 $\mu\text{g/g}$.

*** Droplet Digital™ PCR analysis, per milligram of target protein. Detection limit is 1.2 pg/ml.

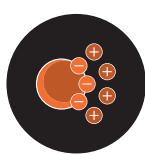
† ELISA, detection limit is 1.6 ng/ml.

CASE STUDY

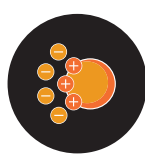
Rapid Single-Step Whey Protein Purification with Nuvia S and Nuvia Q Resins



PROTEIN
PURIFICATION



CATION
EXCHANGE



ANION
EXCHANGE

Whey proteins play a vital role in the formulation of food supplements because of numerous desirable nutritional and functional properties. This case study depicts the establishment of a process workflow for whey purification using Nuvia S and Nuvia Q Ion Exchange Resins. The major whey proteins are α -lactalbumin (ALA), β -lactoglobulin (BLG), bovine serum albumin (BSA), and bovine immunoglobulins (Hahn et al. 1998) (Table 1). Minor proteins such as lactoperoxidase, lactoferrin, and proteose-peptone account for the other protein components in whey.

Table 1. Protein composition of bovine whey.

Protein	Average Concentration, g/L	Proportion in Whey, %	Molecular Weight	Isoelectric Point
ALA	1.5	22	14,200	4.7–5.1
BLG	3–4	51	18,400	5.2
BSA	0.3–0.6	6.6	65,000	4.9
IgG, IgA, IgM	0.6–0.9	11	150,000–900,000	5.8–7.3
Lactoperoxidase	0.06	<1	78,000	9.6
Lactoferrin	0.05	<1	78,000	8.0
Proteose-peptone	0.5	7	4,000–20,000	—

The objective of this study was to develop a separation process to fractionate usable proteins from whey (Figure 1). First, total whey protein isolate (WPI) was chosen for purification because of its high-end functional advantages, as evidenced by a demand from nutrition and health segments. Second, ALA and BLG were purified because of their value in infant formula and in confections, respectively.

Protocol

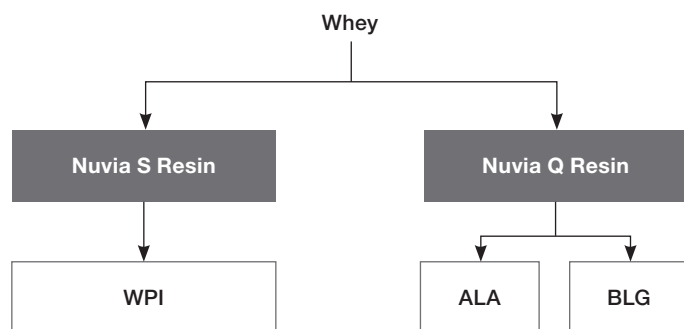


Fig. 1. Workflow for protein purification from whey with Nuvia S and Nuvia Q Resins.

Method

CEX Chromatography Using Nuvia S Resin

Column: 1 ml, 0.7 x 2.7 cm

Step	Buffer	CV	Flow Rate, cm/hr
Equilibration	0.04 M sodium lactate, pH 4.0 (buffer A)	3	150
Sample	Whey concentrate diluted 1:10 in buffer A, 1.2 mM filtered	20	150
Wash 1	Buffer A	10	150
Elution	0.04 M sodium phosphate, pH 8.0	20	150
Wash 2	1 M NaCl	3	150
Wash 3	1 M NaOH	3	150
Equilibration	0.2 M sodium lactate, pH 4.0	3	150
Re-Equilibration	Buffer A	5	150

AEX Chromatography Using Nuvia Q Resin

Column: 1 ml, 0.7 x 2.7 cm

Step	Buffer	CV	Flow Rate, cm/hr
Equilibration	0.02 M sodium phosphate, pH 6.0 (buffer A)	3	150
Sample	Whey concentrate diluted 1:10 in buffer A, 1.2 mM filtered	5	150
Wash	Buffer A	4	150
Elution 1	Gradient of buffer A to buffer A + 0.1 M NaCl	10	150
Elution 2	Buffer A + 0.1 M NaCl	5	150
Elution 3	Gradient of buffer A + 0.1 M NaCl to buffer A + 0.3 M NaCl	10	150
Elution 4	Buffer A + 0.3 M NaCl	5	150
Elution 5	Gradient of buffer A + 0.3 M NaCl to buffer A + 0.9 M NaCl	10	150
Elution 6	Buffer A + 0.9 M NaCl	10	150

Results

Preparation of Whey Protein Isolate

As defined by the American Dairy Products Institute (adpi.org), WPI is a highly purified form of whey protein obtained by removing nonprotein constituents from whey, resulting in a finished dry product containing at least 90% protein on a dry matter basis, with minimal or no lipids and lactose. Since the majority of the proteins have isoelectric points >4.7, loading of whey at pH 4.0 onto Nuvia S Resin resulted in the capture of these proteins (Etzet 2004, Hahn et al. 1998). Lipids, lactose, and unbound proteins remain in the flowthrough and the wash. As shown in Figure 2, a single peak was collected upon elution with pH 8.0 elution buffer. Capacity of the Nuvia S Column, based on the measurement of the protein content in the eluate pool, was 60 ± 2 g/L (60 CV/hr, $n = 4$). As shown in Figure 3, proteins in WPI and in whey are similar. Interestingly, densitometry of the major species, including ALA, BLG, and BSA, in whey and WPI showed nearly identical distributions.

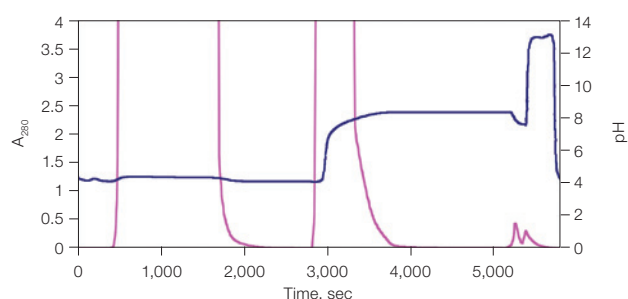


Fig. 2. CEX chromatography of 80 mg of crude whey on a Nuvia S Column. Column: 7 x 27 mm; 2 ml fractions were collected; flow rate: 60 CV/hr; buffer A: 0.04 M Na lactate, pH 4.0; buffer B: 0.04 M Na phosphate, pH 8.0. A_{280} (—); pH (—).

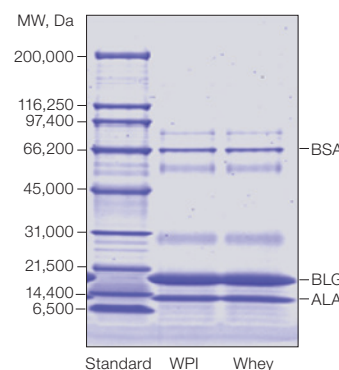


Fig. 3. SDS-PAGE analysis of WPI and whey. Whey protein and WPI (10 mg) along with Bio-Rad SDS-PAGE standards were run on a 4–20% Criterion TGX Precast Gel. Proteins were stained using Bio-Safe Coomassie Stain.

Anion Exchange Resin Selection and Optimization

ALA and BLG account for about 75% of the proteins in whey. These two model proteins were therefore selected for further experimentation. In an initial screen, Nuvia Q Resin was compared to two resins with high binding capacity (resin A and resin B). The resins were tested using a small mixture of ALA and BLG (10% CV) that was eluted using a shallow NaCl gradient (0–0.3 M). This permits rapid and uniform evaluation of process parameters, but further optimization will be needed for the selected resin. The resulting chromatograms from this screening experiment were compared (Figure 4). All three resins produced two equally well-resolved peaks. Peaks of resin A eluted sooner, suggesting weaker AEX properties than Nuvia Q Resin and resin B. The bead size of resin A is the smallest of the three resins, which may lead to higher column pressure upon scale-up. Additionally, rigidity, cleanability, capacity, and cost will have to be factored in when choosing the appropriate resin for large-scale purification.

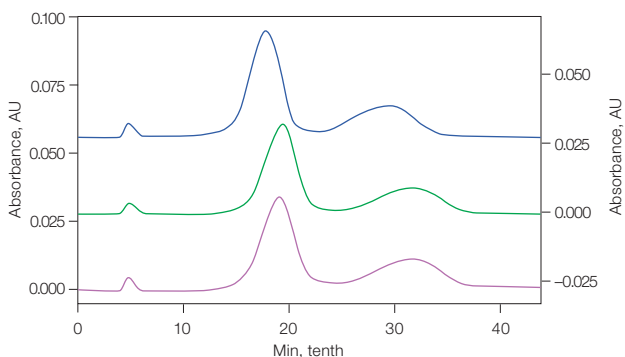


Fig. 4. ALA and BLG purification chromatograms on three anion exchangers. Comparison of Nuvia Q Resin (—), resin A (—), and resin B (—) performance.

Purification of ALA and BLG from Whey

To minimize processing costs, a purification scheme for ALA and BLG from whey ideally should require a single chromatographic step to purify both proteins. Initial studies (Kim et al. 2003, Gerberding and Byers 1998) demonstrated the removal of the major whey proteins (ALA, BSA, and BLG) in a segmented gradient (data not shown). Further work was performed to optimize these conditions. The purification profile of each protein was first identified by injecting the purified protein alone and then a mixture containing the three purified proteins. As expected, separation of the mixture paralleled that of the individual proteins. When whey was used as the feed stock, two major pools were resolved upon adsorption and elution with the segmented gradient (Figure 5).

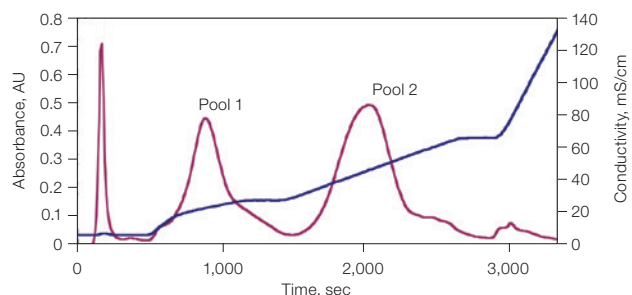


Fig. 5. AEX chromatography of 8 mg of crude whey on a 7 x 27 mm Nuvia Q Column. Fractions of 2 ml were collected; flow rate: 60 CV/hr; buffer: 0.02 M Na phosphate, pH 6.0; segmented gradient: 0–0.1 M NaCl, 0.1–0.3 M NaCl, and 0.3–0.9 M NaCl. A_{280} (—); conductivity (—).

SDS-PAGE analysis (Figure 6) of the fractions showed two peaks, one containing predominantly ALA (Pool 1) and the other containing predominantly BLG (Pool 2). The corresponding ALA and BLG standards are also shown in the gel scan. Purities of the two whey-derived proteins in each peak were estimated by densitometry to be at least 85%.

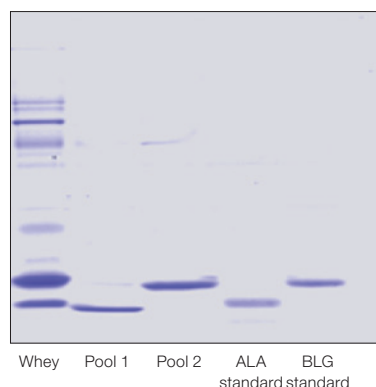


Fig. 6. SDS-PAGE analysis of whey, purified ALA (Pool 1), and purified BLG (Pool 2) against protein standards.

Summary

Nuvia S and Nuvia Q Resins, two novel high-capacity resins for IEX chromatography, allow efficient enrichment of proteins from crude mixtures. The feasibility to prepare WPI, purified ALA, and purified BLG has been demonstrated. Because of the high-throughput properties of Nuvia Resins, a rapid single-step purification of proteins from whey is possible.

Get references and complete details of this case study in Workflow for Protein Purification from Whey Using Nuvia Ion Exchange Media, bulletin 6128.



Visit [bio-rad.com/ResinSample](https://www.bio-rad.com/ResinSample) to request a sample of the resins used in this workflow.

Ion Exchange Resin Comparison Tables

Anion Exchange Resins

Property	Nuvia Q	Nuvia HP-Q	UNOsphere Q	Macro-Prep High Q	Macro-Prep High Q-3HT	Macro-Prep DEAE
Type of ion exchanger	Strong anion	Strong anion	Strong anion	Strong anion	Strong anion	Weak anion
Functional group	$-\text{N}(\text{CH}_3)_3^+$	$-\text{N}(\text{CH}_3)_3^+$	$-\text{N}(\text{CH}_3)_3^+$	$-\text{N}(\text{CH}_3)_3^+$	$-\text{N}(\text{CH}_3)_3^+$	$-\text{N}^+(\text{C}_2\text{H}_5)_2$
Base matrix composition	Polyacrylamide	Polyacrylamide	Polyacrylamide	Polymethacrylate	Polymethacrylate	Polymethacrylate
Mean particle size	85 μm	50 μm	120 μm	70 $\mu\text{m}^\dagger$	75 $\mu\text{m}^\dagger$	65 $\mu\text{m}^\dagger$
Total ionic capacity	100–170 $\mu\text{eq/ml}$	48–88 $\mu\text{eq/ml}$	120 $\mu\text{eq/ml}$	325–475 $\mu\text{eq/ml}$	400–520 $\mu\text{eq/ml}$	100–250 $\mu\text{eq/ml}$
Dynamic binding capacity	>170 mg/ml at 300 cm/hr*	>50 mg/ml at 100 cm/hr**	≥ 180 mg/ml at 150 cm/hr*** ≥ 125 mg/ml at 600 cm/hr***	≥ 37 mg BSA/ml	≥ 37 mg BSA/ml	≥ 30 mg BSA/ml
Recommended linear flow rate	50–600 cm/hr	50–300 cm/hr	50–1,200 cm/hr	50–300 cm/hr	50–300 cm/hr	50–300 cm/hr
Compression factor (settled bed volume/packed bed volume)	1.10–1.15	~1.2	1.15–1.20	NA	NA	NA
pH stability (short term)	2–14	2–10	1–14	1–10	1–10	1–10
pH stability (long term)	4–12	2–10	1–14	1–10	1–10	1–10

* 10% BT capacity determined with 5 mg/ml BSA in 20 mM Tris-HCl, pH 8.5.

** 10% BT capacity determined with 1.1 mg/ml thyroglobulin in 20 mM Tris HCl, pH 8.0.

*** 10% BT capacity determined with 2.0 mg/ml BSA in 1.1 x 10 cm column.

† Mean particle size after derivatization. Underivatized bead size is 50 μm .

Cation Exchange Resins

Property	Nuvia S	Nuvia HR-S	UNOsphere S	Macro-Prep High S	Macro-Prep CM
Type of ion exchanger	Strong cation	Strong cation	Strong cation	Strong cation	Weak cation
Functional group	$-\text{SO}_3^-$	$-\text{SO}_3^-$	$-\text{SO}_3^-$	$-\text{SO}_3^-$	$-\text{COO}^-$
Base matrix composition	Polyacrylamide	Polyacrylamide	Polyacrylamide	Polymethacrylate	Polymethacrylate
Mean particle size	85 μm	50 μm	80 μm	60 $\mu\text{m}^\dagger$	60 $\mu\text{m}^\dagger$
Total ionic capacity	90–150 $\mu\text{eq/ml}$	100–180 $\mu\text{eq/ml}$	219–319 $\mu\text{eq/ml}$	120–200 $\mu\text{eq/ml}$	170–250 $\mu\text{eq/ml}$
Dynamic binding capacity	>110 mg/ml at 300 cm/hr*	>70 mg/ml at 300 cm/hr**	60 mg/ml at 150 cm/hr*** 30 mg/ml at 1,200 cm/hr***	≥ 49 mg IgG/ml	≥ 37 mg hemoglobin/ml
Recommended linear flow rate	50–300 cm/hr	50–200 cm/hr	50–300 cm/hr	50–300 cm/hr	50–300 cm/hr
Compression factor (settled bed volume/packed bed volume)	1.15–1.20	1.20–1.30	1.15–1.20	NA	NA
pH stability (short term)	2–14	2–14	1–14	1–10	1–10
pH stability (long term)	4–13	4–13	1–14	1–10	1–10

* 10% BT capacity determined with 4.5 mg/ml hlgG in 40 mM Na acetate + 30 mM NaCl, pH 5.0.

** 10% BT capacity determined with 5.0 mg/ml hlgG in 20 mM Na acetate, pH 5.0.

*** 10% BT capacity determined with 4.5 mg/ml hlgG in 1.1 x 10 cm column.

† Mean particle size after derivatization. Underivatized bead size is 50 μm .

Mixed-Bed Resins

Property	AG 11 A8 Resin	AG 501-X8 and AG 501-X8(D)	Bio-Rex MSZ 501(D)
Type of ion exchanger	Strong anion and weak cation	Strong anion and strong cation	Strong anion and strong cation
Functional group (ligand)	$-\text{N}^+(\text{CH}_3)_3$ and $-\text{COO}^-$	$-\text{N}^+(\text{CH}_3)_3$ and $-\text{SO}_3^-$	$-\text{N}^+(\text{CH}_3)_3$ and $-\text{SO}_3^-$
Counterions	OH^- and H^+	OH^- and H^+	OH^- and H^+
Base matrix composition	Styrene-divinylbenzene	Styrene-divinylbenzene	Styrene-divinylbenzene
Particle size	180–425 μm (full particle size)	300–1,180 μm (full particle size)	600 $\pm$ 100 μm (median)
Mesh	40–80	20–50	20–50
Crosslinkage	8%	8%	8 and 10%
Approximate MW exclusion	1,000 Da	1,000 Da	1,000 Da
Minimum capacity	$-\text{N}^+(\text{CH}_3)_3$: ≥ 1.2 meq/ml $-\text{COO}^-$: ≥ 1.1 meq/ml	~1.5 meq/ml	~1.5 meq/ml
Nominal density	0.70 g/ml	0.75 g/ml	0.75 g/ml
pH stability	4–10	0–14	0–14

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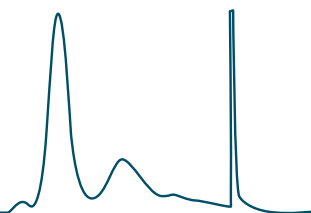
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Nuvia aPrime 4A Resin is covered by U.S. Patent Number 9,669,402 and foreign counterparts.

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